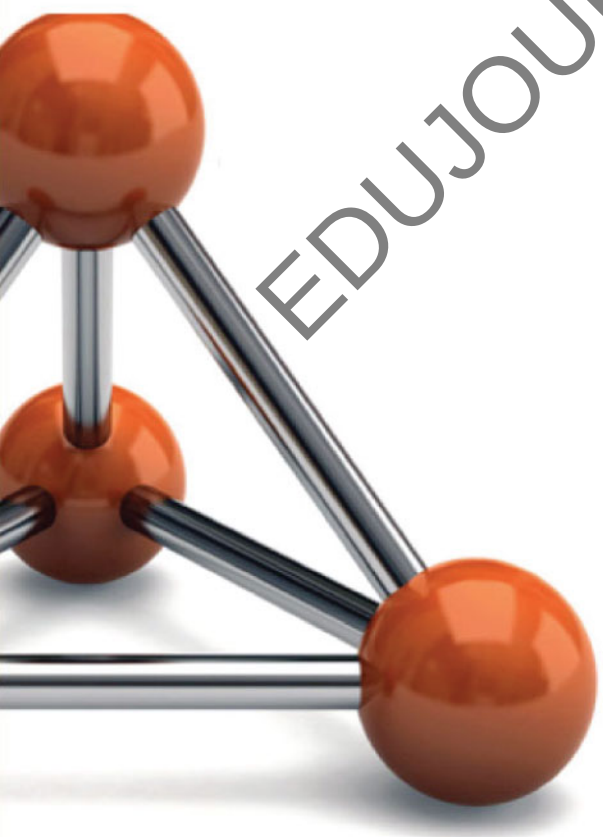
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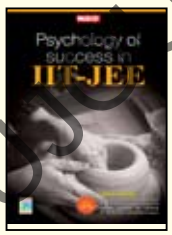
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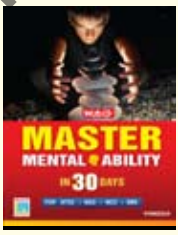
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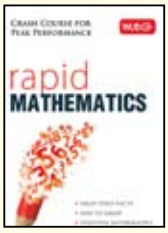
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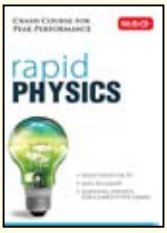
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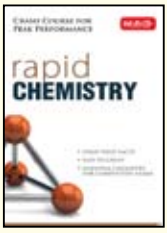
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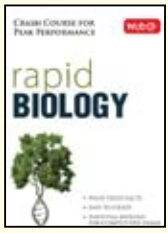
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PREFACE

"Good things come in small packages."

Rapid Chemistry (High Yield Facts Book) is designed for people who have only enough time to glance at a book-literally. Our goal is to create an effective memory aid for those who wish to review chemistry.

The book covers complete syllabus in points form. The quality of the writing leaves the reader with the potential to achieve a good understanding of a given topic within a short period of learning. It gives a concise overview of the main principles and reactions of chemistry, for students studying chemistry and related courses at undergraduate level. Based on the highly successful and student friendly "at a glance" approach, the material developed in this book has been chosen to help the students grasp the essence of chemistry, ensuring that they can confidently use that knowledge when required.

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The books has been crafted extremely well for a very specific purpose: review. A person who has been away from chemistry (but who understood it very well at the time) can use this book effectively for a rapid review of any basic topic. The book is so highly compressed that every page is like a food with a rich sauce that needs to be slowly savored and slowly digested for maximum benefit. You may only glance into the book, but you can think about the chemistry for much longer.

All the best!

Catalysed by : **Zarrin Khan**
M.Sc. Chemistry

Karabi Ghosh
M.Tech.

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1

CHAPTER

atoms, molecules & chemical arithmetic

Significant figures

The total number of digits in the number is called the number of significant figures (S.F.). It equals the number of digits written including the last one, even though its value is uncertain. The following rules should be followed in counting of (S.F.) in a given measured quantity.

(i) All digits are significant except zero at the beginning of the number.

(ii) The zeros to the right of the decimal point are significant.

(iii) The above rules purpose that the numbers are expressed in scientific notation. In this term, every number is written as $N \times 10^n$,
where N = a number with a single non-zero digit to the left of the decimal point,
n = an integer
We can write 20,000 in scientific notation as

2×10^4	2.0×10^4	2.00×10^4
having 1 S.F.	2 S.F.	3 S.F.

(iv) Zero at the end of a number and before the decimal point may or may not be significant.

Any number can be conveniently transformed to scientific notation by moving the decimal point in the number to obtain a new number, A greater than or equal to 1 and less than 10. If the decimal point is moved to the left, you multiply A by 10^n, when n equals the number of places moved. If the decimal point is moved to the right, multiply A by 10^{-n}.

Digit	S.F.	Rule
101	3	i
0.101	3	i
0.0101	3	i
101.0	4	ii
101.00	5	ii

S.F. in numerical calculations

To express the result of an experiment, we have to often add, subtract, multiply or divide the numbers obtained in different measurements.

Rule I : S.F. rule in multiplication/division

The result of multiplication and/or division may carry no more S.F. than the least precisely known quantity in the calculation. In the following multiplication, the result should be in three S.F. : $14.79 \times 12.11 \times 5.05 = 904.48985 = 904 = 9.04 \times 10^2$
(4 S.F.) (4 S.F.) (3 S.F.) (3 S.F.)

In the following division, the result should be reported in two S.F.

$$\frac{0.18\,(2\text{ S.F})}{2.487\,(4\text{ S.F})} = 0.072371 = 0.072 = 7.2 \times 10^{-2}$$
(2 S.F.) (2 S.F.)

and the following in three S.F.

$$\frac{5.28 \times 0.156 \times 3.00}{0.0428} = 57.71458 = 57.7$$

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Rule II : S.F. rule in addition/subtraction

The result of addition and/or subtraction must be expressed with the same number of decimal places as the term carrying the smallest number of decimal places.

$$22.2 + 2.22 + 0.222 = 24.642 = 24.6$$

(One least, Two, Three, Three, One)

$$5.2748 - 5.2722 = 0.0026$$

(Four, Four, Three)

Rule III : S.F. rule for each exact number

Exact numbers can be considered to have an unlimited number of S.F.

"Rounding off" the numerical results

To three S.F., we should express 15.453 as 15.5 and 14755 as 1.48×10^4. It is called "Rounding off" the result.

- If the first digit removed is less than 5, round down by dropping it and all following digits. Thus, 5.663507 becomes 5.66 when rounded off to three S.F. because first of the dropped digits (3) is less than 5.
- If the first digit removed is 6 or greater than 6 round off by adding 1 to the digit on the left.
- Thus 5.663507 becomes 5.7 when rounded off to two S.F.
- If the first digit removed is 5 and there are more non-zero digits following round up. Thus, 5.663507 becomes 5.664 when rounded off to four S.F.
- If the digit removed is 5 and there is no digit after, then add one to the preceeding digit if it is odd, otherwise write as such if it is even.

Thus, 4.7475 becomes 4.748 when rounded off to four S.F.

(odd digit before 5)

and 4.7465 becomes 4.746 when rounded off to four S.F.

(even digit before 5)

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S.I. Units (International System of Units)

The S.I. has seven base units (table 1) from whom all other units are derived. The standard prefixes, which allow us to reduce or enlarge the base units are given in table 2.

Table 1 : The seven basic units

Physical quantity	Unit	Unit symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Amount of substance	mole	mol
Electric current	ampere	A
Luminous intensity	candela	cd
Plane angle*	radian	rad
Solid angle*	steradian	sr

* These are two other fundamental quantities with dimensionless units.

Table 2 : SI Prefixes

Multiple	Prefix	Symbol	Submultiple	Prefix	Symbol
10^{24}	yotta	Y	10^{-1}	deci	d
10^{21}	zetta	Z	10^{-2}	centi	c
10^{18}	exa	E	10^{-3}	milli	m
10^{15}	peta	P	10^{-6}	micro	μ
10^{12}	tera	T	10^{-9}	nano	n
10^{9}	giga	G	10^{-12}	pico	p
10^{6}	mega	M	10^{-15}	femto	f
10^{3}	kilo	k	10^{-18}	atto	a
10^{2}	hecto	h	10^{-21}	zepto	z
10^{1}	deca	da	10^{-24}	yocto	y

Problem solving

The conversion factor method (dimensional analysis)

A number of quantities must be derived from measured value of the SI base quantities. Two sets of derived units are given, those whose names follow directly [table 3 (a)] from the base units and those that are given special names [table 3(b)].

Table 3(a) : Derived units

Physical quantity	Unit	Symbol
Area	square metre	m^2
Volume	cubic metre	m^3
Velocity	metre per second	$m\ s^{-1}$
Acceleration	metre per second square	$m\ s^{-2}$
Density	kilogram per cubic metre	$kg\ m^{-3}$
Molar mass	kilogram per mole	$kg\ mol^{-1}$
Molar volume	cubic metre per mole	$m^3\ mol^{-1}$
Molar concentration	mole per cubic metre	$mol\ m^{-3}$

Table 3(b) : Derived units

Physical quantity	Unit	Symbol	In terms of SI unit
Frequency	hertz	Hz	s^{-1}
Force	newton	N	$kg\ ms^{-2}$
Pressure	pascal	Pa	Nm^{-2}
Energy	joule	J	$kg\ m^2\ s^{-2}$
Power	watt	W	Js^{-1}, $kg\ m^2s^{-3}$
Electric charge	coulomb	C	$A\ s^{-1}$
Electric potential difference	volt	V	$J\ A^{-1}\ s^{-1}$
Electric resistance	ohm	Ω	VA^{-1}

Many of the calculations of general chemistry simply require that we convert quantities from one set of units to another. We can do this by using **Conversion Factor** (C.F.), A C.F. must always have the numerator and denominator representing equivalent quantities.

Information sought = information given × C.F.

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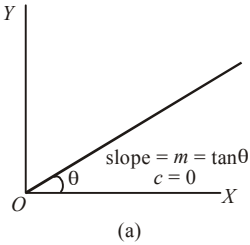
Table 4 : Conversion factors

1 m	= 39.37 inch	1 year	= 365 days
1 inch	= 2.54 cm		= 3.1536×10^7 s
1 litre	= 1000 mL = 1000 cm^3	1 curie (Ci)	= 3.7×10^{10} dps or Bq
	= 10^{-3} m^3 = 1 dm^3	1 rutherford (Ru)	= 1×10^6 dps or Bq
1 gallon	= 3.785412 L	1 debye (D)	= 1×10^{-18} esu cm
1 lb	= 453.59237 g	1 mol of a gas	= 22.4 L at STP
1 newton	= 1 kg m s^{-2}	1 mol of a substance	= N_0 molecules
1 J	= 1 Nm = 1 kg m^2 s^{-2}	1 g atm	= N_0 atoms
1 cal	= 4.184 J	t (°F)	= $\frac{9}{5}t$ (°C) + 32
	= 2.613×10^{19} eV	1 g cm^{-3}	= 1000 kg m^{-3}
1 eV	= 1.602189×10^{-19} J	equivalent wt. of an acid	= $\frac{\text{molecular weight}}{\text{basicity of the acid}}$
	= 3.827×10^{-20} cal	equivalent wt. of a base	= $\frac{\text{molecular weight}}{\text{acidity of a base}}$
1 eV/atom	= 96.485 kJ mol^{-1}	equivalent wt. in a redox reaction	= $\frac{\text{mol. wt. (stoichiometric)}}{\text{change in O.N. per atom}}$
1 amu	= $1.6605653 \times 10^{-27}$ kg		
	= 931.5016 MeV		
1 kilo watt hour (kWh)	= 3600 kJ		
1 horse power (hp)	= 746 watt		
1 joule	= 10^7 erg		
1 e.s.u	= 3.3356×10^{-10} C		
1 dyne	= 10^{-5} N		
1 atm	= 101325 Nm^{-2}		
	= 101325 Pa		
1 bar	= 1×10^5 N m^{-2}		
1 litre atm	= 101.3 J = 24.21 cal		

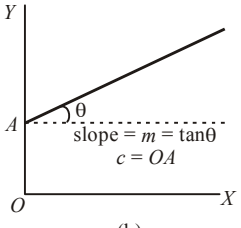
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Graphical analysis

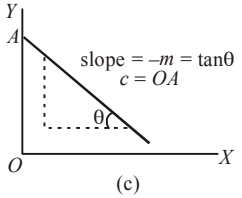
It requires less time to give more information than lengthy and tedious calculations. For this one has to compare the results with a standard graph. Generally we make a graph between y (along y-axis) and x (along x-axis) to have a straight line; its nature varies equation to equation.



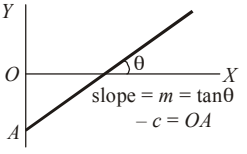
(a)



(b)



(c)



(d)

Graph	Equation	Slope	Intercept
(a)	$y = mx$	m	0
(b)	$y = mx + c$	m	$+c = OA$
(c)	$y = -mx + c$	$-m$	$+c = OA$
(d)	$y = mx - c$	m	$-c = OA$

Unit analysis

Unit of the final result is also the important part like its numerical value. If proper units are not used for the given parameters, unit of the final result can be absurd.

General data and fundamental constants

Quantity	Symbol	Commonly used value
Speed of light (in vacuum)	c	3.0×10^{8} m s^{-1}
Elementary charge	e	1.6×10^{-19} C
Avogadro number	N_0	6.02×10^{23} mol^{-1}
Faraday	$F = eN_0$	96500 C mol^{-1}
Gas constant	$R = kN_0$	8.314 mol^{-1} K^{-1}
		8.21×10^{-2} dm^3 atm mol^{-1} K^{-1}
		62.36 L torr mol^{-1} K^{-1}
Boltzmann constant	k	1.381×10^{-23} JK^{-1}
Planck constant	h	6.63×10^{-34} Js
	$\hbar = \frac{h}{2\pi}$	1.05×10^{-34} Js
Atomic mass unit	u	1.66×10^{-27} kg $atom^{-1}$
Mass of electron	m_e	9.1×10^{-31} kg
Mass of proton	m_p	1.67×10^{-27} kg
Mass of neutron	m_n	1.67×10^{-27} kg
Bohr radius	a_0	5.29×10^{-11} m
Rydberg constant (in hydrogen)	R_H	1.09677×10^{7} m^{-1}
Standard acceleration of free fall	g	9.8 m s^{-2}
Atmospheric pressure	P	1.01×10^{5} N m^{-2}
Molar volume of a gas at S.T.P.	V_m	0.0224 m^3

Laws of chemical combination

(i) *Law of Conservation of Mass.*
Total mass of reactants = Total mass of products

(ii) *Law of Constant Composition/Definite Proportions.*

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For the same compound, obtained by different methods, calculate the percentage of each element. This should be same in each case.

(iii) *Law of Multiple Proportions* - For two elements combining to form two or more compounds, calculate the weights of one element which combine with the fixed weight (1 g or 100 g) of the other. They should be in a simple whole number ratio.

(iv) *Law of Reciprocal Proportions* - Calculate the ratio of the weights of two elements *A* and *B* which combine with a fixed weight of the third element *C*. Also calculate the ratio of the weights of *A* and *B* which combine directly with each other. The two ratios should be same or simple multiple of each other.

(v) *Gay Lussac's law of gaseous volumes* - When gases react together, they always do so in volumes which bear a simple ratio to one-another and to the volumes of products, if gaseous at same temperature and pressure conditions.

Mole concept

(i) 1 Mole of atoms = Gram atomic mass (or 1 g atom)
= 6.022×10^{23} atoms

(ii) 1 Mole of molecules = Gram molecular mass (or 1 g molecule)
= 6.022×10^{23} molecules
= 22.4 L at STP.

(iii) 1 Mole of ionic compound = Gram formula mass
= 6.022×10^{23} formula units.

Calculation of molecular weight

(i) Molecular mass = 2 × Vapour density

(ii) Molecular mass = Mass of 22.4 L of vapour at STP

(iii) Rates of diffusion, $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$

Calculation of equivalent weight

(i) Eq. wt. of metal $= \frac{\text{wt. of metal}}{\text{wt. of } H_2 \text{ displaced}} \times 1.008$

or $= \frac{\text{wt. of metal}}{\text{Vol. of } H_2 \text{ in ml displaced at STP}} \times 11200$

(ii) Eq. wt. of metal $= \frac{\text{wt. of metal}}{\text{wt. of oxygen combined}} \times 8$

or $= \frac{\text{wt. of metal}}{\text{wt. of } O_2 \text{ displaced/combined in ml at STP}} \times 5600$

(iii) Eq. wt. of metal $= \frac{\text{wt. of metal}}{\text{wt. of chlorine combined}} \times 35.5$

or $= \frac{\text{wt. of metal}}{\text{Vol. of } Cl_2 \text{ combined in ml at STP}} \times 11200$

(iv) $\frac{\text{wt. of metal added to a salt solution}}{\text{wt. of metal displaced}} = \frac{\text{Eq. wt. of metal added}}{\text{Eq. wt. of metal displaced}}$

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(v) $$\frac{\text{wt. of salt } AB \text{ added to salt } CD \text{ (in solution)}}{\text{wt. of ppt. } AD \text{ formed}} = \frac{\text{Eq. wt. of radical } A + \text{Eq. wt. of radical } B}{\text{Eq. wt. of radical } A + \text{Eq. wt. of radical } D}$$

(vi) Eq. wt. = wt. deposited by 1 Faraday (96500 coulombs)

(vii) On passing the same quantity of electricity through two different electrolytic solutions,

$$\frac{\text{wt. of } X \text{ deposited}}{\text{wt. of } Y \text{ deposited}} = \frac{\text{Eq. wt. of } X}{\text{Eq. wt. of } Y}$$

(viii) Eq. wt. of an acid = wt. of the acid neutralized by 1000 cc of 1N base solution
Eq. wt. of a base = wt. of the base neutralized by 1000 cc of 1N acid solution

(ix) For an organic acid (RCOOH)

$$\frac{\text{Eq. wt. of silver salt (RCOOAg)}}{\text{Eq. wt. of silver (108)}} = \frac{\text{wt. of silver salt}}{\text{wt. of silver}}$$

Eq. wt. of acid (RCOOH) = Eq. wt. of RCOOAg – 107.

(x) For a compound (I) being converted into another compound (II) of the same metal

$$\frac{\text{wt. of compound I}}{\text{wt. of compound II}} = \frac{\text{Eq. wt. of metal} + \text{Eq. wt. of anion of compound I}}{\text{Eq. wt. of metal} + \text{Eq. wt. of anion of compound II}}$$

(xi) $$\text{Eq. wt. of an acid} = \frac{\text{Mol. wt. of the acid}}{\text{Basicity}}$$

$$\text{Eq. wt. of a base} = \frac{\text{Mol. wt. of the base}}{\text{Acidity}}$$

$$\text{Eq. wt. of a salt} = \frac{\text{Mol. wt. of the salt}}{\text{Total positive valency of the metal atoms}}$$

(xii) $$\text{Eq. wt. of oxidizing/reducing agent} = \frac{\text{Mol. wt. of the substance}}{\text{No. of electrons gained/lost by one molecule}}$$

Calculation of atomic weight

(i) Atomic wt. × specific heat = 6.4 approx
(Dulong and Petit's law for solids)

$$\therefore \quad \text{Approx. atomic weight} = \frac{6.4}{\text{Sp. heat}}, \quad \text{Valency} = \frac{\text{Approx. At. wt.}}{\text{Eq. wt.}}$$

Exact atomic weight = Eq. wt. × valency

(ii) $$\text{Valency of the metal whose chloride is volatile} = \frac{2 \times \text{V.D. of metal chloride}}{\text{Eq. wt. of metal} + 35.5}$$

Atomic weight = Eq. wt. × valency.

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End

2

CHAPTER

states of matter

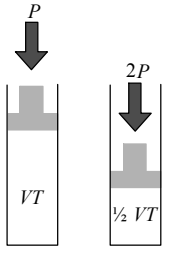
GASEOUS STATE

- Gases are easily compressed by application of pressure to a movable piston fitted in the container.
- The volume of the container is the volume of the gas sample and it is usually given in litre (L) or millilitre (mL).
- Pressure of a gas is measured with a manometer and is equal to the difference in levels of Hg in the two limbs with a closed limb manometer and is equal to atm pressure minus difference in levels in case of an open limb manometer.
- The pressure of a gas is defined as the force exerted by the impacts if its molecules per unit surface are in contact.
- The unit of pressure, millilitre of mercury is also called torr.
- For gases S.T.P. conditions are 273 K (0°C) and 1 atm-pressure.
- **Boyle's law** states that at a constant temperature (T), the pressure (P) of a given mass (or moles, n) of any gas varies inversely with the volume (V).

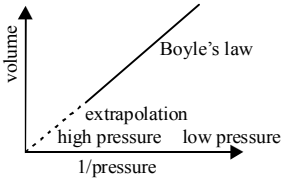
  i.e. $P \propto \frac{1}{V}$ (for given n and T)

  $PV = K$ = constant or $P_1V_1 = P_2V_2$

  In a container with a movable piston the product volume times pressure is constant. If the pressure is doubled the volume decreases to half of its original value.



- Boyle's law suggests that the graph $V = f(1/P)$ is a straight line. The extrapolation to $1/P = 0$ (infinite pressure) yields $V = 0$ which obviously is impossible. At elevated pressure Boyle's law is not valid.



- **Charles' law** : n and P are constant.

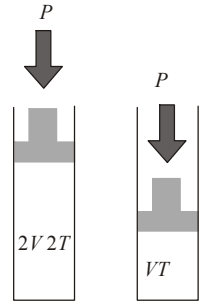
  $T_1/V_1 = T_2/V_2$

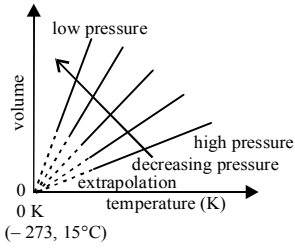
  Example : If 1 litre of gas at 300 K (27°C) and at a pressure of 1 bar is heated at constant pressure to 600 K (327°C) its volume raises to two litres.

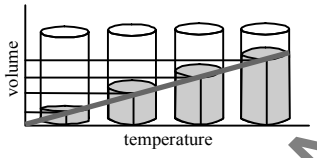
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Charles' law suggests that the volume of a gas depends linearly on its temperature. For every sample the plot $V = f(T)$ should be a straight line with the origin $V = 0$ m^3 at $T = 0$ K.

Obviously the extrapolation of Charles' law to very low temperatures does not make sense.







If a gas is heated at constant pressure it expands. Doubling the temperature (K) causes the volume to double.

- Combined gas law can be stated as for a fixed mass of gas, the volume is directly proportional to Kelvin temperature and inversely proportional to the pressure. If k be the proportionality constant,

$$V = \frac{kT}{P} \text{ } (n \text{ constant}) \quad \text{or,} \quad \frac{PV}{T} = k \text{ } (n \text{ constant}).$$

- At one condition, for a given mass of a gas P_1, V_1 and T_1 are pressure, volume and temperature and at some other condition P_2, V_2 and T_2 are new pressure, volume and temperature, then

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

- Gay Lussac's law states that the pressure of a given mass of any gas is directly proportional to the absolute temperature at constant volume. $P \propto T$ (for constant n and V)

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

- Gay Lussac's law of combined volume states that when measured at same temperature and pressure, the ratios of volumes of the gases that combined, and of gases that were products (in a chemical reaction), were always some whole numbers.

$$\underset{2 \text{ volumes}}{2H_{2\,(g)}} + \underset{1 \text{ volume}}{O_{2\,(g)}} \rightarrow \underset{2 \text{ volumes } (2:1:2)}{2H_2O_{\,(g)}}$$

- The temperature at which solid ice, liquid water and water vapour i.e. all the three states of the substance exist together is called **triple point**.

- **Avogadro's law :** P and T are constants.

$$n_1V_1 = n_2V_2$$

At constant temperature and pressure the number of moles of a gas in a container is proportional to the container's volume. Or, at a given temperature and pressure equal volumes contain an equal number of moles, independent of the kind of gas.

Example : At a given temperature and pressure 9 atoms and 9 tri-atomic molecules occupy the same volume.

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At a pressure of 1 bar and 0°C, all gases (to the point where they are "ideal gases" under these conditions) have the same volume of 22.4 litre. 22.4 litre is the molar volume of gases under the so called "STP" (standard temperature pressure) conditions.

- **Rule of thumb** : At 25°C and 1 bar one mol of gas has a volume of 25 litre.
- 1 mole contains 6.023×10^{23} molecules (a number called as Avogadro's number).
- Universal gas law or ideal gas law may be stated as the volume of a given amount of gas is directly proportional to the number of moles of gas, directly proportional to the temperature and inversely proportional to the pressure.

$$V = R\frac{nT}{P}$$

This is called **ideal gas equation** and constant R is called **gas constant**.

- In the international system (SI) the unit of pressure is the Pascal (1 Pa = 1 $N.m^{-2}$) and the unit of volume is the cubic meter, m^3. In the SI system the value of the gas constant R is 8.3145 J mol^{-1} K^{-1}. Sometimes it is more convenient to use units of everyday life: bar for pressure (1 bar ~ 1 atmosphere) and litre for volume (dm^3).
  R = 0.083145 bar lit. mol^{-1} K^{-1}
  Explanation : 1 J = 1 Nm
  1 Pa = 1 Nm^{-2} = 1 J m^{-3}
  1 bar = 10^5 J m^{-3} = 10^2 J dm^{-3}
  Therefore, 1 J = 10^{-2} bar dm^3.
- Gases which obey the ideal gas equation are called as **ideal gases**. Ideal gases obey the gas equation at all temperatures and pressures.
- Ideal gases do not show any cooling or heating effect on adiabatic expansion because there are no intermolecular forces of attraction present in them.
- The gases which deviate from gas equation are called as real gases. These gases obey gas laws only at low pressure and high temperature.
- **Partial pressure :** Actually the pressure of a gas is due to the elastic shocks of the molecules with the walls of the container.
- In a gas mixture, the partial pressure exerted by one component is proportional to its concentration.

$$P_{tot} = P_1 + P_2 + P_3 + ... = \left(\frac{n_1}{V_{tot}} + \frac{n_2}{V_{tot}} + \frac{n_3}{V_{tot}} + ...\right) R \cdot T$$

The total pressure is the sum of the partial pressures.
This is called **Dalton's law**.

- Dalton's law of partial pressure is applicable only to non-reacting gases.
- The **partial pressure** is defined as the pressure of a gas would exert if it was alone in the container at the same temperature and pressure conditions.
  Example: Air consists of 78 % N_2, 21 % O_2, 1 % Ar and 0.03 % CO_2 (percent of volume = percent of moles). When the pressure of the air is 1 bar, the partial pressures are: $p(N_2)$ = 0.78 bar, $p(O_2)$ = 0.21 bar, p(Ar) = 0.01 bar and $p(CO_2)$ = 0.0003 bar.
- Process of mixing of gases by random motion of the molecules is called **diffusion**.

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- **Graham's law of diffusion** states that under the same conditions of temperature and pressure, the rates of diffusion of different gases are inversely proportional to the square roots of their molecular masses. Mathematically the law can be expressed as $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$ where r_1 and r_2 are the rates of diffusion of gases 1 and 2, while M_1 and M_2 are their molecular masses.
- A gas confined to a container at high pressure than the surrounding atmosphere will escape from a small hole which is opened in the container until the pressure outside and inside have been equalised. This process is called **effusion**.
- The rate of diffusion of a gas is also proportional to the pressure of gas (or number of molecules) at a given temperature. In that case, the rate of diffusion is given as
$$r \propto \frac{P}{\sqrt{d}}$$
- If two gases 1 and 2 at different pressures P_1 and P_2 are allowed to effuse through a small hole in a container then the ratio of rates of diffusion of two gases is given by
$$\frac{r_1}{r_2} = \frac{P_1}{P_2}\sqrt{\frac{d_2}{d_1}} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}}$$
- When a real gas is allowed to expand adiabatically through a porous plug or a fine hole into a region of low pressure, it is accompanied by cooling (except for hydrogen and helium which get warmed up). This effect is known as **Joule-Thomson effect**.
- **Kinetic theory of gases** : The macroscopic behaviour of gases can be explained by a model containing three hypotheses.
  (i) A gas is an ensemble of particles in continuous, fast random motion, moving in straight lines until they collide.
  (ii) The particles are infinitely small and (on the average) far from each other. (volume of the particles <<< volume of the gas).
  (iii) The particles do not influence one another except during collisions. The collisions of the particles with each other and with the wall of the container are elastic, *i.e.* the kinetic energy of the particles is maintained (there is no transformation of kinetic energy into heat of friction).

  The pressure of a gas exerted on the walls of the container is caused by the collision of the gas particles with the wall.
- $PV = (1/3)\,mNu^2$ is the fundamental equation of the kinetic molecular theory of gases. It is called **kinetic gas equation**.
- The average translational kinetic energy of one molecule of an ideal gas will be given by
$$E_t = \frac{K.E.}{N_A} = \frac{(3/2)RT}{N_A} = \frac{3}{2}kT$$ where (R/N_A) is **Boltzmann constant**.
- At absolute zero (*i.e.* $T = 0$), kinetic energy is zero. In other words, thermal motion ceases completely at absolute zero.
- In 1860, James Clark Maxwell derived the following equation for the distribution of molecular velocities.

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$$\frac{dN_C}{N} = 4\pi\left(\frac{M}{2\pi RT}\right)^{3/2} e^{-\frac{MC^2}{2RT}} C^2 dC$$

dN_C = number of molecules having velocities between C and $(C + dC)$, N = total number of molecules, M = molecular mass, T = temperature on absolute scale (K).
The above relation is called **Maxwell's law of distribution of molecular velocities**.

- The **average velocity** of a gas is given by the arithmetic mean of the different velocities possessed by the molecules of the gas at a given temperature.

  Average velocity or $C_{av} = \frac{C_1 + C_2 + C_3 + ... + C_N}{N}$

  Also $C_{av} = \sqrt{\frac{8RT}{\pi M}}$ (M is in kg)

- The **root mean square velocity** is defined as the square root of the mean of the squares of different velocities possessed by molecules of a gas at a given temperature.

  $$C_{rms} = \sqrt{\frac{C_1^2 + C_2^2 + C_3^2 + ...}{N}}, \quad C_{rms} = \sqrt{\frac{3RT}{M}}$$

- The **most probable velocity** is defined as the velocity possessed by maximum number of molecules of a gas at a given temperature.

  $$v_{mp} = \sqrt{\frac{2RT}{M}}$$

- Relation between average velocity, RMS velocity and most probable velocity is given as
  Average velocity = 0.9213 × RMS velocity
  Most probable velocity = 0.8165 × RMS velocity

- The variation of volume V with temperature T, keeping pressure P constant is called the **coefficient of thermal expansion** or the **coefficient of isobaric expansion** or simply **expansivity, α** of the fluid. Thus, $\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P$.

- The variation of V with P, keeping T constant is called **coefficient of isothermal compressibility** or simply **compressibility, β** of the fluid. Thus, $\beta = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T$

- The distance between the centres of two molecules at the point of their closest approach is known as **collision diameter** and it is represented by σ.

- **Collision number** gives the number of collisions suffered by a single molecule per unit time per unit volume of the gas. Thus $N_C = \sqrt{2}\pi\sigma^2 \langle C \rangle n$.

- The mean distance travelled by a molecule between two successive collisions is called the **mean free path**. It is denoted by λ.

  $$\lambda = \eta\sqrt{\frac{3}{Pd}}$$

  where P = pressure of the gas
  d = density of the gas and
  η = coefficient of viscosity of the gas.

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- The mean free path is directly proportional to the absolute temperature and inversely proportional to the pressure of a gas at constant temperature.
- **Collision frequency** is the number of molecular collisions occurring per unit time per unit volume of the gas.
- The collision frequency of a gas increases with increase in temperature, molecular size and the number of molecules per c.c.
- Collision frequency is given by $Z = \frac{\pi C \sigma^2 n^2}{\sqrt{2}}$
- Collision frequency is directly proportional to the square root of absolute temperature and also directly proportional to the square of the pressure of the gas.
- The process of separation of two gases on the basis of their different rates of diffusion due to difference in their densities is called **atmolysis**. It has been applied with success for the separation of isotopes and other gaseous mixtures.
- **Specific heat** of a substance is defined as the amount of heat required to raise the temperature of 1 g of the substance through 1°C. It is represented in calories.
- **Molar heat** of a substance is defined as the quantity of heat required to raise the temperature of one mole of the substance through 1°C. Evidently,
  Molar heat = specific heat × molar mass of the substance.
- One **calorie** is defined as the amount of heat required to raise the temperature of 1 g of water through 1°C.
- **Specific heat at constant volume** is the amount of heat required to raise the temperature of one gas through 1°C while the volume is kept constant and the pressure is allowed to increase. It is denoted by the symbol C_V.
- **Specific heat at constant pressure** is defined as the amount of heat required to raise the temperature of one gram of gas through 1°C, the pressure remaining constant while the volume is allowed to increase. It is written as C_P.
- The pressure exerted by the water vapour at a particular temperature is called **aqueous tension** at that temperature. It depends only on temperature.
- **Values of molar heat capacities (in S.I. units).**

Gas (mol^{-1})	C_V (JK^{-1})	C_P	$C_P - C_V = R$	$C_P/C_V = \gamma$	Atomicity
Helium	12.6	20.9	8.3	1.659	1
Argon	12.5	20.8	8.3	1.664	1
Mercury vapour	12.5	20.8	8.3	1.664	1
Hydrogen	20.4	28.8	8.4	1.412	2
Oxygen	20.9	29.3	8.4	1.402	2
CO	21.0	29.3	8.3	1.395	2
CO_2	28.7	37.2	8.5	1.294	3
Ethylene	34.3	42.8	8.5	1.247	*n*

- The **degrees of freedom** of a molecule are defined as the independent number of parameters required to describe the state of the molecule completely.

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- The **centre of gravity** of any molecule has three translational degrees of freedom.
- All linear molecules such as CO_2 and C_2H_2 have two rotational degrees of freedom because their rotational motion is similar to that of a diatomic molecule.
- Non-linear molecules such as H_2O, H_2S, CH_4, C_6H_6 can undergo rotation about the three Cartesian axes so that they have three rotational degrees of freedom.
- There are $3n - 6$ vibrational degrees of freedom for a non-linear molecule and $3n - 5$ vibrational degrees of freedom for a linear molecule.
- A normal mode of vibration is defined as a molecular motion in which all the atoms in the molecule vibrate with the same frequency and all the atoms pass through their equilibrium positions simultaneously.
- CO_2 molecule (linear) has 4 vibrational degrees of freedom.
- In the solid state, the solids possess only vibrational degrees of freedom. The translational and rotational motion in solids are converted into vibrational motion where the atoms in the lattice vibrate about their equilibrium positions.
- According to the Born-Oppenheimer approximation, the total energy of a molecule is given by

  $$E_{total} = E_{tr} + E_{rot} + E_{vib} + E_{el}$$

  where E_{tr} is the translational energy, E_{rot} is the rotational energy, E_{vib} is the vibrational energy and E_{el} is the electronic energy.
- The variation of pressure with altitude is given by the barometric formula

  $$P = P_0\, e^{(-Mmgx/RT)}$$
- van der Waal's equation for n moles of a gas is given by

  $$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

  where n = number of moles, a and b are new empirical constants varying for different gases.
- The **critical temperature**, T_C of a gas may be defined as that temperature above which it cannot be liquefied no matter how great the pressure applied.
- The **critical pressure**, P_C is the minimum pressure required to liquefy the gas at its critical temperature.
- The **critical volume**, V_C is the volume occupied by one mole of the gas at the critical temperature and critical volume.
- The numerical values of critical constant derived from van der Waal's equation are

  $$T_C = \frac{8a}{27Rb};\ P_C = \frac{a}{27b^2};\ V_C = 3b$$
- The temperature at which a real gas behaves like an ideal gas over an appreciable pressure range is known as **Boyle's temperature (T_B)**. Boyle's temperature of a gas is always higher than its critical temperature (T_C).

  $$T_B = \frac{a}{Rb}.$$

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- The *P-V* curves of a gas at constant temperature are called **isotherms** or isothermals.
- When a gas under high pressure is permitted to expand into a region of low pressure, it suffers a fall in temperature. This phenomenon is known as Joule Thomson effect.
- The constant *a* in van der Waal's equation is a measure of intermolecular forces of attraction. Greater the value of *a*, more easily the gas can be liquefied.
- The constant *b* in van der Waal's equation is related to the volume of the molecules and takes into account the fact that the space actually occupied by the molecules themselves is unavailable for the molecules to move in and hence must be subtracted from the total volume of the gas (*V*). *b* is also called co-volume or excluded volume.
- At low pressures, van der Waal's equation is written as

$$\left(P + \frac{a}{V_m^2}\right)V_m = RT \qquad \text{or,} \qquad Z = \frac{PV_m}{RT} = 1 - \frac{a}{V_m RT}$$

  where *Z* is known as the **compressibility factor**.
- At higher pressures, the gas equation is written as

$$P(V_m - b) = RT \qquad \text{or,} \qquad Z = \frac{PV_m}{RT} = 1 + \frac{Pb}{RT}$$

IMPORTANT FACTS ABOUT GAS CONSTANT *R*

1. In ideal gas equation, $PV = nRT$, *R* is known as **universal gas constant**.
2. The value of *R* depends on the units of measurement of *P*, *V* and *T*.
3. *R* has the dimensions of energy.
4. For one mole of an ideal gas $PV = RT$ (since $n = 1$).
5. Only very few gases such as H_2, He and N_2 show some ideal behaviour.
6. Real gases show ideal behaviour at low pressure and high temperatures.
7. Gas constant for single molecule is known as Boltzmann constant *k*. ($R/N = k$).
   $k = 1.38 \times 10^{-16}$ erg/degree-molecule
   $= 1.38 \times 10^{-23}$ Joule/degree-molecule.
8. $R = 0.0821$ lit-atom/deg.mole $= 8.314$ Joule/degree-mole
   $= 1.987$ cal/degree-mole $= 82.1$ mL-atm/degree-mole
   $= 8.314 \times 10^7$ erg/degree-mole $= 62.4$ lit-mm/degree-mole
   $= 6.24 \times 10^4$ mL-mm/degree-mole $= 0.002$ k.cal/degree-mole
   $= 5.28 \times 10^{19}$ eV/degree-mole.

LIQUID STATE

- Liquid state is intermediate between gaseous and solid states. They possess fluidity like gases but incompressibility like solids.
- In terms of kinetic molecular model, the nature of the liquid state is described as follows:

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(*i*) Liquids are composed of molecules.

(*ii*) The molecules of liquids are held together by appreciable intermolecular forces.

(*iii*) Due to weak intermolecular forces, the molecules are in constant random motion.

(*iv*) The average kinetic energy of molecules in a given sample is proportional to the absolute temperature.

Properties of Liquids

- **Shape :** Liquids have no shape of their own but assume the shape of the container in which they are kept.
- **Volume :** In liquids the intermolecular forces are strong and therefore, they do not expand to occupy all the sapce available (as gases do).
- **Density :** The higher densitites of liquids than gases are due to the fact that molecules of liquids are more closely packed than gases. In general, the density of the liquids decreases with increase in temperature.
- **Compressibility :** The molecules in a liquid are held in such close contact by their mutual attractive forces that the volume of any liquid decreases very little with increased pressure. Thus, liquids are relatively incompressible compared to gases.
- **Diffusion :** The diffusion of liquids is defined as the process of intermixing of the molecules of two or more liquids to form a homogenous mixture solution. However, the rate of diffusion of a liquid can be increased by raising the temperature the temperature which increases the kinetic energy of the molecules.
- **Evaporation :** The process of change of liquid into vapour state below its boiling point is termed evaporation.

  The liquids having low intermolecular forces evaporate faster incomparision to the liquids having high intermolecular forces.

  Rate of evaporation $\propto$ surface area

  Rate of evaporation $\propto$ temperature
- **Heat of vaporisation :** The amount of heat required to evaporate 1 mole of a given liquid at a constant temperature is known as the heat of evaporation or heat of vaporisation.

  The value of heat of vaporisation generally decreases with increase in temperature. It becomes zero at the critical temperature.
- **Vapour Pressure :** "The pressure exerted by the vapour in equilibrium with liquid, at a given temperature, is called the vapour pressure."
- **Boiling point** is the temperature at which the vapour pressure of a liquid becomes equal to the atmospheric pressure.
- **Freezing point :** The temperature at which the vapour pressures of solid and liquid forms of a substance become equal is termed as freezing point.
- **Surface tension** may be defined as force per unit length acting perpendicular to the tangential line on the surface.

  The units of surface tension are force per unit length i.e., dynes cm^{-1}. In S.I. the unit is Nm^{-1}. For example, the surface tension of water is 72.75×10^{-3} Nm^{-1} and that of mercury is 47.5×10^{-2} Nm^{-1}.

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- Surface tension decreases with rise in temperature.

  I. Let γ_1 and d_1 be the surface tension and density of water and γ_2 and d_2 be surface tension and density of the liquid whose surface tension is to be determined.

$$\frac{\gamma_1}{\gamma_2} = \frac{n_1 d_2}{n_2 d_1}$$

- **Viscosity :** The internal resistance to flow in liquids which one layer offers to another layer trying to pass over it is called viscosity.

  Force of friction '*f*' between two cylindrical layers each having area '*A*' sq. cm separated by a distance '*x*' cm and having a velocity difference '*v*' cm/sec is given by

$$f \propto A\frac{v}{x}; \qquad f = \eta A\frac{v}{x}$$

- Viscosity is generally determined by Ostwald's method.

$$\frac{\eta}{\eta_w} = \frac{d \times t}{d_w \times t_w}$$

where η_w and η = coefficient of viscosity of water and liquid respectively, d = density of liquid d_w = density of water ; t = time of flow of liquid ; t_w = time of flow of water.

SOLID STATE

- Solids are characterized by their high density, low compressibility, definite shape, considerable mechanical strength and rigidity. These properties are due to the existence of very strong forces of attraction amongst the molecules, atoms or ions of the solids.
- The solid state represents the physical state of matter in which the constituent molecules, atoms or ions have no translatory motion although they vibrate about the fixed position, that they occupy in a crystal lattice.

  Crystalline solids have definite shape and volume. They are rigid, incompressible, *anisotropic*, *i.e.* their mechanical, electrical properties depend on the direction along which these are measured. Amorphous solids (like plastic, glass) however, are *isotropic*.

- **Crystalline solids**

– have definite geometrical shape, sharp melting point.
– are anisotropic
– have definite order close-packed structure.
– can be ionic, covalent, molecular and metallic.

- **Amorphous solids**

– do not have definite shape (glass, plastic, rubber, wood) and melting point
– are isotropic
– have short-range order packing.

- **Intermolecular forces**

  Solids have been classified based on type of intermolecular forces existing in them.

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– **Dispersion forces or London forces**
When distribution of electrons around the nucleus is not symmetrical then there is formation of instantaneous electric dipole. Field produced due to this distorts the electron distribution in the neighbouring atom or molecule so that it acquires a dipole moment itself. The two dipole will attract and this makes the basis of London forces or dispersion forces. These forces are attractive in nature and the interaction energy due to this is proportional to $(1/r^6)$. Thus, these forces are important at short distances. This force also depends on the polarisability of the molecule.

– **Dipole-dipole forces**
These type of forces occur between molecules having permanent electric dipole. In HCl, bond between H and Cl is formed by sharing of electrons, but shared electron pair is near to chlorine because of higher electronegativity compared to H-atom. Thus bond formed is said to be polar and a partial charge is developed on both atoms.

– **Dipole-induced dipole forces**
Attractive forces are operative not only between the two molecules with permanent dipoles but also between a molecule having a dipole moment and a molecule without any dipole moment like CH_4. As the size of the atom increases the influence of the electric dipole on it also increases. The electron cloud of the molecule is deformed in the electric field of the permanent dipole. This causes a shift in the centre of gravity of the negative charge relative to the nuclear charge and leads to the formation of an induced dipole moment.

• **Crystal**
Crystal is made of a number of unit cells, each possessing a definite geometry and bound by plane faces. *Face* is a planar surface arranged on a definite plane which binds the crystals. Faces in a crystal may be like or unlike, all faces together constitute a *form*.
Edges are the intersection of two adjacent faces and the angle between the normals to the two intersecting faces is called *interfacial angle*.

• **Law of crystallography**
– *Law of constancy of interfacial angles* - A crystal may have different shapes according to the number and size of the faces, but the angle of intersection of two adjacent faces is always constant.
– *Hauy's law of rationality of indices* - The intercepts of any face of a crystal on a suitable axes can be expressed by small multiples of three unit distances a, b, c or their integral multiples (*m, n, p*).
– *Law of symmetry* - All crystals of the same substance possess same elements of symmetry. A crystal can have three types of symmetry.
– *Plane of symmetry* is present in a crystal when an imaginary plane passing through its centre gives two parts which are mirror images.

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– *Axes of symmetry* is an imaginary line passing through the crystal such that when the crystal is rotated about it, it gives the same appearance more than once in a complete revolution.
– If similar appearance occurs twice, thrice and so on the axis is respectively called a diad, triad, tetrad, hexad, etc.
– *Centre of symmetry* is an imaginary point within a crystal such that any line passing through it intersects the surface of the crystal at equal distances on each side of the point.

Nature of solids

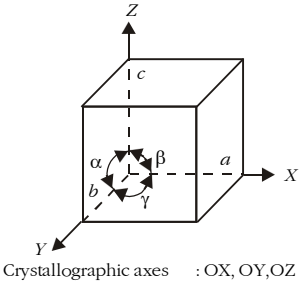
Types of solid	Constituents	Bonding	Examples	Physical nature	Melting point	Electrical conductivity
Ionic	Ions	Coulombic	NaCl, KCl, CaO, MgO	Hard but brittle	High $\simeq$ 1000 K	Conductor (in molten state and in aq. solution)
Covalent	Atoms	Electron sharing	SiO_2 (quartz), SiC C (diamond), C (graphite)	Hard	Very high $\simeq$ 4000 K	Insulator
Molecular	Simple covalent molecules	Molecular interactions (intermolecular forces),	I_2, S_8, P_4, CO_2, CCl_4	Soft	Low ($\simeq$ 300 K to 600 K)	Insulator
		Hydrogen bonding	starch, sucrose, water, ice	Soft	Low ($\simeq$ 273 K to 400 K)	Insulator
Metallic	Positive ions and electrons	Metallic	sodium, magnesium, metals and alloys	Ductile malleable	High ($\simeq$ 800 K to 1000 K)	Conductor

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- **Unit cell**

  Unit Cell is the smallest repeating unit in a three dimensional space or crystal lattice.

- **Characteristics of a unit cell**
  (a) Primitives or the three sides, *a, b, c* of a unit cell are also known as characteristic intercepts.
  (b) Crystallographic axes are lines drawn parallel to lines of intersection of any three faces of the unit cell which are not in the same plane.
  (c) Interfacial angles α, β, γ are made between the three crystallographic axes.



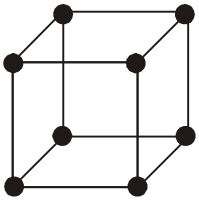
Crystallographic axes : OX, OY,OZ
Interfacial angles : α, β, γ
Primitives : *a, b, c*

- **Space lattice or Crystal lattice** is a three dimensional arrangement of points showing the particles (atoms,

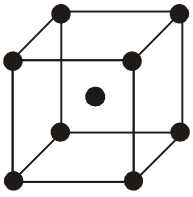
molecules, or ions) in a definite orderly distribution.

Bravais also showed that there are basically four types of unit cells depending on the manner in which they are arranged in a given shape.

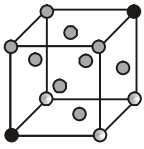
These are: Primitive, Body Centred, Face Centred and End Centred.



Primitive cubic unit cell



Body centred cubic unit cell



Face centred cubic unit cell

- **Atoms per unit cell**

There are three kinds of lattice points in a unit cell, points at the corners and face centres are shared by other cells, whereas point within the cell is not shared.

Based on the location of points and their contribution to each cell we can calculate the number of atoms per unit cell.

A point that is at the corners of a unit cell is shared by eight unit cells and contributes 1/8 to each cell.

A point on an edge is shared by four unit cells and contributes 1/5 to each cell.

A face centred point is shared between two unit cells only and contributes 1/2 to each.

A point in the centre of the body is not shared and contributes wholly to the unit cell.

- **Co-ordination number**

It is number of nearest neighbours or spheres in contact with the sphere under consideration.

Co-ordination number of a crystal depends upon its structure.

- simple cubic structure has CN = 6
- face centred cubic structure (fcc) has CN = 12
- body centred cubic structure (bcc) has CN = 8

Density of lattice matter is the ratio of mass per unit cell to the total volume of unit cell:

$$D = \frac{\text{mass per unit cell}}{\text{volume of unit cell}} = \frac{n \times \text{at. wt.}}{\text{Av. no.} \times \text{Volume of unit cell}}$$

n = no. of atoms per unit cell, a^3 = volume for cubic crystal systems

Packing-fraction or density of packing is the ratio of volumes occupied by atoms in a unit cell (ν) to the total volume of the unit cell (V).

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Packing fraction = v/V.

The density of packing can show how closely the atoms are packed in a unit cell. Calculations reveal that close packing in cubic crystal system follows the order: fcc > bcc > sc.

- **Dimensions of unit cells** are actually the inter atomic distances in a unit cell and can be calculated for different crystal structures if its density, molecular weight and Avagadro's number are known.

  Simple cubic structure, $a = \sqrt[3]{\frac{1 \times A}{\text{Av. No} \times \rho}}$

  Face centred cubic structure, $a = \sqrt[3]{\frac{4 \times A}{\text{Av. No} \times \rho}}$

  Body centred cubic structure, $a = \sqrt[3]{\frac{2 \times A}{\text{Av. No} \times \rho}}$

  A = at. weight, ρ = density

- Density of a unit cell $= \frac{z \times M}{N_0 \times V} = \frac{n \times M}{N_0 \times a^3}$ (for a cube)

  where z is the number of atoms in a unit cell and V is the volume of unit cell. For a cube $V = a^3$ where a is the edge length of the cubic unit cell.

- Packing fraction or density of packing

$$= \frac{\text{Volume occupied by atoms in unit cell}}{\text{Total volume of the unit cell}} = \frac{v}{V}$$

For simple cubic structure

V = volume of the unit cell is a^3 and since one atom is present in a unit cell,

$$\therefore \text{Volume } V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi \left(\frac{a}{2}\right)^3 = \frac{\pi a^3}{6}$$

$$\therefore \text{Packing fraction} = \frac{v}{V} = \frac{\pi a^3/6}{a^3} = \frac{\pi a^3}{6} \cdot \frac{1}{a^3} = \frac{\pi}{6} = 0.52$$

$$\therefore \quad \text{Percentage efficiency} = 52\,\%$$

For *fcc* structure, there are four atoms present in a unit cell, therefore total volume is

$$V = 4 \times \left(\frac{4}{3}\pi r^3\right)$$

$$\text{for } fcc \quad r = \frac{a}{2\sqrt{2}}$$

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$$\therefore V = \frac{16}{3}\pi \left(\frac{a}{2\sqrt{2}}\right)^3 \Rightarrow V = \frac{\pi}{3\sqrt{2}}a^3$$

$$\text{Packing fraction} = \frac{\pi \; a^3}{3\sqrt{2}a^3} \quad [\because V = a^3]$$

$$= \frac{\pi}{3\sqrt{2}} = 0.74$$

$\therefore$ Percentage efficiency is 74% *i.e.* 74% of the unit cell is occupied by atoms and 26% is empty.

In bcc structure, there are two atoms present in unit cell, therefore, their volume is

$$V = 2 \times \frac{4}{3}\pi \; r^3$$

$$\text{for } bcc \;\; r = \frac{\sqrt{3}}{4}a$$

$$\therefore \;\; V = 2 \times \frac{4}{3}\pi\left(\frac{\sqrt{3}}{4}a\right)^3 = \frac{\sqrt{3}}{8}\pi a^3$$

Since the volume of unit cell, $V = a^3$

$$\therefore \;\; \text{packing fraction} = \frac{\sqrt{3}}{8}\frac{\pi \; a^3}{a^3} = \frac{\sqrt{3}}{8}\pi = 0.68$$

i.e. 68% of the unit cell is occupied by atoms and 32% is empty.

- **Bragg's equation**

Bragg's Equation interprets the diffraction pattern resulting from scattering of X-rays by regular arrangement of atoms or ions:

$n\lambda = 2d \sin\theta$

n = 1, 2, 3 ... (diffraction order)

λ = wavelength of X-rays incident on crystal

d = distance between atomic planes

θ = angle at which interference occurs

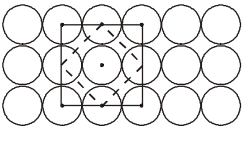
Radius Ratio is the ratio of the radii of positive and negative ions in a crystal:

$$\text{Radius ratio} = \frac{\text{radius of cation}}{\text{radius of anion}} = \frac{r^+}{r^-}$$

- **Packing of constituents in crystals**

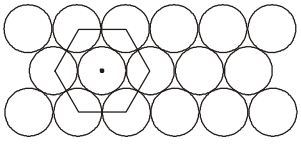
Constituents of a crystal have a tendency to pack as closely as possible to have maximum density and stability.

- **Square Close Packing** system has spheres of adjacent rows one over the other, showing a vertical as well as horizontal alignment to form a square.

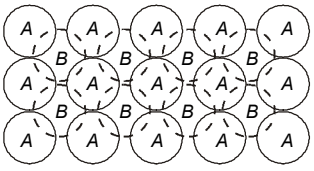


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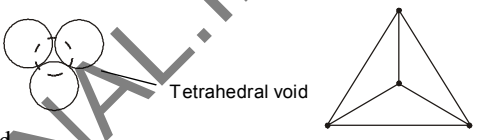
- **Hexagonal Close Packing** system has spheres of energy second row placed in the depression between spheres of the first row. Subsequently, spheres of third row are vertically aligned with those of the first row.



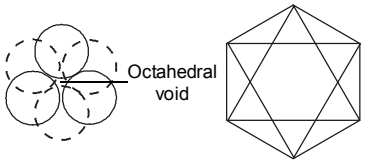
- **Body Centred Cubic** arrangement (bcc) is not the closest system, it is obtained when the spheres of the first row (layer) are slightly open and not in contact with each other.



- **Void space or holes**
  In a unit cell some empty space exists between spheres, this is called void space or hole, also called interstitial void, or interstices.
- **Tetrahedral Voids** are holes or interstices surrounded by four spheres present at the corners of a tetrahedron.
  CN of tetrahedral void is 4



- **Octahedral Voids** are holes surrounded by six spheres located on a regular tetrahedron.
  CN of octahedral void is 6.



Structures of Simple Ionic Compounds

- **AB Type Structures**

  The three common types of structure sodium chloride, caesium chloride and zinc sulphide are described below

  1. **Sodium chloride or rock salt structure :** NaCl has *fcc* structure with each sodium atom surrounded by 6 Cl^- ions and vice versa. Therfore, there will be one Na^+ ion for every Cl^- ion. Thus, the ratio of Na^+ and Cl^- ions in this structure is 1 : 1.

     In this octahedral arrangement, coordination number of both Na^+ and Cl^- is 6.
  2. **Caesium chloride structure :** It is a body centred cubic structure. Cs^+ ion is surrounded by 8 Cl^- ions which are disposed towards the corners of a cube. Cl^- ion is also surrounded by 8 Cs^+ ions. Thus the coordination number of Cs^+ and Cl^- is 8 : 8.
  3. **Zinc sulphide or sphalerite structure :** In this face centered cubic lattice each zinc ion is surrounded by four sulphide ions. Similarly, each S^{2-} ion is surrounded by four Zn^{2+} ions. Therefore, there is one zinc ion for every sulphide ion. Thus, the compound has the formula ZnS.

- **A_2B and AB_2 Type Structures**

  The common example of AB_2 type structure is calcium fluoride (CaF_2) called fluorite structure and of A_2B type structure is sodium oxide (Na_2O) called antifluorite structure.

  In CaF_2 the Ca^{2+} ions are located at face centred cubic lattice points and therefore

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have cubic closed packed arrangement. The F^- ions occupy all the eight tetrahedral voids. In this structure, each F^- ion is surrounded by four Ca^{2+} ions, while Ca^{2+} ion is surrounded by eight F^- ions. Thus, the coordination number of Ca^{2+} and F^- ions are 8 : 4.

In the antifluorite structure *i.e.* in A_2B structure the position of cations and anions are reversed. In this structure, the small cation (Na^+) occupy the positions of fluoride ions and the larger anions (O^{2-}) occupy the positions of the calcium ions in the fluorite structure.

The coordination number of Na^+ ions is 4 and that of O^{2-} ions is 8. Thus Na_2O has 4 : 8 coordination. There are several oxides and sulphides which have antifluorite structure such as Li_2O, K_2O, Rb_2O and Rb_2S.

- **Crystal defects**

– In a crystalline solid the atoms, ions or molecules are arranged in a definite repeating pattern, but some defects may occur in the pattern. Deviations from perfect arrangement may occur due to temperature changes or presence of additional particles.

- Less commonly, some atoms or ions in a crystal may occupy positions, called interstitial sites, that are located between the regular positions for atoms.

Nature of defects in crystals

Defect	Nature of defect
1. Schottky	Atom or ion missing from the lattice point and thus giving a vacancy. Density of the crystal is lowered.
2. Interstitial	Atom or ion in a vacant void, also called hole, (or interstitial site).
3. Frenkel	This is a hybrid type of defect produced from the combination of (1) and (2). Atom or ion at the lattice point displaced to an interstitial site creating a vacancy.
4. F-centre	Electron trapped in an anionic vacancy is called F-centre. If the concentration of F-centres is high, colourless crystals (like KCl, LiCl, NaCl) develop some colour.
5. Dislocation	Line defects are called dislocations.
6. Non-stoichiometric	It is in cases where the compounds contain the combining elements in a ratio different from that required by their stoichiometric formulae. VO_x (x = 0.6 to 1.3), $Fe_{0.95}O$.

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End

3

CHAPTER

atomic structure

- Until the 19th century and the development of the Bohr model, it was believed that atoms were tiny, indivisible particles. An atom is a microscopic structure found in all ordinary matter around us.
- **John Dalton** (1809) regarded the atom as a hard dense and smallest indivisible particle of matter.
- When an electric discharge from a high potential source is passed through a gas contained in a Geissler discharge tube at a very low pressure of the order of a few millimeters, invisible rays are emitted from the cathode of the discharge tube and are known as **cathode rays**.
- The mass of an electron equals to 5.5×10^{-4} amu or 9.1019×10^{-28} g or 9.1019×10^{-31} kg. This is called the rest mass of the electron, *i.e.* the mass which it possesses when it is moving with velocity much smaller than that of light. At high speeds, the mass of the electron in accordance with the theory of relativity, is given by

$$m' = \frac{m}{\left(1-(v^2/c^2)\right)^{1/2}}$$

(where m' is the mass of the electron moving with velocity v, c is the velocity of light and m is the rest mass of the electron).
- In 1932, **James Chadwick** discovered the neutron.
- The mass of neutron 1.675×10^{-24} g is slightly greater than that of a proton (= 1.673×10^{-24} g).
- **Mosley** postulated the frequency of the X-rays was related to the charge present on the nucleus of the atom of the element used as anticathode and found that $\sqrt{\upsilon} = a(z-b)$, where υ is the frequency, z is the nuclear charge and a and b are constants.
- The number of unit positive charges carried by the nucleus of an atom is called the **atomic number** of the element.
- An atom consists of minute positively charged body located at its centre, called the **nucleus** and contains protons and neutrons.
- The nucleus has a diameter of the order of 10^{-15} m while atom has the diameter of the order of 10^{-10} m.
- The sum of the number of protons and neutrons is called the **mass number**.

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- The nucleus of the atom was discovered by **Rutherford** in 1911.
- The number of protons present in the nucleus determines the net positive charge on the nucleus and is equal to the atomic number of the element.
- **Yukawa** (1935) suggested that a pair of nucleons is held together by continuously exchanging their charge through the agency of particles called mesons (π) which may be electrically neutral, positive or negative.
- **Mesons** are unstable particles about 200 times as heavy as an electron and are found to exist in cosmic rays.
- The rays which proceed from the cathode and move away from it at right angles in straight lines are called **cathode rays**.
- Radius of the nucleus of an atom is proportional to the cube root of the mass number of an atom (*i.e.* the number of nucleons in the atom). If r_0 denotes the radius of the nucleus, then $r_0 = [1.33 \times 10^{-15}] A^{1/3}$ m [A = atomic number]
- Cathode rays produce X-rays when they strike a metallic target.
- **Avogadro number**, the number of atoms in one gram atom of any element is 6.023×10^{23}.
- An atom of hydrogen is 1835 times as heavy as an electron.
- An **electron** is a subatomic particle which bears charge -1.60×10^{-19} coulomb and has mass 9.1×10^{-28} g.
- **E. Goldstein** (1886) discovered proton in the discharge tube containing hydrogen.
- The actual mass of proton is 1.672×10^{-28} gram. On the relative scale, proton has mass 1 atomic mass unit (amu).
- A **proton** is defined as a subatomic particle which has a mass of 1 amu and charge +1 elementary charge unit.
- A **neutron** is a subatomic particle which has a mass almost equal to that of a proton and has no charge.
- **J.C. Maxwell** in 1864, suggested that an alternating current of high frequency is capable of radiating energy in the form of waves which travel in space with the same speed as light. He called these waves as electromagnetic waves or electromagnetic radiations.
- The arrangement of the various types of electromagnetic radiations in the order of their increasing (or decreasing) wavelengths is known as **electromagnetic spectrum**.
- A **black body** is defined as an object that absorbs all the radiation falling on it.
- The **wavelength** is defined as the distance between two successive crests or troughs of a wave.
- The **frequency** is the number of waves which pass a given point in one second.
- The **speed** or **velocity** of a wave is the distance through which a particular wave travels in one second.
- The **wave number** is the number of wavelengths per unit length covered.
- Cathode rays produce fluorescence when strike the glass walls of the discharge tube.

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- According to Planck's quantum theory, the radiant energy is emitted or absorbed not continuously but discontinuously in the form of small discrete packets of energy. Each such packet of energy is called a **quantum**. In case of light, the quantum of energy is called **photon**.
- The energy, E of each quantum is directly proportional to the frequency υ of the radiation.
  i.e. $E \propto \upsilon$ or, $E = h\upsilon$
  where h is a proportionality constant called **Planck's constant** and its value is approximately equal to 6.626×10^{-34} Js.
- de Broglie equation is $\lambda = \frac{h}{mv}$ or, $\lambda = \frac{h}{p}$
  where $mv = p$ is the momentum of the particle.
- Louis de Broglie's concept of wave nature of electron was experimentally verified by Davisson and Germer in 1927.
- According to Heisenberg's uncertainty principle, it is impossible to measure simultaneously the position and momentum of a small particle with absolute accuracy or uncertainty.
- The product of the uncertainty in the position (Δx) and the uncertainty in the momentum ($\Delta p = m \cdot \Delta v$) where m is the mass of the particle and Δv as the uncertainty in velocity is always constant and is equal to or greater than $h/4\pi$ where h is Planck's constant *i.e.* $\Delta x \cdot \Delta p \geq h/4\pi$
- When white light composed of all visible wavelengths is passed through the cool vapour of an element certain wavelengths may be absorbed. These absorbed wavelengths are thus found missing in the transmitted light. The spectrum thus obtained consists of a series of dark lines which is referred to as atomic **absorption spectrum**.
- The atoms of hydrogen in gas discharge tube emit radiations whose spectrum shows line characteristics (line spectra) and lies in the infra-red, visible and ultraviolet region of the electromagnetic spectrum.
- The set of lines in the visible region are known as Balmer series, those in ultra-violet as Lyman series and there are three sets of lines in infra-red region: Paschen, Brackett and Pfund series.
- In 1885, Balmer discovered a relation between the wave number, $\bar{v}$ (reciprocal of wavelength) and the position of lines in the series. The relation is

$$\bar{v} = \frac{1}{\lambda} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

  where R is a constant called Rydberg constant and n_1 and n_2 are integers having values of 1, 2, 3, 4, 5, 6, etc., $R = 1{,}09{,}678$ cm^{-1} and $n_2 > n_1$.
- **Ritz combination principle** states that the wave number of any line in the hydrogen spectrum of a particular series can be represented as a difference of the two terms one of which is a constant and the other varies throughout the series.
  Mathematically, $\bar{v} = R\left[\frac{1}{x^2} - \frac{1}{y^2}\right]$

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R = Rydberg constant, x and y are integers and y is always greater than x.

- The atomic spectra of hydrogen is simplest of line spectra.
- According to Bohr's theory, electrons travel around the nucleus in specific permitted circular orbits called stationary states.
- The angular momentum of an electron orbiting around the nucleus is an integral multiple of Planck's constant divided by 2π.

  Angular momentum = $mvr = n\frac{h}{2\pi}$

  where m = mass of electron, v = velocity of the electron, r = radius of the orbit, n = 1, 2, 3 etc. and h = Planck's constant.
- The energy of a photon emitted or absorbed is given by using Planck's relation, $E = h\upsilon$.

  $$\Delta E = E_2 - E_1 = h\upsilon = \frac{hc}{\lambda}$$

  where E_2 = energy of any higher energy state
  E_1 = energy of any lower energy state
  h = Planck's constant
  υ = frequency of radiation emitted or absorbed.
- The radius (Bohr's) r of the nth orbit (which is written as r_n) is given by

  $$r_n = \frac{n^2h^2}{4\pi^2Ze^2m} = 0.53 \times \frac{n^2}{Z} \text{ m}$$

  where Z is the atomic number, e is the charge on an electron and m is mass of an electron.
- The velocity of an electron in the nth state (v_n) is

  $$v_n = \frac{2\pi kZe^2}{nh} = 2.165 \times 10^6 \frac{Z}{n} \text{ m/s.}$$
- The energy of an electron in the nth orbital, E_n is given by

  $$E_n = -\frac{2\pi^2Kme^4Z^2}{n^2h^2} = \frac{-2.178 \times 10^{-18} Z^2}{n^2} \text{ J/atom}$$

  or, $$E_n = \frac{-2.178 \times 10^{-18} Z^2 \times 6.023 \times 10^{23}}{n^2} \text{ J/mol}$$

  or, $$E_n = \frac{-1311.8\, Z^2}{n^2} \text{ kJ/mol} \quad \text{or,} \quad E_n = \frac{-313.3Z^2}{n^2} \text{ kcal/mol}$$
- Also $E_n = -\frac{13.6 \times Z^2}{n^2}$ eV (1 eV = 1.6×10^{-19} J)
- When an electron jumps from one outer orbit (higher energy) n_2 to an inner orbit (lower energy) n_1, then the energy emitted in the form of radiation is given by

  $$\Delta E = E_{n_2} - E_{n_1} = \frac{2\pi^2Kme^4Z^2}{n^2}\left(\frac{1}{{n_1}^2} - \frac{1}{{n_2}^2}\right)$$

  $$\Rightarrow \quad \Delta E = -2.178 \times 10^{-18} Z^2 \left(\frac{1}{{n_1}^2} - \frac{1}{{n_2}^2}\right) \text{ J/atom}$$

  Also, $$\Delta E = 13.6\, Z^2 \left(\frac{1}{{n_1}^2} - \frac{1}{{n_2}^2}\right) \text{ eV/atom}$$

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- $E = h\upsilon$; $c = \nu\lambda$ and $\bar{\nu} = \frac{1}{\lambda}$, $\bar{\nu} = \frac{\Delta E}{hc} = \frac{2\pi^2 K^2 me^4 Z^2}{ch^3}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

  $$\bar{\nu} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), \text{ where } R = \frac{2\pi^2 K^2 me^4}{ch^3} \text{ and } \bar{\nu} = \text{wave number.}$$

  R - Rydberg constant and its value is $1.097 \times 10^7\ m^{-1}$.
- When an electron jumps from any of the higher states to the ground state or Ist state ($n = 1$), the series of spectral lines emitted lies in ultraviolet region and are called as Lyman series. The wavelength (or wave number) of any line of the series can be given by using the relation

  $$\bar{\nu} = RZ^2\left(\frac{1}{1^2} - \frac{1}{n_2^2}\right),\ n_2 = 2,\ 3,\ 4,\ 5\ldots.$$
- The Balmer series results when the electron jumps from third, fourth, fifth etc., energy levels to second energy level. The wave number of any spectral line can be given by using the relation

  $$\bar{\nu} = RZ^2\left(\frac{1}{2^2} - \frac{1}{n_2^2}\right),\ n_2 = 3, 4, 5,\ldots.$$
- When an electron jumps from any of the higher states to the state with $n = 3$, the series of lines emitted lies in the near infra-red region and are known as Paschen series. The wave number of any spectral line can be given by the relation.

  $$\bar{\nu} = RZ^2\left(\frac{1}{3^2} - \frac{1}{n_2^2}\right),\ n_2 = 4, 5, 6,\ldots.$$
- Bohr's model of circular orbits was extended by Sommerfeld by introducing the concept of elliptical orbits. The principal quantum number, n is used by Bohr and azimuthal quantum number K used by Sommerfeld are related to one another as

  $$\frac{n}{K} = \frac{\text{length of major axis of elliptical orbit}}{\text{length of minor axis of elliptical orbit}}$$
- When monochromatic X-rays are allowed to fall on some light element *i.e.* X-rays interact with the electrons, the scattered X-rays have longer wavelength or less frequency or less energy than the incident rays. This effect is called **Compton effect**.
- Erwin Schrodinger in 1927 described the behaviour of electrons around the nucleus by a mathematical equation known as Schrodinger wave equation.

  $$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$$

  where x, y and z are the three space coordinates, m is the mass of the electron, h is Planck's constant, E is the total energy and V is the potential energy of the electron, ψ is called the wave function and $\frac{\partial^2\psi}{\partial x^2}$ refers to the second derivative of ψ with respect to x only and so on.
- The square of the wave function *viz.* ψ^2 gives the probability of finding an electron of a given energy E, from place to place in a given region around the nucleus.

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- An atomic orbital represents a definite region in three-dimensional space around the nucleus where there is a high probability of finding an electron of a specific energy E.
- Zeeman in 1896, found that when a strong magnetic field is applied to a source of a spectrum, each spectral line gets splits up into a number of separate lines. This phenomenon is known as **Zeeman effect**.
- **Schrodinger wave equation** for hydrogen atom is given by

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{8\pi^2 m}{h^2}\left(E+\frac{e^2}{r}\right)\psi=0$$

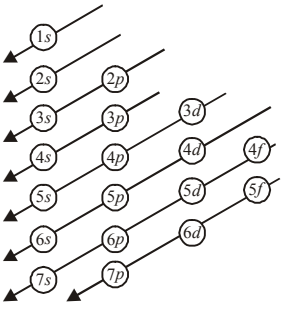
- Quantum numbers may be defined as a set of four numbers with the help of which one can get complete information about all the electrons in an atom *i.e.* location, energy, the type of orbital occupied, shape and orientation of orbital, etc.
- The probability of locating the electron at different distances from the nucleus can be represented graphically by plotting probability (ψ^2) against the distance (r) from the nucleus of the atom. Such a plot of probability versus distance is known as **probability distribution curve**.
- The radial probability distribution of the electron is obtained by plotting the function $4\pi r^2\psi^4$ against the distance r from the nucleus.
- In the plots of radial probability versus distance from the nucleus, number of peaks, *i.e.* regions of maximum probability = $n - l$ (number of radial nodes = $n - l - 1$).
- A nodal plane is the plane on which the probability of finding the electron is zero.
- The distance of maximum probability for 1s electron of hydrogen atom is 0.53 Å and it is equal to Bohr's radius for the first orbit.
- **Electronic configuration :** The distribution of electrons in an atom is known as electronic configuration. Spectroscopic studies, which are used in elucidating electronic configuration show that four numbers known as quantum numbers are required to characterize each electron in an atom.
- *Principal Quantum number* (n) : It represents the main electronic energy shells from nucleus. It can take only intergal values like 1, 2, 3, 4 ....etc. The corresponding shells are also known as K, L, M, N shells respectively. In the absence of any external field, it mainly decides the energy of the electron in the orbit. It also gives the number of the electrons that may be accommodated in each shell, the capacity of each shell being given as $2n^2$. Principal quantum number decides the size of a shell.
- *Azimutal Quantum number* (l) : It represents the number of sub-shells in an orbit. The number of sub-shells in any shell can have the values 0 to ($n - 1$). It gives the shape of the shell.
- When $l = 0$, the sub-shell is called s sub-shell. Similarly when $l = 1$, 2 and 3 the sub-shells are called p,d, and f sub-shells respectively. When $n = 1$, l can have only one value of zero. Hence in 1st orbit, there is one sub-orbit which may be represented as 1s. It takes only integral values from 0 to ($n - 1$). When $n = 2$, l can have 2 values namely 0 and 1, which means that second shell has two sub-shell represented as 2s and 2p respectively.

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Likewise, when $n = 3$, we have three sub-shells designated as 3*s,* 3*p* and 3*d* with corresponding *l* values 0, 1, and 2.

The 4th shell ($n = 4$) has four subshells (4*s*, 4*p*, 4*d* and 4*f*) with *l* values 0, 1, 2 and 3. The total number of sub-shells in any shell is same as the principal quantum number.

- *Magnetic Quantum Number* (*m*) : This gives the number of orbitals in a sub-shell. It takes only integral values from $-l$ to $+ l$ through zero. $m = 2l + 1$ for any value of *l*. *e.g.* $l = 0$, $m = 1$. *m* decides the orientation of electrons. In *s* sub-shell there is only one orbital. [Since $l=0$, $(2l+1) = 1$].
- In *p* sub-shell there are 3 orbitals –1, 0 and +1. [$l = 1$, $(2l + 1) = 3$].The 3 orbital are designated as p_x, p_y and p_z where *x, y* and *z* refer to the axes perpendicular to each other in space.
- In *d* sub-shells there are 5 orbitals –2, –1, 0, + 1 and +2. [$l = 2$, $(2l + 1) = 5$]. The five orbitals are represented as d_{xy}, d_{yz}, d_{zx}, $d_{x^2-y^2}$ and d_{z^2} .
- In *f* sub-shells there are 7 orbitals –3, –2, –1, 0, +1, +2 and +3. [$l = 3$, $(2l + 1) = 7$].
- *Spin Quantum number* (*s*) : When an electron rotates around a nucleus, it also spins about its own axis. If the spin is clockwise, its spin quantum number is +1/2 and it is represented by ↑. If the spin is anticlockwise, its value is –1/2 and is represented by ↓ . If *s* value is +1/2, then by convention, we take that electron as the first electron in that orbitals and if *s* value is –1/2, it is taken as the second electron.
- **Pauli's Exclusion Principle :** No two electrons in an atom can have all the four quantum numbers identical or in any orbital maximum number of electrons can be two with opposite spins.
- Maximum number of electrons in :
  (*i*) *s* sub-shell can be 2
  (*ii*) *p* sub-shell can be 6
  (*iii*) *d* sub-shell can be 10
  (*iv*) *f* sub-shell can be 14.
- **Aufbau Principle :** An electron enters the sub-shell that has the least energy. The sub-shells are filled in the increasing order of energy.
  $1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f <$ 5d....



The ascending order is 1*s*, 2*s*, 2*p*, 3*s*, 3*p,* 4*s,* 3*d*, 4*p,* 5*s*, 4*d*, 5*p*, 6*s*, 4*f*, 5*d*, 6*p*, 7*s*, 5*f*, 6*d*,....

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- To write the electronic configuration of elements, go on putting the electrons in various sup-shells in increasing order according to maximum capacity till all the electrons are over.
- **Hund's rule :** Electron pairing in any *s, p, d* or *f* orbital is not possible until all the available orbitals of the same sub-shell contain one electron each. It means an electron occupies a vacant orbitals in the same sub-shell and pairing can start when all orbitals are filled up. Pairing takes place only after filling 3, 5 and 7 electrons in *p*, *d* and *f* orbitals respectively.
- **Shapes of orbitals :** *s* orbitals is spherical in shape, *p* orbital is dumbbell shaped, *d* orbital is leaf like and the shape of *f* orbital is complicated.
  It may be noted that half filled and fully orbitals are more stable. This factor also should be taken into consideration in arriving at the electronic configuration of elements. The discrepancy in the configuration of $_{24}Cr$ and $_{29}Cu$ is in agreement with this.
  ($_{23}V$ has the outermost configuration $3d^3 4s^2$ and according to Hund's rule instead of $_{24}Cr$ having a structure $3d^4 4s^2$ the actual configuration is $3d^5 4s^1$).

End

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4

CHAPTER

chemical bonding

- The process of combination of atoms, called chemical bonding involves the union of two or more atoms through redistribution of electrons in their outermost shell to acquire the stable electronic configuration of noble gases having a state of minimum energy.
- All atoms attract one another at small distances; the universal attractive interactions known as van der Waals forces exist between all matter, and play an important part in determining the properties of liquids and solids. These attractions are extremely weak, however, and they lack specificity: they do not lead to aggregates having any special structure or composition.
- A molecule is an aggregate of atoms that possesses distinctive observable properties.
- Chemical bonding gives the idea of the existence of an aggregate (assembly) of atoms that is sufficiently stable to possess a characteristic structure and composition.
- Chemical combination takes place due to tendency of atoms to acquire noble gas configuration and minimum energy.
- **Octet rule** : All atoms except (H, Li and Be) try to achieve noble (inert) gases configuration, *i.e.*, 8 electrons in their outermost orbit.
  $_{10}Ne = 2, 8$; $_{18}Ar = 2, 8, 8$
  While H, Li, Be achieve configuration of He = 2.
  The tendency of atoms to achieve eight electrons in their outermost shell is known as octet rule.
- Chemical bonding mainly depends on the number of valence electrons *i.e.* electrons present in outermost level. Valence electrons in atoms are shown in terms of Lewis symbol.
  The structures are written as the element symbol surrounded by dots that represent the valence electrons. The Lewis structures for the elements in the first two periods of the periodic table are shown below.

H•	Lewis dot structures						He:
Li•	Be•	•B•	•Ċ•	•N:	:Ö:	•F:	:Ne:

- According to "Electronic theory of valency", the valency of an element is the number of electrons that its atoms can gain, lose or share to acquire stable nearest noble gas configuration.

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Ionic bonding

- Bonding between metals and nonmetals
- Metal atoms have a low number of valence electrons and a low electronegativity.
- Non-metal atoms have numerous valence electrons.

(a) Metals

- lose valence electrons
- achieve a stable valence shell (usually 8 electrons)
- gains a positive charge, *i.e.* a positive ion

examples :

Na 2, 8, 1 → Na^+ 2, 8
Li 2, 1 → Li^+ 2
K 2, 8, 8, 1 → K^+ 2, 8, 8
Mg 2, 8, 2 → Mg^{2+} 2, 8
Ca 2, 8, 8, 2 → Ca^{2+} 2, 8, 8
Al 2, 8, 3 → Al^{3+} 2, 8

The charge is the number of valence electrons it has to lose, e.g. Mg 2, 8, 2 loses 2, Mg^{2+}.

(b) Non-metals

- gain valence electrons
- achieve a stable valence shell (usually 8 electrons)

examples :

F 2, 7 → F^- 2, 8
Cl 2, 8, 7 → Cl^- 2, 8, 8
Br 2, 8, 8, 7 → Br^- 2, 8, 8, 8
I 2, 8, 18, 18, 7 → I^- 2, 8, 18, 18, 8
O 2, 6 → O^{2-} 2, 8
S 2, 8, 6 → S^{2-} 2, 8, 8
N 2, 5 → N^{3-} 2, 8
P 2, 8, 5 → P^{3-} 2, 8, 8
H 1 → H^- 2.

The charge = 8 – group number.
e.g. nitrogen, group 5, charge 8 – 5 = N^{3-}.

- **Ionic lattice**
  Positive and negative ions attract each other to form a three dimensional continuous lattice structure.
  Each +ve ion is surrounded by a number of –ve ions.
  Each –ve ion is surrounded by a number of +ve ions.
  The ratio of +ve and –ve ions in the lattice is determined by the charges of the ions.

Ions	Formula
Na^+ Cl^- +1 –1	NaCl 1 : 1
Mg^{2+} Cl^- +2 –1	$MgCl_2$ 1 : 2
Na^+ S^{2-} +1 –2	Na_2S 2 : 1

- This table shows properties of ionic lattices (compounds) and explanations of these properties.

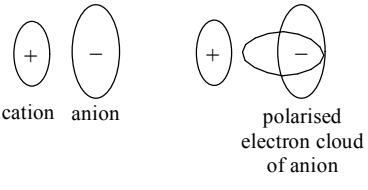
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Property	Explanation
Melting point and boiling point	The melting and boiling points of ionic compounds are high because a large amount of thermal energy is required to separate the ions which are bound by strong electrical forces.
Electrical conductivity	Solid ionic compounds do not conduct electricity when a potential is applied because there are no mobile charged particles. No free electrons causes the ions to be firmly bound and cannot carry charge by moving.
Hardness	Most ionic compounds are hard; the surfaces of their crystals are not easily scratches. This is because the ions are bound strongly to the lattice and aren't easily displaced.
Brittleness	Most ionic compounds are brittle; a crystal will shatter if we try to distort it. This happens because distortion cause ions of like charges to come close together then sharply repel.

- Method of writing formula of an ionic compound:
  (a) Write the symbols of the ions side by side in such a way that positive ion is at the left and negative ion at the right as AB.
  (b) Write their electrovalencies in figures on the top of each symbol as $A^x B^y$.
  (c) Divide their valencies by HCF.
  (d) Now apply criss-cross rule as $A^{x} \times B^{y}$ *i.e.*, formula is $A_y B_x$.

  e.g. magnesium nitride $Mg^{2} \times N^{3} = Mg_3N_2$.

- Conditions for the formation of ionic bond
  (i) Ionisation energy of electropositive atom should be low.
  (ii) Electron affinity of electronegative atom should be high.
  (iii) Lattice energy of the ionic crystal should be high.
- **Variable valency :** Heavier *p*-block elements (with high atomic number), transition and inner transition elements exhibit more than one valency, e.g. iron exhibits an electrovalency of 2 and 3, tin 2 and 4. These elements are said to possess variable valency.
- Variable electrovalency is actually due to unstable electronic configuration of ion and inert pair effect (some of heavier representative elements of III, IV and V groups having configurations of the outermost shell $ns^2\ np^1$, $ns^2\ np^2$ and $ns^2\ np^3$ show valencies with a difference of 2 *i.e.* (1, 3), (2, 4), (3, 5) respectively. Thus the two *s* electrons (ns^2) in the valency shell tend to remain inert and do not participate in formation of bonds. This is called inert pair effect).
- **Fajan's rule :** In ionic bond, some covalent character is introduced because of the tendency of the cation to polarise the anion. In fact cation attracts the electron cloud of the anion and pulls electron density between two nuclei.



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According to Fajan's rule, the magnitude of covalent character in the ionic bond depends upon the extent of polarisation caused by cation. In general,

(i) smaller the size of cation, larger is its polarising power.
(ii) Larger the anion, more will be its polarisability.
(iii) More charge on cation and anion.
(iv) Presence of a non-polar solvent.
(v) Cation with an electronic configuration other than noble gas.

Covalent bonding

- Bonding between non-metals.
- All atoms included have fairly high electronegativity and few vacancies in valence energy levels. When they bond, they gain electrons to achieve stable configuration. Hence, electrons are shared.
- Sharing produces low energy (stable) electron arrangements. i.e. full outer shell (e.g. He 2 ; Ne 2, 8)
  8 electrons (4 pairs) in outer shell (e.g. Ar 2, 8, 8)
- ***Covalency*** is the number of electrons an atom needs to gain to produce a stable outer shell.
- The number of shared pairs (covalent bonds) of electrons an atom forms. e.g. Hydrogen. H needs 1 additional electron.
- Hydrogen bonds 1 pair of electrons shared between 2 atoms - covalent bond.
- Bonding pairs of electrons orbits both nuclei - attracts both nuclei - provides bonding force.
- Hydrogen molecule consists of 2 covalently bonded hydrogen atoms which have no tendency to bond further (both have achieved a stable outer shell).
- Each molecule exists independently.
  e.g. Fluorine F 2, 7 ; covalency = 1
- This table shows properties of covalent molecular compounds and explanations of these properties.

Property	Explanation
Do not conduct electricity	No mobile charged particles. Molecules are not charged. Electrons tightly bound to atoms or shared by atoms in covalent bonds.
Melting and boiling points low	During melting/boiling, molecules become separated. Forces of attraction between molecules are weak and little thermal energy is required to separate them.
Soft	Molecules weakly attracted to each other and are easily displaced.

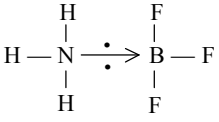
- In covalent bonds, electron pairs are shared equally between atoms of equal electronegativity.

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- If the atoms in a covalent bond have differing electronegativities, the atoms with the higher electronegativity has > 50% of the shared pairs of electrons and the atoms with low electronegativity has 50% of the shared pairs of electronegativity.
  - The atom tending to gain electrons acquires a slight negative charge (delta –ve)
  - The atom tending to lose electrons acquires a slight positive charge (delta +ve)
  - The bond is polar.
- **Multiple bonds** : For every pair of electrons shared between two atoms, a single covalent bond is formed. Some atoms can share multiple pairs of electrons, forming multiple covalent bonds. For example, oxygen (which has six valence electrons) needs two electrons to complete its valence shell. When two oxygen atoms form the compound O_2, they share two pairs of electrons, forming two covalent bonds.

O_2 :Ö: :Ö: or :Ö = Ö:

- **Polar and non-polar covalent bonding :** There are, in fact, two sub-types of covalent bonds. The H_2 molecule is a good example of the first type of covalent bond, the non-polar bond. Because both atoms in the H_2 molecule have an equal attraction (or affinity) for electrons, the bonding electrons are equally shared by the two atoms, and a non-polar covalent bond is formed. Whenever two atoms of the same element bond together, a non-polar bond is formed.
- A polar bond is formed when electrons are unequally shared between two atoms. Polar covalent bonding occurs because one atom has a stronger affinity for electrons than the other (yet not enough to pull the electrons away completely and form an ion). In a polar covalent bond, the bonding electrons will spend a greater amount of time around the atom that has the stronger affinity for electrons. A good example of a polar covalent bond is the hydrogen-oxygen bond in the water molecule.
- The polarity of a molecule is indicated in terms of dipole moment, which is defined as the product of the distance separating charges of equal magnitude and opposite sign with the magnitude of the charge.
- **Coordinate or dative bond :** It was proposed by Sidgwick. In this type of combination both the electrons needed for sharing are contributed only by one atom. The atom which contributes the pair of electrons (lone pair) is known as donor and the atom which accepts these electrons is called acceptor. The coordinate bond is usually represented by an arrow pointing towards the acceptor. e.g.



Coordinate bond is found in the compounds like SO_2, SO_3, O_3, NH_4^+, H_3O^+, NH_4Cl, SO_4^{2-} and H_2SO_4 etc.

- **Characteristics of co-ordinate compounds :** These are usually insoluble in water but soluble in organic solvents. They usually do not conduct electricity. The melting point and boiling point of these compounds are higher than covalent compounds but lesser than the ionic compounds. The coordination bond is directional so these compounds exhibit isomerism.

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Examples of the compounds in which all the three ionic, covalent and dative bonds are present are : $CuSO_4$, NH_4X, $K_4[Fe(CN)_6]$ and $[Cu(NH_3)_4]SO_4$.

- **Hybridisation** : It is a linear combination of atomic orbitals of approximately the same energy, yielding hybridised or hybrid orbitals. Hybridised orbitals are of the same energy and have a special geometrical arrangement.
- **Hydrogen bond** : It is the force of attraction that exists between the hydrogen atom covalently bonded to highly electronegative atom (N or F or O) in a molecule and the electronegative atom of the same or neighbouring molecule, the bond is represented by a dotted line as shown below:

- - - H — F - - - H — F - - - H — F - - -

- The hydrogen bond is of two types, inter molecular (formed between H atom of one molecule with electronegative atom of neighbouring molecule). Intra molecular hydrogen bonding (H-atom and electronegative atom of the same molecule).

Type of Molecule	Total No. of electron pairs	No. of bond pairs	No. of lone pairs	Type of hybri-dization involved	Geometry of molecule	Examples
AB_2	2	2	0	sp	Linear	BeF_2, $[Ag(NH_3)_2]^+$
AB_3	3	3	0	sp^2	Trigonal planar	BF_3, $AlCl_3$
AB_2L	3	2	1	sp^2	V-shaped	$SnCl_2$, $PbCl_2$
AB_4	4	4	0	sp^3	Tetrahedral	CH_4, SiF_4, CCl_4
AB_3L	4	3	1	sp^3	Trigonal pyramidal	NH_3, PX_3 (X = F, Cl, Br, I)
AB_2L_2	4	2	2	sp^3	V-shaped	H_2O, OF_2, SCl_2
AB_5	5	5	0	sp^3d	Trigonal bipyramidal	PF_5, PCl_5, $SbCl_5$,
AB_4L	5	4	1	sp^3d	See saw	SF_5, $TeBr_4$
AB_3L_2	5	3	2	sp^3d	T-shaped	ClF_3, $XeOF_2$
AB_2L_3	5	2	3	sp^3d	Linear	XeF_2, ICl_2^-, I_3^-
AB_6	6	6	0	sp^3d^2	Octahedral	SF_6, $[SbF_6]^-$
AB_5L	6	5	1	sp^3d^2	Square pyramidal	IF_5, ClF_5, $[SbF_5]^{2-}$
AB_4L_2	6	4	2	sp^3d^2	Square planar	SF_4, XeF_4, ICl_4^-
AB_7	7	7	0	sp^3d^3	Pentagonal bipyramidal	lF_7, XeF_6

- **Important features of H-bonding**

(i) Hydrogen bonding decreases the volatility and increases the viscosity and surface tension of a substance.

(ii) Order of strength of hydrogen bonding.

Energy H.F. > H.O. > H.N.

10 Kcal/mole 7 Kcal/mole 2 Kcal/mole

H-bond is much stronger than van der Waals forces but weaker than a covalent bond and an ionic bond.

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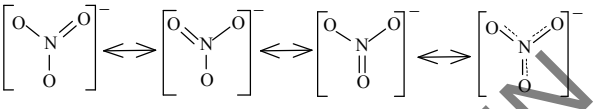
The order of strength is therefore as:
van der Waals forces < H-bond < covalent bond < ionic bond.

(iii) Because of hydrogen bonding water (H_2O) has higher boiling point than that of H_2S.

(iv) Ice has lower density than water due to the formation of open cage like structure because of formation of hydrogen bonds in water (H_2O) molecules.

- **Resonance**

There is a rather large class of molecules for which one has no difficulty writing Lewis structures; in fact we can write more than one valid structure for a given molecule.



These structures differ only in which oxygen atom is attached by the double bond. Since there is no reason to prefer one over another, the NO_3^- ion is regarded as a superposition, or hybrid of these three structures. The term resonance has been used to describe this phenomenon, which is indicated in the above structures by the double-ended arrows.

- **Important features of resonance**

The contributing structures should not differ in atomic arrangement. The contributing structures should have same number of unpaired electrons. Hybrid structure is more stable than any of the resonating structure. The difference between the energy of actual structure and most stable resonating form is termed as resonance energy.

- **Metallic bonding**

Bonding between atoms with low electronegativity *i.e.* 1, 2 or 3 valence electrons, therefore there are many vacancies in valence shell. When electron clouds overlap, electrons can move into electron cloud of adjoining atoms. Each atom becomes surrounded by a number of others in a 3-D lattice, where valence electrons move freely from 1 valence shell to another. Delocalised valence electrons moving between nuclei generate a binding force to hold the atoms together.

- The table shows metallic properties and the explanations of these properties.

Property	Explanation
Metals are dense	The particles present in metals are tightly packed in the lattice
Metals have high melting and boiling points	Strong forces of attraction exist between particles. A large amount of thermal energy is required to overcome the strong electrical forces between the positive ions and the delocalised electrons. These forces operate throughout the lattice.
Metals are good conductors of heat	Delocalised electrons transmit the energy of vibrations of 1 positive ion to its neighbours.
Metals are good conductors of electricity	Mobile delocalised electrons within the lattice. Electrons flow in at one end, and the same number flow out the other end.
Metals are malleable and ductile	The distortion does not disrupt the metallic bonding.
Metals are lustrous	The presence of free electrons causes most metals to reflect light (non-metals are transparent)

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Types of bonding in solids

- **Ionic bonding :** Solid containing ionic bonds consists of an array or a net work of positive and negative ions arranged systematically in a characteristic pattern. The binding forces are strong electrostatic bonds between positive and negative ions.
  Examples : Compounds of elements of group 1 and 2 with elements of group 16 and 17. e.g. NaCl, ZnS etc.
- **Covalent bonding :** The solid containing covalent bonding consists of an array of atoms that share electrons with their neighbouring atoms. The atoms are linked together by strong covalent bonds extending into three dimensional structure.
  Examples : Diamond, silicon carbide, silicon dioxide etc.
- **Molecular bonding :** The solid containing molecular bonding consists of symmetrical aggregates of discrete molecules. However, these molecules are further bound to other molecules by relatively weak force such as dipole-dipole forces, dispersion forces or H-bonds depending upon the nature of molecules.
  Examples : Iodine, solid CO_2, ice, solid hydrogen etc.
- **Formal charge** -The formal charge of an atom is computed as the difference between the number of valence electrons that a neutral atom would have and the number of electrons that belong to it in the Lewis structure. Electrons in covalent bonds are split equally between the atoms involved in the bond. The total of the formal charges on an ion should be equal to the charge on the ion, and the total of the formal charges on a neutral molecule should be equal to zero.
  - Formal charge equation is based on the comparing the number of electrons in the individual atom with that in the structure:

$$\text{Formal charge} = \left(\begin{matrix}\text{Group}\\ \text{number}\end{matrix}\right) - \frac{1}{2}\left(\begin{matrix}\text{number of electrons}\\ \text{in covalent bonds}\end{matrix}\right) - \left(\begin{matrix}\text{number of electrons}\\ \text{in lone pairs}\end{matrix}\right)$$

number of valence electrons in the neutral atom

remember that these electrons are shared

$$\text{Formal charge} = \left(\begin{matrix}\text{Group}\\ \text{number}\end{matrix}\right) - \left(\begin{matrix}\text{number of}\\ \text{covalent bonds}\end{matrix}\right) - \left(\begin{matrix}\text{number of electrons}\\ \text{in lone pairs}\end{matrix}\right)$$

Assign the formal charge to the nitrogen in the compound:

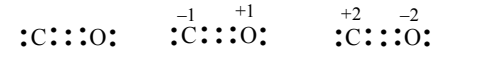
– Nitrogen is in group five; thus, it has five valence electrons.
– Number of non-bonding electrons : there are no non-bonding electrons, so this is zero.
– Number of shared electrons: there are four bonds, and there are two electrons in each bond, so this number is eight.

Formal charge = 5 – 0 – 0.5 (8) = +1

Thus, the nitrogen has a formal charge of +1.

- **Oxidation number** : In contrast to formal charge, in which the electrons in a bond are assumed to be shared equally, oxidation number is the electric charge an atom would have if the bonding electrons were assigned exclusively to the more electronegative atom.
- The following diagram compares the way electrons are assigned to atoms in calculating formal charge and oxidation number.

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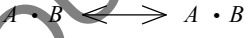


Lewis structure Formal charge Oxidation number

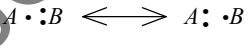
- **Bond energies**
  The bond energy is the amount of work that must be done to pull two atoms completely apart. This is almost, but not quite the same as the bond dissociation energy actually required to break the chemical bond; the difference is the very small.
- **Odd electron bond**
  There are some stable molecules in which double bonds are formed by sharing of an odd number of electrons, i.e. one, three, five etc. between two bonded atoms. The bonds of this type are called odd electron bonds.

  Properties of odd electron bonds

  (i) The odd electron bonds are generally established either between two like atoms or between different atoms which have not more than 0.5 difference in their electronegativities.
  (ii) Odd electron bonds are approximately half as strong as a normal covalent bond.
  (iii) Molecules containing odd electrons are extremely reactive and have the tendency to dimerise.
  (iv) Bond length of one electron bond is greater than that of a normal covalent bond. Whereas the bond length of a three electron bond is intermediate between those of a double and a triple bond.
  (v) One electron bond is a resonance hybrid of the following two structures.



Similarly, a three electron bond is a resonance hybrid of the following two structures.



VSEPR theory

The VSEPR model is an extraordinarily powerful one, considering its great simplicity. Its application to predicting molecular structures can be summarised as follows.

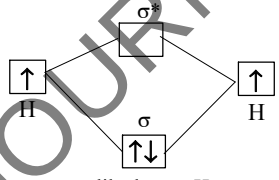
- Electron pairs surrounding a central atom repel each other; this repulsion will be minimized if the orbitals containing these electron pairs point as far away from each other as possible.
- The coordination geometry around the central atom corresponds to the polyhedron whose number of vertices is equal to the number of surrounding electron pairs (coordination number). Except for the special case of 5, and the trivial cases of 2 and 3, the shape will be one of the regular polyhedron.
- If some of the electron pairs are nonbonding, the shape of the molecule will be simpler than that of the coordination polyhedron.
- Orbitals that contain non-bonding electrons are more concentrated near the central atom, and therefore offer more repulsion than bonding pairs to other orbitals.

While VSEPR theory is quite good at predicting the general shapes of most molecules, it cannot yield exact details. For example, it does not explain why the bond angle in H_2O is 104.5°, but that in H_2S is about 90°. This is not surprising, considering that the emphasis is on electronic repulsions, without regard to the detailed nature of the orbitals containing the electrons, and thus of the bonds themselves.

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Molecular orbital theory

- According to this theory, all the atomic orbitals of the atoms participating in molecule formation get disturbed when the concerned nuclei approach nearer. They all get mixed up to give rise to an equivalent number of new orbitals that belong to the molecule now. These are called molecular orbitals.
- When two atomic orbitals overlap they can be in phase (added) or out of phase (subtracted). If they overlap in phase, constructive interaction occurs in the region between two nuclei and a bonding orbital is produced. When they overlap out of phase, destructive interference reduces the probability of finding an electron in the region between the nuclei and antibonding orbital is formed.
- The energy of an antibonding orbital is higher (less stable) than the energies of the combining atomic orbitals.
- This scheme of bonding and antibonding orbitals is usually depicted by a molecular orbital diagram such as the one shown here for the dihydrogen ion H_2^+. Electrons fill the lower-energy molecular orbitals before the higher ones, just as is the case for atomic orbitals. Thus, the single electron in this simplest of all molecules goes into the bonding orbital, leaving the antibonding orbital empty. Since any orbital that can hold a maximum of two electrons, the bonding orbital in H_2^+ is only half-full.



dihydrogen H_2

H – H bond energy 452 kJ

- Bond order = $\frac{1}{2}$[number of bonding electrons – number of antibonding electrons]
- If bond order is zero for any molecule then it will not exist.
- **Paramagnetism and diamagnetism :** If all the electrons in a species (atom, ion or molecule) are paired, the substance is diamagnetic (repelled by magnetic field).
  If some of the electrons in a species (atom, ion or molecule) are unpaired, the substance is paramagnetic (attracted by magnetic field).
- **Molecular orbital configuration of some molecules/ions**

1. H_2 : σ_{1s^2}; Bond order = 1/2(2 – 0) = 1 (diamagnetic)
2. H_2^+ : σ_{1s^1}; Bond order = 1/2(1 – 0) = 1/2 (paramagnetic)
3. H_2^- : $\sigma_{1s^2}\sigma^*_{1s^1}$; Bond order = 1/2(2 – 1) = 1/2 (paramagnetic)
4. He_2 : $\sigma_{1s^2}\sigma^*_{1s^2}$; Bond order = 1/2(2 – 2) = 0 (does not exist)
5. N_2 : $\sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\pi_{2p_x^2}\pi_{2p_y^2}\sigma_{2p_z^2}$
   Bond order = 1/2(10 – 4) = 3 (diamagnetic)
6. N_2^+ : $\sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\pi_{2p_x^2}\pi_{2p_y^2}\sigma_{2p_z^1}$
   Bond order = 1/2(9 – 4) = 2.5 (diamagnetic)

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7. N_2^- : $\sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\pi_{2p_x^2}\pi_{2p_y^2}\sigma_{2p_z^2}\pi^*_{2p_x^1}$
   Bond order = 1/2(10 – 5) = 2.5 (paramagnetic)
8. N_2^{2-} : $\sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\pi_{2p_x^2}\pi_{2p_y^2}\sigma_{2p_z^2}\pi^*_{2p_x^1}\pi^*_{2p_y^1}$
   Bond order = 1/2(10 – 6) = 2 (paramagnetic)
9. O_2 : $\sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\sigma_{2p_z^2}\pi_{2p_x^2}\pi_{2p_y^2}\pi^*_{2p_x^1}\pi^*_{2p_y^1}$
   Bond order = 1/2(10 – 6) = 2 (paramagnetic)
10. O_2^+ : $\sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\sigma_{2p_z^2}\pi_{2p_x^2}\pi_{2p_y^2}\pi^*_{2p_x^1}$
    Bond order = 1/2(10 – 5) = 2.5 (paramagnetic)
11. O_2^- : $\sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\sigma_{2p_z^2}\pi_{2p_x^2}\pi_{2p_y^2}\pi^*_{2p_x^2}\pi^*_{2p_y^1}$
    Bond order = 1/2(10 – 7) = 1.5 (paramagnetic)
12. O_2^{2-} : $\sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\sigma_{2p_z^2}\pi_{2p_x^2}\pi_{2p_y^2}\pi^*_{2p_x^2}\pi^*_{2p_y^2}$
    Bond order = 1/2(10 – 8) = 1 (diamagnetic)
13. Ne_2 : $\sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\sigma_{2p_z^2}\pi_{2p_x^2}\pi_{2p_y^2}\pi^*_{2p_x^2}\pi^*_{2p_y^2}\sigma^*_{2p_z^2}$
    Bond order = 1/2(10 – 10) = 0 (does not exist)
14. CN: $\sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\pi_{2p_x^2}\pi_{2p_y^2}\sigma_{2p_z^1}$
    Bond order = 1/2(9 – 4) = 2.5 (paramagnetic)
15. CN^- : $\sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\pi_{2p_x^2}\pi_{2p_y^2}\sigma_{2p_z^2}$
    Bond order = 1/2(10 – 4) = 3 (diamagnetic)
16. NO : $\sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\pi_{2p_x^2}\pi_{2p_y^2}\sigma_{2p_z^2}\pi^*_{2p_x^1}$
    Bond order = 1/2(10 – 5) = 2.5 (paramagnetic)
17. NO^+ : $\sigma_{1s^2}\sigma^*_{1s^2}\sigma_{2s^2}\sigma^*_{2s^2}\pi_{2p_x^2}\pi_{2p_y^2}\sigma_{2p_z^2}$
    Bond order = 1/2(10 – 4) = 3 (diamagnetic)

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End

5

CHAPTER

solutions

- A **solution** is defined as a homogeneous mixture of two (or more) substances, the composition of which may vary within limits.
- A solution consisting of two components is called a **binary solution**.
- A **solvent** is that component of the solution which is present in larger amount by weight than the other component, termed as **solute**.
- The point at which a liquid cannot dissolve more of the solute at a constant temperature is called saturation point and such a solution is called **saturated solution**.
- A solution may sometimes contains more solute than would be necessary to saturate it at a given temperature. Such a solution is called **super saturated solution**.
- A solution in which more of the solute can be dissolved at a given temperature is called **unsaturated solution**.
- **Solubility** of a substance is defined as the amount of solute dissolved in 100 grams of a solvent to form a saturated solution at a given temperature.
- The **molarity** of a solution gives the number of moles of the solute present in one litre (dm^3) of the solution.
- The **molality** of a solution gives the number of moles of the solute present in 1000 g of the solvent.
- The **normality** of a solution gives the number of gram equivalents of the solute present in one litre of the solution.
- The mole **fraction** of a solute in a solution gives the ratio of the number of moles of the solute present in the solution to the total number of moles of the solute and the solvent present in the solution.
- **Formality** of a solution is defined as the number of gram formula masses of the solute dissolved per litre (or dm^3) of the solution. It is denoted by *F*.

$$\text{Formality} = \frac{\text{No. of gram formula masses of solute}}{\text{volume of solution in litres}}$$

- When a solute is present in very small amounts its concentration is expressed in **parts per million** (10^6). It can be defined as the number of parts by mass of the solute per million parts by mass of the solution. It is abbreviated as ppm.

$$\text{Parts per million} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$$

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- Molarity and normality are related to each other as follows:

$$\text{Normality} = \text{molarity} \times \frac{\text{molecular mass of solute}}{\text{equivalent mass of solute}}$$

- Percent by weight is the weight of the solute as a per cent of the total weight of the solution. That is

$$\%\text{ by weight of solute} = \frac{\text{wt. of solute}}{\text{wt. of solution}} \times 100$$

- A solution obtained by dissolving one mole of the solute in 1000 g of solvent is called **one molal solution**.
- Unit of molarity is **mol litre^{-1}**.
- **Dilution :** Adding more solvent to a solution to decrease the concentration is known as dilution. Starting with a known volume of a solution of known molarity, we should be able to prepare a more dilute solution of any desired concentration.
- **Titration :** We can measure the concentration of a solution by a technique known as a titration. The solution being studied is slowly added to a known quantity of a reagent with which it reacts until we observe something that tells us that exactly equivalent numbers of moles of the reagents are present. Titrations are therefore dependent on the existence of a class of compounds known as indicators.
  The endpoint of an acid-base titration is the point at which the indicator turns colour. The equivalence point is the point at which exactly enough base has been added to neutralize the acid.
- **Polar compounds** are highly soluble in polar solvents but only sparingly soluble in non-polar solvents.
- **Non-polar** compounds are highly soluble in non-polar solvents but only sparingly soluble in polar solvents.
- Larger the lattice energy of the crystal of a solute, the smaller is its solubility. (**Reason :** *Ionic solids consist of positively and negatively charged ions lying close to one another. It is the force of attraction between the oppositely charged ions in a crystal which gives rise to lattice energy of the crystal. It is the lattice energy which opposes the tendency of a solute to dissolve).*
- Ionic solids dissolve to a larger extent in a solvent having a high dielectric constant than in a solvent having a low dielectric constant.
- For ionic solids, the lattice energy describes the attractive forces between the solute molecules (i.e. ions).
- For an ionic solid to dissolve in water, the water-solute attractive forces has to be strong enough to overcome the lattice energy.
- The process known as **solvation** is where the solute-solvent interactions are strong enough separate, surround and disperse a solute. If the solvent is H_2O, then solvation is referred to as hydration.
- The process of solvation involves energy changes also, known as the enthalpy of solvation (ΔH_{solv}). It is a physical process, not chemical.
- Processes in which the overall heat energy of the system decreases tend to be spontaneous (i.e. a negative value for the overall ΔH indicates spontaneity).

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- However, the hydration of ammonium nitrate is spontaneous, but has a slightly positive value for the overall enthalpy (it has absorbed heat energy). Clearly, something else is going on that forces the hydration of ammonium nitrate to occur.
- In the case of the solvation of ammonium nitrate we start with a crystal of the ionic solid being placed in a container with water.
  - When the crystal initially dissolves all the ions, though hydrated, are not randomly distributed throughout the water in the container-they are concentrated in the vicinity of the original crystal.
  - The solvated ions are therefore initially not in a random distribution throughout the container.
  - They are initially in a more organized state or ordered state (i.e. concentrated in one location in the contanier) and will naturally want to become more disordered or randomly distributed.
- Process in which the disorder of the system increases tend to occur spontaneously.
- Thus the increase in disorder is a driving force that can overcome the slight enthalpy associated with the hydration of ammonium nitrate.

- Solubility of a gas in a liquid generally decreases with increase in temperature.
- A substance is generally soluble if it dissolves to give a solution of concentration greater than 1 g per 100 ml and insoluble if it dissolves to give a solution of concentration less than 0.001 g per 100 ml.
- Substances with solubilities less than 1 g but more than 0.001 g per 100 ml are called **sparingly soluble**.
- The solubility of a gas in a liquid (called the solubility coefficient) is the volume of the gas in cc's which will dissolve in 1 cc of the liquid to form a saturated solution, the volume of the gas being measured at the temperature and pressure at which the measurement of the solubility is made.

Factors Affecting Solubility :

There are three factors that explain why solubilities will vary even within the same solution.

All nitrates are soluble.

All acetates are soluble.

All chlorides, bromides and iodides are soluble except those of Ag^+, Hg^{2+} and Pb^{2+}. $PbCl_2$ and $PbBr_2$ are sparingly soluble in cold water.

Most sulphates are soluble. Some exceptions are $SrSO_4$, $BaSO_4$ and $PbSO_4$, $CaSO_4$, Ag_2SO_4 and Hg_2SO_4 are sparingly soluble.

Almost all salts of Na^+, K^+ and NH_4^+ are soluble.

All normal carbonates and phosphates are insoluble except those of the group 1 elements (H^+, Li^+, Na^+, K^+ etc.) and NH_4^+.

Hydroxides are insoluble except those of the group 1 elements, [$Sr(OH)_2$, $Ba(OH)_2$ and $Ca(OH)_2$ are slightly soluble].

Sulphides other than those of group 1 and 2 elements and NH_4^+ are insoluble.

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- **Solute/Solvent interactions**- The molecular size of the solute molecules will affect the solubility. The larger the solute molecules the more likely the solubility will diminish. In addition, the polar nature of the solute molecules compared to the solvent will also alter the solubility. For aqueous solutions where polar water is the solvent, the more polar the solute molecules are the more likely the solubility will be higher. Non-polar hydrocarbons have a very low solubility in polar water. So do such things as carbon tetrachloride, oxygen gas, nitrogen gas, and other non-polar molecular substances. On the other hand polar solutes like ionic salts and polar molecular substances will have higher solubilities in water. There is an old addage "Like dissolves Like". If we were to use a non-polar solvent instead of polar water then the non-polar solutes above would have higher solubilities in that solvent.
- **Temperature** - Generally speaking the water solubility of a liquid or solid will increase with increasing temperature.

  Solubility $\propto$ temperature

  Exception : Some solutes like solid $Ce_2(SO_4)_3$ will have a decreasing water solubility with increasing temperature. This depends upon the thermodynamics of the solution process. According to Le Chatelier's principle, when a stress is applied externally to an equilibrium, the equilibrium is disrupted temporarily and will shift in such a way as to undo the stress that had been applied. One such stress that can be applied is temperature change. According to the principle, increasing the temperature of an equilibrium will always favour the endothermic process of an equilibrium since it is the endothermic process that can absorb the added energy resulting from the increase in temperature. That effectively counteracts the temperature increase. Since most liquid and solid solutes dissolved in water have the solution formation process endothermic, that would be favoured when the temperature was increased resulting in an increase in the solubility limit. However some solutes like $Ce_2(SO_4)_3$ have an exothermic heat of solution. Since increasing the temperature will always favour the endothermic process the dissolution (solution breakdown) process endothermic) will be favoured and the solubility will decrease. Gaseous solutes always have an exothermic heat of solution.
- Consequently, the solubility of all gases in water decrease with increasing temperature. That is why carbonated drinks that have carbon dioxide gas dissolved in them will become "flat" tasting when heated. The sparkle of the drink will have disappeared along with the carbon dioxide gas.
- **Pressure**- Pressure changes above the solution do not affect the solubility limits of solids or liquids dissolved in water. However gaseous solutes are affected. If the pressure of the gas is increased above the gaseous solution then the solubility will be increased in a linear fashion. This was investigated by Henry and resulted in Henry's law.
- **Henry's law** states that "the mass of a gas dissolved per unit volume of a solvent is proportional to the pressure of the gas in equilibrium with the solution at constant pressure". Mathematically, $m \propto P$ or, $m = kP$
  where m = the mass of a gas dissolved per unit volume of a solvent, P = the pressure of the gas in equilibrium with the solution, k = constant called Henry's constant characteristic of the nature of the gas, the nature of the solvent and the temperature.

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- Henry's law can also be stated as "the volume of a gas dissolved in a solvent at a given temperature is independent of the pressure".
- The temperature at which one form of the substance changes into another is called transition temperature.
- For a mixture of gases simultaneously in equilibrium with a liquid, Henry's law states that "the mass of each gas dissolved is directly proportional to its partial pressure".
- **Vapour pressure** of a liquid/solution is the pressure exerted by the vapours in equilibrium with the liquid/solid solution at a particular temperature.
- Liquids which have weak intermolecular forces are more volatile and have a higher vapour pressure. Thus diethyl ether has greater vapour pressure than ethyl alcohol.
- **Raoult's law** states that in a solution, the partial pressure of a component at a given temperature is equal to the mole fraction of that component in the solution multiplied by the vapour pressure of that component in the pure state.

  For a solution containing components A and B,

  $p_A = X_A \times p^0_A$ and $p_B = X_B \times p^0_B$

  $\therefore \quad P_{total} = p_A + p_B = X_A\, p^0_A + X_B\, p^0_B = (1 - X_B)p^0_A + X_B p^0_B$

  or, $P_{total} = (p^0_B - p^0_A)\, X_B + p^0_A$

  We know, $X_A + X_B = 1$

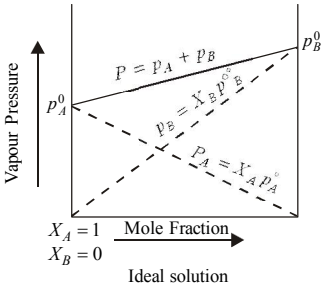
  Substituting number of moles,

  $$P_{total} = p^0_A \frac{n_A}{n_A + n_B} + p^0_B \frac{n_B}{n_A + n_B}$$

  or, substituting weights and molecular weights,

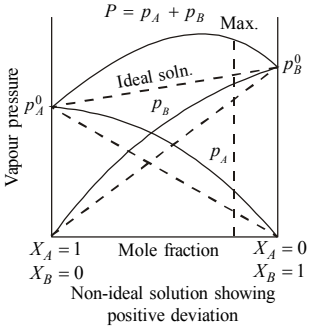
  $$P_{total} = p^0_A \frac{w_A / m_A}{\frac{w_A}{m_A} + \frac{w_B}{m_B}} + p^0_B \frac{w_B / m_B}{\frac{w_A}{m_A} + \frac{w_B}{m_B}}.$$

  Raoult's law is valid only for ideal solutions.



- **Ideal Solution** is a solution in which the interactions between A and B are of the same magnitude as in the pure components, or is a solution which obeys Raoult's law at all temperatures and concentrations. An ideal solution will show no change in volume or enthalpy on mixing. *i.e.*

  $$\Delta V_{mixing} = 0,\ \Delta H_{mixing} = 0$$



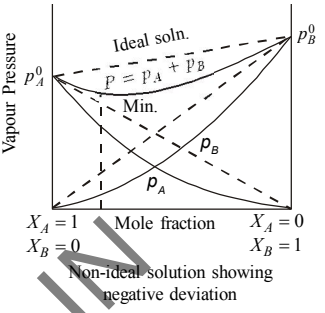
- **Non-Ideal Solution** is a solution in which $A - B$ interactions are of different magnitudes than those in pure components, or $\Delta V_{mixing} \neq 0$ and $\Delta H_{mixing} \neq 0$. These solutions do not obey Raoult's law.
- **Non-Ideal Solutions showing positive deviation** *i.e.* the total vapour pressure determined experimentally is higher than that calculated from Raoult's law. This is due to weaker (lower) interactions than in pure components, *e.g.* ethanol and cyclohexane, the inter-

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molecular H-bonding in ethanol is reduced on adding cyclohexane. Solution showing +ve deviation boils at a lower temperature than expected, also

$$\Delta V_{\text{mixing}} = +\text{ve and } \Delta H_{\text{mixing}} = +\text{ve}.$$

- **Non-Ideal Solutions showing negative deviation** *i.e.* the total vapour pressure determined experimentally is lower than that calculated from Raoult's law. This is due to stronger interactions than in pure components, *e.g.* chloroform and acetone, show –ve deviation due to new H-bonding between chloroform and acetone molecules.
  Solution showing –ve deviation boils at a relatively higher temperature than expected, also show $\Delta V_{\text{mixing}} = -\text{ve and } \Delta H_{\text{mixing}} = -\text{ve}$.



- Some liquids on mixing form azeotropes which are binary mixtures having same composition in liquid and vapour phase and boil at a constant temperature.

Colligative properties

- Dilute solutions containing non-volatile solute exhibit some physical properties which depend only upon the number of solute particles present in the solution irrespective of their nature. These properties are termed as colligative properties.
- **Lowering in vapour pressure :** When a non-volatile solute is added to a solvent the vapour pressure is lowered due to the following reasons:
  (i) Percentage surface area occupied by the solvent decreases. Thus the rate of evaporation and vapour pressure decreases. The solute molecules occupy the surface and so the percent surface area occupied by the solvent decreases.
  (ii) According to Graham's law of evaporation,

  $$\text{rate of evaporation} \propto \frac{1}{\sqrt{\text{density}}}$$

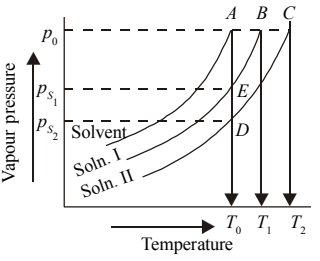
- When a non-volatile solute is dissolved in a liquid, its density increases. Thus, both rate of evaporation and vapour pressure are lowered.
- If p_0 is the vapour pressure of pure solvent and p_S is the vapour pressure of the solution, the difference $(p_0 - p_S)$ is termed lowering in vapour pressure and the ratio $\left[\frac{p_0 - p_S}{p_0}\right]$ is termed relative lowering in vapour pressure.
- According to Raoult's law, the relative lowering in vapour pressure of a dilute solution is equal to mole fraction of the solute present in the solution.

$$\frac{p_0 - p_S}{p_0} = \frac{n}{n + N}.$$

- **Boiling point** is the characteristic temperature of a liquid at which its vapour pressure becomes equal to the atmospheric pressure.
- **Elevation in boiling point** is the increase in boiling point when a non-volatile solute is added to the solvent. Addition of the solute lowers the vapour pressure of solvent, hence more heat is required to increase the vapour pressure upto the atmospheric pressure.

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If boiling point of pure solvent = T_0
Boiling point of solution I = T_1
Boiling point of solution II = T_2



∴ Increase in boiling point of solution I

$$\Delta T = T_1 - T_0 = \Delta T_{b_1}$$

Increase in boiling point of solution II

$$T_2 - T_0 = \Delta T_{b_2}$$

(concentration of solution II > concentration of solution I)

∴ Elevation in boiling point,

$$\Delta T_b \propto \frac{p_0 - p_S}{p_0} \propto \frac{n}{N} \text{ (as in Raoult's law)}$$

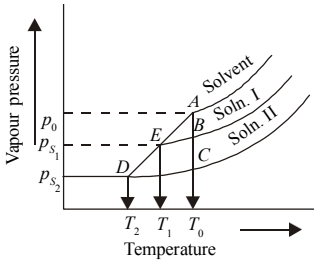
$$\therefore \quad \Delta T_b \propto \frac{w}{m} \times \frac{M}{W} \propto \frac{w \times 1000}{m \times W}$$

(mol. wt. of solvent is constant)

∴ $\Delta T_b \propto$ molality

or $\Delta T_b = K_b \times$ molality $= K_b \times \frac{w \times 1000}{m \times W}$.

- K_b is the *molal elevation constant* or the *molal ebullioscopic constant*. It is defined as the elevation in boiling point produced when 1 mole of solute is dissolved in 1 kg solvent. Unit of K_b = K molality^{-1} or K mol^{-1} kg.
- **Freezing point :** Freezing point of a solvent is the characteristic temperature at which its vapour pressure in the liquid and solid phase becomes same.
- **Depression in freezing point** is the decrease in freezing point when a non-volatile solute is added to the solvent. Addition of a non volatile solute lowers the vapour pressure of the solvent, hence the solid form separates out at a lower temperature.



If freezing point of solvent = T_0
Freezing point of solution I = T_1
Freezing point of solution II = T_2

∴ decrease in freezing point of solution I

$$\Delta T = T_0 - T_1 = \Delta T_{f_1}$$

Decrease in freezing point of solution II

$$T_0 - T_2 = \Delta T_{f_2}$$

(concentration of soln. II > concentration of solution I)

∴ Depression in freezing point,

$$\Delta T_f \propto \frac{p_0 - p_S}{p_0} \propto \frac{n}{N} \text{ (as in Raoult's Law)}$$

$$\therefore \quad \Delta T_f \propto \frac{w \times M}{m \times W}; \propto = \frac{w \times 1000}{m \times W}$$

(mol. wt. of solvent is constant)

∴ $\Delta T_f \propto$ molality or $\Delta T_f = K_f \times$ molality

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or $\Delta T_f = \dfrac{K_f \times w \times 1000}{m \times W}$

- K_f is the *molal depression constant* or the *molal cryoscopic constant*. It is defined as the depression in freezing point produced when 1 mole of solute is dissolved in 1 kg solvent. Unit is the same as K_b; *i.e.* K mol^{-1} kg.
- A specially designed Beckmann's thermometer is used to determine ΔT as its least count is 0.01° and gives more accurate values for small changes.
- Boiling point and freezing points on Beckmann's thermometer should not be taken as actual temperatures as the scale is different.
  Thermodynamic derivation of the molal constants gives another relationship:

$$K = \frac{RT^2}{1000l}, \text{ where } K = K_b \text{ or } K_f$$

R = molar gas constant, T = boiling point or freezing point, l = latent heat of vapourisation or fusion.

- **Osmosis** is the spontaneous movement of solvent molecules from a less concentrated solution to a more concentrated solution, through a semi-permeable membrane.
- **Osmotic pressure** is the equilibrium hydrostatic pressure of the column set up as a result of osmosis.

$$\pi = h \times d \times g$$

- It is also defined as the minimum external pressure applied on the solution in order to prevent osmosis. Two solutions of different substances having same osmotic pressure at same temperature are called *isotonic solutions*. *Hypotonic solution* has lower osmotic pressure than the other solution. *Hypertonic solution* is a solution having higher osmotic pressure than the other solution.
- Osmotic pressure is proportional to the molarity, M of the solution at a given temperature T.

$$\pi = MRT = \frac{n}{V}RT = \frac{w}{mV}RT$$

- **Reverse osmosis :** Osmosis continues till osmotic pressure becomes equal to hydrostatic pressure or osmosis can be stopped by applying external pressure equal to osmotic pressure of solution. If external pressure greater than osmotic pressure is applied, the flow of solvent molecules can be made to proceed from solution towards pure solvent, *i.e.*, in reverse direction of the ordinary osmosis. This is termed as *reverse osmosis*.
- **Abnormal molecular weights** : Abnormal molecular weights and colligative properties are observed in some cases where the experimental and theoretical values differ considerably. They can be explained due to:

– **Dissociation of solute** in water (solvent) increases the number of particles in the solution, resulting in increase in experimental values.

Since colligative property $\propto \dfrac{1}{\text{molecular weight}}$

$\therefore$ Experimental molecular weight < normal molecular weight

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An electrolyte A_xB_y dissociates as:

$$A_xB_y \rightleftharpoons xA^{+y} + yB^{-x}$$

Before dissociation	1	0	0
After dissociation	$1 - \alpha$	$x\alpha$	$y\alpha$

$\therefore$ total number of particles after dissociation

$= (1 - \alpha + x\alpha + y\alpha)$ (α is degree of dissociation)

$$\alpha = \frac{\text{moles dissociated}}{\text{moles present initially}}$$

Since normal colligative property $\propto$ number of particles before dissociation and experimental colligative property $\propto$ number of particles after dissociation

$$\therefore \text{ van't Hoff factor } (i) = \frac{\text{experimental colligative property}}{\text{normal colligative property}}$$

$$= 1 - \alpha + x\alpha + y\alpha$$

$$\text{or } (i) = \frac{\text{normal molecular weight}}{\text{experimental molecular weight}}$$

$$= 1 - \alpha + x\alpha + y\alpha$$

$$\alpha = \frac{i - 1}{(x + y - 1)} = \frac{i - 1}{n - 1} \quad (x + y = n; \text{ if 1 molecule furnishes } n \text{ particles})$$

– **Association of solute particles** in a solvent results in a decrease in number of particles, thereby showing a decrease in the experimental values of colligative properties.

Experimental molecular weight > normal molecular weight

Solute shows association as,

$$nA \rightleftharpoons A_n$$

Before association	1	0
After association	$1 - \alpha$	α/n

$\therefore$ total number of particles after association

$= 1 - \alpha + (\alpha/n)$ (α is degree of association)

$\therefore$ Vant Hoff factor (i)

$$= \frac{\text{experimental colligative property}}{\text{normal colligative property}} = 1 - \alpha + \frac{\alpha}{n}$$

$$= \frac{\text{normal molecular weight}}{\text{experimental molecular weight}} = 1 - \alpha + \frac{\alpha}{n}$$

van't Hoff factor (i)

> 1 for solutes showing dissociation

< 1 for solutes showing association

= 1 for solutes showing neither dissociation nor association

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Value of van't Hoff factor for some common solutes

Solute	Example	Ionisation / association (x – degree)	no. of products (y)	van't Hoff factor $i = [1 + (y - 1)x]$	Abnormal mol. weight
non-electrolyte	urea, glucose etc.	none	1	1	normal mol. wt. (m_1)
Binary electrolyte $A^+ B^-$	NaCl, KCl, CH_3COOH, etc.	$AB \rightleftharpoons A^+ + B^-$ $1-x \quad x \quad x$	2	$(1 + x)$	$\frac{m_1}{(1+x)}$
Ternary electrolyte A_2B, AB_2	K_2SO_4, $K_2[PtCl_6]$, etc.	$A_2B \rightleftharpoons 2A^+ + B^{2-}$ $1-x \quad 2x \quad x$ $AB_2 \rightleftharpoons A^{2+} + 2B^-$ $1-x \quad x \quad 2x$	3	$(1 + 2x)$	$\frac{m_1}{(1+2x)}$
Quarternary electrolyte A_3B, AB_3	$AlCl_3$, $K_3[Fe(CN)_6]$	$A_3B \rightleftharpoons 3A^+ + B^{3-}$ $1-x \quad 3x \quad x$ $AB_3 \rightleftharpoons A^{3+} + 3B^-$ $1-x \quad x \quad 3x$	4	$(1 + 3x)$	$\frac{m_1}{(1+3x)}$
Associated solute	benzoic acid in benzene forming dimer any solute forming polymer A_n	$2A \rightleftharpoons A_2$ $(1-x)$ $A \rightleftharpoons 1/2\ A_2$ $x/2$ $nA \rightleftharpoons A_n$ $(1-x)$ $A \rightleftharpoons 1/n\ A_n$ (x/n)	1/2 1/n	$\left(1-\frac{x}{2}\right)=\left(\frac{2-x}{2}\right)$ $\left[1+\left(\frac{1}{n}-1\right)x\right]$	$\frac{2m_1}{(2-x)}$ $\left[\frac{m_1}{1+\left(\frac{1}{n}-1\right)x}\right]$
General	one mole of solute giving y mol of products	$A \rightleftharpoons yB$	y	$[1 + (y - 1)x]$	$\frac{m_1}{[1+(y-1)x]}$

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End

6

CHAPTER

energetics

- The study of the flow of heat or any other form of energy into or out of a system as it undergoes a physical or chemical transformation is called **thermodynamics**.
- A **system** may be defined as any specified portion of matter under study which is separated from the rest of the universe with a bounding surface.
- The part of the universe other than the system is known as **surroundings**.
- A system which can exchange neither energy nor matter with its surroundings is called an **isolated system**.
- A system which can exchange energy but not matter with its surroundings is called a **closed system**.
- A system which can exchange matter as well as energy with its surroundings is called an **open system**.
- The properties associated with a macroscopic system (*i.e*. consisting of large number of particles) are called **macroscopic properties**. These properties are pressure, volume, temperature, density etc.
- A system is said to be **homogeneous** if it consists of one phase only and **heterogeneous** if it consists of more than one phase.
- A system consisting of two or more immiscible liquids or a solid in contact with a liquid in which it does not dissolve is a heterogeneous system.
- When macroscopic properties of a system have definite values, the system is said to be in a definite state.
- A system in which the macroscopic properties do not undergo any change with time is said to be in **thermodynamic equilibrium**.
- A system is said to be in **thermal equilibrium** if there is no flow of heat from one portion of the system to another. This is possible if the temperature remains the same throughout in all parts of the system.
- A system is said to be in **mechanical equilibrium** if no mechanical work is done by one part of the system on another part of the system. This is possible if the pressure remains the same throughout in all parts of the system.
- A system is said to be in **chemical equilibrium** if the composition of the various phases on the system remains the same throughout.

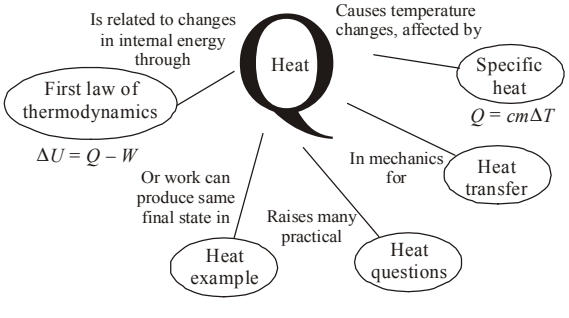
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- The properties of a system which depend on the quantity of matter contained in it are called **extensive properties**. Mass, volume, energy are extensive properties.
- The properties of a system which are independent of the quantity of matter present in it are called **intensive properties**. Temperature, pressure, viscosity, specific heat are intensive properties.
- A physical quantity is said to state function if its value depends only upon the state of the system and does not depend upon the path by which this state has been attained.

A thermodynamic process may be defined as the energetic evolution of a thermodynamic system proceeding from an initial state to a final state. Typically, each thermodynamic process is distinguished from other processes, in energetic character, according to what parameters, as temperature, pressure, or volume, etc., are held fixed. The five most common thermodynamic processes are shown below:

- An isobaric process occurs at constant pressure.
- An isochoric process occurs at constant volume.
- An isothermal process occurs at constant temperature.
- An isentropic process occurs at constant entropy.
- An adiabatic process occurs without loss or gain of heat.

- A process carried out infinitesimally slowly so that the driving force is only infinitesimally greater than the opposing force is called a **reversible process**.
- Any process which does not take place infinitesimally slowly is said to be an **irreversible process**.
- In C.G.S. system, the unit of energy is **erg**. It is defined as the work done when a resistance of 1 dyne is moved through a distance of 1 centimetre.
- If a system absorbs heat q from the surroundings, q is positive and if the system gives out heat q to the surroundings, q is taken as negative.
- **Heat energy** is measured by the product of temperature (intensity factor) and the heat capacity (capacity factor) of the system.
- Electrical work done = E.M.F. × quantity of electricity.



- The **first law of thermodynamics** states that energy can neither be created nor destroyed although it can be transformed from one form to another. This is also known as law of conservation of energy.

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- **Internal energy** of a substance or a system is a definite quantity and it is a function only of the state (*i.e.*, chemical nature, composition, temperature, pressure and volume) of the system at the given moment, irrespective of the manner in which that state has been brought about.
- The **change of internal energy** ΔE is given as
$$\Delta E = E_B - E_A = q - W$$
where E_A and E_B is the energy of a system in its state A and state B respectively, q is the heat absorbed by the system while undergoing change from state A to state B, and the work done is equal to W. The above equation is mathematical statement of first law of thermodynamics.
- **Enthalpy** of a system may be defined as the sum of its internal energy and pressure-volume (PV) energy. It is denoted by H. Thus $H = E + PV$
where E is the internal energy and P and V are pressure and volume of the system respectively.
- $\boldsymbol{\Delta H = \Delta E + \Delta n_g RT}$
where ΔH = change in enthalpy, ΔE = change in internal energy, R is gas constant, and T is the temperature on Kelvin scale. Δn_g is the change in number of moles of gaseous products and gaseous reactants.
- For a reaction (or a process) taking place at constant temperature and at constant pressure the enthalpy change is equal to the amount of heat evolved or absorbed (q_P) in it.
$$\Delta H = q_P$$
- The total amount of work done by the isothermal reversible expansion of the ideal gas from V_1 to V_2 is
$$W = -nRT \ln \frac{V_2}{V_1}$$
- **Enthalpy of formation** is defined as the enthalpy change in accompanying the formation of one mole of a compound from its constituent element at a given temperature and pressure. It is denoted by ΔH_f.
- The standard state of an element is the pure element in its stable form or more common form under standard conditions of 1 atm and 298 K.
- The standard state of oxygen, carbon, mercury and sulphur are oxygen gas, graphite, liquid mercury and rhombic sulphur at 1 atm pressure at 298 K.
- The enthalpy of formation of any element in the standard state is taken as zero, *i.e.*, $\Delta H_f^\circ = 0$.
- The standard heat of formation of graphite is 0.0 whereas that of diamond is not zero but equal to 1.896 kJmol^{-1}.
- The **enthalpy of combustion** of a substance is defined as the amount of heat evolved when 1 mole of the substance is completely burnt or oxidised.
- The change in enthalpy (ΔH) when a liquid changes into vapour state or when vapour changes into liquid state is known as **enthalpy of vaporisation**.
- The standard enthalpy of a reaction (ΔH°) is the difference of the standard enthalpies of all the products and standard enthalpies of all the reactants.

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$\Delta H^\circ = \Sigma\, \Delta H_f^\circ$ (products) – $\Sigma\, \Delta H_f^\circ$ (reactants)

- **Heat capacity** of a system between any two temperatures is defined as the quantity of heat required to raise the temperature of the system from the lower to the higher temperature of the divided by the temperature difference.

$$C_{(T_2, T_1)} = \frac{q}{T_2 - T_1} = \frac{q}{\Delta T}$$

- Heat capacity at constant volume is defined as the rate of change of internal energy with temperature at constant volume.

$$C_V = \left(\frac{\partial E}{\partial T}\right)_V$$

- The heat capacity at constant pressure is defined as the rate of change of enthalpy with temperature at constant pressure.

$$C_P = \left(\frac{\partial H}{\partial T}\right)_P$$

- For one mole of the heat capacities at constant volume and at constant pressure are denoted by C_V and C_P respectively. These are termed molar heat capacities. Thus for one mole of the gas

$$C_V = \left(\frac{\partial E}{\partial T}\right)_V, \quad C_P = \left(\frac{\partial H}{\partial T}\right)_P$$

- The difference between the molar heat capacity of a gas at constant pressure (C_P) and at constant volume (C_V) is equal to the gas constant *R viz.* 1.987 cal or 8.314 J.

$$C_P - C_V = R$$

- When a gas is heated at constant volume, no external work is done by the gas. But when a gas is heated at constant pressure, the gas will expand and do some external work. Hence the molar heat capacity of a gas at constant pressure (C_P) must be greater than at constant volume (C_V).
  i.e. $C_P > C_V$.

- Ratio of two specific heats, *i.e.* $\frac{C_P}{C_V} = \gamma$

  $\gamma = 1.66$, the gas is monoatomic *e.g.* He, Ne, Ar, Kr, Xe
  $\gamma = 1.40$, the gas is diatomic e.g. O_2, H_2, Cl_2, N_2 etc.
  $\gamma = 1.33$, the gas is polyatomic e.g. SO_3, O_3, CO_2 etc.

- For an isothermal process, ΔT, ΔE and ΔH are zero.
- In an isothermal expansion, the work is done at the expense of the heat absorbed.

$$q = W$$

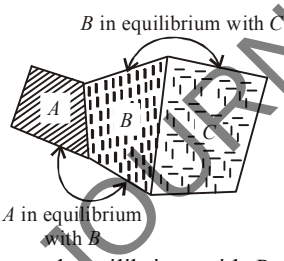
- The temperature below which a gas becomes cooler on expansion is known as the **inversion temperature**.
- The phenomenon of change of temperature produced when a gas is made to expand adiabatically from a region of high pressure to a region of extremely low pressure is known as the **Joule-Thomson effect**.

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- The number of degrees temperature produced per atmosphere drop in pressure under constant enthalpy conditions on passing a gas through the porous plug is called **Joule-Thomson coefficient**.

  Thus, $\mu = \dfrac{dT}{dP}$

- Joule-Thomson effect is zero in an ideal gas. *i.e.* when an ideal gas expands in vacuum, there is neither absorption nor evolution of heat. *i.e.* $q = 0$.
- When an ideal gas undergoes expansion under adiabatic conditions in vacuum, no change takes place in its internal energy. *i.e.* $\left(\dfrac{\partial E}{\partial V}\right)_T = 0$.
- The quantity $\left(\dfrac{\partial E}{\partial V}\right)_T$ is called the **internal pressure**. Thus internal pressure of an ideal gas is zero.
- **Zeroth law of thermodynamics** states that if two systems are at the same time in thermal equilibrium with a third system, they are in thermal equilibrium with each other.



Therefore *A* and *C* are in thermal equilibrium. If they were brought in contact, there would be no net heat transfer.

If *A* and *C* are in thermal equilibrium with *B*, then *A* is in thermal equilibrium with *B*. Practically this means that all three are at the same temperature, and it forms the basis for comparison of temperatures. It is so named because it logically preceds the first and second law of thermodynamics.

- Work done in reversible compression of an ideal gas is given by

$$W = nRT \ln\left(\frac{V_1}{V_2}\right) = nRT \ln\left(\frac{P_2}{P_1}\right).$$

- In the case of expansion of a gas the work is done by the system on the surroundings and W has a positive sign.
- In the case of compression of a gas, the work is done by the surroundings on the system and W has a negative sign.
- When a gas expands freely, *i.e.*, when it expands against vacuum such $p_{ext} = 0$, no work is done by the system.

  i.e. work of expansion, $W = -p_{ext}\,\Delta V \quad [\because p_{ext} = 0]$

  $W = 0$.
- For an isochoric process, *i.e.* for which $\Delta V = 0$, change in internal energy, $\Delta E = q_V$.
- For an isobaric process, *i.e.* for which $\Delta P = 0$, change in enthalpy, $\Delta H = q_P$, where q_P is the amount of heat changes at constant pressure.

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- For an adiabatic process, *i.e.* for which $q = 0$, work done, $W = \frac{nR(T_2 - T_1)}{\gamma - 1}$

  where R is gas constant (8.314 $JK^{-1}mol^{-1}$) and $\gamma = C_P/C_V$ (where C_P and C_V are molar heat capacities at constant pressure and constant volume). T_2 and T_1 are final and initial temperatures on Kelvin scale.
- The integral enthalpy of solution is defined as the enthalpy change when one mole of solute is dissolved in a definite quantity say '*n*' moles of solvent to get a solution of specified concentration.
- **Enthalpy of dilution** is defined as the enthalpy change that occurs when a solution containing one mole of a solute is diluted from one concentration to another concentration.
- **Enthalpy of sublimation** is defined as the amount of heat required to change one mole of the solid completely into vapour at a constant temperature.
- The **enthalpy of neutralisation** is defined as the enthalpy change which accompanies the complete neutralisation of one gram equivalent of an acid by a base.
- $\Delta H_{\text{neutralisation}}$ is constant for strong acids and base neutralisation.
  ΔH = –13.7 kcal/mole = –57.27 kJ/mol.
- Heat of neutralisation for weak acids (HCN, CH_3COOH, benzoic acid) and weak bases (NH_4OH, amines) is lower than that for strong acids and bases. The reason is that heat is absorbed in complete ionisation of weak acids and bases (unlike in case of strong acids and bases where no heat is required).
- **Hess's law** states that "the enthalpy change in a chemical or a physical process is same whether the process is carried out in one step or in several steps".
  $\Delta H = \Delta H_1 + \Delta H_2$.
- Hess's law is useful in determining the enthalpies of transition of allotropic modifications such as graphite to diamond, rhombic sulphur to monoclinic sulphur, yellow phosphorus to red phosphorus etc.
- **Bond energy** is the energy released when gaseous atoms form molecules.
- **Bond dissociation energy** can be defined as the energy required to break one mole of a particular type of bonds in gaseous molecules so as to get the separated atoms in gaseous state.
- The branch of chemistry which deals with energy changes in chemical reactions is called **thermochemistry**.
- Heat of reaction at constant volume and at a given temperature is given by the difference in the internal energies of the products and the reactants, the quantities of the products and the reactants being the same.
  $\Delta E = E_P - E_R = q_V$ = heat of reaction at constant volume
- Heat of reaction at constant pressure and at a given temperature is given by the difference in the enthalpies of the products and the reactants, the quantities of the products and reactants being the same as represented by the chemical equilibrium.
  $\Delta H = H_P - H_R = q_P$ = heat of reaction at constant pressure
- Variation of heat of reaction with temperature is given by Kirchoff's equation *viz*.

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$$\left(\frac{\partial(\Delta H)}{\partial T}\right)_P = \Delta C_P \quad \text{or,} \quad \frac{\Delta H_2 - \Delta H_1}{T_2 - T_1} = \Delta C_P$$

where $\Delta C_P = \Sigma$ heat capacities of product $- \Sigma$ heat capacities of reactants.

- If the solubility of a substance is known at two different temperatures, the mean molar enthalpy of solution over this temperature range can be calculated by applying an equation similar to van't Hoff equation (relating equilibrium constant with temperature). The equation is

$$\log \frac{C_2}{C_1} = \frac{\Delta H}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right).$$

where C_1 and C_2 are solubilities at temperatures T_1 and T_2 respectively.

- A process which proceeds of its own accord without any outside assistance is termed a **spontaneous** or **natural process**.
- The process which has no natural tendency or an urge to occur is said to be a **non-spontaneous process**.
- The tendency to attain minimum energy *i.e.* a negative value of enthalpy change ΔH might be responsible for a process or a reaction to be spontaneous or feasible.
- Dissolution of common salt in water, evaporation of water in an open vessel and flow of water down a hill are spontaneous processes.
- Flow of water up a hill, flow of heat from a cold body to a hot body and dissolution of sand in water are non-spontaneous process.
- **Enthalpy** is a measure of randomness or disorder of the system.
- For a given substance, the crystalline solid state has the lowest entropy, the gaseous state has the highest entropy and the liquid state has the entropy between the two.
- The greater the randomness of a system, higher is its entropy.
- At absolute zero, a perfectly crystalline substance has zero entropy.
- The entropy change, ΔS for a chemical reaction is equal to the sum of entropies of the product minus sum of entropies of reactants.

Entropy is

- a state variable whose change is defined for a reversible process at T where Q is the heat absorbed.
- a measure of the amount of energy which is unavailable to do work.
- a measure of the disorder of a system.
- a measure of the multiplicity of a system.

$$\Delta S = Q/T$$

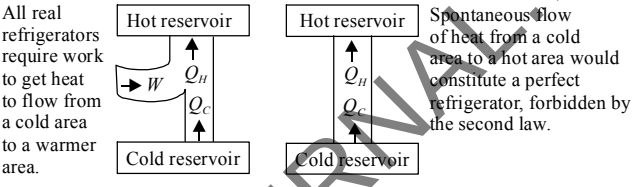
 low entropy  high entropy

- **Entropy change** during a process is defined as the amount of heat (q) absorbed isothermally and reversibly (infinitesimally slowly) divided by the absolute temperature (T) at which the heat is absorbed.

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$$\Delta S = \frac{q_{\text{rev, iso}}}{T}.$$

- Entropy of fusion may be defined as the entropy change taking place when one mole of the substance changes from solid state into liquid state at its melting point.
- **Entropy of vaporisation** may be defined as the entropy change taking place when one mole of the substance changes from liquid state into vapours at its boiling point.
- **Claussius statement** of the second law is given as "The energy of the universe remains constant, the entropy of the universe tends towards a maximum".
- **Second law in refrigerator :** It is not possible for heat to flow from a colder body to a warmer body without any work having been done to accomplish this flow. Energy will not flow spontaneously from a low temperature object to a higher temperature object. This precludes a perfect refrigerator. The statements about refrigerators apply to air conditioners and heat pumps, which embody the same principles. This is the "second form" or Clausius statement of the second law.



- In a reversible process, the entropy of the system and the surroundings taken together remains constant while in an irreversible process, the entropy of the system and the surroundings increases.
- Suppose one mole of a substance melts reversibly at the fusion point T_f, at constant pressure. Let H_f be the molar heat of fusion. The entropy change of the process, ΔS_f will be then given by

$$\Delta S_f = \frac{\Delta H_f}{T_f}.$$

- Suppose 1 mole of a substance changes from liquid to vapour state reversibly at its boiling point T_b under a constant pressure. If ΔH_v is the molar heat of vaporisation, then the entropy change accompanying the process will be given by

$$\Delta S_v = \Delta H_v / T_b.$$

- If we consider the change of state from vapour to liquid or from liquid to solid, ΔH_v and ΔH_f will be both negative and hence the process of condensation of vapour or freezing of a liquid is accompanied by decrease of entropy.
- The change in entropy when 1 mole of a solid substance undergoes change of state from one crystalline form (say rhombic form) to another crystalline form (say monoclinic form) at the transition temperature T, is given by

$$\Delta S_T = \frac{\Delta H_t}{T}$$

where ΔH_t is the molar heat of transition of the substance.

- **Molar heat of transition** of the substance ΔH_t is the amount of heat absorbed or evolved by one mole of a substance when it undergoes change of state from one crystalline form to another at transition temperature T.

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- Entropy change of an ideal gas is given by

$$\Delta S = nC_V \ln\left(\frac{T_2}{T_1}\right) + nR \ln\left(\frac{V_2}{V_1}\right).$$

- Entropy change for n moles of an ideal gas is given by

$$\Delta S = nC_P \ln\left(\frac{T_2}{T_1}\right) - nR \ln\left(\frac{P_2}{P_1}\right).$$

- **Entropy of mixing** is defined as the difference between the entropy of the mixture of gases and the sum of entropies of the separate gases, each at a pressure P.
- For a total of 1 mole of the gaseous mixture, the entropy of mixing is given by $\Delta S_{mix} = -R \Sigma x_i \ln x_i$
  Since x_i is a fraction, the entropy of mixing is always positive.
- The total entropy change (ΔS) of A and B involving transfer of heat from a body B at a higher temperature T_2 to a body A at a lower temperature T_1, is given by

$$\Delta S = \frac{q_{rev}(T_2 - T_1)}{T_1 T_2}$$

- The entropy change of a chemical reaction is given by the difference between the sum of the entropies of all the products and the sum of entropies of all the reactants.

$$\Delta S = \Sigma S_{products} - \Sigma S_{reactants}.$$

- Entropy of 1 mole of a substance in pure state at one atmospheric pressure and 25°C is termed as **standard entropy** of that substance and is denoted as $S°$.

$$\Delta S° = \Sigma S°_{products} - \Sigma S°_{reactants}.$$

- Entropy is regarded as a measure of the disorder of a system.
- **Gibb's free energy** is defined as the amount of energy available from a system that can be put into useful work. Mathematically, free energy G is defined by the following relation, $G = H - TS$
  where H = enthalpy of the system
  S = entropy of the system
  T = temperature on Kelvin scale.
- **Gibb's-Helmholtz equation** is $\Delta G = \Delta H - T\Delta S$
  where $\Delta H = H_2 - H_1$ is the enthalpy of the system
  $\Delta S = S_2 - S_1$ is the entropy of the system
  T = absolute temperature
  $\Delta G = G_2 - G_1$ is the change in free energy of the system
- (i) If ΔG is negative, the process will be spontaneous.
  (ii) If ΔG is zero, the process is in equilibrium.
  (iii) If ΔG is positive, the direct process is non-spontaneous; the reverse process may be spontaneous.
- The standard free energy change is defined as the **free energy change** for a process in which reactants in their standard state are converted into the products in their standard state. It is denoted by the symbol $\Delta G°$.

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- The standard free energy of formation of a compound is defined as the free energy change which takes place when one mole of the compound is formed from its elements taken in their standard states.
- For a reaction in equilibrium, the standard free energy change is related to the equilibrium constant of the reaction according to the relation

$$\Delta G^\circ = -RT \ln K \quad \text{or} \quad \Delta G^\circ = -2.303RT \log K$$

where R is the gas constant.

- Entropy change of an ideal gas (for 1 mole),

$$\Delta S = C_V \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1} = C_P \ln \frac{T_2}{T_1} + R \ln \frac{P_1}{P_2}$$

At constant temperature (isothermal process),

$$\Delta S_T = R \ln \frac{V_2}{V_1} = R \ln \frac{P_1}{P_2}$$

At constant volume (isochoric process),

$$\Delta S_V = C_V \ln \frac{T_2}{T_1}$$

At constant pressure (isobaric process),

$$\Delta S_P = C_P \ln \frac{T_2}{T_1}$$

- $\Delta G^\circ_{\text{reaction}} = \Sigma\, \Delta G_f^\circ(\text{products}) - \Sigma\, \Delta G_f^\circ(\text{reactants})$
- For elementary substances, $\Delta G_f^\circ = 0$.
- The entropy of a perfectly crystalline solid is zero at the absolute zero of temperature.
- Entropy of every substance (element or compound) in the standard state is not equal to zero.
- **Nernst** in 1906 formulated the **third law of thermodynamics** which states that "at absolute zero the entropy of a perfectly crystalline substance is zero".
- In a perfect crystal, at absolute zero temperature, each atom must be at a crystal lattice point and it must have lowest energy. This means that this particular state is of perfect order, *i.e.*, zero disorder and hence of zero entropy.
- In case of solids,

$$\Delta S = \int_0^T \frac{C_P dT}{T} = C_P \ln T = 2.303 C_P \log T$$

where C_P is the heat capacity of the substance at constant pressure and is supposed to remain constant in the range 0 to T K.

- If a system returns to its original state after undergoing a number of successive changes, it is said to be a **cyclic process**.
- The fraction of the heat absorbed by a machine that is converted into work is called the **efficiency of the machine**. It is given by

$$\eta = \frac{W}{Q_2} = \frac{Q_2 - Q_1}{Q_2} = \frac{T_2 - T_1}{T_2}.$$

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where Q_2 = heat absorbed from the source temperature T_2, Q_1 = heat rejected to the sink at temperature T_1.

- **Criteria for spontaneity of reaction**

ΔH	ΔS	ΔG	Reaction Characteristics	Example
–	+	Always negative	Reaction is spontaneous at all temperatures	$2O_{3(g)} \rightarrow 3O_{2(g)}$
+	–	Always positive	Reaction is non - spontaneous at all temperatures	$3O_{2(g)} \rightarrow 2O_{3(g)}$
–	–	Negative at low temperature but becomes non - spontaneous at high temperature	Reaction is spontaneous at low temperature	$CaO_{(s)} + CO_{2(g)} \rightarrow CaCO_{3(s)}$
+	+	Positive at low temperature but negative at high temperature	Reaction is non-spontaneous at low temperature but becomes spontaneous at high temperature	$CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$

End

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7

CHAPTER

equilibrium

- **Chemical equilibrium** may be defined as the state of a reversible reaction when the two opposing reactions occur at the same rate and the concentrations of reactants and products do not change with time.
- At equilibrium, both reactions *i.e.* forward and backward continue to perform and such a state of equilibrium where both opposing forces balance each other and molecular activity still continues is known as **dynamic equilibrium**.
- At equilibrium, the **Gibb's free energy** (G) is minimum and any change taking place at equilibrium proceeds without change in free energy *i.e.* $\Delta G = 0$.
- According to **law of mass action**, the rate of a chemical reaction is proportional to the product of the active masses of the reactants.
- A **catalyst** can hasten the approach of equilibrium but does not alter the state of equilibrium.
- For a reaction $A + B \rightleftharpoons C + D$,

$$\frac{k_1}{k_2} = K = \frac{[C][D]}{[A][B]}$$

where K is known as the equilibrium constant.

- The ratio between the products of molar concentration of the products to that of the reactants with each concentration term raised to a power equal to its stoichiometric coefficient is known as **equilibrium constant**.
- Equilibrium constant (K) has no unit, means it is dimensionless, if the total number of mole of the product is exactly equal to the total number of mole of reactants.
- If the number of moles of products and reactants are not equal, then equilibrium constant (K) has specific units.
- The magnitude of the equilibrium constant is a measure of the completion of a reversible reaction.
- Larger the value of K, the greater will be the equilibrium concentration of the components on the right hand side of the reaction relative to the left hand side.

- When the coefficient of a given reaction equations are multiplied by 1/2, then K for new reaction equation is the square root of K.
  e.g. $H_{2\,(g)} + I_{2\,(g)} \rightleftharpoons 2HI_{\,(g)}$; $K_1 = 48$

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$\frac{1}{2}H_{2\,(g)} + \frac{1}{2}I_{2\,(g)} \rightleftharpoons HI_{\,(g)}$; $K_2 = \sqrt{48}$

- When the coefficient of a given reaction equations are doubled, then K for the new equation is the square of the original K.
  e.g. $H_{2\,(g)} + I_{2\,(g)} \rightleftharpoons 2HI_{\,(g)}$; $K_1 = 48$
  $2H_{2\,(g)} + 2I_{2\,(g)} \rightleftharpoons 4HI_{\,(g)}$; $K_2 = 48^2$
  In general, when we multiply the terms of an equation by a certain value, we must change the K, for the equation to a power equal to that value.
  e.g. $nH_{2\,(g)} + nI_{2\,(g)} \rightleftharpoons 2nHI_{\,(g)}$; $K_2 = (K_1)^n$
- When a given reaction equation is reversed, then the K for the reverse reaction equation is the reciprocal of the K for the original equation.
  e.g. $H_{2\,(g)} + I_{2\,(g)} \rightleftharpoons 2HI$; $K_1 = 48$
  $2HI \rightleftharpoons H_{2\,(g)} + I_{2\,(g)}$; $K_2 = 1/48$

- The value of the equilibrium constant (K) is independent of the original concentration of reactant and has a definite value for every reaction at a particular temperature.
- For a general gaseous reaction, $aA + bB + \ldots \rightleftharpoons lL + mM + \ldots$,

  the equilibrium constant K_P is given by $K_P = \frac{[P_L]^l \times [P_M]^m \ldots}{[P_A]^a \times [P_B]^b \ldots}$

  i.e. partial pressure is taken in place of molar concentration.
- Equilibrium constants K_P and K_C are related as
  $K_P = K_C \times (RT)^{\Delta n_g}$
  where $\Delta n_g = (l + m) - (a + b)$ is the difference of the sum of the coefficients for the gaseous products and reactants.
- **van't Hoff's isotherm** is given by the equation $\Delta G = \Delta G^\circ + RT \ln J$
  where ΔG° is the free energy change of the reaction when the products and the reactants are all in their respective standard states and J stands for the reaction coefficient of partial pressures of the products and reactants.
- $\Delta G^\circ = -RT \ln K_P$ is the equation of the law of chemical equilibrium and it permits calculations of ΔG° of a reaction from the known value of its equilibrium constant and vice versa.
- $\frac{d \ln K_P}{dt} = \frac{\Delta H^\circ}{RT^2}$ is known as **van't Hoff's equation**.
  where ΔH° is the standard enthalpy change for the reaction at constant pressure when the reactants as well as the products are in their standard states.
- $\log \frac{K_P''}{K_P'} = \frac{\Delta H^\circ}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$
  where K_P'' = equilibrium constant at temperature T_2, K_P' = equilibrium constant at temperature T_1.
- In **homogeneous equilibrium**, all the reactants and products are in the same phase.

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- In **heterogeneous equilibrium**, the reactants and products are present in two or more phases.
- For gaseous reactions in which the number of molecules remains the same, for example $H_{2\,(g)} + I_{2\,(g)} \rightleftharpoons 2HI_{(g)}$
  the value of K_P and K_C are identical because $\Delta n_g = 0$;
  $\therefore\ K_P = K_C$.
- The equilibrium constant is independent of the presence of catalyst because the catalyst affects rate of forward and backward reaction equally.
- The temperature dependence of the equilibrium constant is given by the expression $\log \frac{K_2}{K_1} = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$.
- If an equilibrium is subjected to a change in concentration, pressure or temperature etc. equilibrium shift in such a way so as to undo the effect of a change imposed and this is known as **Le - Chatelier's principle**.
- When an inert gas is added to the equilibrium at constant volume, the total pressure will increase. But the concentration of reactant and product will not change. Hence there will be no effect on the equilibrium.
- Increase in concentration of reactants $\xrightarrow{\text{shifts the equilibrium to}}$ forward reaction.
  Increase in concentration of products $\xrightarrow{\text{shifts the equilibrium to}}$ backward reaction.
- The effect of temperature on equilibrium system may be summed up as

  increase in temperature $\xrightarrow{\text{shifts the equilibrium to}}$ endothermic reaction.

  Decrease in temperature $\xrightarrow{\text{shifts the equilibrium to}}$ exothermic reaction.
- In most cases, formation of solution (solute in solvent) is an endothermic process. In such cases increasing temperature increases the solubility of solutes. In cases, where dissolution of solute is followed by evolution of heat, increasing temperature lowers the solubility of solutes.
- **Ionic equilibrium** is the study of equilibrium in the reactions where formation of ions take place in aqueous solution.
- According to **Arrhenius theory**, an acid is a compound that releases H^+ ions in water and base is a compound that releases OH^- ions in water.
- According to **Bronsted and Lowry concept**, an acid is any molecule or ion that can donate a proton (H^+) and base is any molecule or ion that can accept a proton.
- The acid (HA) and its conjugate base (A^-) that are related to each other by donating and accepting a single proton are said to constitute a conjugate acid-base pair.
- A **weak base** has strong conjugate acid and a weak acid has a strong conjugate base.

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- A strong acid has a weak conjugate base and a strong base has a weak conjugate acid.
- **Monoprotic acids** are capable of donating one proton and **monoprotic bases** can accept one proton.
- **Polyprotic acids** are capable of donating two or more protons and **polyprotic bases** can accept two or more protons.
- Molecules or ions that can behave both as Bronsted acid and base are called **amphiprotic substances**.
- The strength of a Bronsted acid depends upon its tendency to donate a proton. The strength of a Bronsted base depends upon its ability to accept a proton.
- Some common acids have been arranged in the following order of their acid strengths.
  $HClO_4 > HBr > H_2SO_4 > HCl > HNO_3$
- According to **G . N. Lewis**, an acid is an electron-pair acceptor and base is an electron-pair donor.
- For the dissociation of a weak monobasic acid HA in water, represented by the equation
  $HA + H_2O \rightleftharpoons H_3O^+ + A^- \; [H_3O^+ = H^+]$
  Applying the law of chemical equilibrium,
  $$K_a = \frac{[H^+][A^-]}{[HA]}$$
  where K_a is called **acid dissociation constant**.
- The value of K_a for a particular acid is a measure of its acid strength or acidity.
- The value of acid dissociation constant is large for a strong acid while it is small for a weak acid.
- According to Ostwald's dilution law for weak electrolytes the degree of dissociation is inversely proportional to the square root of concentration *i.e.* $\alpha \propto \frac{1}{\sqrt{C}}$ or, $\alpha = \sqrt{\frac{K_a}{C}}$
- For two weak acids of dissociation constant K_{a_1} and K_{a_2} at the same concentration,
  $$\frac{\alpha_1}{\alpha_2} = \sqrt{\frac{K_{a_1}}{K_{a_2}}} \qquad \left[\because \alpha = \sqrt{\frac{K_a}{C}}\right]$$
  where α_1 and α_2 are the respective degrees of dissociation of the two acids.
- **Degree of dissociation** of an acid is a measure of its capacity to furnish hydrogen ions and hence a measure of its strength.
  $$\frac{\text{strength of one acid, } HA_1}{\text{strength of another acid, } HA_2} = \sqrt{\frac{K_{a_1}}{K_{a_2}}}$$
  Thus, relative strengths of any two weak acids at the same concentration are given by the ratio of the square-roots of their dissociation constants.
- The **hydrogen ion concentration** in a solution of a weak acid in water at a given concentration is directly proportional to the square root of the dissociation constant of the acid.

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$$[H^+] = C\alpha = C\sqrt{\frac{K_a}{C}} = \sqrt{CK_a}$$

- The **strength of a base** is defined as the concentration of OH^- ions in its aqueous solution at a given temperature.
- The dissociation constant K_b of the base is given by

$$K_b = C\alpha^2 \quad \text{or,} \quad \alpha = \sqrt{\frac{K_b}{C}} \quad \text{or,} \quad [OH^-] = C\alpha = C\sqrt{\frac{K_b}{C}} = \sqrt{CK_b}.$$

- $K_w = [H^+]\,[OH^-]$ where K_w is called the dissociation constant or more commonly the ionic product of water.
- An aqueous solution in which hydrogen ion concentration is greater than 1×10^{-7} mole per litre is said to be **acidic**.
- An aqueous solution in which hydrogen ion concentration is less than (*i.e.* hydroxyl ion concentration is greater than) 1×10^{-7} mole per litre is said to be **alkaline**.
- The **pH of a solution** is the negative logarithm of the concentration (in moles per litre) of hydrogen ions which it contains. $pH = -\log[H^+]$.
- If pH value of a solution is 7, it is neutral, if pH value is less than 7, the solution is acidic and if it is more than 7, the solution is alkaline.
- The suppression of the dissociation of a weak acid or a weak base on the addition of an electrolyte containing its own ion is called **common ion effect**.
- A **buffer solution** is one which can resist change in its pH value on the addition of an acid or base.
- The capacity of a solution to resist alternation in its pH value is known as its **buffer capacity**.
- The scale on which pH values are computed is called the **pH scale**.
- The lower the pH, higher is the $[H^+]$ or acidity.
- For any aqueous solution at 25°C, pH + pOH = 14.00.
- A weak acid together with a salt of the same acid with a strong base are called **acid buffers**.e.g. $CH_3COOH + CH_3COONa$
- A weak base and its salt with a strong acid are called **basic buffers**. e.g. $NH_4OH + NH_4Cl$.
- $pH = pK_a + \log\frac{[salt]}{[acid]}$ is known as **Henderson's** equation and enables the calculation of pH values of buffer solutions made by mixing known concentrations of a weak acid and its salt.
- The phenomenon of the interaction of anions and cations of the salt with the H^+ and OH^- ions furnished by water yielding acidic or alkaline or sometimes even neutral solution is known as **salt hydrolysis**.
- Salts of strong acids and bases do not undergo hydrolysis. e.g. KCl.
- A buffer solution is assumed to be destroyed if an addition of strong acid or base, changes its pH by 1 unit. *i.e.* pH (new) = $pK_a \pm 1$. This means the ratio

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$\frac{[\text{salt}]}{[\text{acid}]}$ or $\frac{[\text{salt}]}{[\text{base}]} = 10$ or, $\frac{1}{10}$.

- The aqueous solution of the salt of a weak acid and a strong base is alkaline because of hydrolysis.
- The aqueous solution of the salt of weak base and strong acid is acidic because of hydrolysis.
- The Henderson-Hasselbalch equation for a basic buffer is given as

$$\text{pOH} = \text{p}K_b + \log\frac{[\text{salt}]}{[\text{base}]}$$

- The reaction of an anion or cation with water accompanied by cleavage of O – H bond is called **hydrolysis**.
- The **solubility product** of a sparingly soluble salt forming a saturated solution in water is given by the product of the concentrations of the ions each raised to a power equal to the number of ions produced on dissociation of one molecule of the electrolyte.
- **Degree of hydrolysis** is defined as the fraction of the total salt that has undergone hydrolysis on the attainment of the equilibrium.
- The hydrolysis constant K_h of the salt varies inversely with the dissociation constant K_a of the weak acid.

$$K_h = K_w/K_a.$$

- Weaker the acid, the greater is the hydrolysis constant of the salt.
- The degree of hydrolysis increases when concentration decreases *i.e.* dilution increases.
- The degree of hydrolysis of a salt of a weak acid and strong base at any concentration C of the salt can be calculated using the relation

$$h = \sqrt{\frac{K_w}{K_a \times C}}$$

provided the dissociation constant K_a of the acid is known.

- For salts of weak bases and strong acids, degree of hydrolysis is given as

$$h = \sqrt{\frac{K_w}{K_b \times C}}$$

- For salts of weak acids and weak bases, degree of hydrolysis is given as $h = \sqrt{\frac{K_w}{K_a K_b}}$
- The weaker the base, the greater is the hydrolysis constant of the salt and hence greater is the degree of hydrolysis.

$$K_h = K_w/K_b$$

- The pH of a salt of a weak acid and strong base can be calculated using the relation

$$\text{pH} = 7 + \frac{1}{2}[\text{p}K_b + \log C]$$

- The pH of salts of weak bases and strong acids can be calculated by using the relation

$$\text{pH} = 7 - \frac{1}{2}[\text{p}K_b - \log C]$$

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- The pH of a salt of a weak acid and weak base is given by
  at 25°C, $pH = 7 + \frac{1}{2}(pK_a - pK_b)$
- An **acid-base indicator** is an organic dye that signals the end-point by a visual change in colour.
- A plot of pH against the volume of the solution being added is known as **pH curve** or **titration curve**.
- A **suitable indicator** for a given titration may be defined as one which has a narrow pH range as possible that lies entirely on the upright part of the titration curve.
- Both methyl orange and phenolphthalein are suitable indicators for strong acids and strong bases titrations.
- The pH range of methyl orange is (3.1 – 4.4) and phenolphthalein (8.3 – 10.0).
- For a weak acid-strong base titration, **phenolphthalein** is a suitable indicator while methyl orange is not.
- Methyl orange and methyl red are suitable indicators for strong acids/weak base titrations.
- The **solubility** (S) of a substance in a solvent is the concentration in the saturated solution.
- **Molar solubility** is defined as the number of moles of the substance per litre (L) of the solution.
- Whenever the product of concentrations of ions of a substance present in solution exceeds the solubility product of that substance, the substance gets precipitated.
- The compound with the lower solubility product gets precipitated in preference.
- Silver iodide has lower solubility product than silver chloride, and so the former gets precipitated in preference to the later.

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End

8

CHAPTER

redox reactions

- According to *classical concept* **oxidation** involves addition of either oxygen or some electronegative radical or removal of either the hydrogen or some electropositive radical. On the other hand, **reduction** involves addition of either hydrogen or some electropositive radical or removal of either the oxygen or some electronegative radical.
- According to *valency concept*, oxidation is a reaction in which positive valency of species is increased or the negative valency is decreased.
- According to *modern concept*, oxidation is a process in which a chemical substance loses electrons and reduction is a process in which a chemical substance gains electrons.
- **Oxidants :** Oxidants are substances which oxidise others and in process get reduced themselves. They show gain of electrons and decrease in oxidation number during a change.
  (i) Molecules of most electronegative elements
  e.g. O_2, O_3, halogens.
  (ii) Compounds having either of an element (under lines) in their highest oxidation state *e.g.* $K\underline{Mn}O_4$, $K_2\underline{Cr_2}O_7$, $H_2\underline{S}O_4$, $H\underline{N}O_3$, $\underline{Fe}Cl_3$, $\underline{Hg}Cl_2$, $K\underline{Cl}O_3$, $Na\underline{N}O_3$ etc.
  (iii) Oxides of metals and non metals *e.g.* MgO, CaO, $Cr\underline{O}_3$, H_2O_2, CO_2, SO_3 etc.
- **Reductants :** Reductants are substances which reduce others and gets oxidize themselves. They show loss of electrons and show an increase in oxidation number during a change.
  (i) All are metals *e.g.* Na, Al, Zn etc.
  (ii) Some are non metals *e.g.* C, S, P, H_2 etc.
  (iii) Halogen acids *e.g.* HI, HBr, HCl.
  (iv) Metallic hydrides *e.g.* NaH, LiH, CaH_2 etc.
  (v) Compounds having either of an element (under lined) in their lowest oxidation state e.g. $\underline{Fe}Cl_2$, $\underline{Fe}SO_4$, $\underline{Hg_2}Cl_2$, $\underline{Sn}Cl_2$, $\underline{Cu_2}O$ etc.
  (vi) Some are organic compounds *e.g.* HCOOH, aldehydes, oxalic acid, tartaric acid etc.

LEO	The Lion Goes	GER
Loose Electrons, Oxidize		**Gain Electrons, Reduce**

- Oxidation and Reduction must both occur in a redox reaction.

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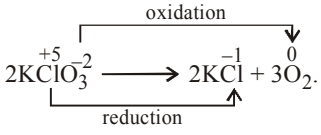
- **Types of Redox Reaction**

  [A] Intermolecular redox reaction: When oxidation and reduction take place separately in two compounds, called intermolecular redox reaction.

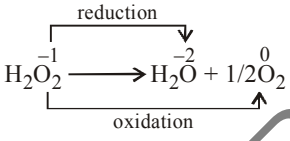
  $SnCl_2 + 2FeCl_3 \rightarrow SnCl_4 + 2FeCl_2$

  $Sn^{2+} \rightarrow Sn^{4+}$ (oxidation), $Fe^{3+} \rightarrow Fe^{2+}$ (reduction)

  [B] Intramolecular redox reaction: During the chemical reaction, if oxidation and reduction take place in a single compound then reaction is called intramolecular redox reaction.



  [C] Disproportionation reaction: When reduction and oxidation take place on same element of a compound then it is called disproportionation reaction.



- **Classification of Redox Reactions**

  1. **Direct redox reactions:** The reactions in which oxidation and reduction take place in the same vessel are called *direct redox reactions*.
  2. **Indirect redox reactions:** The reactions in which oxidation and reduction take place in different vessels are called *indirect redox reactions*.

- The basic difference between the two is that in *the direct redox reaction, energy is liberated in the form of heat energy whereas in indirect redox reaction, energy is liberated in the form of electrical energy*. Thus, indirect redox reactions lead to the production of electrical energy. The arrangement for carrying out indirect redox reactions is called electrochemical cell. Thus, an **electrochemical cell** is *a device used to convert chemical energy produced in a redox reaction into electrical energy*.

- **Oxidation and reduction**

	Oxidation	***Reduction***
(1)	During oxidation a substance donates one more electrons.	During reduction, a substance accepts one or more electrons.
(2)	The process is called de-electronation.	The process is called electronation.
(3)	It indicates increase in oxidation number and loss of electrons.	It indicates decrease in oxidation number and gain of electrons.
(4)	Oxidation is caused by an oxidising agent.	Reduction is caused by a reducing agent.
(5)	Oxidation occurs at anodein electrochemical cell.	Reduction occurs at cathode in electrochemical cell

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- **Oxidation number :** Oxidation number of an element in a particular compound represents the number of electrons lost or gained by an element during its change from free state into that compound. **or**
- Oxidation number of an element in a particular compound represents the extent of oxidation or reduction of an element during its change from free state into that compound.
- Oxidation number is given positive sign if electrons are lost and negative sign if electrons are gained. It represents real charge in case of ionic compounds, however in covalent compounds it represents imaginary charge. An *oxidation state* of an atom is defined as oxidation number per atom.
  e.g. in K_2MnO_4, oxidation number of Mn = +6.
  $\therefore$ Oxidation state of Mn = Mn^{+6}.
- An *oxidation-reduction* reaction is one in which one or more atoms change oxidation number, implying that there has been a transfer of electrons.

Oxidation number rules

Rule	Applies to	Statement
1.	Elements	The oxidation number of an atom in an element is zero.
2.	Monoatomic ions	The oxidation number of an atom in a monoatomic ion equals the charge on the ion.
3.	Oxygen	The oxidation number of oxygen is –2 in most of its compounds (an exception is O in H_2O_2 and other peroxides, where oxidation number is –1) and oxygen fluoride (OF_2), it is +2.
4.	Hydrogen	The oxidation number of hydrogen is +1 in most of its compounds. (The oxidation number of hydrogen is –1 in binary compounds with a metallic hydride such as CaH_2, NaH_2).
5.	Halogens	The oxidation number of fluorine is –1, in all of its compounds. Each of other halogens (Cl, Br, I) has an oxidation number of –1 in binary compounds, except when the other element is another halogen above it in the periodic table or the other element is oxygen.
6.	Compounds and ions	The sum of the oxidation numbers of the atoms in a compound is zero. The sum of the oxidation number of the atoms in a polyatomic ion equals the charge on the ion.
7.	Alkali metals	The oxidation number of alkali (Na, K, Li, etc) metals in compounds is +1.
8.	Alkaline earth metals	The oxidation number of alkaline (Mg, Ca, Ba, Sr, etc.) earth metals in compounds is +2.
9.	Sulphides	In all sulphides the oxidation number of sulphur is –2.

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Rule	Applies to	Statement
10.	Transition elements and *p*-block elements	Variable oxidation number is most commonly shown by transition elements as well as *p*-block elements. e.g. Fe (+2 & +3), Cu (+1 & +2), Mn (+7, +6, +5, +4, +3, +2, +1) etc. *p*-block : As (+3 and +5), Sb (+3 and +5), Sn (+2 and +4) etc.

- **Evaluation of oxidation number**

  Determine oxidation number of the element underlined in each of the following.

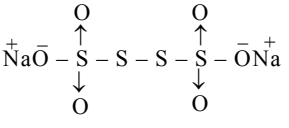
  (a) **$K\underline{Mn}O_4$:** Let a be the oxidation number of Mn.
  Oxidation number of K + oxidation number of Mn + 4(oxidation number of O) = 0
  $+1 + a + 4(-2) = 0$ or, $+1 + a - 8 = 0$
  or, $a - 7 = 0$ or, $a = +7$.
  Oxidation number of Mn in $KMnO_4$ is +7.

  (b) **$K_4\underline{Fe}(CN)_6$:** Let a be the oxidation number of Fe.
  4(oxidation number of K) + oxidation number of Fe + 6(oxidation number of CN) = 0
  $4(+1) + a + 6(-1) = 0$ or, $+4 + a - 6 = 0$
  or, $a - 2 = 0$ or, $a = +2$.
  Oxidation number of Fe in $K_4Fe(CN)_6$ is +2.

  (c) **$\underline{S}O_4^{2-}$ ion :** Let a be the oxidation number of S.
  Oxidation no. of S + 4(oxidation no. of O) = –2
  $a + 4(-2) = -2$ or, $a - 8 = -2$
  or, $a = -2 + 8 = +6$.
  Oxidation number of S in SO_4^{2-} ion is +6.

- **Special examples of oxidation number or oxidation state determination**

  1. **Oxidation state of sulphur in $Na_2S_4O_6$:** It is only average oxidation number of sulphur. Let us see the structure of $Na_2S_4O_6$.



  From the structure, it is clear that the sulphur atoms acting as donor atoms have +5 oxidation number (each). On the other hand, the sulphur atoms involved in pure covalent bond formation have zero oxidation number.

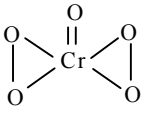
  2. **Fe in its oxides, FeO, Fe_2O_3 and Fe_3O_4 :**
  In FeO $\rightarrow x - 2 = 0$ or $x = +2$
  In $Fe_2O_3 \rightarrow 2x - 6 = 0$ or $x = +3$
  In $Fe_3O_4 \rightarrow 3x - 8 = 0$ or $x = +8/3$.

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Here in Fe_3O_4, oxidation number is the average of those in FeO and Fe_2O_3.
$FeO + Fe_2O_3 = Fe_3O_4$.

Average oxidation number of Fe in $Fe_3O_4 = \frac{+2 + 2(+3)}{3} = +\frac{8}{3}$.

3. **Oxidation state of chromium in CrO_5 :** CrO_5 has butterfly structure having two peroxo bonds.



Peroxo oxygen has (–1) oxidation number.

Let oxidation number of chromium be x.
$x + 4(-1) + (-2) = 0$ or $x - 4 - 2 = 0$
or $x - 6 = 0$ or $x = +6$.

4. **Oxidation number of chlorine in bleaching powder:** Bleaching powder has two chlorine atoms having different oxidation states.

Ca^{2+} (OCl^-) Cl^-
(hypochlorite ion) (chloride ion)
Chlorine in +1 state Chlorine in –1 state

5. **Oxidation state of carbon and nitrogen in HCN and HNC:** It depends upon the following facts:
   (a) Single covalent bond contributes one unit for oxidation number.
   (b) Negative oxidation number is assigned to more electronegative atom and positive oxidation number to less electronegative atom.
   (c) Co-ordinate bond is represented by an arrow from donor atom to acceptor atom.

$A \rightarrow B$
Donor Acceptor

If donor atom is less electronegative and acceptor is more, then +2 state is given to donor and –2 state is given to acceptor. But it should be noted that if the donor is more electronegative than the acceptor, then contribution of co-ordinate bond for both atoms regarding oxidation state is neglected.

e.g. (i) $H - C \equiv N$

$+1 + a - 3 = 0$ or $a - 2 = 0$ or $a = +2$.
Carbon is in +2 state and nitrogen is in –3 state. Each bond contributes –1 state to more electronegative atom.

(ii) $H - N \rightrightarrows C$

Oxidation state of H = +1
Oxidation state of N = (–1) + (–2) + (0) = –3.

Covalent bond with hydrogen contributes (–1) and covalent bond with carbon contributes (–2) and there is zero contribution of co-ordinate bond. Let the oxidation state of carbon be x.

$+1 - 3 + x = 0$ or $x - 2 = 0$, $x = +2$.

- **Oxidation numbers (states) in different types of elements**

Zero group elements have zero oxidation number (state) as they do not show chemical activity while other elements have at least two oxidation states: zero when they exist

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in free state and positive or negative when they exist in compound. Many elements show different oxidation states in different compounds. In the case of representative elements, the highest positive oxidation number (state) of an element is the same as its group number while the highest negative oxidation state is equal to 8 – group number with negative sign, with a few exceptions.

Group	outershell configuration	Common oxidation numbers (states) except zero in free state
I A	ns^1	+1
II A	ns^2	+2
III A	$ns^2\ np^1$	+3, +1
IV A	$ns^2\ np^2$	+4, +3, +2, +1, –1, –2, –3, –4
V A	$ns^2\ np^3$	+5, +3, +1, –1, –3
VI A	$ns^2\ np^4$	+6, +4, +2, –2
VII A	$ns^2\ np^5$	+7, +5, +3, +1, –1

- Transition metals exhibit a large number of oxidation states due to involvement of $(n-1)d$ electrons besides *ns* electrons.
- **Balancing equation :** Two methods are used to balance a redox reaction.
  (i) Ion electron method or half reaction method.
  (ii) Oxidation state method
- **Ion electron method**

This method was developed by Jette and LaMev in 1927. It involves three sets of rules depending upon the nature of the medium (*i.e.* neutral, acidic or alkaline) in which reaction occurs.

(a) *Neutral medium*

e.g. $H_2C_2O_4 + KMnO_4 \rightarrow CO_2 + K_2O + MnO + H_2O$

Step-1 : Select the oxidant, reductant atoms and write their half reactions, one representing oxidation and other reduction.

$C_2^{3+} \rightarrow 2C^{4+} + 2e,\ \ 5e + Mn^{7+} \rightarrow Mn^{2+}$

Step-2 : Balance the number of electrons and add the two equations.

$$5C_2^{3+} \rightarrow 10C^{4+} + 10e$$
$$10e + 2Mn^{7+} \rightarrow 2Mn^{2+}$$
$$5C_2^{3+} + 2Mn^{7+} \rightarrow 10C^{4+} + 2Mn^{2+}$$

Step-3 : Write complete molecule of the reductant and oxidant from which respective redox atoms were obtained.

$$5H_2C_2O_4 + 2KMnO_4 \rightarrow 10CO_2 + 2MnO$$

Step-4 : Balance other atoms if any (except H and O). In above example, K is unbalanced, therefore

$$5H_2C_2O_4 + 2KMnO_4 \rightarrow 10CO_2 + 2MnO + K_2O + 5H_2O$$

(b) *Acidic medium*

e.g. $NO_3^- + H_2S \xrightarrow{H^+} NH_4^+ + HSO_4^-$

Proceed like neutral medium for steps 1 to 4.

Step-1 : $8e + N^{5+} \rightarrow N^{3-},\ S^{2-} \rightarrow S^{+6} + 8e$

Step-2 : $N^{5+} + S^{2-} \rightarrow N^{3-} + S^{+6}$

Step-3 : $NO_3^- + H_2S \rightarrow NH_4^+ + HSO_4^-$

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Step-4 : No other atom (except H and O) is unbalanced and thus, no need for this step.

Step-5 : Balance O atom by using H_2O and H^+ ions. Add desired molecules of H_2O on the side deficient with oxygen atom and double H^+ on opposite side. Therefore,

$H_2O + NO_3^- + H_2S \rightarrow NH_4^+ + HSO_4^- + 2H^+$

Step-6 : Balance charge by H^+

$3H^+ + H_2O + NO_3^- + H_2S \rightarrow NH_4^+ + HSO_4^- + 2H^+$

$\therefore$ The balanced equation is $H^+ + H_2O + NO_3^- + H_2S \rightarrow NH_4^+ + HSO_4^-$

(c) *Alkaline medium*

e.g. $Fe + N_2H_4 \xrightarrow{OH^-} Fe(OH)_2 + NH_3$

Proceed like neutral medium for steps 1 to 4.

Step-1 : $Fe \rightarrow Fe^{2+} + 2e$, $2e + N_2^{2-} \rightarrow 2N^{-3}$

Step-2 : $Fe + N_2^{2-} \rightarrow Fe^{2+} + 2N^{3-}$

Step-3 : $Fe + N_2H_4 \rightarrow Fe(OH)_2 + 2NH_3$

Step-4 : No other atom (except H and O) is unbalanced and thus, no need for this step.

Step-5 : Balance O atom by using H_2O and OH^- ions. Add desired molecules of H_2O on the side rich with O atoms and double OH^- on opposite side. Therefore,

$$4OH^- + Fe + N_2H_4 \rightarrow Fe(OH)_2 + 2NH_3 + 2H_2O$$

Step-6 : Balance charge by H^+.

$4OH^- + 4H^+ + Fe + N_2H_4 \rightarrow Fe(OH)_2 + 2NH_3 + 2H_2O$

$\therefore$ The balanced equation is $2H_2O + Fe + N_2H_4 \rightarrow Fe(OH)_2 + 2NH_3$

- ***Spectator ions***

Species that are present in the solution but do not take part in the reaction that occurs and are omitted in writing the net ionic reaction.

$$Zn + 2H^+ + 2Cl^- \rightarrow Zn^{2+} + 2Cl^- + H_2$$

Cl^- ions are omitted. These omitted ions are called as spectator ions or bystander ions, in order to indicate that they do not take part in the reaction. The spectator ions appear on the reactant as well as on the product side.

- **Oxidation state method**

In a balanced redox reaction, total increase in oxidation number must be equal to the total decrease in oxidation number. This equivalence provides the basis for balancing redox reactions. The general procedure involves the following steps:

(a) Write the skeleton equation representing the chemical change.

(b) Assign oxidation numbers to the atoms in the equation and find out which atoms are undergoing oxidation and reduction. Write separate equations for the atoms undergoing oxidation and reduction.

(c) Find the change in oxidation number in each equation. Make the change equal in both the equations by multiplying with suitable integers. Add both the equations.

(d) Complete the balancing by inspection. First balance those substances, which have undergone change in oxidation number and then other atoms except hydrogen and oxygen. Finally balance hydrogen and oxygen by putting H_2O molecules wherever needed. The final balanced equation should be checked to ensure that there are as many atoms of each element on the right as there are on the left.

(e) In ionic equations the net charges on both sides of the equation must be exactly

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the same. Use H^+ ion/ions in acidic reactions and OH^- ion/ions in basic reactions to balance the charge and number of hydrogen and oxygen atoms.

$$K_2Cr_2O_7 + HCl \rightarrow KCl + CrCl_3 + H_2O + Cl_2$$

Writing oxidation numbers of all the atoms.

$$\overset{+1}{K_2}\overset{+6}{Cr_2}\overset{-2}{O_7} + \overset{+1}{H}\overset{-1}{Cl} \rightarrow \overset{+1}{K}\overset{-1}{Cl} + \overset{+3}{Cr}\overset{-1}{Cl_3} + \overset{+1}{H_2}\overset{-2}{O} + \overset{0}{Cl_2}$$

The oxidation number of Cr has decreased while that of chlorine has increased.

$$K_2\overset{+6}{Cr_2}O_7 \rightarrow 2\overset{+3}{Cr}Cl_3 \quad \text{... (i)}$$

$$H\overset{-1}{Cl} \rightarrow \overset{0}{Cl_2} \quad \text{... (ii)}$$

Decrease in oxidation number of Cr = 6 units per molecules of $K_2Cr_2O_7$.
Increase in oxidation number of Cl = 1 unit per molecule of HCl.
Equation (ii) is multiplied by 6. $K_2Cr_2O_7 + 6HCl \rightarrow 2CrCl_3 + 3Cl_2$
To balance chlorine and potassium, 14 molecules of HCl are required.
$K_2Cr_2O_7 + 14HCl \rightarrow 2CrCl_3 + 3Cl_2$
To balance hydrogen and oxygen $7H_2O$ are added to R.H.S. Hence balanced equation is $K_2Cr_2O_7 + 14HCl \rightarrow 2KCl + 2CrCl_3 + 3Cl_2 + 7H_2O$.

Characteristics of Oxidation Reduction

- Oxidation or de-electronation is a process which liberates electrons.
- Reduction or electronation is a process which gains electrons.

	Oxidation	**Reduction**
	$M \rightarrow M^{+n} + ne$	$M^{+n} + ne \rightarrow M$
$n_2 > n_1$	$M^{+n_1} \rightarrow M^{+n_2} + (n_2 - n_1)e$	$M^{+n_2} + (n_2 - n_1)e \rightarrow M$
	$A^{-n} \rightarrow A + ne$	$A + ne \rightarrow A^{-n}$
$n_1 > n_2$	$A^{-n_1} \rightarrow A^{-n_2} + (n_1 - n_2)e$	$A^{-n_2} + (n_1 - n_2)e \rightarrow A^{-n_2}$

- Oxidants are substances which
  (a) oxidize other.
  (b) are reduced themselves.
  (c) show gain of electrons.
  (d) show a decrease in oxidation number during a change.
  (e) has higher oxidation number in a conjugate pair of redox.
- Reductants are the substances which
  (a) reduce other.
  (b) are oxidized themselves.
  (c) show loss of electrons.
  (d) show an increase in oxidation number during a change.
  (e) has lower oxidation number in a conjugate pair of redox.
- A redox change involves the process in which a reductant is oxidized to liberate electron, which are then taken up by an oxidant to get itself reduced.

  $M_1 \rightarrow M_1^{+n} + ne$ Oxidation
  (Reductant)
  $M_2^{+n} + ne \rightarrow M_2$ Reduction
  (Oxidant)
  $M_1 + M_2^{+n} \rightarrow M_1^{+n} + M_2$ Redox
- A redox change occur simultaneously.

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End

9

CHAPTER

electrochemistry

- Electrochemistry is the study of relationship between electrical energy and chemical energy, produced in a redox reaction, and how one can be converted into another.
- Conductors are substances which allow the passage of current whereas insulators do not allow electric current to pass through them.
- Electrolytes are the aqueous solutions of compounds which conduct electricity and are decomposed by the passage of current.
- A compound whose aqueous solution does not conduct electricity is called **non-electrolyte**.
- Electrolysis is a chemical reaction brought about by the passage of electric current through an electrolyte.

  For example when current is passed through molten NaCl, the Na^+ and Cl^- ions move towards the oppositely charged electrodes. The electrochemical reactions may take place as

  $Na^+ + e^- \rightarrow Na$ - reduction at cathode

  $Cl^- \rightarrow \frac{1}{2} Cl_2 + e^-$ - oxidation at anode
- **Electrolysis of NaCl (*aq*) using Pt electrodes**

  $NaCl \rightarrow Na^+ + Cl^-$

  $2H_2O \rightleftharpoons H_3O^+ + OH^-$ (slight dissociation)

  At anode : Both Cl^- and OH^- ions move towards the anode, but Cl^- ions get discharged preferentially due to their low discharge.

  At cathode : Both Na^+ and H_3O^+ ions move towards the cathode but, H_3O^+ ions get discharged preferentially due to their low discharge potential.

  $H_3O^+ + e \rightarrow \frac{1}{2} H_2 + H_2O$

Electrical units

- *Coulomb* : It is the amount of electricity which will deposit 0.001118 gram of silver from a 15% solution of silver nitrate in a coulometer.
- *Ampere* : It is that current which will deposit 0.001118 gram of silver in one second. In other words, an ampere is a current of one coulomb per second.
- *Ohm* : It is the resistance offered at 0°C to a current by a column of mercury 106.3 cm long of about 1 sq. mm cross-sectional area and weighing 14.4521 grams.

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- *Volt* : It is the difference in electrical potential required to send a current of one ampere through a resistance of one ohm.

Faraday's law of electrolysis

- *First law :* The amount of any substance that is deposited or liberated at an electrode is directly proportional to the quantity of electricity passed through the electrolyte such as

  $$m \propto Q \quad \text{or} \quad m = zQ$$

  Q = quantity of electricity, z = electrochemical equivalent
  m = mass (in gm) of the substance

- *Second law :* When the same quantity of electricity flows through different electrolytes, the amount of different substances produced at the same electrodes are directly proportional to their equivalent mass such as

  $\frac{m_1}{m_2} = \frac{E_1}{E_2}$; E_1 and E_2 are equivalent mass.

- **Electrochemical equivalent (*z*)** is the weight of the ion deposited by the passage of one ampere current in one second.

  Note : A unit of charge in the honour of Michael Faraday called Faraday (F) was introduced. It is 1.0 mole of electron.

  $F = N_A \times e^- = 6.023 \times 10^{23} \times 1.601 \times 10^{-19}$
  $= 96485 \text{ C mol}^{-1}$ of electron

- **Relation between Faraday, Avogadro's number and charge on an electron**

  1 F of electricity liberates 1 gm equivalent of substances, hence one gm ion contains Avogadro's number of ions (N_A) therefore charge on one ion = $\frac{F}{N_A}$ coulombs.

  If valency of ion is n, then charge on one ion is a multiple: $\frac{nF}{N_A}$.

  $\frac{F}{N_A}$ is a fundamental quantity, it is the charge carried by one electron, therefore,

  $$e = \frac{F}{N_A} \quad \text{or,} \quad F = N_A \times e$$

  i.e. 1 faraday is the charge on one mole of electrons. Value of N_A can be calculated by knowing charge of an electron.

  $$N_A = \frac{96500 \text{ C}}{1.602 \times 10^{-19} \text{ C}} = 6.02 \times 10^{23}.$$

- **Electrochemical cell** is a device in which the free energy of physical or chemical process is converted into electrical energy.

  Following are the types of electrochemical cells.

  1. *Galvanic cells* : Here the electrical energy arises from the chemical reactions which take place in the cells.
  2. *Concentration cells* : Here the electrical energy arises not due to any chemical reaction but due to transfer of matter from one half-cell to other half-cell.

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- In Galvanic cell, two half cells or solutions of the two beakers are connected by an inverted tube filled with an electrolyte, called salt bridge. *Salt bridge* is a U-tube containing a concentrated solution of an inert electrolyte like KCl, KNO_3, NH_4NO_3, K_2SO_4 etc., sometimes taken in agar-agar or gelatin to give a semi-solid mass.
- The salt bridge helps to maintain neutrality of the two solutions by diffusing out the oppositely charged ions into the two cells.
- The anode or oxidation half cell acquires a negative charge due to liberation of electrons as :

$$Zn_{(s)} \rightarrow Zn^{2+}_{(aq)} + 2e$$

- The cathode or reduction half cell is electron deficient as it attracts electrons from external circuit for the reduction of Cu.

$$Cu^{2+}_{(aq)} + 2e \rightarrow Cu_{(s)}$$

- The redox reaction of the cell, or cell reaction is

$$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$$

- An electrochemical cell can be represented as :

$$\underset{\text{anode}}{\text{metal | metal ion (conc.)}} \;||\; \underset{\text{cathode}}{\text{metal ion (conc.) | metal}}$$

- For Daniel cell:

$$Zn;\ ZnSO_4,\ H_2SO_4 \mid CuSO_4 \text{ (satd)};\ Cu$$

When a salt bridge is used two vertical lines (||) as above are made.

- The tendency of an electrode to lose or gain electron when it is in contact with its own ion in solution is called electrode potential.

Electrode potential
- *Reduction potential* (Tendency to gain electrons)
- *Oxidation potential* (Tendency to lose electrons)

- Oxidation potential is the reverse of reduction potential. If the reduction potential of given electrode is +1.5 V, then its oxidation potential is taken as –1.5 V.
- **Single electrode potential or half-cell potential** is impossible to determine the half-cell potential experimentally. It is only the difference of potentials between two electrodes that we can measure by combining them to give a complete cell.
- **Standard electrode potentials :** At the unity concentration of electrode and the temperature of 25°C, the potential of the electrode is termed as the standard electrode potentials ($E°_{cell}$).
  The standard electrode potential of hydrogen electrode is zero. So it is used as reference electrode.
- The values of standard electrode potentials in the decreasing order on hydrogen scale is called electrochemical series.
  $E°$ for some electrodes are given in the table.

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Standard electrode (reduction) potentials at 25°C

	Reduction half-reaction	$E°$ (V)	
stronger oxidising agent	$F_{2\,(g)} + 2e^- \rightarrow 2F^-_{(aq)}$	2.87	weaker reducing agent
	$H_2O_{2\,(aq)} + 2H^+_{(aq)} + 2e^- \rightarrow 2H_2O_{(l)}$	1.78	
	$MnO_4^-{}_{(aq)} + 8H^+_{(aq)} + 5e^- \rightarrow Mn^{2+}_{(aq)} + 4H_2O_{(l)}$	1.51	
	$Cl_{2\,(g)} + 2e^- \rightarrow 2Cl^-_{(aq)}$	1.36	
	$Cr_2O_7^{2-}{}_{(aq)} + 14H^+_{(aq)} + 6e^- \rightarrow 2Cr^{3+}_{(aq)} + 7H_2O_{(l)}$	1.33	
	$O_2\,(g) + 4H^+_{(aq)} + 4e^- \rightarrow 2H_2O_{(l)}$	1.23	
	$Br_{2\,(l)} + 2e^- \rightarrow 2Br^-_{(aq)}$	1.09	
	$Ag^+_{(aq)} + e^- \rightarrow Ag_{(s)}$	0.80	
	$Fe^{3+}_{(aq)} + e^- \rightarrow Fe^{2+}_{(aq)}$	0.77	
	$O_{2\,(g)} + 2H^+_{(aq)} + 2e^- \rightarrow H_2O_{2\,(aq)}$	0.70	
	$I_{2\,(s)} + 2e^- \rightarrow 2I^-_{(aq)}$	0.54	
	$O_{2\,(g)} + 2H_2O_{(l)} + 4e^- \rightarrow 4OH^-_{(aq)}$	0.40	
	$Cu^{2+}_{(aq)} + 2e^- \rightarrow Cu_{(s)}$	0.34	
	$Sn^{4+}_{(aq)} + 2e^- \rightarrow Sn^{2+}_{(aq)}$	0.15	
	$2H^+_{(aq)} + 2e^- \rightarrow H_{2\,(g)}$	0	
	$Pb^{2+}_{(aq)} + 2e^- \rightarrow Pb_{(s)}$	– 0.13	
	$Ni^{2+}_{(aq)} + 2e^- \rightarrow Ni_{(s)}$	– 0.26	
	$Cd^{2+}_{(aq)} + 2e^- \rightarrow Cd_{(s)}$	– 0.40	
	$Fe^{2+}_{(aq)} + 2e^- \rightarrow Fe_{(s)}$	– 0.45	
	$Zn^{2+}_{(aq)} + 2e^- \rightarrow Zn_{(s)}$	– 0.76	
	$2H_2O_{(l)} + 2e^- \rightarrow H_{2\,(g)} + 2OH^-_{(aq)}$	– 0.83	
	$Al^{3+}_{(aq)} + 3e^- \rightarrow Al_{(s)}$	– 1.66	
	$Mg^{2+}_{(aq)} + 2e^- \rightarrow Mg_{(s)}$	– 2.37	
	$Na^+_{(aq)} + e^- \rightarrow Na_{(s)}$	– 2.71	
weaker oxidising agent	$Li^+_{(aq)} + e^- \rightarrow Li_{(s)}$	– 3.04	stronger reducing agent

- Substances with stronger reducing power are placed above hydrogen and those with weaker reducing power are placed below hydrogen.
- **Applications of electrochemical series**
  - (i) It is useful in predicting the relative strengths of oxidising and reducing agents, or the ease of reduction or oxidation. Greater the value of $E°$, more is the tendency of element to get reduced, hence acts as a strong oxidising agent.
  - (ii) Reactivity of an element can be predicted.
  - (iii) It is useful to predict whether a metal will liberate hydrogen gas from an acid or not.
  - (iv) To predict feasibility of a redox reaction by calculating standard EMF of cell: $E°_{\text{cell}} = E°_R - E°_L$
    If EMF of cell is positive, reaction is spontaneous.
  - (v) It is useful for determining standard free energy change of the reaction: $-\Delta G° = nFE°$
  - (vi) Knowing the value of $E°_{\text{cell}}$, equilibrium constant (K_C) can be calculated.

$$E°_{\text{cell}} = \frac{0.0591}{n}\log K_C \text{ (at 298 K).}$$

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- **Electromotive force and equilibrium constant of a cell reaction**
  The reversible reaction may be represented as follows:
  $$aA + bB \rightleftharpoons cC + dD$$
  Decrease in free energy ($-\Delta G$) can be written as
  $$-\Delta G^\circ = nFE^\circ.$$
  We know that, $-\Delta G^\circ = RT \ln K$.
  If the standard EMF (E°) is known as the equilibrium constant of the cell reaction can be calculated by the use of the relation
  $$-\Delta G^\circ = RT \ln K = nFE^\circ$$
  $$\ln K = \frac{nFE^\circ}{RT} \; ; \; \log K = \frac{nFE^\circ}{2.303RT} = \frac{nE^\circ}{0.0591} \text{ at } 25^\circ\text{C}.$$
  Reaction quotient, Q, is simply expression like that of equilibrium constant but its value is that of before equilibrium is reached i.e. $Q = K_{eq}$ at equilibrium.
- **Nernst equation**
  The equation shows the relationship of electrolyte concentration on electrode potential of the cell.
  If we write the electrode reaction in general, as
  $$\text{oxidised state} + ne^- \rightleftharpoons \text{reduced state}$$
  Then electrode potential
  $$E_{cell} = E^\circ_{cell} - \frac{RT}{nF} \ln \frac{[\text{reduced state}]}{[\text{oxidised state}]}$$
  $$= E^\circ_{cell} + \frac{RT}{nF} \ln \frac{[\text{oxidised state}]}{[\text{reduced state}]}$$
- Electrolyte solutions conduct electrical currents through them by movement of the ions to the electrode. The power of electrolytes to conduct electrical currents is termed conductivity or conductance.
- **Ohm's law** relates the current in ampere (i) to potential difference (E) applied across the conductor and the resistance (R) of the conductor as $i = E/R$.
- The SI unit of electric resistance is *ohm* (Ω). At a given temperature resistance is directly proportional to length (l) and inversely proportional to area of cross section (a) of the conductor, i.e.
  $$R \propto \frac{l}{a} \quad \text{or} \quad R = \frac{\rho \times l}{a}$$
  ρ is a constant, called *specific resistance* or *resistivity*, if l and a are unity then, $R = \rho$
- **Conductance** (C) is a term which implies the case with which current flows through a conductor. It is defined as the reciprocal of resistance;
  $$C = \frac{1}{R}$$
  Its unit is ohm^{-1} or mho. The SI unit is Siemens (S).
- **Specific Conductance** (κ)
  Since $C = \frac{1}{R}$ and $R = \frac{\rho \times l}{a}$ $\quad \therefore C = \frac{1}{\rho} \times \frac{a}{l} = \kappa \frac{a}{l}$.
  κ is called specific conductance and is equal to the reciprocal of specific resistance. Specific conductance of a conductor is the conductance of one unit cube of conductor.

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Specific conductance of an electrolyte is the conductance of 1 cm^3 solution between electrodes placed 1 cm apart.

$$\therefore \quad \kappa = C \times \frac{l}{a}$$

l/a is called *cell constant* and depends upon the dimensions of the cell. Unit of κ is $ohm^{-1}\ cm^{-1}$ or mho cm^{-1} or $\Omega^{-1}cm^{-1}$. In SI unit it is Sm^{-1}

- **Equivalent Conductance**, (Λ) is more useful in companing the conducting power of different electrolyte solutions, having different concentrations of ions.
  It is defined as the conducting power of all the ions produced by one gram equivalent of an electrolyte in the solution.
  If 1 cm^3 solution contains 1 gm equivalent weight of electrolyte then conductance of the solution = specific conductance = equivalent conductance
  If 1 gm equivalent weight is present in V cc. of solution then,
  Λ = sp. conductance $\times$ $V = \kappa \times V$
  If concentration, C is given in gm equivalents per litre, then,
  $$V = \frac{1000}{C} \quad \text{or,} \quad \Lambda = \kappa \times \frac{1000}{C};$$
  C is gm eq/litre = normality
  Unit of L is $ohm^{-1}\ m^2\ g\ eq^{-1}$ or $\Omega^{-1}\ m^2\ g\ eq^{-1}$
- **Molar conductance (Λ_m)** or molar conductivity is defined as the conducting power of all the ions produced by one mole of the electrolyte in solution.
  $\therefore \quad \Lambda_m = \kappa \times V_m$, Where V_m = volume of solution containing 1 mole of electrolyte.
  If C_m is concentration in moles $litre^{-1}$ then,
  $$\Lambda_m = \kappa \times \frac{1000}{C_m};\ C_m = \text{molarity}$$
  In SI system, concentration C_m is expressed as moles/m^3 and volume V_m as m^3 per mole, hence
  $$V_m = \frac{1}{C_m}, \qquad \therefore \Lambda_m = \kappa \times V_m = \frac{K}{C_m}.$$
  Unit of Λ_m is $ohm^{-1}\ cm^{-2}\ mol^{-1}$, in SI system it is $ohm^{-1}\ m^2\ mol^{-1}$ or $\Omega^{-1}\ m^2\ mol^{-1}$
  For univalent electrolytes like NaCl, $AgNO_3$, KCl etc. equivalent conductance and molar conductance are equal, $\Lambda = \Lambda_m$
  For bivalent electrolytes like $MgSO_4$, molar conductance is
  $$\Lambda_m = 2 \times \Lambda.$$
  It Z is total positive or negative charge per formula unit of electrolyte then,
  $$\Lambda_m = Z \times \Lambda\ .$$
  "At infinite dilution, when dissociation is complete, each ion makes a definite contribution towards molar conductance of the electrolyte irrespective of the nature of the other ion with which it is associated and that the molar conductance of any electrolyte at infinite dilution is given by the sum of the contributions of the two ions". This is called *Kohlrausch's law*.
  $$\Lambda_m^\infty = \lambda_+^\infty + \lambda_-^\infty$$
  λ_+^∞ and λ_-^∞ are molar ionic conductances at infinite dilution for cations and anions.

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- **Applications of Kohlrausch's law:**

– It is useful in the calculation of Λ_m^∞ for weak electrolytes, e.g. $\Lambda^\infty_{m CH_3COOH} = \Lambda^\infty_{m CH_3COONa} + \Lambda^\infty_{m HCl} - \Lambda^\infty_{m NaCl}$

– In the calculation of degree of dissociation,

e.g. $\alpha = \dfrac{\text{molar conductivity at conc. } C}{\text{molar conductivity at infinite dilution}} = \dfrac{\Lambda_m^C}{\Lambda_m^\infty}$.

– Useful in calculation of solubility of a sparingly soluble salt. The solution of a sparingly soluble salt becomes saturated at infinite dilution, then $\Lambda_m = \Lambda_m^\infty$ and molarity = solubility,

hence $\Lambda_m = \dfrac{\kappa \times 1000}{\text{molarity}}$

or solubility (moles L^{-1}) $= \dfrac{\kappa \times 1000}{\Lambda_m^\infty}$.

- **Primary Cells** are those in which the reaction occurs only once and cannot be used again, i.e. cannot be recharged, e.g. dry cell, mercury cell.
- **Secondary Cells** are those which can be recharged by reversing the reaction, e.g. lead storage battery and Ni–Cd cell.
- **Fuel Cells** are specially designed to convert energy produced from combustion of fuels, directly into electrical energy, e.g. $H_2 - O_2$ cell.
- Any device which consists of two electrode with electrolyte solution are called cell. Cells are of two types.
  1. **Electrolytic Cell :** In electrolytic cells electrical energy is passed through the electrodes to the system (i.e., the cell) which converts into the chemical energy.
  2. **Electrochemical Cell :** In this cell the chemical energy comes out through the electrode in the form of electrical energy. Electrochemical cells are of two types:

  (i) Chemical cell : This cell consists of two electrodes of different nature and both of them are dipped into their individual salt solutions. The circuit is completed through a salt bridge e.g. Daniel Cell.
  Note : Salt bridge is a *U* type tube which is filled with those salts for which ionic mobility and transport number are equal. Such salts are called Agar-Agar Salts.

  (ii) Concentration Cell : In this cell the emf arises not from any chemical reaction (difference from chemical cell) but from a transfer of matter from one electrode to another, which occurs due to infinitesimal difference in concentration of the same species at the two electrodes.
- **EMF of a Galvanic Cell :** Every galvanic or voltaic cell is made up of two half-cells, the oxidation half-cell (anode) and the reduction half-cell (cathode). The potentials of these half-cells are always different. On account of this difference in electrode potentials, the electric current moves from the electrode at higher potential to the electrode at lower potential, *i.e.,* from cathode to anode. The direction of the flow of electrons is from anode to cathode.

$$\text{Anode} \underset{\text{Flow of current}}{\overset{\text{Flow of electrons}}{\rightleftharpoons}} \text{Cathode}$$

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The difference in potentials of the two half-cells is known as the electromotive force **(emf) of the cell or cell potential.**

- The emf of the cell or cell potential can be calculated from the values of electrode potentials of the two half-cells constituting the cell. The following three methods are in use:

(i) When oxidation potential of anode and reduction potential of cathode are taken into account:

$$E^\circ_{cell} = \text{Oxidation potential of anode} + \text{reduction potential of cathode}$$
$$= E^\circ_{ox\ (\text{anode})} + \text{E}^\circ_{red\ (\text{cathode})}$$

(ii) When reduction potentials of both electrodes are taken into account:

$$E^\circ_{cell} = \text{Reduction potential of cathode} - \text{reduction potential of anode}$$
$$= E^\circ_{cathode} - E^\circ_{anode} \quad = E^\circ_{right} - E^\circ_{left}$$

(iii) When oxidation potentials of both electrodes are taken into account:

$$E^\circ_{cell} = \text{Reduction potential of anode} - \text{oxidation potential of cathode}$$
$$= E^\circ_{ox}(\text{anode}) - E^\circ_{ox}(\text{cathode})$$

- We conclude that in case, the electron accepting tendency of the metal electrode is more than that of a Standard Hydrogen Electrode (S.H.E). Its standard reduction potential gets a **positive sign** and in case the electron accepting tendency of the metal electrode is lesser than that of S.H.E. its standard reduction potential gets a **negative sign.** It must be remembered that according to latest convention, **all standard potentials are taken as reduction potentials.**
- The electrode at which **reduction** occurs with respect to S.H.E. has **+ve reduction potential.**
- The electrode at which **oxidation** occurs with respect to S.H.E. has **–ve reduction potential.**

Electrode Potential - Factors Affecting it

$$E = E^\circ - \frac{2.303\,RT}{nF}\log Q$$

- Electrode potential of oxidation half-cell M/M^{n-} (aq) with reaction $M(s) \rightarrow M^{n+}(aq) + ne^-$ is

$$E_{ox} = E^\circ_{ox} - \frac{2.303RT}{nF}\log[M^{n+}]$$

E_{ox} depends on

(i) $[M^{n-}]$, concentration of ionic species

If $[M^{n+}]$ increases, E_{ox} decreases

We can say electrode is reversible with respect to M^{n+}.

(ii) Temperature T

If T increases at constant $[M^{n+}]$, E_{ox} decreases.

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- Electrode potential of reduced half-cell $M^{n+}(aq)/M$ with a reaction $M^{n+}(aq)+ne^- \rightarrow M(s)$ is

$$E_{red} = E^{\circ}_{red} - \frac{2.303RT}{nF}\log\frac{1}{[M^{n+}]}$$

E_{red} depends on

(i) $[M^{n+}]$, If $[M^{n+}]$ increases, E_{red} increases, electrode is reversible to M^{n+}.

(ii) Temperature T

If T increases at constant $[M^{n+}]$, E_{red} increases

Therefore, the Nernst equation for the general reduction reaction at 25°C is :

$$E = E^{\circ} - \frac{0.059}{n}\log\frac{1}{[M^{n+}(aq)]}$$

- **Electrode Signs**

The signs of the anode and cathode in the voltaic or galvanic cells are opposite to those in the electrolytic cells.

	Electrolytic cell		**Voltaic cell or Galvanic cell**	
	Anode	Cathode	Anode	Cathode
Sign	+	–	–	+
Electron flow	out	in	out	in
Half-reaction	oxidation	reduction	oxidation	reduction

- **Difference in electrolytic cell and galvanic cell**

Electrolytic cell	**Galvanic cell**
(i) Electrical energy is converted into chemical energy.	Chemical energy is converted into electrical energy.
(ii) Anode +ve electrode. Cathode –ve electrode.	Anode –ve electrode. Cathode +ve electrode.
(iii) Ions are discharged on both the electrodes.	Ions are discharged only on the cathode.
(iv) If the electrodes are inert, concentration of the electrolyte decreases when the electric current is circulated.	Concentration of the anodic half-cell increases while that of cathodic half-cell decreases when the two electrodes are joined by a wire.
(v) Both the electrodes can be fitted in the same compartment.	The electrodes are fitted in different compartments.

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End

10

CHAPTER

kinetics

- **Chemical kinetics** is the branch of chemistry which deals with rate and mechanism of chemical reaction.
- The speed with which the reactants are converted into products is called **rate of the reaction**.
- The rate of reaction is defined as change in the concentration of any one of the reactants or products per unit time.
- The rate of the reaction may be expressed in either of the two ways:
  (i) The rate of the disappearance or decrease in concentration of reactant with time.
  (ii) The rate of appearance or increase in concentration of product with time.
- The significance of the negative sign in case of expressing rate of reaction in term of reactants is important. We know that rate is always positive, as also obtained from the rate of formation of products. Therefore, minus sign is put before dA/dt so that rate is positive
  e.g., $A \rightarrow B$.
  Rate of reaction $= -\frac{d[A]}{dt} = \frac{d[B]}{dt}$
- The reaction rate cannot be determined by dividing total change in concentration by the time taken as in case of mechanical speed.
- As the reaction progresses, the rate of reaction decreases, becasue the concentration of the reactants decreases.
- The rate of a reaction that does not involve gases, does not depend upon pressure.
- The rate constant of a reaction increases with increase of temperature but not affected by concentration or catalyst.
- The rate does not depend upon the reactant present in large excess.
- The rate of reaction must be expressed with reference to particular moment of time.
- The average rate can be calculated by dividing the concentration difference by the time interval.
- The rate of reaction is not constant but it decreases with time reaching a value zero when the reaction is complete.
- The rate of change of concentration of any one of the reactants or products at a given time is called instantaneous rate.

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- The time taken by a reaction to proceed to a certain definite stage (98%) is called **life time of the reaction**.
- The times, the reaction takes to proceed midway (50%) is called **half life period** ($t_{1/2}$).
- Law of the mass action was given by the two Norwegian Scientist **Guldberg** and **Waage**.
- **Rate law** or rate equation are the mathematical expression which expresses the observed rate of a reaction in terms of the concentration of the reacting species.
  e.g. $aA + bB \rightarrow$ product
  $$\text{Rate} = k[A]^a[B]^b.$$
- Rate constant of a reaction is the rate of the reaction when the molar concentration of each of the reactant is unity.
- **Order of the reaction** is the sum of exponents of the concentration terms in the rate law.
- Order of reaction can be zero, integer or fractional.
- The units of rate constant depend upon the order of reaction.
- The specific rate constant of a first order reaction depends only on temperature.
- A reaction is of first order when the rate is linearly related to the concentration of the reactant.
- For the first order reaction $A \rightarrow$ products, plot of log $[A]$ *vs* time is linear with negative slope.

Examples:

1. *Zero order reaction* : $A \xrightarrow{k_0}$ Product ; $\dfrac{-d[A]}{dt} = k_0$

2. *A first order reaction* : $A \xrightarrow{k_1}$ Product, $\dfrac{-d[A]}{dt} = k_1[A]$ or $k_1 = -\dfrac{1}{A} \times \dfrac{d[A]}{dt}$

3. A second order reaction

   (i) $2A \xrightarrow{k_2}$ product, $\dfrac{-d[A]}{dt} = k_2[A]^2$ or, $k_2 = \dfrac{-1}{[A]^2} \times \dfrac{d[A]}{dt}$

   (ii) $A + B \xrightarrow{k_2}$ product, $\dfrac{-d[A]}{dt} = k_2[A][B]$ or, $k_2 = \dfrac{-1}{[A][B]} \times \dfrac{d[A]}{dt}$

- Certain bimolecular reactions which follow the kinetics of first order are called pseudounimolecular reactions.
- Hydrolysis of ester in presence of alkaline medium is a second order reaction.
- The units of the rate constant can be remembered by this formula ; liter^{n-1} mole^{1-n} sec^{-1} where n is the order of the reaction.

  Molecularity of a reaction is defined as the number of reacting molecules which collide simultaneously to bring about a chemical reaction.

  e.g. $N_2O_5 \longrightarrow N_2O_4 + \frac{1}{2}O_2$ (Unimolecular)

  $H_2 + I_2 \longrightarrow 2HI$ (Bimolecular)

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Difference between Order and Molecularity

	Order	Molecularity
1.	It is an experimental quantity.	It is a theoretical concept.
2.	It is the sum of the powers of the concentration terms in rate law.	It is the number of species which simultaneously collide.
3.	It may have fractional values.	It has only whole number values.
4.	It can be zero.	It cannot be zero.

- The finer the particles, the faster is the rate, hence rate of the chemical reaction is increased when the particle size of the solid substance is decreased.
- According to collision theory of reaction rates, the rate of reaction depends on energy factor and orientation factor.
- The number of collision that takes place per second per unit volume of the reaction mixture is known as **collision frequency** (z).
- The collision which actually produce the products and result in the chemical reaction are called **effective collisions**.
- The minimum amount of energy which the colliding molecules must possess is known as **threshold energy**.
- Collision frequency is proportional to the square root of the absolute temperature. *i.e.* $z \propto \sqrt{T}$.
- Increase in the rate of reaction with the rise in temperature is mainly due to the increase in the number of effective collisions.
- The excess energy (above the average energy of the reactant) required by the reactant to undergo chemical reaction is called **activation energy**.
  Activation energy = threshold energy – average kinetic energy of the reactants.
- Low activation energy $\Rightarrow$ Fast reaction
  High activation energy $\Rightarrow$ Slow reaction.
- **Activation energy barrier** is the energy acquired by the reactant molecules to cross the threshold energy to form products.
- Substances which increase the rate of reactions (both backwards and forwards) and either remain unaltered during the reaction or are regenerated after the reaction are called **catalysts**.
- Although a catalyst speeds up the reactions but it does not shift the position of equilibrium.
- The catalyst does not change ΔE of the reaction means that the addition of catalyst does not change energies of reactant (E_r) and product (E_p) so that ΔE and ($E_p - E_r$) remains same.
- The number of reacting species which collide simultaneously to bring about a chemical reaction is called molecularity.
- Molecularity of reaction cannot be zero. It has a whole number values only. *e.g.*, 1, 2, 3, etc.
- $A \rightleftharpoons B$.
  Net rate of reaction = rate of forward reaction – rate of backward reaction.

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- An endothermic reaction which proceeds with the decrease in volume will give maximum yield of the products at high pressure and temperature.
- The state at which the concentration of reactants and products do not change with time is called a **state of equilibrium**.
- The reaction which takes place in both forward and backward direction is called **reversible reaction**.
  e.g., $H_{2\,(g)} + I_{2\,(g)} \rightleftharpoons 2HI_{\,(g)}$.
- The chemical reaction in which the products formed do not combine to give back the reactant are known as **irreversible reaction**.
- The equilibrium can be approached from either direction and catalyst does not alter the equilibrium point.
- The rate of chemical reaction is directly proportional to the product of the molar concentration of the reactants at a constant temperature. This is known as **law of mass action**.
  e.g. $A + B \rightarrow$ product.
  Rate $\propto [A]\,[B]$
- Photochemical reaction takes place only in presence of light as each reactant molecule absorbs radiant energy.
- H_2 dissociates in presence of light only when Hg vapours are present. This is an example of **photosensitization**.
- Reaction which takes place in fraction of second is called fast reaction such as photosynthetic reaction has half life one pico second (10^{-12} s).
- Change in energy (ΔE) for the exothermic reaction are negative
  $\Delta E = E_a$ (forward) $- E_a$ (backward).
- The change in energy (ΔE) for the endothermic reaction are positive
  $\Delta E = E_a$ (forward) $- E_a$ (backward).
- Molecularity is a theoretical concept whereas order of the reaction is determined experimentally.
- For the feasibility of a reaction, the free energy should decrease (ΔG should be negative).
- The reaction rate can not be determined by dividing the total change in concentration by the time taken, therefore, the reaction must be expressed with reference to a particular moment of time, It is therefore also called instantaneous rate of reaction.
- Rate of radioactive disintegration follows 1st order kinetics.
- The population growth follows the 1st order kinetic, when the death and birth rates are equal.
- Order of the reaction cannot be more than 3rd order, but there is some exception.
- Reaction between iodide ion and iodate ion follow fifth order kinetic in acidic medium.
  Rate $= dx/dt = K\,[I^-]\,[IO_3^-]^2\,[H^+]^2$
  $n = 1 + 2 + 2 = 5$.

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End

11

CHAPTER

nuclear chemistry

- Nuclear chemistry is the branch of chemistry that deals with the study of atomic nucleus and nuclear changes.
- The nuclei of atoms represented by their atomic numbers and mass numbers are called nucleides. ${}^{4}_{2}\mathrm{He}$ represents nucleide of helium.
- Nuclei with same number of protons but different number of nucleons (proton + neutron) are called isotopes. e.g. ${}^{1}_{1}\mathrm{H}$ and ${}^{2}_{1}\mathrm{H}$; ${}^{16}_{8}\mathrm{O}$ and ${}^{17}_{8}\mathrm{O}$ etc.
- Nuclei with same number of neutrons but different number of protons are called isobars.
- Mass defect is the difference between the actual nuclear mass and expected nuclear mass (sum of the individual masses of nuclear particles).
- $E = \Delta mc^2$, where, Δm is mass defect and c is the velocity of light.
- Binding energy is the energy equivalent to mass defect and is responsible for holding the nucleus together. It is known as binding energy of the nucleus.
- The spontaneous emission of radiations by an element or its compound is called radioactivity.
- The spontaneous disintegration of the nuclei of certain naturally occurring elements like uranium, thorium, polonium, radium etc. is known as **natural radioactivity**.
- The spontaneous emission of radiations by many man-made elements, which are radioactive is called **artificial radioactivity**.
- Radioactive elements undergo nuclear disintegration to emit α, β and γ rays (particles).

α-particles are not good projectiles, because they carry double positive charge and hence repelled by the positively charged nucleus. Protons and deuterons, carrying single positive charge are much better projectiles than α-particles, but these are not as good projectiles as the neutrons, which carry no charge. Neutrons are most versatile and widely used projectiles, because being neutral particles, they are not subjected to electrostatic repulsion with positively charged nucleus.

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Comparison of α, β and γ rays			
	α-particle/ray	β-particle/ray	γ-ray
Charge and Mass	2 unit (+ve); 4 unit mass	1 unit (–ve); no mass	Electromagnetic rays, with very short wavelength (approx. 0.05 Å)
Nature	Fast moving $_2He^4$ or He^{++}	Fast moving $_{-1}e^0$	$_0\gamma^0$
Velocity/Path	1/10th of light, straight line	33% to 90% of light crooked	Same as light waves
Magnetic field	Deflected towards the cathode	Deflected towards the anode	Not deflected
Penetrating power	Low, or (0.01 mm of Al foil)	100 times of α-particle	10 times of β-particle
Relative ionizing power	Very high, nearly 100 times of β-rays	Low, nearly 100 times of γ-rays	Vey low
Effect on ZnS plates	They cause luminescence	Very little effect	Very little effect
Range	Very small (8 - 12 cm)	More than that of α-particle	More
Nature of product	Product obtained by the loss of one α-particle has atomic number less by 2 units and mass number by 4 units.	Product obtained by the loss of one β-particle has atomic number more by 1 unit, without any change in mass number.	There is no change in atomic number as well as mass number.

- Slow or fast neutrons may penetrate the nucleus with equal ease. A high speed neutron may strike a nucleus and pass through it. A slow neutron may be captured and absorbed by the nucleus. For example, two different isotopes of uranium are produced when uranium is bombarded with high and slow moving neutrons

$$^{238}_{92}U + ^{1}_{0}n \longrightarrow ^{237}_{92}U + 2\,^{1}_{0}n$$

high speed

$$^{238}_{92}U + ^{1}_{0}n \longrightarrow ^{239}_{92}U$$

slow speed

Type of decay	Atomic number	Mass number
α-decay	decreases by 2 unit	decreases by 4 unit
β-decay	increases by 1 unit	no change
γ-decay	no change	no change

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- **Group displacement law** - The emission of α-rays by an element results in the formation of new element which lies two places left to the parent element and the emission of a β-particle results in the formation of a new element which lies one place right to the parent element in the periodic table.
- There are certain reactions in which a heavy nucleus is rendered unstable by bombarding it with neutrons. As a result, the nucleus breaks up into two parts of comparable masses emitting several neutrons and a huge amount of energy. Such nuclear reactions are called **nuclear fission**.
- Few nuclear reactions are also known where isotopes of very little element, such as deuterium etc. may react with one another to form heavier or more stable nucleus. Very high temperature, of millions of degrees, is required to cause such reactions, which are called **fusion reactions**.
- Instability of nucleus is due to high N/P ratio. The range of N/P value for stability is 1 to 1.56. If a nuclide has N/P ratio greater than 1.5, it emits a β–particle.
- N/P ratio is zero for H. N/P ratio for $^{209}_{83}Bi$ is 1.52. If N/P ratio is greater than 1.56, that nuclide becomes radioactive. There is no naturally occurring nuclide having N/P ratio less than 1. If a nuclide has N/P ratio outside the belt of stability it shows radioactivity. Nuclide with highest N/P ratio of 2.0 is $^{3}_{1}H$.
- Binding energy (MeV) = Mass defect × 931.5.
- Average binding energy = Total binding energy/Mass number
- The larger the binding energy per nucleon, the more stable the nucleus.
- In order for positron emission to occur, the energy equivalent of the difference in mass between parent and daughter element must be greater than 1.022 MeV. If it is less than 1.022 MeV, but greater than zero, only electron capture can occur.

Positron decay increases the N/P ratio,
while beta particle decay decreases the N/P ratio.

- For stable nuclides packing fraction is either zero or negative, while for unstable nuclides packing fraction is positive. A decay process is possible only if accompanied by the release of energy.
- Packing fraction is positive in case of unstable nuclide and negative or zero in case of stable nuclides. The minimum in packing fraction curve is occupied by Fe, Ni, Co, Cr etc. Packing fraction is highest for H (+78) and least for Fe (Negative value).
- Despite the fact that beta emission is the ejection of an electron by a nucleus, an electron can not exist in the nucleus. It is believed that electrons are created at the time of emission, just as photons are from excited atoms. Beta decay corresponds to the following processes in the nucleus.
  $n \rightarrow p + e^{-}$; $p \rightarrow n - e^{-}$
- Calculations have shown that nuclear shells can have different subshells corresponding to 2, 6, 12, 8, 22, 32 and 44 according to the increasing order of energy. Thus the first nuclear shell can have 2 nucleons, the second 2 + 6 = 8 nucleons, the third 8 + 12 = 20, the fourth 20 + 8 = 28, the fifth 28 + 22 = 50, the fifth 50 + 32 = 82 and the seventh 82 + 44 = 126. These numbers are all **magic numbers**. So magic numbers correspond to the completed nuclear shells.

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- Stable nuclei are obtained when either the number of protons (Z) or the number of neutrons N = (A – Z) is equal to one of the figures 2, 8, 20, 28, 50, 82 or 126.
- The first laboratory produced radio nuclide was ^{30}P, which emits positron with a half life of 2.5 min. It was produced by Curies in 1934 by bombarding aluminium with alpha particles from polonium.
- Rate of disintegration is directly proportional to the number of atoms present in a sample at that time. The radioactive disintegration follows the following rate law.

  $N_t = N_0 e^{-\lambda t}$, where,

  N_0 = Number of disintegrating nuclei present initially.

  N_t = Number of disintegrating nuclei present at time *'t'*.

  λ = disintegration constant or decay constant.

  This can also be written as $\log \frac{N_0}{N_t} = \frac{\lambda . t}{2.303}$

  The rate of radioactive decay can be measured by such instruments as scintillating counter, Wilson cloud chamber and Geiger-Muller counter.
- Radioactive constant or disintegration constant (λ) is fraction of total number of atoms disintegrate in one second at any instant of time.

$$\lambda = \frac{dN}{dt} = \frac{2.303}{t} \log \frac{N_0}{N} = \frac{0.693}{t_{1/2}}$$

- The reciprocal of radioactive constant (λ) or disintegration constant is called average life period (τ).

$$\tau = \frac{1}{\lambda} = 1.44\ t_{1/2}$$

- Curie is the amount of radioactive substance which gives 3.7×10^{10} disintegrations per second other units are
  Rutherford (Rd) = 10^6 disintegrations per second
  Becquerel (Bq) = 1 disintegration per second
- A series of nuclear reactions that begins with an unstable nucleus and terminates with a stable one is known as radioactive or nuclear disintegration series.

Series	First member	$t_{1/2}$ for first member in years	Last member	At. masses when divided by 4, the remainder	No. of α-particles emitted	No. of β-particles emitted
Thorium ($4n$)	$_{90}Th^{232}$	1.4×10^{10}	$_{82}Pb^{208}$	0	6	4
Uranium ($4n + 2$)	$_{92}U^{238}$	4.51×10^9	$_{82}Pb^{206}$	2	8	6
Actinium ($4n + 3$)	$_{92}U^{235}$	7.07×10^8	$_{82}Pb^{207}$	3	7	4

- About 42 radioactive nuclides (Z > 82) occur in nature and all these are the decay products of only three long lived nuclides ^{235}U, ^{238}U, and ^{232}Th. All these nuclides are arranged in three decay series and the parent element of the series undergoes series of alpha and

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beta changes and finally give a stable end product of an isotope of lead.

- The neptunium series does not occur in nature because the half life of its longest lived member is three orders of magnitude shorter than the age of universe.
- The first nuclear reaction studied by Rutherford (1919) was,

$$^{14}_{7}N + {}^{4}_{2}He \rightarrow {}^{17}_{8}O + {}^{1}_{1}H$$

- The alpha particles required for the reaction was obtained from natural radioactive source such as polonium. The first transmutation reaction using artificially accelerated particles was

$$^{7}_{3}Li + {}^{1}_{1}H \rightarrow {}^{4}_{2}He + {}^{4}_{2}He$$

- Artificial radioactivity was first discovered by Irene Curie, Juliot and her husband F Juliot (1934), in the course of their researches on neutrons emitted. When light elements, such as boron, aluminium, magnesium etc., were bombarded by α-particles from polonium.
- Artificial radioactivity gives rise to two new types of artificial disintegrations, known as K–electron capture and nuclear isomerism, in addition to the simple process of ejection of positrons and electrons.
- In fission reaction, more than one neutron is released for every neutron bombarded which initiates the fission reaction. The neutrons emitted during fission bombards with more fissionable material and generates more neutrons. In this way, a nuclear chain reaction sets up and a huge quantity of energy is released. For chain reaction to sustain, a sufficient amount of fissionable material is needed.
- The minimum amount of fissionable material that is needed to sustain a chain reaction under the given set of conditions is called critical mass.
  If mass of the material is more than the critical mass, it is called super critical mass and if it is less than the critical mass then it is called sub-critical mass.
- **Atomic bomb** is based on the principle of nuclear chain reaction (nuclear fission). Two or more pieces of fissionable material (U-235 or Pu-239), each of subcritical mass are brought together rapidly with the help of a conventional explosive. When they come together the subcritical mass pieces form a single piece of super critical mass. A chain reaction will start and a huge amount of energy will be released leading to explosion.
- Nuclear reactor is an equipment in which the nuclear chain reaction is carried out in a controlled manner. It consists of

  (i) a fissionable materials [uranium enriched in $^{235}_{92}U$ (2-3%)]
  (ii) a moderator (graphite or heavy water)
  (iii) control rod (boron steel or cadmium rods).
- **Breeder reactor** is a reactor which produces more fissionable nuclei than it consumes.
  e.g. when $^{238}_{92}U$ is bombarded with fast neutrons it produces plutonium-239, a fissionable nuclei as follows :

$$^{238}_{92}U + {}^{1}_{0}n \rightarrow {}^{239}_{92}U \rightarrow {}^{239}_{93}Np + {}^{0}_{-1}e$$

$$^{239}_{93}Np \rightarrow {}^{239}_{94}Pu + {}^{0}_{-1}e$$

- Fissile refers to fissionable nuclide like $^{235}_{92}U$ and $^{239}_{94}Pu$.

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- A nuclide such as $^{238}_{92}U$ which can be converted into a fissionable nuclide is called **fertile nuclide.**
- Nuclear fusion may be defined as a process in which lighter nuclei fuse together to form a heavier nucleus. The amount of energy released in a nuclear fusion is more than that in a nuclear fission e.g.

$$^{2}_{1}H + ^{2}_{1}H \rightarrow ^{4}_{2}He + 23 \times 10^{8} \text{ kJ/mole of He}$$

- Those reactions which need a very high temp. ($>10^6$ K) are known as *thermo nuclear reactions e.g. fusion reactions.*
- Hydrogen bomb or Thermonuclear bomb is based on the principle of fusion reaction. The fusion of two hydrogen nuclei is triggered by an atomic bomb which generates the high temperature required for fusion reaction.
- Natural uranium is a mixture of two isotopes, ^{238}U and ^{235}U containing 99.3% of the heavier isotope and only 0.7% of the lighter one. ^{235}U is fissionable by fast as well as slow neutrons, while ^{238}U breaks up only by fast neutrons. ^{238}U, in general, absorbs neutrons, but instead of undergoing fission, it yields transuranic elements Np and Pu.
- ^{238}U suffers nuclear fission to a small extent when bombarded with fast neutrons and the fission is more symmetrical. The fragments produced in fission of ^{238}U are unsymmetrical.
- It will not be possible to produced chain reaction in natural uranium, because neutrons ejected from fission of ^{235}U will be captured by the much more abundant ^{238}U isotope, which does not undergo fission by slow neutrons.
- Radio tracers are radio-active isotopes which are incorporated in a system in very small quantity (trace amounts). They can trace the fate of particular element or compound containing this element through a series of chemical or physical changes.
- Radio carbon dating process is used to find out the age of objects of animal or vegetable origin (e.g. wood, charcoal, textiles etc.). For this $^{14}_{6}C$ is used. It is based on the fact that all the living matter contains a definite proportion of radioactive $^{14}_{6}C$.
- Radiotherapy is the type of therapy in which radioisotopes are used. Such type of therapy is used in the treatment of cancer etc.
- For calculation of carbon dating,
  (i) Calculate k from $t_{1/2}$
  (ii) m% activity of C - 14 now present means $\frac{a-x}{a} = \frac{m}{100}$.
  (iii) Apply $t = \frac{2.303}{\lambda} \log \frac{a}{a-x}$.

End

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12
CHAPTER

surface chemistry

ADSORPTION

- **Adsorption** is the phenomenon of attracting and retaining the molecules of a substance on the surface of a liquid or solid, resulting in a higher concentration of molecules on the surface.
- *Adsorption* is a surface phenomenon.
- The substance being adsorbed is called *adsorbate* and the substance on whose surface it is being adsorbed is called the *adsorbent*.
- Surface area of adsorbate per unit mass of adsorbent is known as specific surface area.
- **Sorption** is a term used when both absorption and adsorption occur simultaneously.
- **Desorption** is the reverse of adsorption, i.e. the removal of the adsorbed substance from the surface of the adsorbent.
- **Occlusion** is a term used for adsorption of gases on a metal surface.
- Adsorption occurs on the surface of solids due to the presence of unbalanced forces, believed to have developed either during crystallisation of solids or due to the presence of unpaired electrons in *d*-orbitals.
- Adsorption is specific and selective in nature.
- Adsorption is accompanied by decrease in free energy, *i.e.* ΔG = –ve. When $\Delta G = 0$, adsorption equilibrium is established.
- Adsorption is a spontaneous process. According to Gibbs' Helmholtz equation: $\Delta G = \Delta H - T\Delta S$
  ΔG = –ve; ΔH = –ve; ΔS = –ve (because adhering of gas molecules to a surface, lowers randomness).
- Enthalpy of adsorption is the enthalpy change accompanying the adsorption of 1 mole of adsorbate on the adsorbent surface.
- Adsorption isotherm is a graph between amount of adsorption and gas pressure keeping the temperature constant.
- Adsorption isobar is the graph drawn between quantities adsorbed under a constant gas pressure at definite temperatures.
- Adsorption isostere is the plot of temperature versus pressure for a given amount of adsorption.

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- Physical Adsorption or Physisorption is the process in which the adsorbate particles are held on the surface of adsorbent by weak van der Waal's forces.
- Chemical Adsorption or Chemisorption is the process in which the adsorbate molecules are held by the adsorbent by chemical forces.

	Physical Adsorption	Chemical Adsorption
1.	Molecules are attracted to the surface by van der Waal's forces.	Molecules are held to the surface by chemical bonds.
2.	Heat of adsorption is low (20-40 kJ mol^{-1}).	Heat of adsorption is high (40-400 kJ mol^{-1}).
3.	Process is reversible it decreases with temperature.	Process is irreversible first increases, then decreases with temperature.
4.	Non-specific in nature.	Specific in nature.
5.	Forms multi-molecular layer on surface.	Forms unimolecular layer.

- Positive adsorption occurs when concentration of adsorbate is more on surface of the adsorbent than in the bulk.
- Negative adsorption occurs when concentration of adsorbate on the surface of adsorbent is less than that in the bulk.

Factors affecting adsorption of gases on solids

- **Nature of Adsorbent:** Transition metals act as good adsorbents for gases due to vacant or half-filled *d*-orbitals and high charge-size ratio.
- **Surface Area of Adsorbent:** Highly porous substances like silica gel, Fuller's earth, charcoal are very good adsorbents since they have larger surface areas.
- **Nature of Adsorbate:** Easily liquefiable gases like NH_3, HCl, CO_2 etc. are adsorbed to a much greater extent than permanent gases like N_2, H_2 etc.
- **Pressure:** At constant temperature, if pressure is increased, adsorption increases. The increase is much greater if temperature is low.
- **Temperature:** Adsorption is an exothermic process having an equilibrium:
  Gas (Adsorbate) + Solid (Adsorbent) $\rightleftharpoons$ Gas adsorbed + heat
  Increase in temperature decreases adsorption.
- **Activation of solid Adsorbent:** It is done by subdividing the solid into smaller particles or by passing super heated steam to increase its adsorbing power.

Adsorption Isotherms

- The variation of the amount of the gas adsorbed by the adsorbent with pressure at constant temperature can be expressed by means of a curve, which is termed as adsorption isotherm at the particular temperature.
- Freundlich (1909) gave an empirical relationship between the quantity of gas adsorbed by unit mass of solid adsorbent and pressure at a particular temperature.

$$\frac{x}{m} = k \cdot P^{1/n} \quad \text{...(i)}$$

where 'x' is the mass of the gas adsorbed on a mass 'm' of the adsorbent at a pressure P. k and n are constants which depend on the nature of the adsorbent and the gas at a particular temperature.

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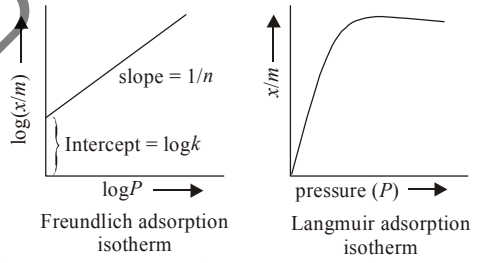
$$\log \frac{x}{m} = \log k + \frac{1}{n} \log P \qquad ...(ii)$$

- Freundlich isotherm explains the behaviour of adsorption in an approximate manner. The factor $1/n$ can have values between 0 and 1. Thus equation (i) holds good over a limited range of pressure
- The factor $1/n$ can have values between 0 and 1.
  when $1/n = 0$, x/m = constant which shows that adsorption is independent of pressure.
  when $1/n = 1$, $x/m = kP$. The adsorption varies directly with pressure.
- **Limitations of Freundlich adsorption isotherm:**
  (i) At higher pressures, it shows deviation.
  (ii) The values of constants k and n change with temperature.
  (iii) It is empirical one and has no theoretical basis.
- According to Langmuir (1916)
  – The layer of the gas adsorbed on the solid adsorbent is one molecule thick.
  – The adsorbed layer is uniform all over the adsorbent.
  – There is no interaction between the adjacent adsorbed molecules.
- If θ is the fraction of the total surface covered by the adsorbed molecules, the fraction of the naked area is (1– θ). The rate of adsorption (R_a) is proportional to the available naked surface (1– θ) and the pressure (P) of the gas.
  Langmuir adsorption isotherm can be derived and represented in short as follows:

$$\frac{x}{m} = \frac{aP}{1 + bP}$$

where a and b are constants. x/m and P are the terms similar to those expressed in Freundlich isotherm.

(a) At low pressure, $\frac{x}{m} = aP$ (b) At high pressure, $\frac{x}{m} = \frac{a}{b}$



Freundlich adsorption isotherm — Langmuir adsorption isotherm

- **Temperature:** Adsorption is an exothermic process having an equilibrium:
  Gas (Adsorbate) + Solid (Adsorbent) ⇌ Gas adsorbed + heat
  Increase in temperature decreases adsorption.
- Gas masks used by miners contain activated charcoal to adsorb poisonous gases like CH_4, CO etc.
- Industrial wastes and toxic chemicals are removed by treating water with an adsorbent.
- Chromatographic technique of separation and purification is based on adsorption.
- In sugar industry to decolourise the crude sugar and ion-exchange resins to purify water.

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- Cleansing action of soaps and detergents, concentration of ores like froth floatation process are all based on adsorption.
- Used in dehumidifiers and deodourisers, in catalytic reactions and formation of stable emulsions.
- Adsorption of reactants on the solid surface of the catalysts affects the rate of reaction between the reactants. The reaction proceeds more rapidly after adsorption.
- Different adsorbates are adsorbed to different extent on the same adsorbent under similar conditions of temperature and pressure.
- If a mixture of gases (or vapours) is allowed to come in contact with a particular adsorbent, the more strongly adsorbable adsorbate is adsorbed to a greater extent, irrespective of its amount present in the mixture.
- Generally, higher the critical temperature of a gas greater is the amount of that gas adsorbed.

CATALYST

- Catalyst is a substance which changes the speed of a reaction, and usually, can be recovered completely at the end of a reaction. However it may take part in a reaction -consumed in one step and regenerated in another.
- If the reaction mixture and the catalyst form a single phase, then it is known as homogeneous catalysis.

  e.g. $2SO_{2\,(g)} + O_{2\,(g)} \xrightarrow{NO\,(g)} 2SO_{3\,(g)}$
- The catalyst process in which the reaction mixture and the catalyst are in different phases is known as heterogeneous catalysis.

  e.g. $N_{2\,(g)} + 3H_{2\,(g)} \xrightarrow{Fe\,(s)} 2NH_{3\,(g)}$
- Activity of a catalyst refers to the ability of a catalyst to accelerate chemical reaction.

  e.g. Pt acts as a catalyst in the reaction

  $H_{2\,(g)} + 1/2\ O_{2\,(g)} \xrightarrow{platinum} H_2O_{(l)}$
- Selectivity of a catalyst refers to the ability of a catalyst to direct reaction to yield a particular product (excluding others), e.g. *n*-heptane selectively gives toluene in presence of platinum as catalyst.
- The catalysis that depends upon the pore-structure of the catalyst and molecular sizes of reactants and product molecules is called *shape selective catalysis*. e.g. zeolites are shape selective catalysts due to their honey-comb structure. ZSM-5 is used for converting methanol to gasoline.
- Zeolites are micro-porous aluminosilicates of the general formula $M_{x/n}[(AlO_2)_z(SiO_2)_y]\cdot mH_2O$. The pore size of the zeolites generally vary between 260 pm and 740 pm.

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	Process	Catalyst
1.	Haber's process for the manufacture of ammonia $N_{2\,(g)} + 3H_{2\,(g)} \rightleftharpoons 2NH_{3\,(g)}$	Finely divided iron Molybdenum as promoter. Conditions: 200 atmo-spheric pressure and 450-500°C temperature
2.	Ostwald's process for the manufacture of nitric acid $4NH_{3\,(g)} + 5O_{2\,(g)} \rightarrow 4NO_{(g)} + 6H_2O_{(g)}$ $2NO_{(g)} + O_{2\,(g)} \rightarrow 2NO_{2\,(g)}$ $4NO_{2\,(g)} + 2H_2O_{(l)} + O_{2\,(g)} \rightarrow 4HNO_{3\,(l)}$	Platinised asbestos Temperature 300°C
3.	Lead chamber process for the manufacture of sulphuric acid $2SO_{2\,(g)} + O_{2\,(g)} \rightleftharpoons 2SO_{3\,(g)}$ $SO_{3\,(g)} + H_2O_{(l)} \rightarrow H_2SO_{4\,(l)}$	Nitric oxide
4.	Contact process for the manufacture of sulphuric acid $2SO_{2\,(g)} + O_{2\,(g)} \rightleftharpoons 2SO_{3\,(g)}$ $SO_{3\,(g)} + H_2SO_{4\,(l)} \rightarrow H_2S_2O_{7\,(l)}$ $H_2S_2O_{7\,(l)} + H_2O_{(l)} \rightarrow 2H_2SO_{4\,(l)}$	Platinised asbestos or vanadium pentoxide (V_2O_5) Temperature 400 - 450°C
5.	Deacon's process for the manufacture of chlorine. $4HCl_{(g)} + O_{2\,(g)} \rightarrow 2H_2O_{(l)} + 2Cl_{2\,(g)}$	Cupric chloride ($CuCl_2$) Temperature 500°C
6.	Manufacture of ethyl alcohol from starch (a) Starch $\xrightarrow{\text{diastase}}$ maltose (b) Maltose $\xrightarrow{\text{maltase}}$ glucose $\xrightarrow{\text{zymase}}$ alcohol	Germinated barley (diastase enzyme) Temperature 50-60°C Yeast (maltase and zymase enzyme) Temperature 25-30°C.

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- Enzymes are complex nitrogeneous compounds which are produced by living plants and animals.

Some enzymatic reactions

	Enzyme	Source	Enzymatic reaction
1.	Invertase	Yeast	Sucrose → glucose and fructose
2.	Zymase	Yeast	Glucose → ethyl alcohol and carbon dioxide
3.	Diastase	Malt	Starch → maltose
4.	Maltase	Yeast	Maltose → glucose
5.	Urease	Soyabean	Urea → ammonia and carbon dioxide.

COLLOIDAL STATE

- The study for this state of matter was initiated by Thomas Graham in 1861.

- Substances whose solutions could pass through filter paper and animal membrane, having higher rate of diffusion are called *crystalloids*.
- Substances whose solutions can pass through filter paper and not animal membrane, also, having slower rate of diffusion are called *colloids*.
- Mixtures of substances in water, which can neither pass through filter paper nor animal membrane are called *suspensions*.

Suspension	Colloid	Solution
size > 10^{-5} cm or 10^3Å or 100 μm	10^{-7} cm to 10^{-5} cm or 10 Å to 10^3 Å or 100 μm	< 10^{-7} cm or 0 Å or 1 μm
Visible with naked eye	Visible with ultra-microscope	Not visible with any of optical means
Does not diffuse Settles under gravity	Diffuses very slowly Does not settle but it may settle under centrifuge	Diffuses rapidly Does not settle
Heterogeneous	Heterogeneous	Homogeneous
Opaque	Generally clear	Clear

Internal phase or Dispersed phase	External phase or Dispersion medium	Colloidal name	Examples
S	S	Solid sols	Alloys, ruby glass, gems or precious stones, marbles, optical and vision glasses.
	L	Sols	Muddy water, gold sol, protein, starch, agar, gelatin in water.
	G	Aerosols of solids	Smoke, storm.
L	S	Gels	Cheese, jams, jellies, curd, plants, fruits, vegetables, cementation, butter.
	L	Emulsions	Milk, blood, cosmetic products e.g. shampoo, creams, emulsified oils polish and medicines.
	G	Aerosols of liquids	Fog, clouds, mist.
G	S	Solid foams	Pumice stone, styrene foam, foam rubber, porous pot, rubber pillows and mattresses.
	L	Foam or froths	Froths, soap suds, air bubble.
	G	Homogeneous system	–

- True solutions are homogeneous while suspensions are heterogeneous systems. The colloidal state is regarded as intermediate between the two.

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- We can say that the colloidal state is a heterogeneous dispersion of solute particles of size ranging between 10 Å to 10^3 Å (10^{-7} cm to 10^{-5} cm) into a solvent.
- Colloid is a heterogeneous state with at least two phases. The phase which is dispersed in the other (medium) is called *dispersed phase* (DP) or *internal phase*, or *discontinuous phase*.
- The phase or medium in which the dispersion is made is called the *dispersion medium* (DM) or *external phase*, or *continuous phase*.
- Colloids refer to substances in the intermediate state between true solution and suspension.
- Particle size varies between 10 Å to 1000 Å or 1 nm to 100 nm.
- Particles can only be seen in an ultra-microscope when they scatter light.
- Colloidal particles do not settle under force of gravity even on keeping for long, but may settle under centrifuge.
- Particles always carry a charge, either positive or negative.
- **Types of colloidal systems**

– Two gases cannot form a colloidal system as they diffuse and form a homogeneous mixture.

– Colloidal systems which have a fluid like appearance are called *sols*. Depending on the nature of dispersion medium the colloids (sols) are given special names:

Water – hydrosols or aquasols Alcohol – alcosols
Benzene – benzosols Gases – aerosols

– **Lyophilic colloids** or **sols** are those in which the particles have a great affinity for the DM and readily form a colloid. They are also called *intrinsic colloids*.

– **Lyophobic colloids** or **sols** are those in which the particles have very little or no affinity for the DM. Their sols are prepared by indirect methods and are also called *extrinsic colloids*.

- **Differences between lyophilic and lyophobic sols are given in the table:**

Property	Lyophilic sols	Lyophobic sols
1. Nature	DP has more affinity for DM, if DM is water then hydrophilic	Less affinity, if DM is water; hydrophobic
2. Preparation	As soon as DP comes in contact with DM, sols are formed	Special methods are required
3. Concentration of sol	More concentration of DP in sol	Less concentration of DP in sol
4. Stability	More stable	Less stable
5. Size of sol particle	Small	Large
6. Viscosity	More viscous than DM	Same as of DM
7. Surface tension	Much less than DM	Same as of DM
8. Reversibility	Reversible with temperature	Irreversible
9. Charge	The charge on sol particles depends upon pH of medium.	Independent of pH
10. Tyndall effect	Less scattering	More scattering
11. Solvation	Higher degree of solvation	Lower degree of solvation

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- Some substances on warming with suitable liquid pass into colloidal solution readily. These are called **intrinsic colloids**. e.g. gum-arabic, glue, starch, gelatin etc. The colloidal solution of such substances are lyophilic sols.
- Substances which do not pass into colloidal solution even on heating are called **extrinsic colloids**. e.g. silver, gold sol.
- **Multimolecular colloids** refer to those colloidal systems in which the dispersed phase is constituted by large aggregates of atoms or molecules with diameters less than 1 nm which are formed as a result of aggregating properties of the dispersing particles. e.g. gold sol, hydrated ferric oxide sol etc.
- **Macromolecular colloids** refer to those colloidal systems in which dispersed phase is constituted by large molecules which are called macromolecules (usually polymers). e.g. starch, cellulose, rubber etc.

- Lyophilic sols are prepared directly by mixing the substance with the dispersion medium.
- Lyophobic sols are prepared indirectly by any of the following methods:
- **Dispersion or Disintegration methods**

- Electro-dispersion by *Bredig's arc* method is used to prepare sols of metals like Ag, Cu, Au, Pt etc. A direct current is passed through electrodes of the metal, the electric arc vapourizes the metal and vapours condense in the medium to form a sol.
- **Peptization** involves the conversion of a freshly prepared precipitate into colloidal size particles by shaking with a suitable electrolyte, *e.g.* freshly prepared $Fe(OH)_3$ is treated with $FeCl_3$ or AgI with $AgNO_3$ etc. The electrolyte used is called a peptizing agent. Peptization is generally carried out by following methods:
  - **(i) By electrolytes:** Freshly formed precipitates can usually be peptized by electrolytes providing ions common with the precipitates. e.g. AgCl by $AgNO_3$, $Fe(OH)_3$ by $FeCl_3$. In this case peptization is due to the preferential adsorption of one type of ions (common ions) furnished by the electrolyte added.
  - **(ii) By adding another colloid:** Peptization of lamp black is carried out by adding gums. Lamp black peptized this way is used under the name 'Indian ink'.
  - **(iii) By washing a precipitate:** If precipitates of CuS, $BaSO_4$ or Prussian blue are washed continuously with water, a stage is reached in each case when the wash water begins to take precipitate also through the filter paper.

- **Condensation or Aggregation methods:** These methods involve the joining together of smaller particles to form colloidal size particles.
- Chemical methods involve different chemical reactions yielding a sol, e.g.

Double decomposition: $As_2O_{3(aq)} + 3H_2S_{(aq)} \rightarrow \underset{\text{sol}}{As_2S_3} + 3H_2O$

Reduction: $2AuCl_{3(aq)} + 3\ SnCl_{2(aq)} \rightarrow \underset{\text{sol}}{2Au} + 3SnCl_4$

Oxidation: $Br_{2(aq)} + H_2S_{(aq)} \rightarrow \underset{\text{sol}}{S} + 2HBr$

Hydrolysis : $FeCl_{3(aq)} + 3H_2O \rightarrow Fe(OH)_3 + 3HCl$

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- Exchange of solvent is the method used to prepare sols of substances less soluble in water, e.g. water is added to a solution of sulphur or phosphorus in alcohol to yield a sol.
- By change of state of a substance, e.g. mercury or sulphur are vapourized and the vapours passed in cold water containing a stabilizing agent (ammonium salt).
- By controlled condensation of certain insoluble substances in presence of a protective colloid, e.g. carbon sol is prepared in presence of gum arabic, or Prussian blue sol is obtained in presence of starch.

- **Dialysis** is the separation of ions or particles of crystalloids from colloids by passing them through a parchment paper or animal membrane.
- **Electrodialysis** is a fast method as ionic impurities are removed under the influence of electric field.
- **Ultra filtration** is the process of separating colloidal particles from the solvent and other solute particles using specially prepared filters or filter papers.
- **Ultra centrifugation** involves placing of a colloidal sol in a high speed centrifuge when colloidal particles settle down and can be separated to form a purified sol.
- **Physical Properties:**
  - Heterogeneous in nature.
  - Diffusion through parchment membrane is slow.
  - Pass through normal filter papers.
  - Sol particles do not settle down due to gravity.
  - Viscosity and surface tension of lyophobic colloids are almost similar to those of pure solvent but, for lyophilic colloids viscosity is higher and surface tension is lowerthan that of solvent.
- **Colligative Properties** like osmotic pressure, elevation in boiling point or depression in freezing pt., lowering of vapour pressure etc. are much lesser than true solutions due to number of particles being relatively lower.
- **Optical Properties** : All colloidal particles are capable of scattering light in all directions, giving rise to a bright glowing cone when seen sideways, this is known as *Tyndall effect*. Scattering of light depends on the wavelength of light used, size of particles and difference in refractive indices ($\Delta\mu$) of DP and DM.
- **Kinetic Properties** : Brownian motion or irregular chaotic motion is observed in sol particles upto particle size of 0.5 microns. This motion depends on temperature and offers an explanation for the stability of colloids.
- **Electrical Properties** : The dispersed phase particles carry either +ve or –ve charge and dispersion medium has an equal and opposite charge. The particles repel one another and hence do not coagulate, thus making the colloid stable.
- Cataphoresis or Electrophoresis is the movement of colloidal particles either towards cathode or anode, depending on their charge, under the influence of an electric field.
- Electro-Osmosis is the movement of only the molecules of dispersion medium towards oppositely charged electrode under the influence of electric field, whereas the colloidal particles are not allowed to move.

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- Coagulation or flocculation is the process of bringing colloidal particles closer so that they aggregate to form larger particles that precipitate and settle down or float on the surface. It is usually done by addition of an electrolyte.
  - *Hardy-Schulze rule* states that "greater is the valency of the oppositely charged ion of electrolyte being added, faster is the coagulation", e.g. for a negatively charged sol, the order is: $Al^{3+} > Ba^{2+} > K^{+}$, for a positively charged sol the order is: $Fe(CN)_6^{4-} > PO_4^{3-} > SO_4^{2-} > Cl^{-}$.
  - *Coagulating power* is the minimum amount of electrolyte required to coagulate a definite amount of sol.
- **Protection of colloids**

– Lyophilic sols are more stable than lyophobic sols hence they are used as *protective colloids* to increase the stability of lyophobic sols, e.g. addition of gums, gelatin etc. to certain metal sols.

– Protective action of lyophilic sols is explained due to formation of a thin layer around the lyophobic sol particles, thus preventing them from coming closer.

– Gold number is a term used to compare protective action of different lyophilic colloids.

– Gold number of a lyophilic sol is the minimum amount of it in milligram, which prevents the coagulation of 10 ml gold sol against 1 ml of 10% NaCl solution.

– Higher the gold number, lesser is the protective power.

– Examples:

Sol	Gold number
Gelatin	0.005-0.01
Casein	0.01-0.02
Gum-arabic	0.15-0.25
Potato Starch	20-25

- **Emulsion** is a colloidal system involving one liquid dispersed in another, provided both are immiscible.
- Emulsifiers or emulsifying agents are substances which help in making the emulsions stable, e.g. soaps, agar, gum etc.
- Examples: milk, butter, milk cream, cold cream, vanishing cream, etc. There are many drugs and medicines which are also in the form of emulsions e.g. various ointments, cod liver oil etc.
- **Oil in water emulsions (*O* in *W*)** are those in which oil is the dispersed phase and water is the dispersion medium e.g. milk.
- **Water in oil emulsions (*W*/*O*)** are those in which water is the dispersed phase and oil is the dispersion medium, e.g. butter, cream etc.
- The process of breaking an emulsion to yield the constituent liquids is known as *demulsification*.
- **Gels** are colloidal systems where liquids are the dispersed phase and solids act as dispersion medium, e.g. curd, cheese etc. Liquid rich systems are called *jellies*.
- Gelling agent is added to stabilize a gel, e.g. gelatin.
- Elastic gels are reversible and can be changed back to original form even after dehydration, e.g. gelatin, agar-agar etc. swell up after absorbing water (imbibition).

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- Non-elastic gels are irreversible and change into a powder on dehydration which does not absorb water, e.g. silicic acid.
- *Syneresis* or weeping of gels is the loss of water (liquid) without disturbing the gel structure.
- Some gels (e.g. gelatin) liquify on shaking in a sol. The sol on standing again changes into a gel. This property is known as **thixotropy**.
- **Surfactants** are substances which possess surface activity, i.e. lower the surface tension or increase surface area.
- Surfactants get preferentially adsorbed at the air-water, oil water and solid-water interfaces, forming an oriented mono layer.
- The hydrophilic groups point towards the aqueous phase and the hydrocarbon chains (hydrophobic) point towards air or the oil phase.
- **Types of surfactants**

(i) *Anionic surfactants* : Soaps and detergents

(ii) *Cationic surfactants* : Quarternary ammonium salts of long chain tertiary amines form detergents which are cationic surfactants. e.g. octadecyl ammonium chloride, $C_{18}H_{37}N^+H_3Cl^-$; cetyl tri-methyl ammonium chloride, $C_{16}H_{33}(CH_3)_3N^+Cl^-$.

(iii) *Non-inorganic surfactants* : This type of surfactants are formed when alcohols react with epoxides.

e.g. $C_nH_{2n+1} - OH + n\underset{\diagdown\ \diagup}{CH_2 - CH_2}\ \longrightarrow\ C_nH_{2n+1} - (O - CH_2 - CH_2 -)_n - OH$ (O)

- **Micelles** are an aggregation of ions or molecules from a sol and are sub-microscopic, e.g. soaps and detergents.
- The minimum concentration of the surfactant at which micelle formation starts is called, *Critical Micelle Concentration*, (CMC).
- Lesser the CMC of surfactant, more is its surface activity and cleaning action (for detergents).
- Shapes of micelles change with change in concentration, e.g. at high concentrations, rod-shaped micelles are formed while spherical micelles are formed near CMC.
- The importance of micelles and their use is based on the fact that their hydrophobic interior can dissolve fat or oil etc. While the water soluble part, makes a hydrophilic surface around this interior, rendering the entire micelle water soluble.
- **Aqua-dag** is a colloidal solution of graphite in water.
- **Oil-dag** is a colloidal solution of graphite in oil.
- **Saturation state** is the state of the system when extent of adsorption becomes constant (x/m value does not change) and does not change with pressure.
- Macromolecules are themselves composed of giant molecules and dissolve in a solvent to yield colloidal solutions directly.
- The dimensions of the macromolecules fall in a range between 10 Å and 10,000 Å.
- Number of average molecular weight is defined as :

$$\bar{M}_n = \frac{\text{total weight, } W}{\text{total number of particles}} = \frac{\sum n_i M_i}{\sum n_i}.$$

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where $n_i M_i$ stands for the weight of macromolecules numbering n_i and having molecular weight M_i.

- Weight average molecular weight is defined as

$$\bar{M}_w = \frac{m_1 M_1 + m_2 M_2 + ...}{m_1 + m_2 + ...} = \frac{\sum m_i M_i}{\sum m_i}$$

where m_1, m_2 etc. represents mass of macromolecules having molecular weights M_1, M_2 etc.

- The **deltas** at the mouths of great rivers are formed by the precipitation of the charged clay particles carried in suspension in the river water, by the action of salts present in sea-water.
- The **chrome tanning of leather** is brought about by the penetration of positively charged particles of hydrated chromic oxide into the leather.

End

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13

CHAPTER

periodic properties

The periodic table displays all chemical elements systematically in order of increasing atomic number i.e. the number of protons in the nucleus.

Historical models of periodic table

1. ***Doebereiner's triads, 1829***

Doebereiner classified elements into a group of three, called triads. In the triads of element the atomic weight of the middle element was the arithmetic mean of the atomic weights of the other two.

Li	Na	K	Ca	Sr	Ba
7	23	39	40	88	137

2. ***Newland's octet law, 1864***

If the elements are arranged in order of their increasing atomic weights, every eighth elements had similar properties to first one like the first and eight note is music.

sa	re	ga	ma	pa	dha	ni	sa
Li	Be	B	C	N	O	F	Na
Na	Mg	Al	Si	P	S	Cl	K

Inert gases were not discovered till then.

3. ***Lothar Meyer's atomic volume curve, 1869***

Lothar Mayer plotted a graph between atomic weight and atomic volume (*i.e.* atomic weight in solid state/density). Elements with similar properties occupied the similar positions on the graph. Strong electropositive elements of IA except Li *i.e.* Na, K, Rb, Cs etc. occupied the top position on the graph. IIA group elements Be, Mg, Ca, Sr, Ba etc. occupied the positions on the ascending part of the graph. Inert gases except He occupied the positions on the descending part of the graph. Halogens also occupied the descending part of the graph.

4. ***Mendeleev's periodic law***

The physical and chemical properties of elements are periodic functions of their atomic weights. If the elements are arranged in the order of their increasing atomic weights, after a regular interval elements with similar properties are repeated. The table is divided into nine vertical columns called groups and seven horizontal rows called periods.

Characteristics of periods

(a) First period is called shortest period and contains only two elements. Second and third periods are called short periods containing eight elements each. Fourth and fifth periods

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are long periods containing eighteen elements each. Sixth period is longest period with thirty-two elements. Seventh period is an incomplete period containing nineteen elements. Numbers 2, 8, 8, 18, 18, 32 are called magic numbers.

(b) Lanthanide and actinide series containing 14 elements each are placed separately under the main periodic table. These are related to sixth and seventh periods of IIIrd group respectively.

(c) Elements of third period from sodium (Na) to chlorine (Cl) are called representative or typical elements.

(d) Valency of an element in a period increases from 1 to 7 with respect to oxygen.

Na_2O	MgO	Al_2O_3	SiO_2	P_2O_5	SO_3	Cl_2O
1	2	3	4	5	6	7

(e) From left to right in a period generally

(i) Atomic weight, effective nuclear charge, ionisation potential, electronegativity and electron affinity of an element increases.

(ii) Atomic radius, electropositive character and metallic character of an element decreases.

(f) Diagonal relationship — Elements of second period Li, Be and B resemble closely with the elements Mg, Al and Si of third period in the next higher group.

Second period	Li	Be	B	C
Third period	Na	Mg	Al	Si

(g) Elements of second period are called bridge elements.

Characteristic of groups

(a) Mendeleef's periodic table contains nine groups. These are represented by Roman numerals I, II, III, IV, V, VI, VII, VIII and zero. Groups I to VII are divided into subgroups A and B, group VIII consists of three sets, each one containing three elements.

(b) Inert gases are present in zero group. These were not discovered till that time.

(c) The valency of an element in a group is equal to the group number.

(d) There is no resemblance in the elements of subgroups A and B of same group, except valency.

(e) The elements of the groups which resemble with typical elements are called normal elements. For example - IA, IIA, IIIA, IVA, VA, VIA, VIIA group elements are normal elements.

(f) Those elements of the groups which do not resemble with typical elements are called transition elements. For example- IB, IIB, IIIB, IVB, VB, VIB, VIIB and VIII group elements are transition elements.

(g) Hydrogen is placed in both IA and VIIA groups.

(h) In a group, from top to bottom in general,

(i) Atomic weight, atomic size, electropositive character and metallic character of an element increases.

(ii) Ionisation potential, electron affinity and electronegativity of an element decreases.

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5. *Long form of the periodic table or Mosley's periodic table*

Mosley (1909) studied the frequency of X-rays produced by the bombardment of a strong beam of electrons on a metal target. He found that the square root of the frequency of X-rays ($\sqrt{\upsilon}$) is directly proportional to the number of nuclear charge (Z) of metal. $\sqrt{\upsilon} = a(Z - b)$ where a and b are constants. Nuclear charge of metal is equal to the atomic number. So Mosley related the properties of elements with their atomic number and gave the new periodic law.

According to him, physical and chemical properties of elements are the periodic functions of their atomic number. If the elements are arranged in order of their increasing atomic number, after a regular interval, element with similar properties are repeated.

6. *Long form of periodic table or Bohr's periodic table*

With better understanding of the role of electrons in the properties of elements and the development of the nature of the electrons in atoms, a better understanding of the periodic properties of elements or the periodic table was possible. The long form of periodic table is also known as Bohr's periodic table.

The long form of periodic table offers the following advantages over the Mendeleev's classification.

(i) Sub-group A and sub-group B elements are placed separately. There is clear demarcation between metals and non-metals.

(ii) The nine elements of group VIII have been placed in separate groups corresponding to d^6, d^7 and d^8 configurations.

(iii) Fourteen lanthanones are not pushed together, but are assigned a separate group for each lanthanon in the *f*-block of elements.

(iv) The uniform trivalent state of the lanthanones can be explained due to the availability of only three electrons in the outer, high energy levels, the differentiating electrons going to the inner chemically inert orbitals.

(v) Uniform bivalence for the transition elements is due to the presence of outer ns^2 electrons, which makes them electropositive in nature.

(vi) The change from a highly electronegative to electropositive character through inert gas structure has been explained on the basis of the long form of periodic table.

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Types of elements

Classification of elements on the basis of their electronic configuration

On the basis of electronic configuration, the elements may be divided into four groups:

(i) *s*-block elements

(a) These are present in the left part of the periodic table.

(b) These are IA and IIA *i.e.* 1 and 2 group elements.

(c) These are metals.

(d) In these elements last electron fills in the *s*-orbital.

(e) Electronic configuration of valence shell is ns^{1-2} (n = 1 to 7).

(ii) *p*-block elements

(a) These are present in right part of the periodic table.

(b) These constitute the groups IIIA to VIIA and zero groups *i.e.* groups 13 to 18 of the periodic table.

(c) Most of these elements are metalloids and nonmetals but some of them are metals also.
(d) The last electron fills in *p*-orbital of valency shell.
(e) The electronic configuration of valence shell is ns^2np^{1-6} (n = 2 to 7).
(f) ns^2np^6 is stable noble gas configuration. The electronic configuration of He is $1s^2$.

(iii) *d*-block elements

(a) These are present in the middle part of the periodic table (between *s* and *p* block element).
(b) These constitute IIIB to VIIB, VIII, IB and IIB *i.e.* 3 to 12 groups of the periodic table.
(c) All are metals.
(d) The last electrons fills in $(n-1)d$ orbital.
(e) The outermost electronic configuration is $(n-1)d^{1-10}ns^{1-2}$ (n = 4 to 7).
(f) There are three series of *d*-block elements as under
3*d* series - Sc (21) to Zn (30)
4*d* series - Y (39) to Cd (48)
5*d* series - La (57), Hf (72) to Hg (80)

(iv) *f*-block elements

(a) These are placed separately below the main periodic table.
(b) These are mainly related to IIIB *i.e.* group 3 of the periodic table.
(c) There are two series of *f*-block elements as under
4*f* series - Lanthanides - 14 elements - Ce (58) to Lu (71)
5*f* series - Actinides - 14 elements - Th (90) to Lr (103)
(d) The last electron fills in $(n-2)$ *f*-orbital.
(e) Their outermost electronic configuration is $(n-2)f^{1-14}(n-1)s^2(n-1)p^6(n-1)d^{0-1}ns^2$ (n = 6 and 7).

Bohr's classification of elements

On the basis of electronic configuration of the incomplete shells, the elements are classified into five main categories-

1. Inert gases
2. Representative elements
3. Transition elements
4. Inner transition elements
5. Transuranium elements.

Although this classification is convenient for understanding of chemical properties of the elements, it overlooks the specific properties of the individual elements.

1. Inert gases

(a) *s* and *p*-orbitals of the outer most shell of these elements are completely filled. The outermost electronic configuration is $ns^2\ np^6$.
(b) He is also inert gas but its electronic configuration is $1s^2$.

2. Representative or normal elements

(a) Outermost shell of these elements is incomplete. The number of electrons in the outermost shell is less than eight.
(b) Inner shells are complete.
(c) *s*- and *p*-block elements except inert gases are called normal or representative elements.

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3. Transition elements

(a) Last two shells of these elements namely outermost and penultimate shells are incomplete.

(b) The last shell contains one or two electrons and the penultimate shell may contain more than eight up to eighteen electrons.

(c) Their outermost electronic configuration is similar to *d*-block elements *i.e.* $(n-1)\,d^{1-10}\,ns^{1-2}$.

(d) According to latest definition of transition elements those elements which have partly filled *d*-orbitals in neutral state or in any stable oxidation state are called transition elements. According to this definition, Zn, Cd and Hg (IIB group) are *d*-block elements but not transition elements because these elements have d^{10} configuration in neutral as well as in stable +2 oxidation state.

4. Inner transition elements

(a) In these elements last three shells *i.e.* last, penultimate and prepenultimate shells are incomplete.

(b) These are related to IIIB *i.e.* group 3.

(c) The last shell contains two electrons. Penultimate shell may contain eight or nine electrons and prepenultimate shell contains more than 18 upto 32 electrons.

(d) Their outermost electronic configuration is similar to *f*-block elements *i.e.* $(n-2)f^{1-14}\,(n-1)s^2(n-1)p^6(n-1)d^{0-1}\,ns^2$.

5. Transuranium elements

Elements of the seventh period after atomic number 93 (*i.e.* actinides) are synthetic elements and are called transuranium elements.

(a) In 1934, an Italian Physicist Enrico Fermi had observed that when an element is bombarded with slow neutrons, the element is transformed into a new element having next higher atomic number. First transuranic element, having atomic number 93 was identified by American Physicist Edwin Mcmilan and Philip H. Abelson. In the next year, element number 94 was discovered in uranium fission products by American Chemist Glenn T. Seaborg and coworkers. The elements 93 and 94 were named Neptunium (Np) and Plutonium (Pu) respectively for Neptune and Pluto, the planets discovered after Uranus.

$_{92}U^{238} + {}_0n^1 \rightarrow {}_{92}U^{239} + \gamma\text{-rays}$

$_{92}U^{239} \rightarrow {}_{93}Np^{239} + {}_{-1}e^0$

$_{93}Np^{239} \rightarrow {}_{94}Pu^{239} + {}_{-1}e^0$.

(b) In 1944, Seaborg and coworkers at the University of California, Berkeley made new elements, 95 and 96 by bombarding uranium and plutonium with accelerated alpha particles. These elements were named Americium (Am) after America and Curium (Cm) after Curies.

$_{92}U^{238} + {}_2He^4 \rightarrow {}_{94}Pu^{241} + {}_0n^1$

$_{94}Pu^{241} \rightarrow {}_{95}Am^{241} + {}_{-1}e^0$

$_{94}Pu^{239} + {}_2He^4 \rightarrow {}_{96}Cm^{242} + {}_0n^1$.

(c) Seaborg and coworkers bombarded elements 95 and 96 to produce element number 97 in 1949 and element number 98 in 1950. These two elements were named Berkelium (Bk) and Californium (Cf) after Berkeley and California. For this work Seaborg and Mcmilan shared the 1951 Nobel Prize in chemistry.

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$_{95}Am^{241} + {}_2He^4 \rightarrow {}_{97}Bk^{245} + \gamma$

$_{96}Cm^{242} + {}_2He^4 \rightarrow {}_{98}Cf^{246} + \gamma$

$_{96}Cm^{242} + {}_2He^4 \rightarrow {}_{98}Cf^{244} + 2\ {}_0n^1$.

(d) Elements number 99 and 100 were produced in the laboratory by Berkeley group by neutron bombardment in 1952. Element number 99 was named as Einsteinium (Es) and 100 as Fermium (Fm) after Albert Einstein and Enrico Fermi respectively.

$_{98}Cf^{252} + {}_0n^1 \rightarrow {}_{98}Cf^{253} + \gamma$

$_{98}Cf^{253} \rightarrow {}_{99}Es^{253} + {}_{-1}e^0$

$_{99}Es^{253} + {}_0n^1 \rightarrow {}_{99}Es^{254} + \gamma$

$_{99}Es^{254} \rightarrow {}_{100}Fm^{254} + {}_{-1}e^0$

$_{99}Es^{253} + {}_2He^4 \rightarrow {}_{101}Md^{256} + {}_0n^1$.

(Mendelevium)

(e) Element number 102 was made in 1958 through a collaboration between university of California, Berkeley and the Nobel institute in Sweden. Using a complicated type of bombardment a small amount element 102 was produced. It was named as Nobelium (No) after Alfred Nobel. In 1961, the Berkeley group reported detection of very small amount element 103. Element 103 was named Lawrencium (Lw or Lr) after E. O. Lawrence.

$$_{96}Cm^{246} + {}_6C^{12} \rightarrow {}_{102}No^{254} + 4\ {}_0n^1$$

$$_{98}Cf^{250} + {}_5B^{11} \rightarrow {}_{103}Lr^{257} + 4\ {}_0n^1$$

(f) Berkeley team also succeeded in making two elements 104 and 105. They named element 104 as Rutherfordium after Ernest Rutherford. Element 105 was named as Dubnium.

(g) For elements beyond atomic number 106, it would require not only more powerful particle accelerator; but also highly sensitive detection and analysis system capable of identifying a few atoms of extremely short-lived elements.

• During 1981-1985 a team of scientists at institute of heavy ion research led by Peter Armbruster used the new technique with the new detector to synthesise and identify elements 107 to 109, which were named as

107 - Bohrium (Bh) after Niels Bohr

108 - Hassium (Hs) after German State of Hasse

109 - Meitnerium (Mt) after Austrian Physicist Lise Meitner.

Elements 110, 111 and 112 are also named as ununnilium, unununium and ununbium.

PERIODICITY IN PROPERTIES

• The term periodicity in properties in the classification of elements means that same properties of the elements reappear at definite intervals when the elements are arranged in order of their increasing atomic numbers. In modern periodic table, these intervals are 2, 8, 8, 18, 18 and 32, *i.e.,* similar properties are observed with elements belonging to the same subgroup which have been arranged in subgroups after the difference of either 2 or 8 or 18 or 32 in atomic numbers as similar valency-shell electronic configuration recur after certain regular intervals of atomic number. **This is the cause of periodicity in properties.**

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Electronic configurations of alkali metals

Element	At. No.	Electronic configuration
Li	3	$1s^2\ 2s^1$
Na	11	$1s^2\ 2s^2\ 2p^6\ 3s^1$
K	19	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$
Rb	37	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 5s^1$
Cs	55	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 5s^2\ 5p^6\ 6s^1$
Fr	87	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 5s^2\ 5p^6\ 6s^2\ 4f^{14}\ 5d^{10}\ 6p^6\ 7s^1$

Thus, the **cause of periodicity** of the properties of elements is the repetition of similar electronic configuration of their atoms in the outermost energy shell (or valence shell) after certain regular intervals.

- **Atomic radii**

– It is usually defined **as the distance between the nucleus and outermost shell where electron or electrons are present.** Three types of radii are commonly used, i.e., (a) covalent radii (b) crystal radii (c) van der Waals' radii. **Covalent radius is defined as half of the distance between the two nuclei of two like atoms bonded together by a single covalent bond.**

– Considering a homonuclear diatomic molecule A_2, bonded together by a single covalent bond, it is assumed that electron clouds of each atom touch each other. Let the bond length be d_{A-A}.

Then $d_{A-A} = r_A + r_A = 2r_A$

So $$r_A = \frac{d_{A-A}}{2}$$

– In a heteronuclear diatomic AB molecule if both atoms are linked by a single covalent bond and have nearly same electronegativity, the bond length d_{A-B} is equal to sum of covalent radii of the two atoms.

$$d_{A-B} = r_A + r_B$$

– If the covalent bond is formed between two elements of different electronegativity then we use the following relation:

$$d_{A-B} = r_A + r_B - 0.09\ (X_A - X_B)$$

where X_A and X_B are electronegativity of A and B respectively. This relation was given by **Stevenson** in 1941.

- **Crystal radii :** It is defined as one half of the distance between the nuclei of two adjacent metal atoms in the metallic closed packed crystal lattice in which metal exhibits a coordination number of 12.

- **van der Waals' radii :** It is half of the distance between the nuclei of two nonbonded neighbouring atoms of two adjacent molecules.

$$r_{\text{covalent}} < r_{\text{crystal}} < r_{\text{van der Waals}}$$

Atomic radius in the nth orbit is given by

$$r_n = \frac{n^2 a_0}{Z^*}$$

where n is principal quantum number (*i.e.*, number of shell), a_0, the Bohr's radius of H-atom (= 0.529Å) and Z^*, the effective nuclear charge.

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- **The screening effect or shielding effect and Effective nuclear charge**

In a multielectron atom, the electrons of the valency shell (outermost shell) are attracted towards the nucleus and repelled by the electrons present in the inner shells. On account of this, the combined effect of this attractive and repulsive force acting on the valence electron is that the valence electron experiences less attraction from the nucleus. This decrease in the force of attraction exerted by the nucleus on the valency electrons due to the presence of electrons in the inner shells, is called screening effect or shielding effect.

Effective atomic number

Due to screening effect the valence electron experiences less attraction towards nucleus. This brings decrease in the nuclear charge (Z) actually present on the nucleus. The reduced nuclear charge is termed **effective nuclear charge** and is represented by Z^*. It is related to actual nuclear charge (Z) by the following formula :

$Z^* = (Z - \sigma)$ where σ is screening constant.

The magnitude of 'σ' is determined by the **Slater's rules.** The contribution of inner electrons to the magnitude of 'σ' is calculated in the following ways:

Slater's rule for Estimating Effective Nuclear Charges, Z^*

(1) Write out the electronic configuration of the element in the following order and grouping: (1*s*), (2*s*, 2*p*), (3*s*, 3*p*), (3*d*), (4*s*, 4*p*), (4*d*), (4*f*), (5*s*, 5*p*) etc.

(2) Electrons in any group higher in the sequence that the electron under consideration contribute nothing to the shielding σ.

(3) Then for an electron in an *ns* or *np* orbital
   (a) all other electrons in (*ns*, *np*) group contribute $\sigma = 0.35$ each
   (b) all electrons in the $n - 1$ shell contribute $\sigma = 0.85$ each
   (c) all electrons in the $n - 2$ or lower shell contribute $\sigma = 1.00$ each

(4) For electron in an *nd* or *nf* orbital, all electrons in the same group contribute $\sigma = 0.35$ each; those in group lying lower in the sequence than the (*nd*) or (*nf*) group contribute $\sigma = 1.00$ each.

- Radius is also dependent on the extent of force of attraction which pulls outer shell inward.

Variation in Period

	Li	Be	B	C	N	O	F	Ne
Z	3	4	5	6	7	8	9	10
σ	1.7	2.05	2.40	2.75	3.10	3.45	3.80	4.15
Z^*	1.30	1.95	2.60	3.25	3.90	4.55	5.20	5.85
n	2	2	2	2	2	2	2	2
r_n (pm)	123	90	80	77	75	74	72	160

In a period, left to right :

- Z (atomic no.) increases (by one unit)
- Z^* also increases (but by 0.65 unit)
- n (number of shells) remains constant

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Thus $r_n \propto 1/Z^*$

In case of Noble gases (as in Ne) there is no covalent bond formation, hence only van der Waals radius is considered. Thus there is high jump in the value of radius from F (72 pm) to Ne (160 pm).

Variation in a Group

Element	Z	σ	Z^*	n	Radius (pm)
Li	3	1.7	1.3	2	123
Na	11	8.8	2.2	3	157
K	19	16.8	2.2	4	203
Rb	37	34.8	2.2	5	216
Cs	55	52.8	2.2	6	235
Fr	87	84.8	2.2	7	—

In a group, top to bottom :

- Z increases
- Z^* almost remains constant
- n increases

Thus $r_n \propto n^2/Z^*$

Hence atomic radius in a group is dependent on the value of n.

- **Ionic radii**

It is defined as the distance between the nucleus and outermost shell of an ion or it is the distance between the nucleus and the point where the nucleus exerts its influence on the electron cloud.

Metal ions are smaller than the atoms from which they are formed.

When a positive ion is formed, the number of positive charges on the nucleus exceeds the number of orbital electrons, and the effective nuclear charge (which is the number of charges on the nucleus to the number of electrons) is increased. This results in the remaining electrons being more strongly attracted by the nucleus. These electrons are pulled in further reducing the size.

A positive ion is thus always smaller than the corresponding atom, and the more electrons which are removed, the smaller the ion becomes.

Thus $Mg > Mg^+ > Mg^{2+}$

$Fe > Fe^{2+} > Fe^{3+}$

The negative ion is always larger than that of the corresponding atom.

– Negative ion is formed by gain of one or more electrons in the neutral atom and thus number of electrons increases but magnitude of nuclear charge remains the same.

– Due to decrease in nuclear charge per electron, there is expansion of outer shell. Thus size of anion is increased.

$O^{2-} > O^- > O$

$I^- > I > I^+$

These can be explained on the basis of Z/e ratio $\left(\frac{\text{Nuclear charge}}{\text{No. of electrons}}\right)$. When Z/e ratio increases, the size decreases and when Z/e ratio decreases, the size increases.

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	Na	Na^+	Cl	Cl^-
Z/e	$\frac{11}{11}=1.0;$	$\frac{11}{10}=1.1;$	$\frac{17}{17}=1.0;$	$\frac{17}{18}=0.95;$
So	$Na^+ < Na$		$Cl^- > Cl$	

For isoelectronic species the size decreases with an increase of atomic number. This is illustrated in the following table:

Atom or Ion	Atomic Number *Z*	No. of electrons *e*	*Z/e* ratio	Size in Å
O^{2-}	8	10	0.8	1.40
F^-	9	10	0.9	1.30
Ne	10	10	1.0	1.12
Na^+	11	10	1.1	0.95
Mg^{2+}	12	10	1.2	0.65

H^+ and Cs^+ are the smallest and largest cations respectively.
H^- and I^- are the smallest and largest anions respectively.

Ionisation potential or Ionisation energy

The minimum amount of energy required to remove the most loosely bound electron from an isolated atom in the gaseous state is known as ionisation potential or ionisation energy or first ionisation potential (I_1) of the element.

$$M \xrightarrow{\text{energy}} M^+ + e$$

$$\text{Ionization potential (eV)} = \frac{\text{Ionization energy in Joule}}{\text{Charge of electron } (1.6\times10^{-19})}$$

The energy required to remove the second electron from the monovalent cation is called second ionisation potential (I_2).

$$M^+ + I_2 \rightarrow M^{2+} + e$$

Similarly, we have third, fourth . . . ionisation potentials.

$$M^{2+} + I_3 \rightarrow M^{3+}\ e$$
$$M^{3+} + I_4 \rightarrow M^{4+} + e$$

It is observed that I_2 is higher than I_1, I_3 is higher than I_2 and so on, *i.e.*, $I_1 < I_2 < I_3 < I_4$. The increase in the values of successive ionisation potentials can be explained on the basis that effective nuclear charge increases from $M_{(g)}$ to $M^{n+}{}_{(g)}$, *i.e.*, force of attraction of the outermost electron towards nucleus increases.

Factors affecting the value of ionisation potential

	Properties	Affect
1.	Atomic Size	Larger the atomic size, smaller is the value of ionisation potential.
2.	Screening Effect	Higher the screening effect, the lesser is the value of ionisation potential.
3.	Nuclear charge	Ionisation potential increases with the increase in nuclear charge.
4.	Penetration effect or Shape of orbtial	Values of ionisation potential for *s, p, d* and *f* electrons are as : $s > p > d > f$.

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If an atom has fully filled or half-filled orbitals, its IE is higher than expected normally from its position in the periodic table.

e.g.	Li	Be	B	C	N	O	F	Ne
	$2s^1$	$2s^2$	$2s^22p^1$	$2s^22p^2$	$2s^22p^3$	$2s^22p^4$	$2s^22p^5$	$2s^22p^6$
eV	5.4	9.3	8.3	11.2	14.5	13.6	17.4	21.6

Be (fully filled 2*s* orbital) and N (half filled 2*p* orbital) have higher values than expected due to stable configurations.

- **Variation of (IE) in a group**

Force of attraction between electrons and nucleus decreases and tendency to remove the valence electron increases. Hence (IE) decreases on moving down the group.

- **Variation of (IE) in a period**

On moving across a period, the atomic size decreases and nuclear charge increases and therefore the force of attraction exerted by the nucleus on the electron in outermost shell increases. Hence **(IE) increases along a period** from left to right.

- The energies required to remove subsequent electrons from the atom in the gaseous state, are known as **successive ionisation energies.** The term first, second, third ..... ionisation energy refers to the removal of first, second, third ..... electron respectively.

Successive ionisation energies are higher : The second ionisation energies are higher than the first ionisation energies. This is mainly due to the fact that after the removal of the first electron, the atom changes into monovalent positive ion. In the ion, the number of electrons decreases but the nuclear charge remains the same. As a result of this, the remaining electrons are held more tightly by the nucleus and it becomes difficult to remove the second electron. Hence, the value of second ionisation energy, IE_2, is greater than that of the first (IE_1).

Electron affinity

- The amount of energy released when an electron is added to an isolated gaseous atom to produce a monovalent anion is called electron affinity or first electron affinity.

$$A + e \longrightarrow A^- + \text{energy}$$

- Most elements have a negative electron affinity. This means they do not require energy to gain an electron; instead, they release energy. Atoms more attracted to extra electrons have a more negative electron affinity. Chlorine most strongly attracts extra electrons; radon most weakly attracts an extra electron.

Although electron affinities vary in a chaotic manner across the table, some patterns emerge. Generally, nonmetals have more negative electron affinities than metals. However, the noble gases are an exception: they have positive electron affinities.

Factors affecting the value of electron affinity

	Properties	Affect
1.	Nucelar charge	Electron affinity increases with the increase in nuclear charge.
2.	Atomic size	With the increase in atomic size, electron affinity decreases.
3.	Electronic configuration	Electron affinities are low or almost zero in cases of stable configurations i.e. half filled or full-filled valence shell.

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	Li	Be	B	C	N	O	F	Ne
	$2s^1$	$2s^2$	$2s^22p^1$	$2s^22p^2$	$2s^22p^3$	$2s^22p^4$	$2s^22p^5$	$2s^22p^6$
eV	–0.61	0.0	– 0.30	–1.25	– 0.20	–1.48	–3.6	0.0

Be, N and Ne have low values due to stable configurations.

- The electron affinities increase across a row (since the radius slightly decreases, because of the increased attraction from the nucleus, and the number of electrons in the top shell increases, helping the atom reach maximum stability) in the periodic table and decrease going down a family (because of a large increase in radius and number of electron that decrease the stability of the atom, repulsing each other).
- Electron affinities are not limited to the elements but also apply to molecules. For instance the electron affinity for benzene is positive, that of naphthalene near zero and that of anthracene positive.

- **Successive electron affinities**

Like ionisation energies, the second and higher electron affinities are also possible. However, second electron is added to a negatively charged ion and the addition is opposed by coulombic repulsions. The energy has to be supplied to force the second electron into the anion.

Electronegativity

- Electronegativity is a measure of the tendency of an element to attract electrons to itself.

$$X_{(g)} + e^- \rightarrow X^-_{(g)}$$

In a molecule, tendency of the atom to attract bonding pair towards itself is its electronegativity. *B* is said to be more electronegative than *A* if it pulls bonding pair towards itself.

$$(A) \overset{\cdot}{\times} (B) \longrightarrow \overset{\delta+}{(A)} - \overset{\delta-}{(B)}$$

- An arbitrary value of 4.0 has been assigned to fluorine (most electronegative element) and the electronegativities of other elements have been calculated against this standard by application of following formula:

$$X_A - X_B = 0.208[E_{A-B} - (E_{A-A} \times E_{B-B})^{1/2}]^{1/2}$$

where X_A and X_B are the electronegativities of two atoms *A* and *B* and E_{A-B}, $E_{(A-A)}$ and E_{B-B} are bond energies of molecules $A - B$, A_2 and B_2, respectively.

- Mulliken regarded electronegativity as the average value ionisation potential and electron affinity of an atom.

$$\text{Electronegativity} = \frac{\text{IP} + \text{EA}}{2}$$

- On Pauling scale, $\text{electronegativity of an atom} = \frac{\text{IP} + \text{EA}}{5.6}$

Values of IP and EA are taken in eV

$$\text{Electronegativity of an atom} = \frac{\text{IP} + \text{EA}}{2 \times 62.5}$$

values of IP and EA are taken in kilo calories per mole

- The **Allred-Rochow scale** in chemistry is a measure of electronegativity. The electrostatic force of attraction between an electron and the nucleus is given by: e^2Z/r^2

where *r* is the distance between the electron and the nucleus (covalent radius) and e is the charge on an electron. *eZ* is the charge effective at the electron due to the nucleus and its surrounding electrons.

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The quantity Z/r^2 correlates well with Pauling electronegativities and the two scales can be made to coincide by expressing the Allred-Rochow electronegativity as:
$AR = 0.744 + 0.359Z/r^2$

- **Electronegativity trends**

Each element has a characteristic electronegativity ranging from 0 to 4 on the Pauling scale. The most strongly electronegative element, fluorine, has an electronegativity of 3.98 while weakly electronegative elements, such as lithium, have values close to 1. The least electronegative element is francium at 0.7. In general, the degree of electronegativity decreases down each group and increases across the periods. Across a period, non-metals tend to gain electrons and metals tend to lose them due to the atom striving to achieve a stable octet. Down a group, the nuclear charge has less effect on the outermost shells. Therefore, the most electronegative atoms can be found in the upper, right hand side of the periodic table, and the least electronegative elements can be found at the bottom left. Consequently, in general, atomic radius decreases across the periodic table, but ionization energy increases.

- **Importance of electronegativity :**

(i) Nature of the bond between two atoms can be predicted from the electronegativity difference of the two atoms.

(a) The difference $X_A - X_B = 0$, *i.e.,* $X_A = X_B$, the bond is purely covalent.

(b) The difference $X_A - X_B$ is small, *i.e.*. $X_A > X_B$, the bond is polar covalent.

(c) The difference $X_A - X_B$ is 1 .7, the bond is 50% covalent and 50% ionic.

(d) The difference $X_A - X_B$ is very high, the bond is more ionic and less covalent.

Percentage ionic character may be calculated as:

Percentage of ionic character = $16 \mid X_A - X_B \mid + 3.5 (X_A - X_B)^2$

where X_A and X_B represents electronegativity of bonded atoms A and B.

(ii) Greater the value of difference $(X_A - X_B)$ more stable will be the bond.

	H — F	H — Cl	H — Br	H — I
$(X_A - X_B)$	1.9	0.9	0.7	0.4

Stability decreases ⟶

Stability of compounds in which $X_A - X_B$ is very small are unstable in nature, NCl_3 (0.0), PH_3 (0), AsH_3 (0.1) are unstable.

In short, periodic properties can be studied as follows:

Properties	along the period	down the group
Ionisation potential	increases	decreases
Electron affinity	increases	decreases
Electronegativity	increases	decreases
Atomic radii	decreases	increases
Ionic radii	iso-electronic ions decrease their radii with increase in atomic number	increases
Atomic volume	decreases upto metals and then increases	increases

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Properties	along the period	down the group
Melting point/ boiling point	increases along the period for metals	decreases
Density	increases for metals	increases
Oxidant-reductant nature	reducing nature decreases	reducing nature of metals increases oxidising nature of non metals decreases
Metallic character	decreases	increases
Electropositive character	decreases	increases
Oxide nature	basic character decreases	basic character increases
Hydride nature	basic character decreases or acidic character increases	basic character increases
Valency	with respect to oxygen increases from 1-7 along the period, with respect to hydrogen increases from 1 to 4 and then decreases to 1.	remains the same

End

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14
CHAPTER

metallurgy

- Metals are substances characterized by a shine, called ***metallic lusture*** having high electrical and heat conductivity and are malleable and ductile.
- The lusture of metals is due to presence of mobile electrons.
- Metals generally have low ionization energies and low electronegativities, easily form cations.
- Metallic elements have oxidation states equal to group number or those in 5th and 6th group also have oxidation states equal to group number minus 2. e.g. Sn, Pb have oxidation states + 4 and +2 respectively.
- Metals form basic oxides, unless metal is in high oxidation state. e.g. chromium VI oxide (CrO_3) is acidic.
- Major source of metals and their compounds is the earth's crust, existing in clay and rocks as silicates.
- Most metallic elements are highly reactive and as such are not found in the native (free) state.
- Metallic elements like gold, silver and copper are a few which occur in the native state.
- The most abundant element present in the earth's crust is oxygen and most abundant metal is aluminium.
- Mineral is a naturally occurring inorganic solid substance or solid solution with a definite crystalline structure. For example Corundum (Al_2O_3).
- The minerals from which metal can be conveniently and economically extracted are called ores.
- All ores are minerals but all minerals are not ores.
- Metals occur in the native form because of their low reactivity.
- Rock is a naturally occurring solid substance composed of one or more minerals. e.g. bauxite.
- An ore is a rock or mineral from which a metal or non-metal can be economically produced. e.g. iron is obtained from haematite (Fe_2O_3).
- Ores can be oxides, sulphides, halides, carbonates or sulphates etc.
- Despite abundance of clay, metals are generally not obtained from silicates as no economical method is known as yet.

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- Most commercial metals are alloys – a mixture of one metal with small quantity of another metal to get desired properties.

Some important ores

IRON	
Ores	Composition
Magnetite	Fe_3O_4
Haematite	Fe_2O_3
Limonite	$Fe_2O_3 \cdot 3H_2O$
Siderite	$FeCO_3$
Iron pyrites	FeS_2
Chalcopyrites	$CuFeS_2$
Chromite	$FeO \cdot Cr_2O_3$

COPPER	
Ores	Composition
Cuprite (Ruby copper)	Cu_2O
Malachite	$CuCO_3 \cdot Cu(OH)_2$
Azurite	$2CuCO_3 \cdot Cu(OH)_2$
Chalcopyrites or Copper pyrite	$CuFeS_2$
Chalcocite or Copper glance	Cu_2S
Bornite	Cu_3FeS_3

ALUMINIUM	
Ores	Composition
Corundum or Ruby or Emerald	Al_2O_3
Diaspore	$Al_2O_3 \cdot H_2O$
Bauxite	$Al_2O_3 \cdot 2H_2O$
Gibbsite	$Al_2O_3 \cdot 3H_2O$
Cryolite	Na_3AlF_6
Alunite	$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 4Al(OH)_3$
Felspar	$KAlSi_3O_8$
Clay or Slate	$Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O$
Turguoise	$AlPO_4 \cdot Al(OH)_3 \cdot H_2O$

MAGNESIUM	
Ores	Composition
Magnesite	$MgCO_3$
Dolomite	$MgCO_3 \cdot CaCO_3$
Kisserite	$MgSO_4 \cdot H_2O$
Epsom salt	$MgSO_4 \cdot 7H_2O$
Carnallite	$KCl \cdot MgCl_2 \cdot 6H_2O$
Asbestos	$[CaMg_3(SiO_3)_4]$
Kainite	$KCl \cdot MgSO_4 \cdot 3H_2O$
Polyhalite	$K_2SO_4 \cdot MgSO_4 \cdot CaSO_4 \cdot 2H_2O$
Talc	$Mg_2(Si_2O_5) \cdot Mg(OH)_2$

LEAD	
Ores	Composition
Galena	PbS
Cerussite	$PbCO_3$
Anglesite	$PbSO_4$

ZINC	
Ores	Composition
Zinc blende	ZnS
Calamine	$ZnCO_3$
Zincite	ZnO
Willemite	Zn_2SiO_4

SILVER	
Ores	Composition
Argentite or Silver glance	Ag_2S
Pyrargyrite or ruby silver or dark red silver ore	$3Ag_2S \cdot Sb_2S_3$ or Ag_3SbS_3
Silver-copper glance	$(Cu \cdot Ag)_2S$
Horn Silver	AgCl

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SOME IMPORTANT MINERALS OF OTHER ELEMENTS

Ores	Composition
Cassiterite (Tin stone)	SnO_2
Cinnabar	HgS
Rutile	TiO_2
Ilmenite	$FeO \cdot TiO_2$
Lime Stone	$CaCO_3$
Fluorspar	CaF_2
Chile saltpetre	$NaNO_3$
Salt petre	KNO_3
Barytes	$BaSO_4$
Gypsum	$CaSO_4 \cdot 2H_2O$
Glauber's salt	$Na_2SO_4 \cdot 10H_2O$
Beryl	$3BeO \cdot Al_2O_3 \cdot 6SiO_2$
Chlor apatite	$3Ca_3(PO_4)_2 \cdot CaCl_2$
Fluor apatite	$3Ca_3(PO_4)_2 \cdot CaF_2$

- ***Metallurgy*** is the scientific study of production of metals from their ores and making of useful alloys.
- Preliminary treatment of ore is done to concentrate it by removing economically worthless matter called ***gangue***.
- Concentration is done by :
  (a) *gravity separation* - *i.e.* washing of ore. e.g. carbonates and oxides.
  (b) *froth floatation process* - used for sulphides
  (c) *magnetic separation* - for magnetic ores
  (d) *electrostatic separation* - e.g. to separate lead sulphide and zinc sulphide.
  (e) *chemical method* - e.g. Baeyer's process used for bauxite.
- The oil used in froth floatation process is pine oil.
- Froth floatation process is based upon preferential wetting of gangue particles by oil.
- $CuSO_4$ acts as an activator in froth floatation process.
- Potassium cyanide is used as a depressant in froth floatation process.
- Gravity separation method is based on the difference in densities of ore particles and impurities.
- Concentrated ore is converted to a compound suitable for production of metal, by either ***roasting*** or ***calcination***.
- Calcination is a process in which ore is heated, generally in the absence of air, to expel water from a hydrated oxide or CO_2 from a carbonate at temperature below their melting points.
- Roasting is a wider term used to denote the process in which ore (usually sulphide) alone or mixed with other materials is heated, usually in the presence of air, at temperatures below their melting points.

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- Metal oxide is subjected to reduction to obtain pure metal:
  (a) *Auto reduction* involves heating in air as for cinnabar (HgS)
  (b) *Reduction by carbon* - Smelting, coal or coke is used. e.g. for iron and zinc. Active metals like Na, Mg, Al or hydrogen gas are also used when carbon cannot.
  (c) *Electrolytic reduction* is used for reactive metals like Li, Na, Mg, Ca, Al. Carbon reduction is not possible for these metals as they form carbides on heating to high temperature.
- Reduction of oxide with carbon at high temperature is known as ***smelting***.
- ***Refining*** is the process of obtaining pure metal from reduced metal. The methods are :
  (a) *Liquation* : When the impurity is less fusible than the metal itself, e.g. Pb, Sn, etc.
  (b) *Distillation* : For these metals which have low boiling points and are easily volatile, e.g. Zn, Cd and Hg.
  (c) *Oxidation* : When the impurities have a greater affinity for oxygen than the metal itself.
  (d) *Electro-refining* : Highly electropositive metals like Al, Cu, Ag, Au, Zn, Sn, Pb, Cr and Ni are refined by this method.
- Higher degree of purity is obtained by :
  (a) *Van-Arkel method* - e.g. titanium and zirconium.
  (b) *Zone refining* - e.g. germanium, silicon etc.
- Magnesium was first obtained by ***amalgamation process***. This process is also employed for extraction of gold and silver.
- ***Dow's process*** is the commercial method of obtaining magnesium from oceans as Mg^{2+} are the third most abundant ions in sea water.
- Electrolysis of $MgCl_2$ is done in presence of fused sodium chloride and calcium chloride to lower fusion temperature and increase conductivity.
- Lighter gangue particles are washed in a current of water by a process called lavigation.
- Neutral refractory material used in furnaces is graphite.
- Hydrometallurgy involves both leaching and precipitation of the metal atom from its solution by adding a precipitating agent.
- Aluminium is obtained by electrolysis of alumina (Al_2O_3) dissolved in fused cryolite (Na_3AlF_6).
- ***Bett's electrolytic method*** is used for refining of lead, the electrolyte used is lead silico fluoride ($PbSiF_6$).
- ***Electrolytic refining*** is also done for crude tin.
- ***Iron*** is obtained by smelting of haematite in a blast furnace.
- ***Cyanide process*** to obtain silver or gold involves formation of soluble sodium cyanide complex from which the metal is precipitated out using zinc powder.
- ***Pattinson's process*** or ***Parke's process*** is used to obtain silver from crude lead.
- Copper is purified by electrolytic method.
- Magnesium is an important structural metal.
- Magnesium burns in CO_2 hence it is used in incendiary bombs as magnesium fire cannot be extinguished easily.

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- Aluminium powder mixed with iron (III) oxide (thermite) is used for welding. After ignition the reaction is self sustaining, hence also forms a basis for certain kinds of incendiary bombs.
- Aluminium cannot be extracted by carbon reduction because it is more electropositive than C, it will react with C to form its carbide.
- Chromium cannot be obtained from electrolysis because it is less electropositive and can be obtained by aluminium reduction of its oxides called as thermite reaction.
- Iron corrodes easily so it is alloyed with tin to give tin plates, used for food containers.
- Aluminium does not corrode and is most commonly used for making containers and packaging materials etc.
- Copper is an important constituent of many useful alloys. e.g. brass, bronze etc.
- Chemical passivity is inertness exhibited by metals under conditions where chemical activity is expected. For example a piece of iron dipped in conc. HNO_3 becomes passive. Cr, Co, Ni and Al also show similar behaviour.
- A substance which combines with gangue to form a fusible mass is called a flux.
- The removal of impurities from an ore by forming molten mass is called slagging.
- ***Case hardening*** - producing a protective thin coating of hardened steel on the surface by first heating mild steel with charcoal and then plunging into oil.
- ***Nitriding*** - Forming a hard coating of iron nitride on the surface by heating steel in an atmosphere of ammonia at 500°C – 600°C for 3 – 4 days.
- Tin disease/tin past/tin plague is the crumbing as powder of white tin to grey tin at low temperatures in cold countries.
- At different temperatures tin changes into different forms.

$$\underset{\text{grey}}{\alpha\text{-Sn}} \underset{}{\overset{15°\text{-}20°C}{\rightleftharpoons}} \underset{\substack{\text{white}\\(\text{most stable})}}{\beta\text{-Sn}} \overset{161°C}{\rightleftharpoons} \underset{\substack{\text{brittle}\\(\text{rhombic})}}{\gamma\text{-Sn}} \overset{232°C}{\rightleftharpoons} \text{liquid Sn}$$

- More than half the lead produced is used to make lead-storage batteries. Since lead resists attack by corrosive substances it is used to make chemical plant/nuclear plant equipment and in manufacture of military and sporting ammunition.

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15
CHAPTER

hydrogen & its compounds

- Hydrogen is the most abundant element in the universe. Some estimates are that 92% of the universe is made up of hydrogen, and 7% helium, leaving only 1% for all of the other elements.
- The abundance of H_2 in the earth's atmosphere is very small. This is because the earth's gravitational field is too small to hold so light an element, though some H_2 is found in volcano gases. In contrast, hydrogen is the tenth most abundant element in the earth's crust (1520 ppm or 0.152% by weight).
- Hydrogen has the simplest atomic structure of all the elements, and consists of a nucleus containing one proton with a charge +1 and one orbital electron. The electronic structure may be written as $1s^1$.
- Hydrogen is the first element in the periodic table, and is unique. There are only two elements in the first period, hydrogen and helium.

GENERAL	
Name, Symbol, Number	hydrogen, H, 1
Chemical series	nonmetals
Group, Period, Block	1, 1, *s*
Appearance	colourless
Atomic mass	1.00794 (7) g/mol
Electronic configuration	$1s^1$
Electrons per shell	1

PHYSICAL PROPERTIES	
Phase	gas
Density	(0°C, 101.325 kPa), 0.08988 g/L
Melting point	14.01 K, (–259.14°C, – 434.45°F)
Boiling point	20.28 K (–252.87°C, – 423.17°F)
Heat of fusion	(H_2) 0.117 kJ/mol
Heat of vapourisation	0.904 kJ/mol
Heat capacity	28.836 J/(mol.K)

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ATOMIC PROPERTIES	
Crystal structure	hexagonal
Oxidation states	+1, –1 (amphoteric oxide)
Electronegativity	2.20 (Pauling scale)
Ionization energies	1st: 1312.0 kJ/mol
Atomic radius	25 pm
Atomic radius (calc.)	53 pm (Bohr radius)
Covalent radius	37 pm
van der Waals radius	120 pm

- The lightest gas is **hydrogen.**
- **Heavy water (D_2O)** is produced by repeated distillation and condensation. One part of D_2O is present in 6000 part of H_2O.
- Ionic hydrides are formed by **elements of very high electropositivity.**
- Hydrogen peroxide is generally prepared on industrial scale by the **electrolysis of 50% H_2SO_4.**
- Heavy water has found application in atomic reactor as **moderator.**
- **Heavy water** posseses high density and different physical properties than those of water.
- Hard water becomes free from Ca^{2+} ions when passed through ion exchange resin containing *R*COOH groups.
- Hydrogen gas is used in the hydrogenation of oils in presence of nickel as a catalyst.
- Hydrogen adsorbed on transition metals such as Pt, Pd, Ni, Os, Ca, Mn, Fe etc. is known as **occluded Hydrogen.**
- In solid hydrogen, the intermolecular bonding is **van der Waals.**
- The conversion of atomic hydrogen into ordinary hydrogen is **exothermic change.**
- The decomposition of H_2O_2 can be slowed down by the addition of small amount of phosphoric acid which act as **inhibitor** (negative catalyst for decomposition of H_2O_2).
- The bleaching properties of H_2O_2 are due to its **oxidising properties.**
- The **ortho and para hydrogen** possess different physical properties but same chemical properties. They have same electronic arrangement but different spin of nuclei.
- **Para hydrogen** is less stable than ortho hydrogen.
- **Tritium atom** $\left({}_1^3H\right)$ has two neutrons and one proton.
- Benzene is oxidised by H_2O_2 in presence of $FeSO_4$ to **phenol.**
- **BaO_2** is peroxide but oxides like MnO_2, PbO_2 and NO_2 do not have O – O bond, *i.e.* peroxide linkage and so are not peroxides.
- Fluorine reacts with water to form **oxygen and ozone.**
- Heavy water is used in nuclear reactors to **slow down the speed of neutrons.**
- The formula of heavy water is **D_2O.**
- The rubber foam is produced by passing oxygen through rubber foaming material. This oxygen is released from **hydrogen peroxide.**

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- The helium nucleus contains **2 neutrons and 2 protons.**
- H_2O_2 gives off O_2 when heated, turns an acid solution of KI violet, and reduces acidified $KMnO_4$.
- **Atomic hydrogen** is obtained by passing silent electric discharge through hydrogen at low pressure.
- The ratio of electron, proton and neutron in tritium is **1 : 1 : 2.**
- **H_2O_2** is stored in plastic container after addition of stabilizer as H_2O_2 easily decomposes into water and oxygen and the decomposition speeds up in the presence of metallic impurities or strong bases and on exposure to light.
- Acidified $KMnO_4$ is decolourised by **Nascent hydrogen** (*i.e.* hydrogen at the moment of generation) is a more powerful reducing agent than ordinary H_2.
- Heavy water freezes at **3.8°C.**
- In aqueous solution, H_2 does not reduce **Zn^{2+}.**
- Hydrogen can be placed in halogens group because **it forms hydrides like chlorides.**
- Acidified solution of chromic acid on treatment with H_2O_2 yields **$CrO_5+H_2O+K_2SO_4$.**
- The melting points of most of the solid subtances increase with an increase of pressure. However ice melts at a temperature lower than its usual melting point when the pressure is increased. This is because **ice is less denser then water.**
- H_2O_2 converts potassium ferrocyanide to ferricyanide. The change observed in the oxidation state of iron is **$Fe^{2+} \rightarrow Fe^{3+}$.**
- Tritium emits **β-particles.** ${}_1H^3 \rightarrow {}_2He^3 + {}_{-1}e^0$
- H_2O_2 acts as a reducing agent in its **reaction with $KMnO_4$ in acid medium.**
- H_2O_2 turns an acidified solution of **TiO_2** to orange red (H_2TiO_4).
- The best method to test whether a clear liquid is water is to **add few drops on anhydrous copper sulphate and look for colour change.**
- **H_2O_2** is prepared in the laboratory when BaO_2 is added to CO_2 bubbling through cold water.
- **Colloidal solution of palladium** can adsorb large volumes of hydrogen gas as it has larger surface area.
- When silicon is boiled with caustic soda solution the gas evolved is **H_2.**
- **Hydrogen is not palced** with the group of alkali metals or halogens because ionization energy of hydrogen is too high for group of alkali metals but too low for halogen group.
- Water is permanently hard when it contains **nitrates of magnesium and calcium.**
- The acidified solution of $FeCl_3$ is reduced by passing **Nascent H.**
- The *n/p* ratio for ${}_1H^1$ is **zero.**
- In periodic table, tritium is placed in group **I.**
- Heavy water was discovered by **Urey band Washburn.**
- In the **hydrogen peroxide molecule,** the four atoms are arranged in a non-linear and non-planar manner.
- Ionic hydrides react with water to give **basic solutions.**
- Maximum density of heavy water is at **11.6°C.**

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- Heavy hydrogen is used **in studying reaction mechanism.**
- In the preparation of hydrogenated oil the chemical reaction involving hydrogen is called **hydrogenation.**
- Ordinary hydrogen has perponderance of **hydrogen atoms.**
- The oxygen atoms of H_2O_2 used for oxidation is bound by **covalent bond.**
- Density of water is maximum at **4°C.**
- Ordinary hydrogen is a mixture of **75% ortho–H_2 + 25% para–H_2.**
- A mixture of hydrazine and 40 to 60 per cent of H_2O_2 solution is used as **rocket fuel.**
- Na_2O_2 gives H_2O_2 on treatment with a dilute acid.
- Calgon is an industrial name given to **sodium hexa meta phosphate $(NaPO_3)_6$ or $Na_2[Na_4(PO_3)_6]$.**
- **Permutit** is hydrated sodium aluminium silicate.
- For the bleaching of hair, the substance used is **H_2O_2.**
- Decolourisation of acidified potassium permanganate occurs when H_2O_2 is added to it. This is due to **reduction of $KMnO_4$.**
- Hydrogen reacts with **F_2** even in the dark.
- When zeolite (hydrated sodium aluminium silicate) is treated with hard water the sodium ions are exchanged with **Ca^{2+} ions.**
- The percentage by weight of hydrogen in H_2O_2 is **5.88.**
- When the same amount of zinc is treated separately with excess of sulphuric acid and excess of sodium hydroxide, the ratio of volumes of hydrogen evolved is **1 : 1.**
- Hydrogen may be prepared by heating a solution of caustic soda with **Zn.**
- The exhausted permutit is generally regenerated by percolating through it a solution of **sodium chloride.**
- **Hydrogen peroxide** has a half open book structure or bent structure.
- Hydrogen peroxide for the first time was prepared by **Thenard.**
- Ortho and para hydrogen differ in the **nature of spins of protons.**
- Hydrogen was discovered by **Cavendish.**
- H_2O_2 acts as an oxidising agent in **acidic as well as in alkaline medium.**
- H_2O_2 acts as antiseptic due to its **oxidising property.**
- Zeolites are extensivley used in **softening of water and catalyst.**
- Sodium zeolite is **$Na_2Al_2Si_2O_8$.**
- The formation of atomic hydrogen from molecular hydrogen will be favoured at **high temperature and low pressure.**
- Hydrogen peroxide is manufactured by the autoxidation of **2-ethylanthraquinol.**
- **Nascent hydrogen** consists of hydrogen atoms with excess energy.
- **Hydrogen molecules** are diatomic and form X^- ions.
- Water acts as excellent solvent due to **high dielectric constant.**
- **30 volume hydrogen** means 1 cm^3 of the solution liberates 30 cm^3 of O_2 at STP.
- An aqueous solution of hydrogen peroxide is **weakly acidic.**

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- Moist hydrogen cannot be dried over concentrated H_2SO_4 because **it is oxidised by H_2SO_4.**
- Hydrogen burns with **blue flame.**
- The most reactive isotope of H is **$_1H^1$.**
- H_2 acts as an oxidant in its reaction with **Ca.**
- The most dangerous method of preparing hydrogen would be by the action of HCl and **K.**
- **Liquid H_2** gas in cold, liquid form expands when it is further cooled.
- The life period of atomic hydrogen is **only one third of a second.**
- The percentage of para hydrogen in ordinary hydrogen increases when **temperature is lowered.**

Strength of H_2O_2 solution

The strength of H_2O_2 solution available in the market is expressed in terms of "volume strength". It is defined as the volume of O_2 liberated in ml at NTP from 1ml of the H_2O_2 solution.

Suppose, 1 ml of a H_2O_2 solution on heating decomposes giving 20 ml of O_2 gas at NTP. Then its volume strength is '20V'. Similarly it may be 10V, 12V, 15V, etc. The relationship between normality and volume strength is:-

$$\text{Normality} = \frac{\text{Volume strength}}{5.6}$$

Again the relationship between % strength and volume strength is :

% strength = volume strength × 0.3035

- The **bond energy of covalent** O—H bond in water is greater than bond energy of hydrogen bond.
- At **absolute zero,** only para hydrogen exists.
- Hydrogen shows **+1, –1 and zero oxidation states.**
- Ozone reacts with H_2O_2 to give oxygen. One volume of ozone gives **two volumes of oxygen.**
- When hydrolith is treated with water it yields **H_2.**
- Hydrogen gas is used on industrial scale in the manufacture of **margarine.**
- Ammonium persulphate solution on heating under reduced pressure gives **H_2O_2.**
- The O—O bond length in H_2O_2 is **1.48 Å.**
- H_2O_2 restores the colour of old lead paintings, blackened by the action of H_2S gas, **by oxidising PbS to $PbSO_4$.**
- Hydrogen has a tendency to gain one electron in order to acquire helium configuration. It thus resembles **halogens.**
- In the case of H_2O_2 the angle between the planes containing the hydrogen atom is **90º.**
- Water contracts on heating **from 0ºC to 4ºC.**
- The geometry of water molecule is same as that of **chlorine oxide.**
- Decomposition of H_2O_2 is accelerated by **finely divided metals.**
- Tritium is obtained by **nuclear reactions.**

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Photohydrogen
"Photohydrogen" is hydrogen produced with the help of artificial or natural light. It is sometimes discussed in the context of obtaining renewable energy from sunlight, by using microscopic organisms such as bacteria or algae.

LH_2
LH_2 is an acronym used in the aerospace industry, which stands for liquid hydrogen. LH_2 is a common liquid fuel for rocket applications. Hydrogen is found naturally in the H_2 form, thus the H_2 part of the name. Hydrogen at normal temperatures is a gas and to exist as a liquid must be cooled to a very low level, 20.268 K (& 8722; 423 °F). Liquid hydrogen has a very low density of 70.8 kg/m³ (at 20 K), so storage tanks for it have to be quite large.

- Hydrogen is evolved by the action of cold dilute HNO_3 on **Mg or Mn.**
- $H_2O_2$when added to a solution containing $KMnO_4$ and H_2SO_4 acts **as a reducing agent.**
- H_2O_2 is **diamagnetic.**
  A saturated solution of CO_2 loses weight on exposure to the atmosphere.
- Smell of H_2O_2 resembles **nitric acid.**
- MnO_2 liberates oxygen from a solution of H_2O_2(the action being catalytic) only if the solution is **acidic.**
- H_2O_2 is concentrated by **distillation under reduced pressure.**
- Deuterium an isotope of hydrogen is **non-radioactive.**
- **Hydrogen peroxide** restores the colour of the lead paintings.
- The ionisation energy of hydrogen is **closer to halogens.**
- If water is boiled for sometime it becomes free from **temporary hardness.**
- Zeolite which shows ion-exchanging ability **is a sodium alumino silicate**.
- Hydrogen combines with O_2 to form H_2O. In this reaction **hydrogen gets oxidised.**
- The boiling point of heavy water is **101.4°C.**
- Atomic hydrogen produces formaldehyde when it reacts with **CO.**
- Nucleus of deuterium contains **one proton and one neutron.**
- **Density of heavy water** is higher than ordinary water.
- Deuterium resembles hydrogen in chemical properties but reacts **slower than hydrogen.**
- The electronic configuration of deuterium is **$1s^1$.**
- Decomposition of H_2O_2 is accompanied by **decrease in free energy.**
- The weight percentage of deutrium in heavy water is **20.**
- One of the most important uses of H_2O_2 is **as rocket fuels.**
- **Pure H_2O_2** is pale blue syrupy liquid.
- Tailing of mercury is a laboratory test for **O_3.** In this test, O_3 reacts with Hg to form Hg_2O which sticks on the walls of glass. This is called tailing of mercury
  $$O_3 + 2Hg \rightarrow Hg_2O + O_2.$$
  The tailing is removed by the action of H_2O_2 on Hg_2O.
  $$H_2O_2 + Hg_2O \rightarrow 2Hg + H_2O + O_2.$$
- A molten ionic hydride on electrolysis **H_2 is liberated at anode.**

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- Experimental evidence for the presence of ortho and para hydrogen was shown by **Melceman and Mcleod.**
- Hydrogen loses its electron to form H^+ ion. In this respect it resembles to **alkali metals.**
- Heavy water is manufactured in India at **Trombay.**
- Oxygen and hydrogen react to form water. This discovery was made by **Cavendish.**
- The catalyst used in Bosch process of manufacture of H_2 is **Fe_2O_3 + Cr_2O_3.**
- The number of radioactive isotopes of hydrogen is **one** as only tritium is radioactive.
- The number of protons, electrons and neutrons respectivley in a molecule of heavy water is **10, 10, 10.**
- Hydrogen is obtained by the action of an alloy of silicon and iron with NaOH. The process is called **silicol process.**
- The O—O bond is present in **peroxide.**
- When different metals like Zn, Sn, Fe are added to dilute sulphuric acid, **H_2 gas,** which burns explosively in air, is evolved.
- The ionisation of hydrogen atom gives **proton.**
- The hybridisation of the orbitals of oxygen in H_2O_2 is **sp^3.**
- High boiling point of water is due to **hydrogen bonding.**
- H_2O_2 on treatment with chlorine gives **oxygen.**
- The most reactive state of hydrogen is **atomic hydrogen.**
- The oxidising property of H_2O_2 is due to the fact that two oxygen atoms in its molecule are **bonded differently.**

Compounds of Hydrogen

Formula of Compound	Name of Compound	Some Uses of Compound
H_2S	Hydrogen sulphide	Hydrogen sulphide, although it has a strong odour of rottten eggs, it is used by chemists to prouce other compounds and analyse the composition of mixtures. Hydrogen sulphide is more often an nuisane than its uses because its odour is present among decaying organic matter, such as garbage and sewage. It is also common for H_2S to be given off during the removal of tarnish from silver, the exhaust of cars and some buses and around some hot springs.
H_2O_2	Hydrogen peroxide	Rocket propellant, sterilize milk industry, used as bleach in paper, wood pulp, textile and food industries, used as antiseptic, germicide and skin cleanser, used to clean, etch and oxidise PCB and semiconductors/other metals and electronics.
H_2SO_4	Sulphuric acid	Processing metals - cleaning/pickling iron and steel before plating them with tin or zinc; production of fertilizers; manufacture of chemicals in making nitric acid, HCl, synthetic detergents, explosives, dyes, pigments, drugs and sulphate salts; in refining of petroleum and the making of rayon.

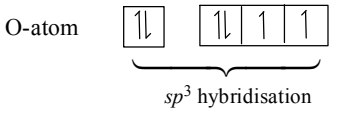
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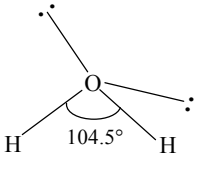
Formula of Compound	Name of Compound	Some Uses of Compound
HCl	Hydrochloric acid or hydrogen chloride	Just like hydrogen peroxide, HCl taken part in pickling/ cleaning metals. Helps to digest food in the walls of the stomach where HCl is present, to separate cotton from wool, manufacture of ammonium chloride, phosphoric acid, dies, pigments in paint, iron, steel, production of corn starch and glucose, make solvents, chloride salts and bleaches, neutralise soap refining, leather tanning, brewing, textiles, waste streams, prevent bacterial in toilet bowls, produces tin and tantalum, used in making glue and gelatin and acts as a starch modifier.
HCN	Hydrogen cyanide	Very dangerous compound that has been used in the past WW1/nazi gas chambers to kill people, used to make the base product, acrylonitrile, for acrylic fibres, plastics, and synthetic rubbers, commercially used as an insecticide and rodenticide and used to make pharmaceuticals.

- **Water**

Water is one of the most plentiful and readily available of all chemicals. It has special importance because of its ability to dissolve so many other substances. Oxygen atom in water molecule is sp^3-hybridised, four hybrid orbitals directed towards the corners of a tetrahedral are formed.

The bond angle is less than the expected angle in tetrahedron due to the presence of two lone pairs of electrons on two uncombined hybrid orbitals which repel each other and the bonded pairs and cause them to come closer and thereby reducing the bond angle from 109° 28′ to 104.5°.





- **Hard and soft water**

The water which lathers easily on shaking with soap solution is called as soft water. While the water which does not produce lather with soap solution readily, rather it forms an insoluble white scum is called as hard water.

- **Cause of hardness of water -** The hardness of water is due to presence of soluble salts of calcium, magnesium and other heavy metals in it. Such water when treated with soap (sodium or potassium salt of higher fatty acid) does not produce lather but produce soluble calcium and magnesium salts of fatty acid, which separate out as scum from water.

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$$2C_{17}H_{35}-COONa \xrightarrow[\text{sodium stearate}]{} \begin{cases} \xrightarrow{CaCl_2} (C_{17}H_{35}COO)_2Ca\downarrow + 2NaCl \\ \xrightarrow{MgCl_2} (C_{17}H_{35}COO)_2Mg\downarrow + 2NaCl \end{cases}$$

Soap will not produce l ather until all the Ca^{2+} and Mg^{2+} ions from water have been removed. Hence the water free from soluble calcium and magnesium salts and containing soluble salts of 'Na' and 'K' is called as soft water.

- **Types of hardness -** It is of two types, such as temporary and permanent hardness. The temporary hardness is due to presence of dissolved bicarbonates of calcium and magnesium and the carbonates of iron. This type of hardness can be easily removed by simply boiling the water, when insoluble carbonates and hydroxides will be formed. Such precipitates are deposited as scale at the bottom of the vessel.

$$Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3\downarrow + H_2O + CO_2\uparrow$$

$$Mg(HCO_3)_2 \xrightarrow{\Delta} MgCO_3\downarrow + H_2O + CO_2\uparrow$$

$$FeCO_3 + 6H_2O + O_2 \xrightarrow{\Delta} Fe(OH)_3\downarrow + 4CO_2\uparrow$$

On the other hand permanent hardness is due to presence of dissolved chlorides and sulphates of Ca, Mg, Fe and other heavy metals. The softening of this type of water can't be done by boiling, rather it needs chemical treatment.

Degree of hardness of water

It is defind as the number of parts by weight of $CaCO_3$ or its equivalent present in one million parts by weight of water. It is shortly expressed in ppm. For example, suppose 12 mg of $MgSO_4$ is present per litre of water sample.

$\Rightarrow$ 12 mg. $MgSO_4$ corresponds to 10^3gm water (Assuming density of water = 1gm/ml)

$\therefore$ 10^6 gm water contains $= 12 \times 10^3$ mg of $MgSO_4$

$= 12$ gm of $MgSO_4$

Again 12 gm $MgSO_4$ = 0.1 mole $MgSO_4$

$\equiv$ 0.1 mole $CaCO_3 \equiv$ 10 gm $CaCO_3$

$\Rightarrow$ 10 gm $CaCO_3$ present per 10^6 gm of water.

$\therefore$ Hardness of water sample is 10 ppm.

- **Softening of hard water**

By boiling- This has been already discussed in types of hardness.

By Clark's process - In this process temporary hardness of water can be removed by adding calculated quantity of slaked lime to it. The slaked lime reacts with soluble calcium and magnesium bicarbonates giving $CaCO_3$ and $Mg(OH)_2$ precipitate, which settle down at the bottom of the tank and can be removed.

$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + 2H_2O$

$Mg(HCO_3)_2 + 2Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + Mg(OH)_2\downarrow + 2H_2O$

Excess amount of slaked lime may cause artificial hardness by forming soluble $Ca(HCO_3)_2$ with CO_2 from air.

By lime soda process- In this process both temporary and permanent hardness of water can be removed by adding a mixture of lime and sodium carbonate to it. Some times NaOH is

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also added to water in order to neutralise any free acid present and to remove magnesium salts as hydroxide.

$MgCl_2 + Na_2CO_3 \rightarrow MgCO_3\downarrow + 2NaCl$

$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + 2H_2O$

$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3\downarrow + Na_2SO_4$

$MgSO_4 + NaOH \rightarrow Mg(OH)_2\downarrow + Na_2SO_4$

By permutit process- In this process water is allowed to pass through zeolite, which is a complex substance containing sodium and aluminium silicate. So that sodium ions are displaced by Ca^{2+}, Mg^{2+} ions forming insoluble calcium and magnesium zeolite. Now a days synthetic zeolites, which are also called as permutit are widely used.

$CaCl_2 + Na_2–Ze \rightarrow Ca – Ze\downarrow + 2NaCl$

$MgSO_4 + Na_2–Ze \rightarrow Mg – Ze\downarrow + Na_2SO_4$

After long use the sodium permutit loses its ability of softening the hard water, as insoluble calcium permutit deposits over sodium permutit.

Ion - exchange resin process - Ion-exchange resins are insoluble, cross linked, long chain organic compounds containing a sulphonic ($–SO_3H$) group or carboxylic (–COOH) group capable of exchanging their H^+ ion with other cations, which is called as cation exchange resin. Where as those containing basic functional groups like amino or quarternary ammonium or quarternary phosphonium, etc. are called as anion exchange resin. These resins after treatment with dilute NaOH solution become capable of exchanging their $OH^–$ ion with the anions present in water.

Dowex-50, Amberlite IR-120 are the cation exchange resins and their function can be represented as :-

Dowex-3, Amberlite - 400 are the anion exchange resins and their function can be represented as :-

$$2RH^+ \text{ (cation resin)} \xrightarrow{2Na^+} 2RNa\downarrow + 2H^+$$

$$2RH^+ \text{ (cation resin)} \xrightarrow{Ca^{2+}} R_2Ca\downarrow + 2H^+$$

$$2RH^+ \text{ (cation resin)} \xrightarrow{Mg^{2+}} R_2Mg\downarrow + 2H^+$$

From the above it is clear that if hard water is passed first through cation exchange resin and then through anion exchange resin, then the resulting water will be free from both cations and anions. Again H^+ ions from first step combines with $OH^–$ ions from second step giving water.

- **Demineralisation of water**

The soft water obtained by any of the above methods is not free from all soluble minerals. It also contains some soluble salts of Na and K. But in ion-exchange resin process all the cations and anions can be removed from water as discussed above. The water obtained in this process is called as demineralised water or de-ionised water. It is different from distilled water, as the former contains some dissolved silica, CO_2, O_2 etc. This type of water is very much useful for high pressure boilers.

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End

16

CHAPTER

s-block elements

ALKALI METALS

Selected data on group I alkali metals					
Chemical symbol, name	**At. No.**	**Electron arrangement**	**Melting point**	**Boiling point**	**Atomic radii (pm)**
Li, Lithium	3	$1s^2, 2s^1$	181°C, 454 K	1347°C, 1620 K	152
Na, Sodium	11	$1s^2, 2s^2\ 2p^6, 3s^1$	98°C, 371 K	883°C, 1156 K	186
K, Potassium	19	$1s^2, 2s^2\ 2p^6, 3s^2\ 3p^6, 4s^1$	64°C, 337 K	774°C, 1047 K	231
Rb, Rubidium	37	$1s^2, 2s^2\ 2p^6, 3s^2\ 3p^6\ 3d^{10}, 4s^2\ 4p^6, 5s^1$	39°C, 312 K	688°C, 961 K	244
Cs, Caesium	55	$1s^2, 2s^2\ 2p^6, 3s^2\ 3p^6\ 3d^{10}, 4s^2\ 4p^6\ 4d^{10}, 5s^2\ 5p^6, 6s^1$	29°C, 302 K	679°C, 952 K	262
Fr, Francium	87	$1s^2, 2s^2\ 2p^6, 3s^2\ 3p^6\ 3d^{10}, 4s^2\ 4p^6\ 4d^{10}, 5s^2\ 5p^6, 6s^2\ 6p^6, 7s^1$	27°C, 300 K	677°C, 950 K	270

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- These elements are collectively called as alkali metals and group I is known as alkali group as the hydroxides of these metals are soluble in water and these solutions are highly alkaline in nature.
- Alkali metals are highly reactive and hence do not occur in the free state but are widely distributed in nature in combined state in the form of halides, oxides, silicates, borates and nitrates. Off all the alkali metals, only sodium and potassium are found in abundance in nature, i.e. they are seventh and eighth most abundant elements by weight in earth's crust. The last member, francium, occurs only in traces as a radioactive decay product because its half life period is very small, i.e. 21 minutes.
- Alkali metals are *s*-block elements, because last electron in them enters the *s*-orbital.

- These metals have only one electron in their outer shell. Therefore they are ready to lose that one electron in ionic bonding with other elements.
- Alkali metals form the first element of the period, with one outer electron, in any period from period 2 onwards. This outer electron similarity makes them behave in a chemically similar way. Some of their physical properties are typical of metals and some are not so typical of metals. Although they all have one outer electron and so similar physical and chemical properties, a characteristic of a periodic table group, but always watch out for trends down a group too.
- As with all metals, the alkali metals are malleable, ductile, and are good conductor of heat and electricity.
- All the group-I elements are silvery-coloured metals. They are soft and can be easily cut with a knife to expose a shiny surface which dulls on oxidation.
- These elements are highly reactive metals. The reactivity increases on descending the group from lithium to caesium.
- There is a closer similarity between the elements of this group than in any other group of the periodic table.
- Caesium and francium are the most reactive elements in this group.
- All alkali metals dissolve in mercury and forming amalgams. This reaction is highly exothermic.
- Fire caused by burning of alkali metals is extinguished by sprinking CCl_4.
- Alkali metals are paramagnetic due to the presence of unpaired electrons. On the other hand, alkali metal ions are diamagnetic and colourless due to their noble gas configuration with no unpaired electrons.
- **Typical metallic properties :** Good conductors of heat and electricity, high boiling points, silvery grey surface (but rapidly tarnished by air oxidation).

  When an alkali metal atom reacts, it loses an electron (oxidation) to form a singly positively charged ion.

  e.g. $Na \rightarrow Na^+ + e^-$.

  In terms of electrons, 2, 8, 1 $\rightarrow$ 2, 8 and so forming a stable ion with a noble gas electron arrangement. They tend to react mainly with non-metals to form ionic compounds which are usually soluble in white solids.
- **Non-typical metallic properties :** Low melting points, low density (first three float on water), very soft (easily squashed, extremely malleable) and so they have little material strength.
- Important trends down the group with increase in atomic number.

  The melting point and boiling point generally decrease.
  - The element gets more reactive.
  - The atoms get bigger (as more electron shells are added).
  - Generally the density increases (although the atom gets bigger, there is greater proportional increase in the atomic mass).
  - Generally the hardness decreases.

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Interesting facts about alkali metals

- Potassium is never found as a pure metal.
  0.07% of the ocean is made up of potassium chloride.
- The National Institute of Standards and Technology has created a caesium fountain atomic clock. It is the Nation's primary time and frequency standard.
- There is atmost one ounce of francium in the whole earth at any given time as a result of the decay of other radioactive elements.

- On moving down in the group from Li to Cs, electropositivity, atomic radii, atomic volume, reactivity, reducing power, conductivity, solubility of salts having small anion and density show an increasing trend. On the other hand, m.p. and b.p., hardness, ionization energy, electronegativity and solubility of salts having large anions (such as SO_4^{2-}, ClO_4^-, etc.) show a decreasing trend.
- The stability and solubility of carbonates, nitrates and bicarbonates increase in the order **:**
  $Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$,
  $LiNO_3 < NaNO_3 < KNO_3 < RbNO_3 < CsNO_3$ and
  $LiHCO_3 < NaHCO_3 < KHCO_3 < RbHCO_3 < CsHCO_3$
- The stability of peroxides and superoxides increase in the order :
  $Na_2O_2 < K_2O_2 < Rb_2O_2 < Cs_2O_2$ and $NaO_2 < KO_2 < RbO_2 < CsO_2$
- The solubility, stability and basic strength of hydroxides increase in the order:
  LiOH < NaOH < KOH < RbOH < CsOH
- The solubility and basic strength of oxides increase in the order :
  $Li_2O < Na_2O < K_2O < Rb_2O < Cs_2O$
- The peroxides of alkaline metals are colourless and diamagnetic while the superoxides are paramagnetic and coloured.
- Most metallic elements in the periodic table are Cs and Fr.
- KOH is better absorber of CO_2 than NaOH because potassium carbonate thus formed is more soluble and does not separate out.
- Li^+ is poor conductor of electricity than Cs^+ because hydrated Li^+ ion is larger in size than hydrated Cs^+ ion.
- Although lithium has the highest ionization enthalpy, yet it is the strongest reducing agent because of its large heat of hydration which is sufficient to overcome its ionization enthalpy.
- Lithium is the lightest metal, least fusible, least dense and least soft of all the alkali metals.
- Degree of hydration of alkali metal ions decreases in the order : $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$. The relative ionic radii in water also decrease in the same order.
- Alkali metals are *paramagnetic* due to the presence of unpaired ns^1 electrons. On the other hand, alkali metal ions are *diamagnetic* and *colourless* due to their noble gas configuration with no unpaired electrons.

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- In biological fluids, many cells tend to accumulate K^+ ions at the expense of Na^+ ions. These concentration gradients can be explained by different mechanisms such as *sodium pump* and *potassium pump*. Development and functioning of nerve cells are controlled by these cation gradients.
- $Na_2SO_4 \cdot 10H_2O$ is called *Glauber's salt*, anhydrous Na_2SO_4 is called *salt cake,* $NaNO_3$ is called *chile salt-petre,* $NaHSO_4$ is called *nitre cake* and KNO_3 is called *Indian salt-petre or nitre.*
- When common salt is fused with a little Na_2CO_3, 5-10% Na_2SO_4 and some sugar, it acquires a dark purple colour and has a characteristic saline taste. It is used in medicine to improve digestion. It is called *Kala namak or black salt or sulemani namak.*
- Although both NaCN and KCN are poisonous but KCN is more poisonous that NaCN.
- An alloy of Na and K is a liquid at room temperature. It is used in special thermometers for recording temperature above the b.p. of mercury (357°C).
- Sodium sesquicarbonate, $Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O$, which is neither deliquescent nor efflorescent,is used for wool washing.
- A mixture of Na_2O_2 and dil. HCl is commercially called oxone and is used for bleaching delicate fibres.
- Potassium superoxide (KO_2) is used as a source of oxygen in submarines, space shuttles and in emergency breathing apparatus. The moisture of the breath reacts with superoxide to liberate the apparatus (*oxygen mask*) to be continuously regenerated.

$$KO_2 + 2H_2O \rightarrow 4KOH + 3O_2\ ;\ KOH + CO_2 \rightarrow KHCO_3$$

- Alkali metals combine with mercury to form compounds (alloys) known as *amalgams.* This reaction is highly exothermic.
- Li^+ ion does not form alums because it is too small to have a coordination number of six.
- Potassium salts of fatty acids are used to make *soft soaps* because they are more soluble than those of sodium salts.
- 28% NaCl solution is called *brine.*
- Sodium hydroxide breaks down the proteins of the skin flesh to a pasty mass and hence it is commonly known as *caustic soda.*
- Lithium is used as a scavenger in metallurgy to remove last traces of oxygen and nitrogen from copper and nickel.
- Fire caused by burning of alkali metals is extinguished by sprinkling CCl_4.
- Francium was discovered by Perey in 1939 in France during nuclear disintegration of actinium-227.

$$_{89}Ac^{227} \rightarrow {}_{87}Fr^{223} + {}_{2}He^{4}$$

- Because of the small size of Li^+ ion, lithium salts usually crystallise from their aqueous solutions in the form of hydrates.
- Only lithium combines directly with carbon to form lithium carbide, Li_2C_2. While other alkali metals react with ethyne to form the corresponding metal carbides.
- All alkali metals dissolve in mercury forming amalgams.

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- Lithium sulphate does not form alums and is also not amorphous with other sulphates.
- Only $LiHCO_3$ exists in solution while all other alkali metal bicarbonates are solids.
- Lithium cannot be stored in kerosene oil since it floats over the surface due to its very low density. Therefore, lithium is usually kept wrapped in paraffin wax.
- **Solubility in liquid ammonia :** All the alkali metals dissolve in liquid ammonia giving deep blue solution when dilute due to the presence of ammoniated (solvated) electrons in the solution but the colour changes to bronze with increasing concentration. If ammonia is evaporated we get back the metal.

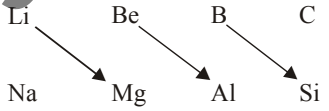
$$M + (x + y)NH_3 \rightarrow [M(NH_3)_x]^+ + [e(NH_3)_y]^-$$

- The solution of alkali metals in liquid ammonia :
  (i) is strongly reducing due to the presence of ammoniated electrons so it is used in Birch reduction as a reducing agent.
  (ii) is conducting due to ammoniated electrons and ammoniated cations, on cooling the conductivity increases.
  (iii) shows paramagnetism due to presence of ammoniated electrons. On increasing the concentration association of ammoniated electrons occur to yield diamagnetic species because of which the colour changes to copper bronze acquiring metallic lusture due to the formation of metal ion clusters.

$$2e^- (NH_3)_y \rightarrow [e^- (NH_3)_y]_2$$

Diagonal relationship between lithium and magnesium

- Lithium shows diagonal relationship with magnesium since they have the same charge/ size ratio i.e. polarising power.
- The first element of group often shows resemblance to the second element of the neighbouring group on the right. This type of behaviour is known as Diagonal Relationship.



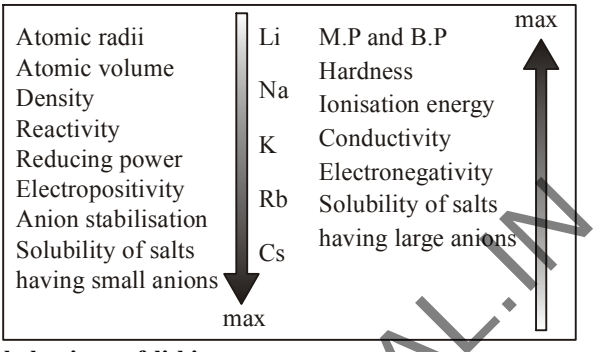
- This similarity between Li and Mg is particularly striking and arises because of their similar ionic sizes (Li^+ = 76 pm; Mg^{2+} = 72 pm)
- Li and Mg show close resemblance in the following :

Nitrides	Li and Mg both form nitrides. Other alkali metals do not. $6Li + N_2 \xrightarrow{\Delta} 2Li_3N$, $3Mg + N_2 \xrightarrow{\Delta} Mg_3N_2$
Carbonates	Like $MgCO_3$, Li_2CO_3 is decomposed by heat (the other alkali carbonates are thermally stable). Both carbonates are insoluble while Na_2CO_3, K_2CO_3... are soluble. $MgCO_3 \xrightarrow{\Delta} MgO + CO_2$, $Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$
Nitrates	$LiNO_3$ decomposes to give Li_2O like $Mg(NO_3)_2$, but other alkali metal nitrates give nitrite. $Mg(NO_3)_2 \xrightarrow{\Delta} MgO + 2NO_2 + 1/2O_2$ $2LiNO_3 \xrightarrow{\Delta} Li_2O + 2NO_2 + 1/2O_2$

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Oxides	Both give their normal oxides, Li_2O, MgO when they burn in oxygen. Na forms peroxide, Na_2O_2, while *K* forms superoxide, KO_2.
Hydration	Both Li^+ and Mg^{2+} are heavily hydrated.

- **Gradation properties of alkali metals**



- **Anomalous behaviour of lithium**

Reason for the anomalous behaviour of lithium is mainly due to its small size and hence it has highest polarizing power. The main points of difference :

(i) Li is harder than any other alkali metal.

(ii) Li combines with O_2 to form monoxide whereas other alkali metals form peroxides and superoxides.

(iii) Li is the only alkali metal which directly reacts with N_2 to form Li_3N.

(iv) Li(OH) decomposes at red heat, however hydroxides of other alkali metals do not decompose.

(v) The bicarbonate of Li is not known in solid state while the bicarbonates of other alkali metals are known in solid state.

(vi) In presence of NH_3, lithium forms imide Li_2NH while other alkali metals form amides, MNH_2.

- **Fajan's rule**

The partial covalent character in ionic compounds results through polarisation of the anion by the cation so that the electron density between the two nuclei increases. The covalent character is favoured by the following factors which are collectively known as Fajan's rules.

(i) **Small cation** : Smaller the cation larger is the covalent character. For example, LiCl is more covalent than KCl.

(ii) **Large anion** : Bigger the anion, larger is the covalent character. For example, amongst KF, KCl, KBr and KI, KI is most covalent.

(iii) **Large charge on the cation or anion** : With increase in the magnitude of charge on the cation or the anion, the covalent character increases. For example, covalent character increases in the order : NaCl, $MgCl_2$, $AlCl_3$, $SiCl_4$ etc.

(iv) **Pseudo inert gas configuration** : For two ions of the same size and charge, one with a pseudo inert gas configuration (transition elements) will be more polarising than a cation with noble gas configuration. For example, AgCl (Ag^+ = 1.26 Å) is more covalent than KCl (K^+ = 1.33 Å).

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MICROCOSMIC SALT, $Na(NH_4)HPO_4.4H_2O$

- It is prepared by dissolving molecular proportions of Na_2HPO_4 and NH_4Cl in hot water and crystallising the contents.

$$Na_2HPO_4 + NH_4Cl \rightarrow Na(NH_4)HPO_4 + NaCl$$

Disod. hydrogen phosphate

It is used for performing 'bead test' for detecting coloured ions (e.g. Cu^{2+}, Fe^{3+}, Mn^{2+}, Ni^{2+}, Co^{2+}) in qualitative inorganic analysis. The bead test is based on the fact that on heating it forms a transparent glassy bead of metaphosphate. The metaphosphate so formed gives coloured beads of orthophosphates when heated with coloured salts (*microcosmic bead test*).

$$Na(NH_4)HPO_4 \rightarrow NH_3 + H_2O + NaPO_3$$

Sod. metaphosphate

$$CuSO_4 \rightarrow CuO + SO_3,\ CuO + NaPO_3 \rightarrow CuNaPO_4\text{(blue bead)}$$

It is especially used for detecting silica which, being insoluble in $NaPO_3$, gives a cloudy bead.

Points to Remember

- **Monoxides, peroxides and superoxides of alkali metals**. All the five alkali metals can be induced to form normal oxides (*i.e.* monoxides), peroxides and superoxides by dissolving the metal in liquid ammonia and bubbling in the appropriate amount of oxygen.
- **Crystal structures of monoxides of alkali metals**. Except Cs_2O which has anti-$CdCl_2$ layer structure, all other monoxides, *i.e.* Li_2O, Na_2O, K_2O and Rb_2O have anti-fluorite structures.
- **Potassium superoxide (KO_2)** is used as a source of oxygen in submarines, space shuttles and in emergency breathing apparatus such as oxygen masks. Such masks are used in rescue work in mines and in other areas where the air is so deficient in oxygen that an artificial atmosphere must be generated.
- **Lithium hydroxide (LiOH)** is used to remove CO_2 from exhaled air in confined quarters like submarines and space vehicles.
- The alkali metals react with halogens and interhalogen compounds forming ionic polyhalide compounds.
- The solution of alkali metals such as Li, Na, or K in liquid ammonia is used for reduction of ethylenic double bonds, acetylenic triple bonds to double bonds and aromatic compounds under the name Birch reduction.
- Lithium is the highest known metal, having density = 0.534 g/cc. Therefore, it cannot be stored in kerosene oil because it floats on the surface. It is kept wrapped in paraffin wax.
- Cs is the most electropositive element due to its lowest ionization energy.
- Lithium cannot be used in making photoelectric cells because out of all the alkali metals, it has highest ionization energy and cannot emit electrons when exposed to light.
- The compounds of alkali metals are colourless (unless the anion is coloured like permanganate or dichromate) and diamagnetic. This is because they have noble gas configuration with no unpaired electron.
- All alkali metals exist as body-centred cubic lattice with a coordination number of 8.
- Due to small size, lithium does not form alums.

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ALKALINE EARTH METALS

- The group II of the periodic table consists of six elements - beryllium, magnesium, calcium, strontium, barium and radium. These elements are known as alkaline earth metals and group II is known as alkaline earth group. Although early chemists gave the name "earths" to a group of naturally occurring substances that were unaffected by heat and insoluble in water, the alkaline earth metals are also usually found in the continental crust. Alkaline earth metals always form divalent cations.

Electronic configuration of alkaline earth metals

Element	At. No.	Electronic configuration	Configuration of the valence shell
Be	4	$1s^2$, $2s^2$	[He] $2s^2$
Mg	12	$1s^2$, $2s^2$, $2p^6$ $3s^2$	[Ne] $3s^2$
Ca	20	$1s^2$, $2s^2$ $2p^6$, $3s^2$ $3p^6$, $4s^2$	[Ar] $4s^2$
Sr	38	$1s^2$, $2s^2$ $2p^6$, $3s^2$ $3p^6$ $3d^{10}$, $4s^2$ $4p^6$, $5s^2$	[Kr] $5s^2$
Ba	56	$1s^2$, $2s^2$ $2p^6$, $3s^2$ $3p^6$ $3d^{10}$, $4s^2$ $4p^6$ $4d^{10}$, $5s^2$ $5p^6$, $6s^2$	[Xe] $6s^2$
Ra	88	$1s^2$, $2s^2$ $2p^6$, $3s^2$ $3p^6$ $3d^{10}$, $4s^2$ $4p^6$ $4d^{10}$, $5s^2$ $5p^6$, $6s^2$ $6p^6$, $7s^2$	[Rn] $7s^2$

Factors responsible for divalent oxidation states

(i) The lattice energy increases as the charge on the ion increases. The increase in the lattice energy on account of the second electron from ns^2 is much more than the energy required (second ionisation energy) to remove it. Hence, the stability of +2 oxidation state is due to high lattice energy.

(ii) The second factor responsible for +2 oxidation state is the hydration energy which is high for M^{2+} ions. On account of the availability of energy, the process does not stop to M^+ state but reach to M^{2+} state readily.
Since the bivalent ions, M^{2+} have an inert gas configuration, it is very difficult to remove the third electron and hence oxidation state higher than +2 is not possible.

- Calcium is present in the soil, plants, bones as $Ca_3(PO_4)_2$ and egg shells etc.
- Gypsum $CaSO_4 \cdot 2H_2O$ is also known as *alabaster*.
- Calcium ions play an important role in *muscle contraction*.
- Magnesium ions are present in *chlorophyll-a* green colouring pigment in plants which absorbs light and is essential for *photosynthesis*.
- The ionization enthalpy or radium is higher than that of barium.
- Melting points of halides decrease as the size of the halogen increases. The correct order is : $MF_2 > MCl_2 > MBr_2 > MI_2$
- Thermal stability of carbonates and sulphates increases down the group from Be to Ba. The correct order is :

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$BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$ and

$BeSO_4 < MgSO_4 < CaSO_4 < SrSO_4 < BaSO_4$

- Basic character of oxides and hydroxides increases in the order :

  $BeO < MgO < CaO < SrO < BaO$ and

  $Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$
- Solubility of sulphates and carbonates decreases in the order:

  $BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$ and

  $BeCO_3 > MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$

  Whereas the solubility of hydroxides increases in the order:

  $Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$
- Solubility of chlorides, bromides and iodides decreases in the order :

  $MgX_2 > CaX_2 > SrX_2 > BaX_2$ (where X = Cl, Br or I)
- BeF_2 is soluble whereas fluorides of Mg, Ca, Sr and Ba are insoluble in water. Solubility decreases in the order :

  $BeF_2 > MgF_2 > CaF_2 > SrF_2 > BaF_2$
- Anhydrous $CaCl_2$ is a good desiccant but it cannot be used to dry alcohol and ammonia as it forms addition products with them.
- Aqueous $Ba(OH)_2$ is known as *baryta water*.
- Only Mg displaces hydrogen from a very dilute HNO_3.
- Like alkali metals, alkaline earth metals also dissolve in liquid ammonia giving coloured solutions which are good conductors of electricity.
- The hydride of Be can be prepared indirectly by reducing $BeCl_2$ with lithium aluminium hydride

  $2BeCl_2 + LiAlH_4 \rightarrow 2BeH_2 + LiCl + AlCl_3$
- Since the alkaline earth divalent ions have no unpaired electrons, these are diamagnetic and colourless.
- Density of calcium is less than that of magnesium due to the presence of vacant 3*d*-orbitals leading to much increase in atomic volume.
- In alkaline earth metals the properties such as metallic nature, reducing nature, reactivity, electropositive character and ionic nature of compounds increases from Be to Ba whereas the complex formation tendency decreases.
- Fly ash, a waste product from steel industry has properties similar to cement. It can be added to cement to reduce the cost without affecting its quality.
- $BaSO_4$ being insoluble in H_2O and opaque to X-rays is used under the name *barium meal* to scan the X-rays of the human digestive system.
- Bicarbonates of alkaline earth metals do not exist in the solid state but are known in soution only.

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- Most of the kidney stones consist of calcium oxalate, $CaC_2O_4 \cdot H_2O$ which dissolves in dilute strong acids but remains insoluble in bases.
- Magnesium perchlorate, $Mg(ClO_4)_2$ is used as a drying agent under the name anhydrone.
- Mg in powder form is used in *flash bulbs* used in photography and Ca is used as *deoxidiser* as well as *desulphuriser* of metals.
- The anhydrous form of $CaSO_4$ (called *anhydrite*) is called *dead burnt plaster* because it does not set like plaster of paris when moistened with water.
- Mortar used in making buildings is a mixture of lime (CaO) and sand in the ratio 1 : 3 with enough water to make a thick paste. When the mortar is placed between bricks, it slowly absorbs CO_2 from the air and the slaked lime reverts to $CaCO_3$.

  $$Ca(OH)_{2\,(s)} + CO_{2\,(g)} \rightarrow CaCO_{3\,(s)} + H_2O_{\,(l)}$$

  Although the sand in the mortar is chemically inert, the grains are bound together by the particles of calcium carbonate and a hard material results.
- The ions Na^+, K^+, Mg^{2+} and Ca^{2+} are the most abundant metal ions in biochemical systems. Ca^{2+}, for example, is important in the process of muscle contraction and blood coagulation, Mg^{2+} is the metal ion present in chlorophyll, the green colouring pigment of the plants.
- Hydroxyapatite, $Ca_5(PO_4)_3OH$ is the main component of tooth enamel. Cavities in your teeth are formed when acids decompose the weakly basic apatite coating.

  $$Ca_5(PO_4)_3OH_{\,(s)} + 4H^+_{\,(aq)} \rightarrow 5Ca^{2+}_{\,(aq)} + 3HPO_4^{2-}{}_{(aq)} + H_2O_{\,(l)}$$

  This can be prevented, however, by converting hydroxyapatite to a much more-resistant coating, fluorapatite

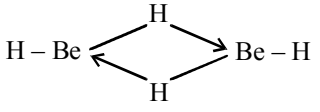
  $$Ca_5(PO_4)_3(OH)_{\,(s)} + F^-_{(aq)} \rightarrow Ca_5(PO_4)_3F_{\,(s)} + OH^-_{\,(aq)}$$

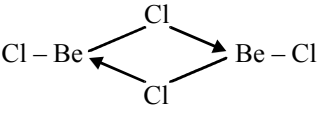
  The source of fluoride ion can be stannous fluoride, sodium fluoride or sodium monofluorophosphate commonly known as MFP in your tooth paste or a soluble fluoride such as NaF in your water supply.
- $CaCl_2 \cdot 6H_2O$ is widely used for melting ice on roads, particularly in very cold countries, because a 30% eutectic mixture of $CaCl_2/H_2O$ freezes at –55°C as compared with $NaCl/H_2O$ at –18°C.
- Anhydrous $CaCl_2$ is a good drying agent (desiccant) due to its hygroscopic nature ($CaCl_2 \cdot 6H_2O$). However it cannot be used to dry alcohols or ammonia/amines since it forms addition products ($CaCl_2 \cdot 6C_2H_5OH$, $CaCl_2 \cdot 6NH_3$ etc.)
- On heating MgC_2 changes into Mg_2C_3 which upon treatment with water gives propyne or allylene.

  $$Mg_2C_3 + 4H_2O \rightarrow 2Mg(OH)_2 + CH_3C \equiv CH$$

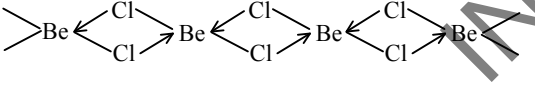
- BeH_2 like $BeCl_2$ is polymeric with the only difference that BeH_2 has three-centre Be.....H.....Be bonds while $BeCl_2$ has halogen bridges in which a halogen atom bonded to one Be atom uses its lone pair of electrons to form a coordinate bond with other Be atom.

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- $BeCl_2$ has a polymeric structure in the solid state but exists as a dimer in the vapour state and as a monomer at 1200 K.



- Magnesium burns with dazzling light even in CO_2 and N_2.
- Except beryllium, all other alkaline earth metals directly combine with hydrogen to form metal hydrides (MH_2).
- Beryllium does not form a peroxide.
- The most abundant alkaline earth metal in the earth's crust is Ca (5th most abundant element) and least abundant is Ra.
- Amongst alkaline earth metals, m.p. of Mg is lowest while density of Ca is the lowest.
- Be and Mg crystallize in *hcp*, Ca and Sr in *ccp* and Ba in *bcc* structures.
- ***Flame colouration :*** Alkaline earth metals impart characteristic flame colour like alkali metals. As we move down the group from Ca to Ba, the ionisation energy decreases hence the flame shows a gradual shift from red to violet. Thus,
  Ca - brick red ; Sr - crimson red ; Ba - apple green ; Ra - crimson
  Be and Mg, due to their high ionisation energies, however, donot impart any characteristic colour to Bunsen flame.

Industrial importance of alkaline earth metals

- **Beryllium** - Used in corrosion resistant alloys.
- **Magnesium** - When alloyed with Al, Mg is widely used as structural material because of its high strength, low density and ease in machining.
- **Calcium** - As an alloying agent to harden aluminium, calcium is the primary constituent of teeth and bones.
- **Strontium** - $SrCO_3$ is used for the manufacture of glass for colour TV picture tubes.
- **Barium** - $BaSO_4$ is used in medicine as a contrast medium for stomach and intestinal X-rays.
- **Radium** - Used in cancer-radiotherapy

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Anomalous behaviour of beryllium

- Beryllium, the first member of alkaline earth metals differs from rest of the metals and shows an anomalous behaviour. The main reasons for this difference are as follows.
- Because of high (IE) and small atomic size it forms compounds which are largely covalent and its salts are easily hydrolysed.
- Beryllium ($1s^2\,2s^2\,2p^0$) can use only $2s$ and three $2p$ orbitals in coordination thus maximum co-ordination number (C.N.) of Be is 4 while other elements can show C.N. of 6 in their compounds by use of *d*-orbitals in addition to *s* and *p* orbitals.
- BeO and $Be(OH)_2$ are amphoteric while other oxides are basic.

Some important properties in which beryllium differs from the rest of the members of its group are as follows:

(i) Beryllium is harder than other members of its group. This is due to the fact that maximum metallic bonding is present on account of smallest size amongst alkaline earth metals.

(ii) It has higher melting and boiling points than the other members due to maximum metallic bonding.

(iii) Be is least reactive as its ionization potential is high. However, it does react with oxygen and nitrogen at high temperature.

(iv) Beryllium forms covalent compounds because of high charge density and hence greater polarising power, whereas other members form ionic compounds.

(v) It dissolves in alkalies with evolution of hydrogen.

$$Be + 2NaOH + 2H_2O \rightarrow \underset{\text{sodium beryllate}}{Na_2BeO_2{\cdot}2H_2O} + H_2$$

Other alkaline earth metals donot react with alkalies.

(vi) Hydroxide of beryllium is amphoteric in nature. The hydroxide is insoluble in water. It is covalent in nature. The hydroxides of other alkaline earth metals are basic, ionic and their solubility increases on moving from $Mg(OH)_2$ to $Ba(OH)_2$.

(vii) Its salts can never have more than four molecules of water of crystallisation as it has only four available orbitals in its valency shell. Other alkaline earth metals can extend their coordination number to 6 by using *d*-orbitals.

- **Diagonal relationship between beryllium and aluminium**

	Group IIA	Group IIIA
Second period	Be	B
Third period	Mg	Al

Beryllium shows some similarities in properties with aluminium, the second typical element of group IIIA of the next higher period. This type of relationship between diagonally placed elements is called diagonal relationship. This is due to the reason that these two elements have the same electronegativity (Be = 1.5, Al = 1.5) and the polarising power *i.e.* charge/radius ratio (Be^{2+} = 2/31 = 0.064 and Al^{3+} = 3/50 = 0.060) of their ions are very similar.

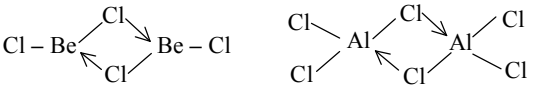
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Some points of similarity are given below.

(i) Both metals have a tendency to form covalent compounds, e.g. the chlorides of both (*i.e.* $BeCl_2$ and $AlCl_3$) being covalent are soluble in organic solvents.

(ii) Both $BeCl_2$ and $AlCl_3$ act as strong Lewis acids.

(iii) Both $BeCl_2$ and $AlCl_3$ have bridged chloride structures in the vapour phase.



(iv) Both the metals dissolve in strong alkalies to form soluble complexes : beryllates $[Be(OH)_4]^{2-}$ and aluminates $[Al(OH)_4]$.

(v) The oxides of both beryllium (BeO) and aluminium (Al_2O_3) are hard high melting insoluble solids.

(vi) Salts of both these elements form hydrated ions, e.g. $[Be(OH_2)_4]^{2+}$ and $[Al(OH_2)_6]^{3+}$ in aqueous solutions.

(vii) Because of similar polarising power both beryllium and aluminium forms complexes. For example, beryllium forms tetrahedral complexes such as $[BeF_4]^{2-}$ and $[Be(C_2O_4]^{2-}$ and aluminium forms octahedral complexes like $[AlF_6]^{3-}$ and $[Al(C_2O_4)_3]^{3-}$.

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Points to Remember

- Amongst alkaline earth metals, melting point of Mg is lowest while density of Ca is the lowest.
- The most abundant alkaline earth metal in the earth's crust is Ca (5th most abundant element) and least abundant is Ra.
- Beryllium does not form a peroxide.
- Except beryllium, all other alkaline earth metals directly combine with hydrogen to form metal hydrides (MH_2).
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- Calcium ions play an important role in muscle contraction.
- Gypsum $CaSO_4 \cdot 2H_2O$ is also known as alabaster.
- Calcium is present in the soil, plants, bones as $Ca_3(PO_4)_2$ and egg shells etc.



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17

CHAPTER

p-block elements

BORON FAMILY

- The boron group is the series of elements in group 13 in the periodic table. These elements are characterised by having three electrons in their outer energy levels (valence layers). Boron is considered metalloid, and the rest are considered metals of the poor metals group. The boron group consists of boron (B), aluminium (Al), gallium (Ga), indium (In) and thallium (Tl).
- Boron is a powerful reducing agent which reduces CO_2 and SiO_2 to C and Si respectively.
- Potash alum $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$ is commonly used for blood coagulation because Al^{3+} ions given by it neutralize the negatively charged albuminoid of blood.
- Aqueous solutions of alums are acidic due to cationic hydrolysis.
- In alums $M_2^{I}SO_4 \cdot M_2^{III}(SO_4)_3 \cdot 24H_2O$, 6 water molecules are held by monovalent cations, 6 water molecules are held by trivalent cations and 12 water molecules are held in the crystal hydrolysis.
- The most electropositive element in group 13 is Al. It also has the lowest density among all the elements of group 13.
- The electron deficient compounds remove their deficiency by dimerisation and by forming coordinate covalent bond.
- BF_3, BCl_3, BBr_3 exist as monomers while $AlCl_3$ exists as a dimer Al_2Cl_6 where each Al atom is sp^2-hybridized.
- $AlCl_3$, $AlBr_3$ are covalent while $Al(NO_3)_3$, $Al_2(SO_4)_3$, AlF_3 are ionic compounds.
- Al displaces hydrogen from acid but B does not.
- Basic nature of oxides and hydroxides follows the order : B < Al < Ga < In < Tl

  Acidic $\rightarrow$ Amphoteric $\rightarrow$ Basic

  $B(OH)_3$ is distinctly acidic and acts as a Lewis acid (and not a proton donor) by accepting OH^- from H_2O

  $$B(OH)_3 + H_2O \rightleftharpoons [B(OH)_4]^- + H^+$$
- The relative strength of Lewis acids of boron trihalides increases in the order :

  $BF_3 < BCl_3 < BBr_3 < BI_3$.
- Reducing nature of group 13 elements follows the order : Al > Ga > In > Tl.
- Stability of hydrides of group 13 elements decreases in the order : B > Al > Ga > In > Tl.
- Stability of +1 oxidation state increases in the order :Ga < In < Tl.

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- Stability of +3 oxidation state decreases in the order : B > Al > Ga > In > Tl.
- Out of all the chlorides of group 13 elements, only $AlCl_3$, sublimes on heating.
- Ruby is red and contains Al_2O_3 and Cr_2O_3. Sapphire is blue and contains Al_2O_3, Fe_2O_3 and TiO_2. Emerald is green and contains Ca, Cr and Al silicates.
- Due to their toxic nature, traces of thalium salts can cause loss of hair.
- **Aluminothermy or thermite process.** Aluminium has a strong affinity for oxygen and hence it can be used to reduce certain metal oxides such as Fe_2O_3, Cr_2O_3 etc. to the corresponding metals.

  $$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe + Heat$$
  $$Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr + Heat$$

  The heat liberated in these reactions is so large that the metal is produced in the molten state which can be used for welding purposes. This reduction of metal oxides by aluminium is called **aluminothermy or thermite process** or **Goldschmidt aluminothermite process.**
- *Ammonal* is a mixture of Al powder and NH_4NO_3 and is used in bombs.
- Al is the chief constituent of *silvery paints*.
- $Al_2(SO_4)_3$ is used for making *fire proof clothes*.
- Boron shows maximum covalency four while Al shows six.
- Thallium is the highly toxic element amongst the group 13 members.
- As Ga remains liquid over a wide range of temperature (303 K to 2510 K), it has been used in quartz thermostats for measuring high temperatures.
- Gallium is a lower melting solid (m.p. = 303 K) and is liquid on a particular warm day. It readily super cools *i.e.* remains liquid even at temperatures several degrees below its melting point.

Anomalous behaviour of boron

Boron shows anomalous behaviour due to its small size, high nuclear charge, high electronegativity and non-availability of *d*-electrons. The main point of differences are :

(1) Boron is a typical non-metal whereas other members are metals.
(2) Boron is a bad conductor of electricity whereas others are good conductors.
(3) Boron alone exhibits allotropy.
(4) Boron forms only covalent compounds whereas aluminium and other elements of group 13 also form some ionic compounds.
(5) Hydroxides and oxides of boron are acidic in nature whereas those of others are amphoteric and basic.
(6) The trihalides of boron (BX_3) exist as monomer. On the other hand, aluminium halides exist as dimers (Al_2X_6).
(7) Borates are more stable than aluminates.
(8) Boron exhibits maximum covalency of four. *e.g.* BH_4^- ion while other members exhibit a maximum covalency of six. *e.g.* $[Al(OH)_6]^{3-}$.
(9) Boron does not decompose steam while other members do so.
(10) Concentrated nitric acid oxidises boron to boric acid but no such action is noticed by other group members.

$$B + 3HNO_3 \rightarrow H_3BO_3 + 3NO_2$$

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[B, Al, Ga, In, Tl ; ns^2 np^1]

INCREASING TRENDS	DECREASING TRENDS	EXCEPTIONS
• Atomic radii (*M*)	• Ionisation energies	• Atomic size of Ga < In
• Ionic Radii		• IE_1 of Tl > In
• Inert pair effect		• IE_1 of Ga ≈ Al
• Tendency to show + 1 oxidation state	• M.P./B.P.	
• Tendency to form ionic compounds	• Tendency to show + 3 oxidation state	
• Electropositive character	• Tendency to form covalent compounds	
• Lewis acid strength of trihalides of B increases from $BH_3 \rightarrow BBr_3$		

- Behaviour of $M(OH)_3$ changes from acidic → amphoteric → basic.
- Compounds of Boron and hydrogen are BORANES which contain 3 centre - 2 electron bond. Compounds of aluminium are ionic as well as covalent.

CARBON FAMILY

- The carbon group is group 14 in the periodic table. Each of the elements in this group has 4 electrons in its outer energy level. The last orbital of all these elements is the p^2 orbital. In most cases, the elements share their electrons. The tendency to lose electrons increases as the size of the atom increases, as it does with increasing atomic number. Carbon alone forms negative ions, in the form of carbide (C^{4-}) ions. Silicon and germanium, both metalloids, each can form +4 ions. Tin and lead both are metals. The group consists of carbon (C), silicon (Si), germanium (Ge), tin (Sn) and lead (Pb).
- Quartz is a crystalline form of SiO_2, zeolite is sodium aluminium silicate, $Na_2Al_2Si_2O_8$ while feldspar is potassium aluminium silicate, $KAlSi_3O_8$. All these are three dimensional networks. Mica is potassium aluminium silicate, $KAl_3Si_3O_{10}(OH)_2$, clay is hydrated aluminium silicate, $Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O$ while talc is hydrated magnesium silicate $3MgO \cdot 4SiO_2 \cdot H_2O$. All these are the examples of sheet silicates. Asbestos is hydrated magnesium silicate, $3MgO \cdot 2SiO_2 \cdot 2H_2O$ which is an example of chain silicate.
- After smelting, the molten tin is drawn into blocks and is called block tin which is of 99.5% purity.
- After leaching and washing tin stone, the heavier particles left behind are called black tin.
- Mixture of massicot (yellow form of PbO) with glycerine is used for joining the broken pieces of stone and glass.
- Lead metal marks the paper. It is so soft that it can be cut with a knife and scratched with the finger nail.

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- Metal fires are put off either by sand or pyrene (CCl_4) or foam type extinguishers.
- Kieselguhr is a mass of hydrated silica (SiO_2) formed from skeletons of minute plants known as diatoms. It is a very porous and absorbent material used in the manufacture of dynamite.
- The reducing character (of M^{2+} species) in dihalides decreases in the order :
  $GeCl_2 > SnCl_2 > PbCl_2$
- Ease of hydrolysis of tetrahalides follows the sequence : $SiCl_4 < GeCl_4 < SnCl_4 < PbCl_4$
- Tetrahalides of Ge, Sn and Pb behave as oxidising agents and the oxidising character of M^{4+} species increases in the order : $GeCl_4 < SnCl_4 < PbCl_4$
- The thermal stability and volatility of tetrahalides with a common central atom decrease with increase in molecular weight of the halide ion $MF_4 > MCl_4 > MBr_4 > MI_4$
- Thermal stability of tetrahalides decreases in the order :
  $CCl_4 > SiCl_4 > GeCl_4 > SnCl_4 > PbCl_4$
- Thermal stability and volatility of hydrides of group 14 elements decrease and reducing nature increases on moving down the group. Thermal stability and volatility vary as :
  $CH_4 > SiH_4 > GeH_4 > SnH_4 > PbH_4$
  Reducing nature varies as : $CH_4 < SiH_4 < GeH_4 < SnH_4 < PbH_4$
- Acidic character of dioxides decreases down the group. Thus,
  CO_2, SiO_2, GeO_2, SnO_2, PbO_2
  Weakly acidic $\rightarrow$ Amphoteric $\rightarrow$ Basic
- Metallic character in case of group 14 elements increases in the order :
  C < Si < Ge < Sn < Pb
  Non-metal $\rightarrow$ Metalloid $\rightarrow$ Metal
- Reactivity of group 14 elements increases in the order : C < Si < Ge < Sn < Pb
- In case of Sn and Pb, due to inert pair effect, Sn (+2) is a strong reducing agent and is oxidising to stable +4 oxidation state while Pb (+4) is a strong oxidising agent and is reduced to stable +2 oxidation state.
- *Wood's* metal is an alloy of Bi (50%), Pb (25%), Sn (12.5%) and Cd (12.5%). It has m.p. 344 K.
- Dry powder extinguishers contain sand and baking soda ($NaHCO_3$).
- *Abrak* (mica) is a naturally occurring aluminium silicate $[KH_2Al_3(SiO_4)_3]$ or $KAl_3Si_3O_{10}(OH)_2$.
- Vaseline is obtained from silicones and is highly useful for low temperature lubrication.
- *Talc* is a pure magnesium silicate in the form of $3MgO{\cdot}4SiO_2{\cdot}H_2O$ and is found in Talcum powder and many face powders. It consists of planar sheets which can slip over one another due to weak forces of attraction. This is the reason that Talcum powder has a slippery touch.
- Tin is not attacked by organic acids and is therefore used in *tinning* of cooking utensils.
- SiC is used as *high temperature semi-conductor* in transistor diode rectifiers.
- **Tetraethyl lead (TEL), $Pb(C_2H_5)_4$.** It is prepared by the action of ethyl chloride on sodium-lead alloy *i.e.*

$$4C_2H_5Cl + 4(Pb{\cdot}Na) \rightarrow (C_2H_5)_4Pb + 3NaCl + 3Pb$$

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- **Chrome yellow ($PbCrO_4$)** is obtained by adding potassium chromate to lead acetate. It is used as a yellow pigment under the name *chrome yellow.* On treatment with NaOH, it gives basic lead chromate, $PbCrO_4 \cdot PbO$ known as *chrome red.*
- **White lead, $Pb(OH)_2 \cdot 2PbCO_3$** is also known as basic lead carbonate. It is prepared in the laboratory by the addition of sodium carbonate solution to any lead salt.

$$3Pb(NO_3)_2 + 3Na_2CO_3 + H_2O \rightarrow Pb(OH)_2 \cdot 2PbCO_3 + 6NaNO_3 + CO_2$$

- Quartz is the most common and purest variety of SiO_2 (silica).
- $SnCl_2$ is ionic, solid, more stable and reducing in character while $SnCl_4$ is covalent, liquid, less stable and oxidising in character.
- U.V. rays can be checked by Crooke's glass which contains CeO_2.
- The material used in solar cells contains Si.
- High strength silicon rubber withstands extreme surface temperatures. Hence *soles* of **Luna boots** were prepared from this rubber and were used by *U.S. Appolo astronauts.*
- *Etching of glass*. Glass is attacked by HF. This property is used in the etching of glass. The glass to be etched is coated with a thin layer of wax and the design to be produced is scratched wth a needle. An aqueous solution of HF is applied to the exposed part. After sometime it is placed in water and wax is removed from the surface. The marks are engraved on the exposed parts.
- Sodium silicate (Na_2SiO_3) is commercially called water glass. Its composition may vary from $Na_2SiO_3 \cdot SiO_2$ to $Na_2SiO_3 \cdot 3SiO_2$. It is soluble in water and its solution is alkaline due to hydrolysis

$$Na_2SiO_3 + 2H_2O \rightarrow 2NaOH + H_2SiO_3$$

Anomalous behaviour of carbon

Carbon differs from rest of the elements of group 14 due to small size and high electronegativity. Some points of difference are :

(i) Its maximum covalency is 4 as *d*-electrons are absent in the valence shell.

(ii) It has maximum property of catenation. It can form multiple bonds *i.e.* double and triple bonds. As a result carbon forms a large number of compounds like alkenes, alkynes etc.

(iii) Because of the smallest atomic radius and lower atomic volume carbon is the hardest element of group 14. It has the highest melting point and boiling point, highest ionisation energy and is the most electronegative element of this group.

(iv) Among group 14, carbon shows a pronounced ability to form $p\pi$- $p\pi$ multiple bonds with itself (in graphite) and with other non-metals especially nitrogen and oxygen.

- $SnCl_4 \cdot 5H_2O$ is called *butter of tin* and is used as a mordant in dyeing.
- Quartz glass used in the manufacture of optical instruments is called *vetreosil.*
- Silicon is added to steel or iron to increase its resistance to attack by acids.
- Glass is soluble in hydrofluoric acid due to the following reactions. Ordinary glass is a mixture of Na_2SiO_3 and $CaSiO_3$.

$$Na_2SiO_3 + 8HF \rightarrow 2NaF + \underset{\text{Hydrofluorosilicic acid}}{H_2SiF_6} + 3H_2O$$

$$CaSiO_3 + 8HF \rightarrow CaF_2 + H_2SiF_6 + 3H_2O$$

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Silicates

Silicates are metal derivatives of silicic acid, H_4SiO_4 or $Si(OH)_4$ silicates are made up of SiO_4^{4-} tetrahedral units in which Si *is* sp^3 hybridised and is surrounded by four oxygen atoms. There are following types of silicates.

(i) *Orthosilicates* : contain single discrete unit of SiO_4^{4-} tetrahedra. *e.g.* zircon $ZrSiO_4$, willemite Zn_2SiO_4, phenacite Be_2SiO_4, olivine or forestrite Mg_2SiO_4.

(ii) *Pyrosilicates* : contain two units of SiO_4^{4-} joined along a corner containing oxygen atom. They contain $(Si_2O_7)^{6-}$ unit. *e.g.* thortveitite $Sc_2Si_2O_7$; hemimorphite $Zn_3(Si_2O_7)\cdot Zn(OH)_2\cdot H_2O$.

(iii) *Cyclic* structure contains $\left[(SiO_3)^{2-}\right]_n$ basic unit which is obtained when two SiO_4^{4-} units share two oxygen atoms (two corners) with each other. *e.g.* catapleite $Na_2ZrSi_3O_9\cdot 2H_2O$, wollastonite $Ca_3Si_3O_9$, beryl $Be_3Al_2Si_6O_{18}$, benitoite $BaTiSi_3O_9$.

(iv) *Chain silicates* are formed by sharing two oxygen atoms by each tetrahedra. Anion of chain silicates have two general formulae,
(a) $(SiO_3)_n^{2n-}$, (b) $(Si_4O_{11})_n^{6n-}$.
e.g. Spodumene $LiAl(SiO_3)_2$, tremolite $Ca_2Mg_5(Si_4O_{11})_2(OH)_2$

(v) *Sheet silicates* are formed when sharing of three oxygen atoms (three atoms) by each tetrahedron results in an infinite two dimensional sheet structure with the general formula $(Si_2O_5)_n^{2n-}$.
e.g. Talc $Mg(Si_2O_5)_2\ Mg(OH)_2$, Kaolin $Al_2(OH)_4\ (Si_2O_5)$.

(vi) *Three dimensional sheet silicates* involve all four oxygen atoms in sharing with adjacent SiO_4^{4-} tetrahedra. They have the general formula $(SiO_2)_n$.
e.g. zeolites, chabazite, feldspars.

[C, Si, Ge, Sn, Pb ; $ns^2\ np^2$]

INCREASING TRENDS	DECREASING TRENDS	EXCEPTIONS
• Atomic radii (*M*)	• Ionisation energies • Electronegativity	• IE of Pb > Sn IE of Ge ≈ Si
• Metallic character	• Non-metallic character	• Only C has ability of $p\pi$-$p\pi$ bonding
• Inert pair effect		• Si and others can form $d\pi$-$p\pi$ bonding
• Tendency to show + 2 oxidation state	• Tendency of +4 oxidation state	• All elements except Pb show allotropy
• Tendency of forming ionic compounds • Reducing character of hydrides (MH_4)	• Tendency of forming covalent compounds • Thermal stability of hydrides • *M* – *M* bond strength • Tendency of catenation	

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- They form oxides of formula MO and MO_2.
- Behaviour of MO_2 change from acidic → amphoteric → weakly acidic.

Important information

- The value of plumbosolvency increases if the water contains nitrates, organic acids and ammonium salts.
- The drawback of the use of white lead is that it turns black by the action of H_2S present in the atmosphere.
- Tin disease or tin plague is the conversion of white tin to grey tin which occurs in cold countries and results in the decrease in density, because of which it is very brittle.
- In the etching of glass the property of glass that it is attacked by hydrofluoric acid is used. The glass is coated with a thin layer of wax and the design to be produced is scratched with a needle. An aqueous solution of HF is applied to the exposed part it is then placed in water and wax is removed from the surface. This process engraves these marks on the exposed part.
- Chrome yellow ($PbCrO_4$) on treatment with NaOH forms basic lead chromate $PbCrO_4 \cdot PbO$ (chrome red).
- Charcoal dissolves slowly in hot dilute HNO_3 forming a brown coloured substance known as artificial tannin.
- Solder is an alloy of tin and lead and is used for soldering purposes.
- Quartz is the most common and purest variety of SiO_2 (silica).
- Tin and lead both form organometallic compounds: $Sn(C_2H_5)_4$ tin tetraethyl and $Pb(C_2H_5)_4$ lead tetraethyl.

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NITROGEN FAMILY

- The nitrogen group elements (group VA) are also known as group 15 (formerly group V) of the periodic table. This group has the defining characteristic that all the component elements have 5 electrons in their outermost shell, that is 2 electrons in the *s* subshell and 3 unpaired electrons in the *p* subshell. They are therefore 3 electrons short of filling their outermost electron shell in their non-ionized state.
- The most important element of this group is nitrogen (N), which in its diatomic form is the principal component of air. Other members of the group include phosphorus (P), arsenic (As), antimony (Sb) and bismuth (Bi).
- The collective name pnicogens (now also spelled pnictogens) is also sometimes used for elements of this group.
- The strength and solubility of oxyacids of group 15 elements decrease rapidly in the order :

  $HNO_3 > H_3PO_4 > H_3AsO_4 > H_3SbO_4$
- In solid state, PCl_5 exists as $[PCl_4]^+$ and $[PCl_6]^-$ having tetrahedral and octahedral structures respectively. : PBr_5 exists in solid state as $[PBr_4]^+ [Br^-]$ while PI_5 exists as $[PI_4]^+ [I^-]$ in solution.
- In case of phosphorus trihalides, the Lewis acid strength decreases in the order :

  $PF_3 > PCl_3 > PBr_3 > PI_3$

and the bond angle increases as the electronegativityof the halogen decreases down the group *i.e.*

$PF_3 < PCl_3 < PBr_3 < PI_3$

- Trihalides of P, As and Sb also behave as Lewis acids and the acid strength shows decreasing trend down the group *i.e.*, $PCl_3 > AsCl_3 > SbCl_3$
- The ease of hydrolysis of trihalides of group 15 elements decrease in the order :

  $NCl_3 > PCl_3 > AsCl_3 > SbCl_3 > BiCl_3$

  NF_3 and PF_3 are not hydrolysed.
- For trihalides of N, the stability decreases in the order : $NF_3 > NCl_3 > NBr_3$

  and Lewis base strength increases in the order: $NF_3 < NCl_3 < NBr_3 < NI_3$
- The m.p. of hydrides increase in the order : $PH_3 < AsH_3 < SbH_3 < NH_3$

  and the increasing order of b.p. is $PH_3 < AsH_3 < NH_3 < SbH_3$
- The reducing power, poisonous nature and covalent nature of hydrides increase in the order: $NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$
- The basic nature, bond angle, thermal stability and dipole moment of hydrides (MH_3) decrease in the order : $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$
- The reactivity of various allotropic forms of phosphorus towards other substances is in the order : White > Red > Black.
- All oxides of P, As and Sb are dimeric. Thus, trioxides and pentoxides are written as M_4O_6 and M_4O_{10} respectively.
- All pentoxides except P_2O_5 show oxidising properties.
- The stability of pentoxides decreases in the order:

  $P_2O_5 < As_2O_5 > Sb_2O_5 > N_2O_5 > Bi_2O_5$
- The acidic strength of oxides of nitrogen increases in the order :

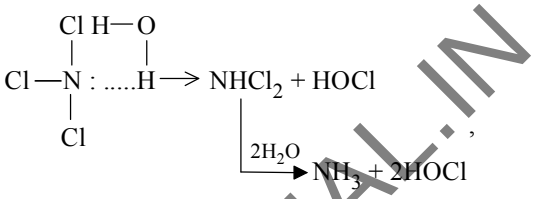
  $N_2O < NO < N_2O_3 < N_2O_4 < N_2O_5$
- The acidic strength of pentoxides decreases in the order :

  $N_2O_5 > P_2O_5 > As_2O_5 > Sb_2O_5 > Bi_2O_5$
- The acidic strength of trioxides decreases in the order : $N_2O_3 > P_2O_3 > As_2O_3$
- The normal oxides and hydroxides of nitrogen and phosphorus are strongly acidic; arsenic is weakly acidic; antimony is amphoteric and bismuth is largely basic.
- Nitrogen and phosphorus behave as non-metals, arsenic and antimony as metalloids and bismuth as a metal.
- Nitrogen and phosphorus show both positive and negative oxidation states but the heavier elements show only positive oxidation states. Nitrogen shows all the oxidation states from –3 to +5.
- BiOCl is called pearl white.
- In tooth paste, $CaHPO_4 \cdot 2H_2O$ is added as *mild abrasive and polishing agent.*
- P_4S_3 is used in strike anywhere matches. The head of a safety match box stick contains $KClO_3$, KNO_2 or red lead (Pb_3O_4) along with grounded glass pieces. Sides of match box

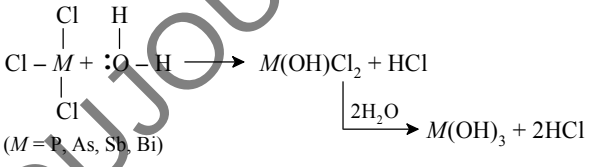
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contain red phosphorus, antimony sulphide (Sb_2S_3) and sand powder. Tips of match stick can also contain a mixture of $K_2Cr_2O_7$, sulphur and white phosphorus.

- Bones and teeth contain 58% P as $Ca_3(PO_4)_2$.
- The substance used in *Holmes signals* of the ship is a mixture of CaC_2 and Ca_3P_2.
- The hydrolysis of NCl_3 is explosive and gives HOCl and NH_3 whereas the hydrolysis of trichlorides of other members of nitrogen family gives HCl and hydroxide of the element. This difference can be attributed to the different mechanistic routes adopted for hydrolysis. In case of NCl_3, the hydrolysis is presumed to proceed through *H-bonding of the lone pair on the N-atom and H-atom of water.* This is due to higher E.N. value of N atom amongst group 15 members.



But as we move downwards in group 15, the E.N. of the elements decreases which decreases the ability to react with water through H-bonding mechanism. The hydrolysis in these cases proceeds through conventional *nucleophilic attack by H_2O molecule because of the presence of vacant d-orbitals of these elements.*

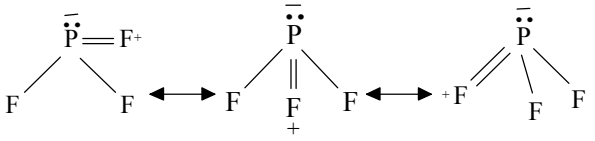


- In case of phosphorus trihalides, PX_3 (X = F, Cl, Br, I), the bond angles increase from PF_3 to PI_3 as the electronegativity of the halogen decreases from F to I. A more electronegative atom will have higher tendency to keep the bonding pair more towards itself with the result, the electron cloud is drawn away from the central atom. This leads to *decrease in bond pair-bond pair repulsions.* Consequently the bond angle increases as the electronegativity of substituents on P decreases. The order of increasing bond angle in PX_3 is

$$\underset{(97°)}{PF_3} < \underset{(100°)}{PCl_3} < \underset{(101°)}{PBr_3} < \underset{(102°)}{PI_3}$$

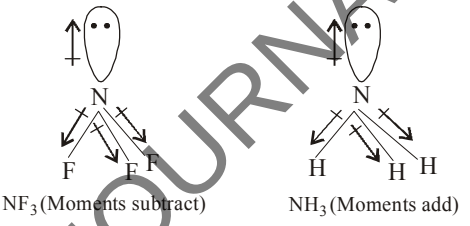
- The bond angle in NH_3 (107°) is greater than the bond angle in NF_3 (102°) whereas the bond angle in PH_3 (94°) is less than the bond angle in PF_3 (97°). The decrease in bond angle from NH_3 to NF_3 can be explained due to the **displacement of electron cloud** of N—F bonds towards the more electronegative F atoms in NF_3 which *reduces the bond pair-bond pair repulsion* and shows a decrease in bond angle. However, the increase in bond angle from PH_3 to PF_3 is attributed to the *enhanced repulsion due to presence of double bond in* PF_3. The molecule PF_3 is expected to acquire partial double bond character due to the resonance forms.

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Thus due to multiple $(p\pi - d\pi)$ bond, the *bond pair-bond pair repulsions increase* to give a higher bond angle. In case of NF_3, such multiple bond is not possible as N does not have *d*-orbitals to accommodate lone pair of electrons from F atoms.

- Both NF_3 and NH_3 have pyramidal structures with FNF and HNH bond angles 102° and 107° respectively but their dipole moments are different, *viz* 0.24 D for NF_3 and 1.48 D for NH_3. The difference is due to the fact that while the dipole moment due to N—F bonds in NF_3 are in *opposite direction to the direction of the dipole moment of the lone pair on* N *atom* which partly cancel out, the dipole moments of N—H bonds in NH_3 are in the *same direction of the dipole moment of the lone pair on* N *atom* which add up as shown below.



NF_3 (Moments subtract) NH_3 (Moments add)

Because of its lower dipole moment, NF_3 is weaker ligand than NH_3.

Anomalous Behaviour of Nitrogen

Nitrogen differs considerably from the rest of the family members because of
(i) Small size (ii) High electronegativity
(iii) Absence of *d*-orbitals in the valence shell
(iv) Tendency to form multiple bonds

The main points of difference are:
(i) Nitrogen is a gas while other members are solids.
(ii) Nitrogen exists as diatomic molecule $(\dot{N}\equiv\dot{N})$ while other elements except bismuth forms tetra-atomic molecules such as P_4, As_4 and Sb_4.
(iii) The catenation property is more pronounced in nitrogen. Chains containing upto eight nitrogen atoms are known but in other elements catenation is limited to two atoms only.
(iv) Nitrogen does not form pentahalides.
(v) Nitrogen exhibits a large number of oxidation states from –3 to +5 *ie* +5, +4, +3, +2, +1, 0, –1, –2 and –3 in N_2O, NO, N_2O_3, NO_2, N_2O_5, N_2, NH_2OH, NH_4NO_3, NH_3 respectively.

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[N, P, As, Sb, Bi ; ns^2 np^3]

INCREASING TRENDS	DECREASING TRENDS	EXCEPTIONS
• Atomic size • Melting/Boiling point increases from N → As • Metal character • Density • Tendency of lower oxidation state + 3 • Reducing character of hydrides (MH_3) • Ionic character of compound dominate towards end	• Ionisation energies • Electronegativity • M.P./B.P. decreases As → Bi • Tendency of covalent bonding • Thermal stability of hydrides • Angle around M in metal hydrides (MH_3) • Basic nature of MH_3 • Acidic character of oxides • Tendency of forming MX_5	• N, P show oxidation state – 3 to + 5 Bi shows oxidation state of + 3 only • Elements except N, Bi exhibit allotropy • B.P. of MH_3 $PH_3 < AsH_3 < SbH_3 < BiH_3$

- Behaviour of oxides change from acidic → amphoteric → basic.
- Their hydrides MH_3 and trihalides (MX_3) are pyramidal but pentahalides (MX_5) are trigonal bipyramidal.

OXYGEN FAMILY

- The chalcogens are the name for the periodic table group 16 in the periodic table. It is sometimes known as the oxygen family. It consists of the elements oxygen (O), sulphur (S), selenium (Se), tellurium (Te) and the radioactive polonium (Po). The compounds of the heavier chalcogens (particularly the sulphides, selenides and tellurides) are collectively known as chalcogenides.
- Oxygen and sulphur are non-metals, and polonium, selenium and tellurium are metalloid semiconductors (*i.e.* their electrical properties are between those of a metal and an insulator). Nevertheless, tellurium, as well as selenium, is often referred to as a metal when in elemental form.
- The acidic nature of oxides of a particular element increases with increase in oxidation number of the central element. For example, $SO < SO_2 < SO_3$.
- The acidic nature of dioxides and trioxides decreases in the order :

  $SO_2 > SeO_2 > TeO_2 > PoO_2$ and

  $SO_3 > SeO_3 > TeO_3$
- In case of hydrides of group 15 elements, melting points and boiling points decrease in the order:

  $H_2O > H_2S > H_2Se > H_2Te$

  Volatility increases in the order :

  $H_2O < H_2Te < H_2Se < H_2S$

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Acidic, covalent and reducing characters increase in the order :

$H_2O < H_2S < H_2Se < H_2Te$

Bond angle, dipole moment and thermal stability decrease in the order :

$H_2O > H_2S > H_2Se > H_2Te$

- Although S_2Cl_2 and Se_2Cl_2 are known, the corresponding Te_2Cl_2 and Po_2Cl_2 are not known due to weaker Te – Te and Po – Po bonds.
- The tetrafluorides act as Lewis base (electron donors) and also as Lewis acid (electron acceptors). For example,

  $F_4S + BF_3 \rightarrow [F_4S \rightarrow BF_3]$

  $SeF_4 + 2F^- \rightarrow [SeF_6]^{2-}$
- SF_4 is a gas, SeF_4 is a liquid and TeF_4 is a solid.
- All elements of group 15 form hexafluorides which have a high degree of covalency and low boiling points. Stability decreases in the order

  $SF_6 > SeF_6 > TeF_6$

  Ease of hydrolysis increases in the order

  $SF_6 < SeF_6 < TeF_6$
- All dioxides (MO_2) of group 15 elements are both oxidising as well as reducing agents. All trioxides (MO_3) are oxidising agents.
- OF_2 is known as oxygen difluoride and not fluorine oxide because fluorine is more electronegative than oxygen. In compounds of oxygen with chlorine, bromine or iodine, since oxygen is more electronegative, these compounds are known as chlorine dioxide, ClO_2 etc.
- Oxygen shows different oxidation states: –2 (in oxides), –1 (in peroxides), –1/2 (in superoxides), zero (in dioxygen), +2 (in oxygen difluoride OF_2) and +1 (in oxygen monofluoride O_2F_2).
- Sulphur is also called as brim stone.
- Lead chamber process for the manufacture of H_2SO_4 was introduced by John Roebuck in 1746.
- The most important application of Se is as a photoconductor in photocopying (xerox) machines.
- With sulphur, fluorine gives SF_6, chlorine forms SCl_4, bromine gives SBr_2 while iodine does not react with sulphur.
- SO_2 gas is dried by bubbling through concentrated H_2SO_4. It is not dried over quick lime as it reacts with it to form calcium sulphite.

  $CaO + SO_2 \rightarrow CaSO_3$
- SO_2 also turns lime water milky due to the formation of calcium sulphite. Milkiness disappears on passing excess SO_2 due to the formation of calcium bisulphite.

$$Ca(OH)_2 + SO_2 \rightarrow \underset{\text{Milkiness}}{CaSO_3} + H_2O$$

$$CaSO_3 + SO_2 + H_2O \rightarrow Ca(HSO_3)_2 \text{ (soluble)}$$

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- SO_2 is anhydride of H_2SO_3 and is called sulphurous anhydride whereas SO_3 is an anhydride of H_2SO_4 and is called sulphuric anhydride.
- Sulphurous acid (H_2SO_3) behaves as both *reducing* as well as *oxidising agent* and also *bleaches* the articles due to *reduction.*
- Sulphuric acid which is also called "*king of chemicals*" is used as a solvent, *an acid*, *an oxidising dehydrating and a sulphonating agent*. It is highly viscous, due to *hydrogen bonding*. Sulphuric acid obtained from Glower tower contains about 80% H_2SO_4 and is known as *Brown oil of vitriol* (BOV) due to its colour. It can be further concentrated and the concentrated acid is called *rectified oil* of *vitriol* (ROV).
- **Wackenroder's liquid or solution.** It is obtained by passing *hydrogen sulphide gas* through *saturated aqueous solution of sulphur dioxide* till its smell disappears and it is turned milky. When H_2S gas is passed through sulphurous acid, the reaction is called Wackenroder's reaction.
- The burning sensation of concentrated H_2SO_4 on skin is due to dehydration of skin.
- H_2S is called *sulphuretted hydrogen.* It is poisonous and its large amount proves fatal. Antidote for H_2S is dilute chlorine solution which destroys the effect of H_2S by oxidising it to sulphur.

$$H_2S + Cl_2 \rightarrow 2HCl + S$$

- SO_2 on reaction with PCl_5, gives *thionyl chloride* ($SOCl_2$) which fumes in moist air and is used in organic chemistry.

$$PCl_5 + SO_2 \rightarrow SOCl_2 + POCl_2$$

- $Na_2S_2O_3 \cdot 5H_2O$ (*hypo*) is also called *photographer's fixer.*
  It is used in *photography to remove* AgBr.

$$2Na_2S_2O_3 + AgBr \rightarrow \underset{\text{(Soluble complex)}}{Na_3[Ag(S_2O_3)_2]} + NaBr$$

  It is used *to remove iodine stains.*

$$2Na_2S_2O_3 + \underset{\text{(Brown Colour)}}{I_2} \rightarrow Na_2S_4O_6 + \underset{\text{(Colourless)}}{2NaI}$$

  It is also used *as an antichlor.*

$$Na_2S_2O_3 + Cl_2 + H_2O \rightarrow Na_2SO_4 + S + 2HCl$$

- SeO_3 exists in *monomeric state in the vapour phase* and is a *cyclic tetramer in the crystalline state.*
- S_2Cl_2 is used *in the vulcanization of rubber while* SF_6 *is used in high voltage transformers* because of its insulating property and inertness.
- The name *sulphur* has been derived from Sanskrit word '*Sulveri*' meaning killer of copper.
- *Vulcanisation of rubber* (*i.e.* heating rubber with sulphur) was discovered by **Charles Goodyear** in 1839.

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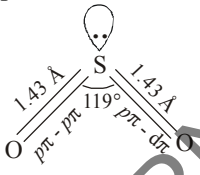
Anomalous behaviour of oxygen

Oxygen differs from the rest of the elements of oxygen family due to
(i) small size,
(ii) high electronegativity and
(iii) non-availability of *d*-orbitals.

Points of difference :
(i) Oxygen is a diatomic gas while others are solids.
(ii) Oxygen exhibits oxidation states of –2, –1 and +2 only while other members show both negative and positive oxidation states like –2, +2, +4 and +6.
(iii) Due to high electronegativity of oxygen, hydrogen bonding is present in water.
(iv) Oxygen is highly non-metallic due to high value of electronegativity.
(v) Oxygen is paramagnetic while others are diamagnetic.

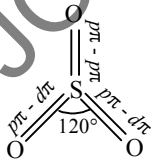
Important compounds of sulphur

- **Sulphur dioxide (SO_2) -**



Uses :
1. It acts both as an oxidising and reducing agent. It is used for manufacture of sulphuric acid.
2. It is used as bleaching agent for delicate articles like wool, silk etc.
3. Used as an antichlor for removing chlorine from the fabric after bleaching.

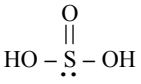
- **Sulphur trioxide (SO_3) -**



Uses :
1. Used for the manufacture of oleum and sulphuric acid.
2. As drying agent for gases.

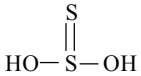
Oxoacids of sulphur

1. **Sulphurous acid** (H_2SO_3) (oxidation state +4)

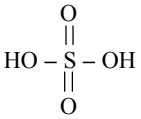


It is reducing in nature and reduces acidified $KMnO_4$ or $K_2Cr_2O_7$ to Mn^{2+} or Cr^{3+}.

2. **Thiosulphurous acid** ($H_2S_2O_2$) -



3. **Sulphuric acid** (oil of vitriol) - H_2SO_4

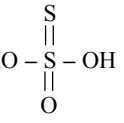


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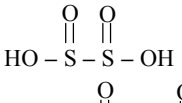
Uses:

(i) in the manufacture of fertilizers like $(NH_4)_2SO_4$ and superphosphate of lime.
(ii) in lead storage batteries
(iii) for tanning of leather
(iv) as dehydrating agent in labs.

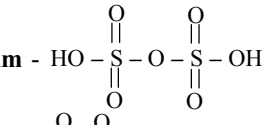
4. **Thiosulphuric acid** ($H_2S_2O_3$) -



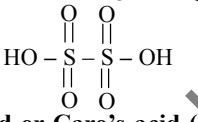
5. **Dithionous acid** ($H_2S_2O_4$) -



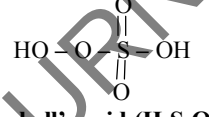
6. **Pyrosulphuric acid or oleum** -



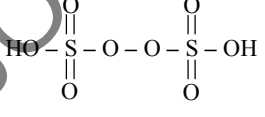
7. **Dithionic acid ($H_2S_2O_6$)** -



8. **Peroxomonosulphuric acid or Caro's acid (H_2SO_5)**



9. **Peroxodisulphuric acid or Marshall's acid ($H_2S_2O_8$)**



[O, S, Se, Te, Po ; ns^2 np^4]

INCREASING TRENDS	DECREASING TRENDS	EXCEPTIONS
• Atomic size • Density • Ionic radius • M.P./B.P. increase, O → Te • Metallic character • Acidic nature of hydrides (H_2M)	• Ionisation energies • Electronegativity • M.P./B.P. ecreases Te → Po • Electron affinity • Thermal stability of H_2M • Bond angle around M • M—M bond strength	• O shows tendency of $p\pi$-$p\pi$ bonding others can form $d\pi$-$p\pi$ bonding. • EA_1 of O < EA_1 of S • S shows some tendency of catenation • M—M bond strength is highest

- B.P. of H_2M : $H_2O > H_2Se > H_2Te > H_2S$
- Atomicity of O is 2 in O_2, 3 in O_3, but those of S, Se, Te is 8
- O shows O.N. of –2, –1, + 1, +2 ; S shows –2, + 2, + 4, + 6; other shows + 2, + 4

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HALOGEN FAMILY

- The halogens are diatomic molecules in their natural form. They require one more electron to fill their outer shells, and so have a tendency to form a singly-charged negative ion. This negative ion is referred to as a halide ion; salts containing these ions are known as halides.
- Halogens are highly reactive, and as such can be harmful or lethal to biological organisms in sufficient quantities. Fluorine is the most reactive element in existence, even attacking glass, and forming compounds with the heavier noble gases. It is a corrosive and highly toxic gas. Chlorine and iodine are both used as disinfectants for such things as drinking water, swimming pools, fresh wounds, dishes and surfaces. They kill bacteria and other potentially harmful microorganisms, a process known as sterilization. Their reactive properties are also put to use in bleaching. Chlorine is the active ingredient of most fabric bleaches and is used in the production of most paper products. The halogens consist of two solids, two gases, and one liquid, which makes the halogens the only group with all three forms of matter (at room temperature).
  They react with each other to form interhalogen compounds. Diatomic interhalogen compounds (BrF, ICl, ClF etc.) bear strong superficial resemblance to the pure halogens.
- HF is more reactive and corrosive than fluorine.
- Fluroine forms only one oxoacid HOF (hypofluorous acid).
- The oxidising power of perhalates (XO_4^-) decreases in the order : $BrO_4^- > IO_4^- > ClO_4^-$.
- Acidity and thermal stability of oxoacids having different halogens with the same oxidation number decrease with increase in atomic number of the halogen. For example, HClO > HBrO > HIO.
- In case of oxoacids of halogens (HXO_n where n = 1 – 4), the greater the number of oxygen atoms, greater is the thermal stability and acidic character and lesser is the oxidising power of the molecule. Thus, the acidic character and thermal stability increase in the order as there is an increase in the oxidation number of the halogen :

  $$HClO < HClO_2 < HClO_3 < HClO_4$$

  and the oxidising power decreases in the order :

  $$HClO > HClO_2 > HClO_3 > HClO_4$$

  The strength of the conjugate bases of these acids follows the order :

  $$ClO^- > ClO_2^- > ClO_3^- > ClO_4^-.$$

  whereas the stability of anions of oxoacids increases in the order :

  $$ClO^- < ClO_2^- < ClO_3^- < ClO_4^-.$$
- Oxides of chlorine, bromine and iodine are acidic and the acidic character increases as the percentage of oxygen in them increases.
- In case of oxides of chlorine the decreasing order of oxidising power is :

  $$Cl_2O > ClO_2 > Cl_2O_6 > Cl_2O_7$$

  and the increasing order of stability is

  $$Cl_2O < ClO_2 < Cl_2O_6 < Cl_2O_7$$

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- HI is least stable of all hydrogen halides and decomposes to H_2 and I_2. This is the reason that a bottle containing HI acquires brown colour due to I_2 after some time.
- In case of (hydrogen halides, H*X*) of group 17 elements, the boiling points increases in the order: HCl < HBr < HI < HF

  and the volatility decreases in the order : HCl > HBr > HI > HF

  The thermal stability, dipole moment and bond strength decrease in the order :

  HF > HCl > HBr > HI

  whereas the acidic strength and reducing character increase in the order :

  HF < HCl < HBr < HI
- In case of diatomic molecules (X_2) of halogens, the bond dissociation energy decreases in the order: $Cl_2 > Br_2 > F_2 > I_2$

  the oxidising power, solubility in water and reactivity decrease in the order :

  $F_2 > Cl_2 > Br_2 > I_2$
- For a particular non-metal atom (*M*), the strength of *M* – *X* bond (covalent) decreases in the order :

  M — F > *M* — Cl > *M* — Br > *M* — I
- For the same metal atom (*M*), the ionic character of *M* – *X* bond, melting point and boiling point of halides decrease in the order :

  M — F > *M* — Cl > *M* — Br > *M* — I
- In case of halide ions (X^-), the heat of hydration and basic character decrease in the order: $F^- > Cl^- > Br^- > I^-$

  and reducing character increases in the order : $F^- < Cl^- < Br^- < I^-$
- In case of group 17 elements (*X*), the electronegativity, reactivity and non-metallic character decrease in the order : F > Cl > Br > I.

  The negative electron gain enthalpy decrease in the order : Cl > F > Br > I

  and the basic character increase in the order : F < Cl < Br < I
- Liquid halogen is bromine while solid halogen is iodine.
- Safety matches are made by dipping the head of a match stick in potassium chlorate paste. The striking surface is made up of red phosphorus and sand.
- Iodex ointment contains iodoform which liberates I_2 slowly.
- Tincture of iodine is a mixture of I_2 and KI dissolved in rectified spirit.
- Halogens react with NH_3 and give different products

  $$\underset{\text{(excess)}}{8NH_3 + 3Cl_2} \rightarrow N_2 + 6NH_4Cl$$

  $$NH_3 + \underset{\text{(excess)}}{3Cl_2} \rightarrow NCl_3 + 3HCl$$

  Br_2 also reacts with NH_3 in the same way as Cl_2. But the reaction of I_2 is different.

  $$2NH_3 + 3I_2 \rightarrow \underset{\text{(explosive)}}{NI_3 \cdot NH_3} + 3HI$$

  $$8NI_3 \cdot NH_3 \rightarrow 5N_2 + 9I_2 + 6NH_4I$$

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- Chlorine and bromine turn starch iodide paper blue. The blue colour is attributed to the formation of starch-iodine complex. Iodine is produced from iodide by its oxidation with bromine or chlorine.

$$2I^- + Cl_2 \rightarrow I_2 + 2Cl^-$$
$$2I^- + Br_2 \rightarrow I_2 + 2Br^-$$
$$I_2 + \text{Starch} \rightarrow \underset{\text{(blue)}}{\text{Starch} - \text{iodine complex}}$$

- Iodine turns starch paper blue. The blue colour is due to the formation of starch-iodine complex. Similarly, bromine turns starch paper yellow.
- **Preparation of pure chlorine.** Pure chlorine may be obtained by heating dry platinic chloride ($PtCl_4$) or gold chloride ($AuCl_3$) in a hard glass tube.

$$PtCl_4 \xrightarrow{374°C} PtCl_2 + Cl_2 \xrightarrow{582°C} Pt + 2Cl_2$$
$$2AuCl_3 \xrightarrow{175°C} 2AuCl + 2Cl_2 \xrightarrow{185°C} 2Au + 3Cl_2$$

- **Chromyl chloride test.** When solid chloride is heated with concentrated H_2SO_4 in presence of solid $K_2Cr_2O_7$ in a dry test tube, deep red vapours of chromyl chloride are evolved.

$$NaCl + H_2SO_4 \longrightarrow NaHSO_4 + HCl$$
$$K_2Cr_2O_7 + 2H_2SO_4 \longrightarrow 2KHSO_4 + 2CrO_3 + H_2O$$
$$CrO_3 + 2HCl \longrightarrow \underset{\text{Chromyl chloride}}{CrO_2Cl_2} + H_2O$$

When these vapours are passed through NaOH solution, the solution becomes yellow due to the formation of sodium chromate.

$$CrO_2Cl_2 + 4NaOH \longrightarrow \underset{\text{Yellow colour}}{Na_2CrO_4} + 2NaCl + 2H_2O$$

The yellow solution is neutralised with acetic acid and on addition of lead acetate gives a yellow precipitate of lead chromate.

$$Na_2CrO_4 + Pb(CH_3COO)_2 \longrightarrow \underset{\text{Yellow ppt.}}{PbCrO_4} + 2CH_3COONa$$

- **Reaction with alkalies.** With cold and dilute NaOH, F_2 gives OF_2 while with hot and concentrated NaOH, it gives O_2.

$$2F_2 + 2NaOH \xrightarrow{\text{cold}} 2NaF + OF_2 + H_2O;$$
$$2F_2 + 4NaOH \xrightarrow{\text{hot}} 4NaF + 2H_2O + O_2$$

Other halogens form *hypohalites* (XO^-) with cold dilute NaOH solution and *halates* (XO_3^-) with hot and concentrated NaOH solution.

$$2NaOH \ \text{(dil)} + X_2 \xrightarrow{\text{cold}} NaXO + NaX + H_2O \text{ where } X = \text{Cl, Br or I}$$
$$6NaOH \ \text{(conc.)} + 3X_2 \xrightarrow{\text{hot}} NaXO_3 + 5NaX + 3H_2O \text{ where } X = \text{Cl, Br or I.}$$

- I_2 reacts with sodium thiosulphate to give sodium tetrathionate

$$2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$$

- Although I_2 does not displace chlorine or bromine from the solution of their salts, yet it displaces them from their oxosalts.

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$$2KClO_3 + I_2 \longrightarrow 2KIO_3 + Cl_2 \ ;$$
$$2KBrO_3 + I_2 \longrightarrow 2KIO_3 + Br_2$$

- *Deep sea weeds of Laminaria variety* are main source of iodine. Their ashes known as *kelp* contain 0.5% of iodine in the form of iodides. *Caliche* or *crude chile salt petre* ($NaNO_3$) is another source of iodine which contains 0.2% of sodium iodate ($NaIO_3$).
- Chlorine is prepared by *Deacon's process* where a mixture of hydrogen chloride and oxygen gases is heated at 673 K in presence of $CuCl_2$ as a catalyst

$$4HCl + O_2 \xrightarrow[673\,K]{CuCl_2} 2Cl_2 + 2H_2O$$

- Fluorine is prepared by *electrolysis* of a mixture of KHF_2 and anhydrous HF using *Monel metal* (alloy of Cu, Ni and Fe) as a catalyst. Two methods used are: *Dennis* and *Whytlaw-Gray.*
- *Reaction of halogens with water.* F_2 reacts with water to form *ozonised oxygen.*

$$2F_2 + 2H_2O \longrightarrow 4HF + O_2$$
$$3F_2 + 3H_2O \longrightarrow 6HF + O_3$$

Chlorine and bromine decompose water in the presence of sunlight to give *hypohalous acids.*

$$Cl_2 + H_2O \longrightarrow HCl + HClO \quad \text{(hypochlorous acid)}$$
$$Br_2 + H_2O \longrightarrow HBr + HBrO \quad \text{(hypobromous acid)}$$

I_2 has a negligible reaction with water

- Cl_2 acts as a *bleaching agent* in the *presence of moisture.* Its bleaching action is permanent and is due to *oxidation*

$$Cl_2 + H_2O \longrightarrow 2HCl + [O]$$

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Anomalous behaviour of fluorine

Fluorine differs from the rest of the members of the halogen family due to following reasons: Small size, Highest electronegativity, Absence of *d*–orbitals in the valence shell, low bond dissociation energy.

The main points of difference are:

(i) Fluorine is most reactive of all the halogens due to low bond dissociation energy of F – F bond.

(ii) Fluorine is the most electronegative so it shows only –1 oxidation state and due to the absence of *d*–orbitals it can not exist in positive oxidation states as other halogens do.

(iii) Due to small atomic size and high electronegativity of F, HF forms strong hydrogen bonds while other halogen acids do not.

(iv) Fluorine does not have vacant *d*–orbitals in valence shell therefore it cannot combine with F^- ions to form polyfluoride ions like Cl_3^-, Br_3^-, I_3^-, I_5^- etc.

(v) Of all the halogens, fluorine has the highest positive electrode potential (F_2 = 2.87, Cl_2 = 1.36, Br_2 = 1.09 and I_2 = 0.53 volt) i.e., it is most easily reduced and hence acts as strongest oxidising agent.

or $Cl_2 + H_2O \longrightarrow HCl + \underset{\text{unstable}}{HClO}$

$HClO \longrightarrow HCl + [O]$

Coloured matter + $[O] \longrightarrow$ colourless

Thus, *chlorine water* acts as an *ink remover.*

- Hypohalites *disproportionate* in aqueous solution on heating to produce *halates*. This reaction is facilitated in the basic medium.

$$3OX^- \overset{300\ K}{\rightleftharpoons} 3X^- + XO_3^-$$

The rate of disproportionation increases in the order : $ClO^- < BrO^- < IO^-$.

- Iodine is purified by **sublimation.**
- Chlorine gas is collected by upward displacement or air as it is heavier than air.

[F, Cl, Br, I, At ; $ns^2\ np^5$]

INCREASING TRENDS	DECREASING TRENDS	EXCEPTIONS
• Atomic size • Ionic radii X^- • Melting/Boiling points • Intensity of colour • Electropositive character • Acidic nature of hydrides (H*X*) • Reducing nature of hydrides (H*X*)	• Ionisation energies • Electronegativity • Electron affinity • Chemical reactivity • E° values • Oxidising power • Thermal stability of H*X* • ΔH_{diss} of H—*X* • Acid strength of HO*X*	• EA_1 of Cl > EA_1 of F • F shows oxidation state of – 1 ; others show oxidation state – 1, + 1, + 3, + 5, + 7.

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NOBLE GASES

- The noble gases are the elements in group 18 (group 0) of the periodic table. It is also called helium family or neon family. They are the most stable due to having the maximum number of valence electrons in their outer shell can hold. Therefore, they rarely react with other elements since they are already stable.
- Noble gases have full valence electron shells. Valence electrons are the outermost electrons of an atom and are normally the only electrons which can participate in chemical bonding.
- The noble gases lack of reactivity can be explained in terms of them having a "complete valence shell". They have little tendency to gain or lose electrons. The noble gases have high ionisation energies and negligible electronegativities. The noble gases have very weak inter-atomic forces of attraction, and consequently very low melting points and boiling points. This is why they are all monoatomic gases under normal conditions, even those with larger atomic masses than many normally solid elements.

- In xenate ion XeO_4^{2-}, Xe is in +6 oxidation state and in prexenate ion XeO_6^{4-}, Xe is in +8 oxidation state.
- **Xenon fluoride complexes.** XeF_2 acts as a fluoride donor and forms complexes with covalent pentafluorides such as PF_5, AsF_5 etc. and transition metal fluorides such as NbF_5, TaF_5 etc.

  $XeF_2 + MF_5 \rightarrow XeF_2 \cdot MF_5$ or $[XeF]^+ [MF_6]^-$

  $XeF_2 + 2MF_5 \rightarrow XeF_2 \cdot MF_5$ or $[XeF]^+ [M_2F_{11}]^-$

  and $2XeF_2 + MF_5 \rightarrow 2XeF_2 \cdot MF_5$ or $[Xe_2F_3]^+ [MF_6]^-$

  XeF_6 can act as a fluoride donor forming complexes such as $XeF_6 \cdot BF_3$ or $[XeF_5]^+ [BF_4]^-$ and also act as a fluoride acceptor such as $XeF_6 + RbF \rightarrow Rb^+ [XeF_7]^-$
- Only He forms interstitial compounds with metals.
- Ar, Kr and Xe form clathrate compounds but He and Ne do not.
- In case of noble gases, the atomic radii, melting and boiling points, ease of liquefaction, heats of vaporization, solubility in water, ease of adsorption on activated charcoal and polari-zability increase in the order : He < Ne < Ar < Kr < Xe

  whereas the enthalpy, ease of diffusion and thermal conductivity decrease in the order : He > Ne > Ar > Kr > Xe
- Clathrates are held by van der Waals' forces. They are not true compounds and have the components as **Clathrate = Organic molecule** or **inorganic molecule (Host) + Inert gas (Guest).**
- Under high energy conditions, several molecular ions, such as He_2^+, HeH^+, HeH^{2+} and Ar_2^+ are formed in discharge tubes. They only survive momentarily and are detected spectroscopically.
- Helium is unique. On cooling, it gives two different liquid phases. Helium I is a normal liquid but helium II is a super fluid. When the temperature of helium gas is lowered to 4.2 K, it liquifies as helium I but continues to boil vigorously. At 2.2 K, the liquid suddenly stops boiling and helium II is formed which has many physical properties different from that of helium I.
- Helium II defines gravity and is able to flow uphill. Thus it behaves like a liquid with gas like properties. Such state is sometimes referred to as the fourth state of matter.
- Noble gases neither act as reducing agents nor as oxidising agents.
- Lord Rayleigh and Sir William Ramsay were awarded noble prizes in 1904 for their discovery of noble gases.
- XeF_2 oxidizes iodine in the presence of fluoride ion acceptor to give IF.

  $$I_2 + XeF_2 \xrightarrow{BF_3} 2IF + Xe$$

  XeF_2 reacts with NO to give nitrosyl fluoride.

  $$2NO + XeF_2 \longrightarrow 2NOF + Xe$$
- Xenon in its compounds *exhibits even valency* from two to eight, +2 in XeF_2, +4 in XeF_4, +6 in XeF_6 and +8 in XeO_4. The compounds of xenon involve only fluorine and oxygen. The only other compound xenon dichloride ($XeCl_2$) is stable at low temperatures.

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- Complexes of KrF_2 are analogous to those of XeF_2 and are confined to cationic species formed with F^- acceptors. Thus, such compounds as $[KrF]^+ [MF_6]^-$, $[Kr_2F_3]^+ [MF_6]^-$, where *M* is As or Sb, are known and more recently $[KrF]^+ [MoOF_5]^-$ and $[KrF]^+ [WOF_5]^-$ have been prepared and characterized.
- Many of the reactions of xenon oxides, fluorides and oxofluorides can be systematized in terms of generalized acid-base theory in which any *acid* (here defined as *an oxide acceptor*) can react with any *base* (*oxide donor*) lying below it in the sequence of descending acidity :

  $XeF_6 > XeO_2F_4 > XeO_3F_2 > XeO_4 >$
  $XeOF_4 > XeF_4 > XeO_2F_2 > XeO_3 \neq XeF_2$.
- XeF_6 cannot be stored in glass vessels because of the following reactions which finally give the dangerously explosive XeO_3

$$2XeF_6 + SiO_2 \longrightarrow 2XeOF_4 + SiF_4$$
$$2XeOF_4 + SiO_2 \longrightarrow 2XeO_2F_2 + SiF_4$$
$$2XeO_2F_2 + \underset{\text{(from glass)}}{SiO_2} \longrightarrow \underset{\text{(explosive)}}{2XeO_3} + SiF_4$$

- Neon lamps are used in botanical gardens and the green houses as it stimulates growth and is effective in the formation of chlorophyll.
- The names helium was derived from Greek word *helios* meaning *sun*, neon from Greek word meaning *new*, argon from *argos* meaning *inert or lazy*, krypton from Greek work meaning *hidden one* and xenon from Greek work meaning *stranger one.*

End

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18

CHAPTER

transition elements

- Elements which have partially filled *d*-sub shell in their elementary form or in their commonly occurring oxidation states are known as transition elements. (These are *d*-Block elements). Sc is the lightest among them and Os is the heaviest. The general electronic configuration for the atoms of *d*-block is : $(n-1)d^{1-10}\ ns^{1-2}$
  where $(n-1)$ stands for inner shell
- The transition elements consist of three complete rows of ten elements and one incomplete row. These rows are called first, second and third **transition series** which involve the filling *of* 3*d*-, 4*d*-, 5*d*-orbitals respectively.
  (*i*) **First transition series** (3*d*-series) includes metals from Sc (Z = 21) to zinc (Z = 30).
  (*ii*) **Second transition series** (4*d*-series) includes elements fromYttrium (Z = 39) to Cadmium (Z = 48).
  (*iii*) **Third transition series** (5*d*-series) includes elements form Lanthanum (Z = 57), Hafnium (Z = 72) to mercury (Z = 80)
  (*iv*) **Fourth transition series** starts from actinium (Z = 89) and is still incomplete.
- **Enthalpies of atomisation** is the heat required to convert 1 mole of a crystal lattice into free atoms. Transition elements have high enthalpies of atomisation.
- **Hydration energy or solvation energy** is the amount of energy released when metal ions get hydrated (or solvated) with water (or solvent) molecules.
- **Enthalpy of sublimation** is the enthalpy change that accompanies the change of 1 mole of solid substance to the vapour state.
- **Electrode potential** is the potential developed on a metal electrode when it is in equilibrium with solution of its ions, leaving electrons from the electrode.
- **Oxidation state** is a measure of the electronic state of an atom in a particular compound. It is equal to the number of electrons it has, more than or less than the number of electrons in free atom. Transition metal ions show variable oxidation state. This is due to the participation of inner $(n-1)$ *d*-electrons in addition to outer *ns*-electrons because, the energies of the *ns* and $(n-1)d$ subshells are almost equal.

Properties with respect to the oxidation states :

- Higher oxidation state ions become less stable across the period.
- Ions in higher oxidation states tend to make good oxidising agents, whereas elements in low oxidation states become reducing agents.

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1. First (3*d*) Transition series (Sc - Zn)

At. No.	21	22	23	24	25	26	27	28	29	30
Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
E.C.	$3d^1\ 4s^2$	$3d^2\ 4s^2$	$3d^3\ 4s^2$	$3d^5\ 4s^1$	$3d^5\ 4s^2$	$3d^6\ 4s^2$	$3d^7\ 4s^2$	$3d^8\ 4s^2$	$3d^{10}\ 4s^1$	$3d^{10}\ 4s^2$

2. Second (4*d*) Transition series (Y - Cd)

At. No.	39	40	41	42	43	44	45	46	47	48
Element	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
E.C.	$4d^1\ 5s^2$	$4d^2\ 5s^2$	$4d^4\ 5s^1$	$4d^5\ 5s^1$	$4d^6\ 5s^1$	$4d^7\ 5s^1$	$4d^8\ 5s^1$	$4d^{10}\ 5s^0$	$4d^{10}\ 5s^1$	$4d^{10}\ 5s^2$

3. Third (5*d*) Transition series (La- Hg)

At. No.	57	72	73	74	75	76	77	78	79	80
Element	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
E.C.	$5d^1\ 6s^2$	$5d^2\ 6s^2$	$5d^3\ 6s^2$	$5d^4\ 6s^2$	$5d^5\ 6s^2$	$5d^6\ 6s^2$	$5d^7\ 6s^2$	$5d^{10}\ 6s^0$	$5d^{10}\ 6s^1$	$5d^{10}\ 6s^2$

4. Fourth (6*d*) Transition series (Ac - Uub)

At. No.	89	104	105	106	107	108	109	110	111	112
Element	Ac	Ku	Ha	Sg (Unh)	Bh (Uns)	Hs (Uno)	Mt (Une)	Uun	Uuu	Uub
E.C.	$6d^1\ 7s^2$	$6d^2\ 7s^2$	$6d^3\ 7s^2$	$6d^4\ 7s^2$	$6d^5\ 7s^2$	$6d^6\ 7s^2$	$6d^7\ 7s^2$	$6d^8\ 7s^2$	$6d^{10}\ 7s^1$	$6d^{10}\ 7s^2$

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- The 2+ ions across the period start as strong reducing agents and become more stable.
- The 3+ ions start stable and become more oxidising across the period.

- The *s*- and *p*-block elements do not have a partially filled *d* shell so there cannot be any *d*-*d* transitions. The energy to promote an *s* or *p* electron to a higher energy level is much greater and corresponds to U.V. light being absorbed. Thus the compound will not be coloured.

- Charge transfer always produces intense colours since the restrictions of selection rules do not apply to transitions between atoms. MnO_4^- ion has an intense purple colour in solution due to charge transfer. In MnO_4^-, an electron is momentarily transferred from O to the metal, thus momentarily changing O^{2-} to O^- and reducing the oxidation state of the metal from Mn (VII) to Mn (VI). Charge transfer requires that the energy levels on the two different atoms involved are fairly close.
- Transition metals are coloured due to *d-d* transition and charge transfer. Colour due to the *d-d* transition is shown by transition metal compounds containing d^1, d^2, d^3, d^4, d^5, d^6, d^7, d^8, d^9 systems. The compounds containing d^0 and d^{10} configurations are coloured due to charge transfer as there is no possibility of *d-d* transitions.

Oxidation state of the hydrated ions	Colour
Sc (III), Ti (IV)	Colourless
Ti (III)	Purple
V (III)	Green
V (II), Cr (III)	Violet
Mn (III)	Violet
Fe (III)	Yellow
Mn (II)	Pink
Fe (II)	Green
Co (II)	Pink
Ni (II)	Green
Cu (II)	Blue
Cu (I), Zn (II)	Colourless

- The property due to which certain substances are repelled by an applied magnetic field is known is diamagnetism. Such substances are called *diamagnetic substances*. They do not have any unpaired electrons *e.g.* Zn, Cd, and Hg.
- **Paramagnetism** is the property of a substance by virtue of which it is attracted into a magnetic field.

  Paramagnetism is due to the presence of unpaired electrons in an atom, ion or molecule *e.g.* Co, Ni, Cr and Mn.
- **Ferromagnetism** is a special type of paramagnetism in which permanent magnetic moment is acquired by substances. Such substances have a large number of unpaired electrons *e.g.* Fe.
- The magnetic moment μ_{eff} of a transition metal can give important information about the number of unpaired electrons present in the atom and the orbitals that are occupied and sometimes indicates the structure of the molecule or complex. If the magnetic moment is due entirely to the spin of unpaired electrons, $\mu_{eff} = \sqrt{4S(S+1)}$ B.M. where S is the total spin quantum number. This equation is related to the number of unpaired electrons n by the equation $\mu_{eff} = \sqrt{n(n+2)}$ B.M.

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Ion	Electronic configuration	Number of unpaired electrons	Observed magnetic moment (Bohr Magneton)
Sc^{3+}	d^0	0	0
Ti^{2+}	d^2	2	2.76
V^{2+}	d^3	3	3.86
Cr^{2+}	d^4	4	4.80
Mn^{2+}	d^5	5	5.96
Fe^{2+}	d^6	4	5.00 - 5.5
Co^{2+}	d^7	3	4.4 - 5.2
Ni^{2+}	d^8	2	2.9 - 3.4
Cu^{2+}	d^9	1	1.8 - 2.2
Zn^{2+}	d^{10}	0	0

- **Complexes** are chemical species in which a central metal atom or ion is surrounded by a number of other molecules or ions which are called ligands. The high tendency of transition metals to form comlexes is due to availability of vacant *d*-orbitals, high nuclear charge and small size of the atoms and ions of transition metals.
- The number of ligands attached to the central metal atom or ion in a complex is called coordination number of the central metal atom or ion in a particular combination.

The Typical Characteristics of Transition Metals

(a) ***Some General Physical Characteristics***

- Generally speaking they are hard, tough and strong (compared with the Group 1 Alkali metals).
- Good conductors of heat and electricity (these have many free electrons per atom to carry thermal or electrical energy).
- They are easily hammered and bent into shape.
- They are typically lustrous/shiny solids (or liquids).

(b) ***High Melting Point and Boiling Point***

- The bonding between the atoms in transition metals is very strong. The strong attractive force between the atoms is only weakened at high temperatures, hence the high melting points and boiling points (again compare with Group 1 alkali metals). Mercury is an another transition metal, but unusually, it has a very low melting point of –39°C.
- For example: iron melts at 1535°C and boils at 2750°C but a Group 1 alkali metal such as sodium melts at 98°C and boils at 883°C.

(c) ***High density***

- Another consequence of the strong bonding between the atoms in transition metals is that they are tightly held together to give a high density.
- For example: iron has a density of 7.9 g/cm^3 and sodium has a density of 0.97 g/cm^3.

(d) (i) ***Form coloured compounds and ions in solution***

They tend to be much less reactive than the alkali metals. They do not react as quickly with water or oxygen so do not corrode as quickly. Transition metals tend to form more coloured compounds more than other elements either in solid form or dissolved in a solvent. The colours of some transition metal salts in aqueous solution are shown next:

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1. Sc - scandium salts, such as the chloride, $ScCl_3$, are colourless and are not typical of transition metals
2. Ti - titanium(III) chloride, $TiCl_3$, is purple
3. V - vanadium(III) chloride, VCl_3, is green
4. Cr - chromium(III) sulphate, $Cr_2(SO_4)_3$, is dark green (chromate(VI) salts are yellow, dichromate(VI) salts are orange)
5. Mn - potassium manganate(VII), $KMnO_4$ is purple (manganese(II) salts *e.g.* $MnCl_2$ are pale pink)
6. Fe - iron(III) chloride, $FeCl_3$, is yellow-orange-brown. Iron(II) compounds are usually light green and iron(III) compounds orange/brown.
7. Co - cobalt sulphate, $CoSO_4$, is pinkish
8. Ni - nickel chloride, $NiCl_2$, is green
9. Cu - copper(II) sulphate, $CuSO_4$, is deep blue. Most common copper compounds are blue and sometimes green.
10. Zn - zinc salts such as zinc sulphate, $ZnSO_4$, are usually colourless and are not typical of transition metals.

(ii) ***Some other odd bits of chemistry***

- Many of the transition metal carbonates are unstable on heating and readily undergo thermal decomposition.
- Many transition metal ions give hydroxide precipitates when mixed with aqueous sodium hydroxide solution.
  - transition metal salt solution + sodium hydroxide ⇒ solid hydroxide precipitate + sodium salt ionically the precipitation reaction is : metal ion hydroxide ion ⇒ hydroxide precipitate
  - $M^{2+}_{(aq)} + 2OH^-_{(aq)} \Rightarrow M(OH)_{2(s)}$
  - *M* can be iron(II), giving dark green iron(II) hydroxide; copper(II), giving blue copper(II) hydroxide;
  - for iron(III): $Fe^{3+}_{(aq)} + 3OH^-_{(aq)} \Rightarrow Fe(OH)_{3(s)}$ giving brown iron(III) hydroxide
  - Also note that iron has two valencies or combining power giving different compound formulae. Multiple valency, hence multiple compound formation, is another characteristic (but not unique) feature of transition metal chemistry.

(e) ***Catalytic Properties***

(1) The metallic elements themselves act as catalysts

- Many transition metals are used directly as catalysts in industrial chemical processes and in the anti-pollution catalytic converters in car exhausts.
  For example iron is used in the Haber Synthesis of ammonia:
  Nitrogen + Hydrogen ⇒ Ammonia (via a catalyst of Fe atoms) or $N_{2(g)} + 3H_{2(g)} \Rightarrow 2NH_{3(g)}$
- Nickel is the catalyst for 'hydrogenation' in the margarine industry. It catalyses the addition of hydrogen to an alkene carbon – carbon double bond (> C ═ C < + H_2 ⇒ > CH – CH <). This process converts unsaturated vegetable oils into higher melting saturated fats.

(2) The compounds of transition metals as catalysts

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- As well as the metals, the compounds of transition metals also act as catalysts. For example manganese dioxide (or manganese(IV) oxide), MnO_2, a black powder, readily decomposes an aqueous solution of hydrogen peroxide:
  Hydrogen peroxide $\Rightarrow$ water + oxygen
  or $2H_2O_{2(aq)} \Rightarrow 2H_2O_{(l)} + O_{2(g)}$
- **Interstitial compounds** are formed by the incorporation of small non-metallic elements (H, B, C, N, etc) in vacant spaces between the metal atoms *e.g.* $Fe_{0.94}O$, $Fe_{0\cdot84}O$, $VSe_{0.98}$ etc.
- **An alloy** is a homogeneous mixture of two or more metals or a metal and a non-metal.
- Alloys of metals with mercury are known as amalgams. These may be solid or liquid.
- **Spinel** are the mixed oxides and in them the oxygen atoms constitute a face-centred cubic lattice *e.g.* $ZnFe_2O_4$ is a normal spinel. In it (normal spinel) the trivalent ions occupy the octahedral holes and divalent ions occupy the tetrahedral holes. In an inverse spinel the trivalent ions occupy the tetrahedral holes *e.g.* $Fe(Fe_2)O_4$.
- **Misch-metals** is an alloy of cerium (approx. 25%) and various other lanthanide metals. It also contains iron upto 5% and traces of sulphur, carbon, silicon, calcium, aluminium. It is a pyrophoric material and is used in lighter flints.
- A mixture of TiO_2 and $BaSO_4$ is called *titanox* while a mixture of ZnS and $BaSO_4$ is called *lithopone.*

- There are many compounds of transition elements which are used as reagents in laboratory/industry. For example,
  Baeyer's reagent — Dilute alkaline solution of $KMnO_4$
  Tollen's reagent — $AgNO_3$ solution + NaOH solution + NH_4OH
  Schweitzer's reagent — $[Cu(NH_3)_4]SO_4$
  Nessler's reagent — Alkaline solution of $K_2[HgI_4]$
  Benedict's reagent — $CuSO_4$ solution + sodium citrate + Na_2CO_3
  Fehling's reagent — $CuSO_4$solution
  Fenton's reagent — $FeSO_4 + H_2O$
  Etard's reagent — CrO_2Cl_2
  Bordeaux mixture — $CuSO_4$ solution + lime
  Lucas reagent — concentrated HCl + anhydrous $ZnCl_2$
  Barfoed's reagent — $Cu(CH_3COO)_2 + CH_3COOH$
  Milon's reagent — Solution of mercuric and mercurous nitrate
- There are many compounds of transition metals which are used as catalyst. For example,
  Adam's catalyst — Pt/PtO
  Brown's catalyst — Nickel boride
  Zeigler-Natta's catalyst — $TiCl_4 + (C_2H_5)_3Al$
  Wilkinson's catalyst — $[Ph_3P]_3RhCl$

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- Finely reduced form of Pt in the form of velvety black powder is called platinum black.
- $TiCl_4$ and TiO_2 are used in smoke screens; tantalum is used in surgical venals and analytical weights; chromium is used in stainless steel and chrome plating ; molybdenum is used in X-rays tubes ; platinum is used in resistance thermometers; zinc is used in galvanising iron sheets while cadmium is used for making joints in jewellery. TiO_2 is also used as white pigment in paints. Cerium is used as a scavenger of oxygen and sulphur in many metals.
- **Nessler's Reagent** is the alkaline solution of the complex formed by dissolving mercuric iodide in aqueous solution of potassium iodide.

$$HgI_2 + 2KI \rightarrow \underset{\substack{\text{(complex)}\\ \text{Potassium tetra–}\\ \text{iodomercurate}}}{K_2[HgI_4]}$$

- **Iodide of Million's base,** (NH_2—Hg—O—Hg—I)

$$HgI_2 + 2NH_3 \rightarrow I—Hg—NH_2 + NH_4I$$

$$NH_2—Hg—I + H—O—H + NH_2—Hg—I \rightarrow NH_2—\underset{\substack{\text{Iodide of Millon's base}\\ \text{(Brown ppt.)}}}{Hg—O—Hg—I} + NH_4I$$

- **Chromyl chloride test.** When potassium dichromate is heated with conc. HCl or with a chloride and conc. H_2SO_4, reddish brown vapours of chromyl chloride (CrO_2Cl_2) are obtained. This test is used to confirm chloride ion.
- The process of depositing a uniform and thin layer of silver on a clean glass surface is called silvering of mirror.
- **Photography** is the process of producing an exact impression of an object on an art paper by using light. It is based on the chemical behaviour of silver halides which undergo decomposition in light and turn black due to liberation of free silver.
- **Hypo** is $Na_2S_2O_3$ (Sodium thiosulphate).
- The elements in which the last electron enters the *f*-orbital of the atom are called *f*-block elements. These are also called inner transition elements as the last electron is added to third to the outermost (called antipenultimate) energy shell *i.e.* $(n - 2)$ *f*. They consist of two series of elements (*i.e.* lanthanides and actinides) placed at the bottom of the periodic table.
- The fourteen elements immediately following lanthanum ($Z = 57$) constitute the first **inner transition series.** These are known as **Lanthanides.** These elements are from cerium ($Z = 58$) to Lutetium ($Z = 71$) in which 4*f*-orbitals are being filled up.
- On moving from Lanthanum (La) to Lutetium (Lu) a gradual decrease in size of Lanthanides is observed with increase in atomic number. This is known as Lanthanide contraction.
- **Actinides** are the elements of second **inner-transition series** and consist of fourteen elements immediately following actinium ($Z = 89$). They include elements from $Z = 90$ (thorium) to Lawrencium ($Z = 103$). In these elements 5*f*-orbitals are being successively filled up.

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- The gradual decrease in the size of the atom or ion, with increase in atomic number, as we move from thorium (Z = 90) to Lawrencium (Z = 103) is called actinide contraction.
- All transition elements are *d*-block elements but all *d*-block elements are not transition elements. To justify this statement we take the example of Zn, Cd and Hg which are the last members of each *d*-block series. These elements are not called transition elements because they have $(n-1)d^{10}ns^2$ type of completely filled electronic configuration and do not show the characteristic properties of transition elements except complex formation.

Additional Information

- **Equivalent mass of $K_2Cr_2O_7$**
  In acidic medium

$$K_2Cr_2O_7 + 4H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + \underset{3\times16=48\text{ parts}}{3O}$$

  or $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

  $\therefore$ Equivalent mass of $K_2Cr_2O_7$

$$= \frac{M}{6} = \frac{294}{6} = 49$$

  (M of $K_2Cr_2O_7$ = 2 × 39 + 2 × 52 + 7 × 16 = 78 + 104 + 112 = 294)

- **Equivalent mass of $KMnO_4$.** Equivalent mass of an oxidising agent is the number of parts by mass of it which give 8 parts by mass of oxygen or it is the molecular mass divided by the number of electrons gained by one molecule of the substance in a redox reaction. If M is the molecular mass of $KMnO_4$ (M = 39 + 55 + 64 = 158), then we have

  (*a*) In acidic medium

$$\underset{2M}{2KMnO_4} + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + \underset{5\times16=80\text{ parts}}{5O}$$

  or $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

  $\therefore$ equivalent mass of $KMnO_4 = \frac{M}{5} = \frac{158}{5} = 31.6$

  (*b*) In neutral and alkaline medium

$$\underset{2M}{2KMnO_4} + 3H_2O \rightarrow 2KOH + 2MnO_2 + \underset{3\times16=48\text{ parts}}{3O}$$

  or $MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^-$

  $\therefore$ equivalent mass of $KMnO_4 = \frac{M}{3} = \frac{158}{3} = 52.67$

End

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CHAPTER

complex compounds

What is a Complex Compound?

"The chemistry of metal ions in solution is essentially the chemistry of their complexes".

A ***coordination complex*** is the product of a Lewis acid-base reaction in which neutral molecules or anions (called ***ligands***) bond to a central metal atom (or ion) by ***coordinate covalent bonds***.

- Ligands are Lewis bases - they contain at least one pair of electrons to donate to a metal atom/ion. Ligands are also called ***complexing agents***.
- Metal atoms/ions are Lewis acids - they can accept pairs of electrons from Lewis bases.
- Within a ligand, the atom that is directly bonded to the metal atom/ion is called the ***donor atom***.
- A coordinate covalent bond is a covalent bond in which one atom (i.e., the donor atom) supplies both electrons. This type of bonding is different from a normal covalent bond in which each atom supplies one electron.
- If the coordination complex carries a net charge, the complex is called a ***complex ion***.
- Compounds that contain a coordination complex are called ***coordination compounds***.

Coordination compounds and complexes are distinct chemical species - their properties and behaviour are different from the metal atom/ion and ligands from which they are composed.

The ***coordination sphere*** of a coordination compound or complex consists of the central metal atom/ion plus its attached ligands. The coordination sphere is usually enclosed in brackets when written in a formula. The ***coordination number*** is the number of donor atoms bonded to the central metal atom/ion.

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- One of the most important properties of transition metals is that they form co-ordination or complex compounds. These compounds play a vital role in our lives. One of the earliest known co-ordination compound is Prussian blue which was accidently prepared in 1704 by a Berlian colour maker, Diesbach by strongly heating animal wastes and sodium carbonate in an iron container.

Addition or molecular compounds

- When solutions of two or more simple stable salts are mixed together in simple molecular proportion and the solution thus obtained is allowed to evaporate, crystals of a new compound are obtained. This new compound is called addition or molecular compound.

Simple compounds	*Addition compounds*
$KCl + MgCl_2 + 6H_2O \rightarrow$	$KCl \cdot MgCl_2 \cdot 6H_2O$ carnallite
$K_2SO_4 + Al_2(SO_4)_3 + 24H_2O \rightarrow$	$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$ potash alum
$(NH_4)_2SO_4 + FeSO_4 + 6H_2O \rightarrow$	$FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ Mohr's salt

- The molecular or addition compounds are of two types:
  - **(i) Double salts or lattice compounds :** The addition compounds which are stable in solid state only but are broken down into individual constituents when dissolved in water are called double salts or lattice compounds. Their solution have the same properties as the mixture of individual compounds.
  - **(ii) Coordination or complex compounds :** The addition compounds in which some of the constituent ions or molecules lose their identity and when dissolved in water they do not break up completely into individual ions, are called coordination compounds. The properties of their solutions are different than those of their constituents.
- A metal ion to which one or more neutral molecules or ions are attached to give new reasonably identifiable entity is called as central ion; the neutral molecules or ions are called ligands and newly formed entity is called complex ion.
- Ligand is a complexing group in co-ordination chemistry. Generally the entity from which electrons are donated.

(i) Double salts	**(ii) Complex salts**
1. These retain their identity in solid state but lose their identity in solution state.	**1.** These retain their identity in solid as well as in solution state.
2. *e.g.* alum is a double salt $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$ or $K_2[Al_2(SO_4)_4] \cdot 24H_2O$.	**2.** *e.g.* Potassium ferro-cyanide is a complex salt. $K_4Fe(CN)_6$.
3. These are characterized by complete dissociation in solution state and therefore give the reactions of all the ions present in them. *e.g.* alum in solution gives sulphate ions as well. $K_2[Al_2(SO_4)_4] \rightarrow 2K^+ + 2Al^{3+} + 4SO_4^{2-}$	**3.** These salts do not give all the ions of their constituents in solution state. *e.g.* $K_4Fe(CN)_6$ gives test of K^+ and Fe $(CN)_6^{4-}$ ions only (fairly stable). It does not the test of K^+, Al^{3+} and give test of Fe^{2+} or CN^- ions. $K_4Fe(CN)_6 \rightarrow 4K^+ + Fe(CN)_6^{4-}$

- There is a difference between central ion and ligand. *Central ion acts as Lewis acid, i.e. electron pair acceptor but the ligands act as Lewis base, i.e.* electron pair donor. A liquid possesses electron pairs available for donation. A liquid may or may not carry charge.

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Mol. formula	Lewis base/ligand	Lewis acid	Donor atom	Coordination number
$[Ag(NH_3)_2]^+$	NH_3	Ag^+	N	2
$[Zn(CN)_4]^{2-}$	CN^-	Zn^{2+}	C	4
$[Ni(CN)_4]^{2-}$	CN^-	Ni^{2+}	C	4
$[PtCl_6]^{2-}$	Cl^-	Pt^{4+}	Cl	6
$[Ni(NH_3)_6]^{2+}$	NH_3	Ni^{2+}	N	6

- **Classification of ligands :** The ligands are classified on the basis of the number of electron pairs it donates to metal ion as well as on the basis of charge they bear on.

Classification depending upon number of co-ordination sites (number of electron pair donated)

1. **Monodentate :-** Donate one electron pair.
   e.g. $\ddot{N}H_3$, $\ddot{P}H_3$, $H_2\ddot{O}$, $\ddot{C}N^-$, etc.
2. **Bidentate :-** Two N atoms are donor, two O atoms are donor.
   e.g. $H_2\ddot{N}\cdot CH_2\cdot CH_2\cdot \ddot{N}H_2$ (ethylenediamine), $\ddot{C}_2O_4^{2-}$, etc.
3. **Tridentate :-** Donate three electron pairs (three N atoms are donor). *e.g.* Diethylenetriamine $NH_2CH_2CH_2NHCH_2CH_2NH_2$.
4. **Quadradentate (or tetradentate) :-** Donate four electron pairs (four N atoms are donor).
   e.g. Triethylenetetramine
   $NH_2CH_2CH_2NH_2CH_2CH_2NH_2CH_2CH_2NH_2$.
5. **Quinquedentate (or pentadentate) :-** Two N atoms and three O atoms are donor.
   e.g. Ethylenediaminetriacetate anion
   $(^-O_2CCH_2)_2N\cdot CH_2\cdot CH_2\cdot NH(CH_2CO_2^-)$
6. **Hexadentate :-** Donate six electron pairs (two N and four O are donor atoms).
   e.g. Ethylenediaminetetraacetate anion (EDTA)
   $(^-O_2CCH_2)_2N\cdot CH_2\cdot CH_2\cdot N(CH_2CO_2^-)_2$.

- ***Classification depending upon nature of charge on ligands***

  (i) Neutral ligands :- They contain no charge.
  e.g. H_2O, NH_3, PH_3, etc.

  (ii) Monoatomic anionic ligands :- Contain negative charge.
  e.g. F^-, Cl^-, Br^-, I^-, etc.

  (iii) Polyatomic anionic ligands :- Contain negative charge.
  e.g. CN^-, OH^-, NO_2^-, etc.

  (iv) Polyatomic cationic ligands :- Contain positive charge.
  e.g. NO^+, $NH_2 - NH_3^+$.

- **Co-ordination number :** The number of atoms of the ligands that are directly bonded to the central metal atom or ion by co-ordinate bond is known as co-ordination number of the metal atom or ion.
- **Co-ordination sphere :** The central metal atom or ion and the ligand that are directly attached to it are enclosed in a square bracket. This has been called co-ordination sphere

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or first sphere of attraction. It behaves like single unit because the ligands present in the co-ordination sphere are held tightly by the metal ion.

- **Effective atomic number (EAN) :** EAN is the resultant number of electrons with the metal atom or ion after gaining electrons from the donor atom of the ligands. The effective atomic number generally coincide with the atomic number of next inert gas in some cases.

  EAN is calculated by the following relation:

  EAN = atomic number of the metal – number of electrons lost in ion formation + number of electrons gained from the donor atoms of the ligands

- Co-ordination compounds are generally prepared by substitution reaction, redox reactions and by direct combination of reactant molecules. *e.g.*
  (i) An aqueous solution of $CuSO_4$ in presence of an excess of NH_3 give deep blue $[Cu(NH_3)_4]^{2+}$ species.
  (ii) Some metal salts give metal amines complex with liquid ammonia.

  $$CuSO_{4\ (aq)} + 4NH_{3\ (excess)} \rightarrow [Cu(NH_3)_4]SO_4$$
  $$NiCl_2 + 6NH_{3\ (l)} \rightarrow [Ni(NH_3)_6]Cl_2$$

  (iii) $K_4Fe(CN)_6$ is formed by the action of $KCN_{(aq)}$ on ferrous sulphate $_{(aq)}$.

  $$FeSO_4 + 2KCN \rightarrow Fe(CN)_2 + K_2SO_4$$
  $$Fe(CN)_2 + 4KCN \rightarrow K_4Fe(CN)_6.$$

- **Detection of formation of a complex**

  The formation of complex compound can be detected by:
  (a) *Solubility* : For example, AgCl is soluble in NH_4OH because of the formation of complex $[Ag(NH_3)_2Cl]$.
  (b) *Change in colour* : The copper sulphate solution, *e.g.* turns deep blue when excess of ammonia is added. This is also due to the formation of $[Cu(NH_3)_4]SO_4$.
  (c) *Change in electrical conductivity* : Generally conductivity of the solution decreases because of complex formation as the number of ions decreases.
  (d) *Change in pH* : When Ca^{2+} or Mg^{2+} forms complexes with EDTA, the pH of the solution decreases.
  (e) *Change in chemical properties* : $Al(OH)_3$ and $Zn(OH)_2$ are soluble in NaOH because of the formation of complexes, $Na_3[Al(OH)_6]$ and $Na_2[Zn(OH)_4]$ respectively.

- **Nomenclature of coordination compounds**

This nomenclature is based upon IUPAC system.

1. The positive ion is named first followed by the negative ion. The names of cation and anion are separated by a space.
2. When writing the name of a complex, the ligands are quoted in alphabetical order, regardless of their charge (followed by the metal).
3. When writing the formula of complexes, the complex ion should be enclosed by square brackets. The metal is named first, then the coordinated groups are listed in the order: negative ligands, neutral ligands, positive ligands (and alphabetically according to the first symbol within each group).

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(a) The names of negative ligands end in –O, for example:

F^-	fluoro	H^-	hydrido	HS^-	mercapto
Cl^-	chloro	OH^-	hydroxo	S^{2-}	thio
Br^-	bromo	O^{2-}	oxo	CN^-	cyano
I^-	iodo	O_2^{2-}	peroxo	NO_2^-	nitro

(b) Neutral groups have no special endings. Examples include NH_3 ammine, H_2O aqua, CO carbonyl and NO nitrosyl. The ligands N_2 and O_2 are called dinitrogen and dioxygen. Organic ligands are usually given their common names, for example ethylenediamine, phenyl, methyl, pyridine triphenyl phosphine.

(c) Positive groups end in -ium, *e.g.* $NH_2 – NH_3^+$ hydrazinium.

4. Where there are several ligands of the same kind, we normally use the prefixes di, tri, tetra, penta and hexa to show the number of ligands of that type. An exception occurs when the name of the ligand includes a number. e.g. dipyridyl or ethylenediamine. To avoid confusion in such cases, bis, tris and tetrakis are used instead of di, tri and tetra and the name of the ligand is placed in brackets.
5. The oxidation state of the central metal is shown by a Roman numeral in brackets immediately following its name (*i.e.* no space, *e.g.* titanium(III)).
6. Complex positive ions and neutral molecules have no special ending but complex negative ions end in -ate.
7. If the complex contains two or more metal atoms, it is termed polynuclear. The bridging ligands which link the two metal atoms together are indicated by the prefix μ-. If there are two or more bridging groups of the same kind, this is indicated by di-μ-, tri-μ- etc. Bridging groups are listed alphabetically with the other groups unless the symmetry of the molecule allows a simpler name. If a bridging group bridges more than two metal atoms it is shown as μ_3, μ_4, μ_5 or μ_6 to indicate how many atoms it is bonded to.
8. Sometimes a ligand may be attached through different atoms. Thus MNO_2 is called nitro and M – ONO is called nitrito. Similarly the SCN group may bond M – SCN thiocyanato or M – NCS isothiocyanato. These may be named systematically thiocyanato-S or thiocyanato-N to indicate which atom is bonded to the metal. This convention may be extended to other cases where the mode of linkage is ambiguous. These ligands are called ambidentate ligands.
9. If any lattice components such as water or solvent of crystallisation are present, these follow the name, and are proceeded by the number of these groups in Arabic numerals.

Examples:

1. $K_4Fe(CN)_6$ – Potassium hexacyanoferrate(II)
2. $[Co(NH_3)_6]Cl_3$ – Hexaamminecobalt(III) chloride
3. $[Co(NH_3)_3{\cdot}NO_2{\cdot}Cl{\cdot}CN]$ – Chlorocyanonitrotriammine cobalt(III)
4. $K_3[Al(C_2O_4)_3]$ – Potassiumtrioxalato aluminate(III)
5. $[Ni(CN)_4]^{-4}$ – Tetracyanonickelate(0) ion

Werner's co-ordination theory

- According to Werner's co-ordination theory, each metal ion possesses two types of valencies. *e.g.* primary or principal valencies or ionisable valencies and secondary, or subsidiary or non-ionisable valencies.
- Primary valencies are satisfied by anions only. The number of primary valencies depends upon the oxidation state of the central metal. They are denoted by ***dotted lines***.

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- The secondary valencies are satisfied only by the ions or the neutral electron pair donor molecules. These are represented by ***solid lines***.
- Every central ion tends to satisfy both its primary and secondary valencies.
- The ions attached by the secondary valencies do not ionise when the complex is dissolved in a solvent.
- The secondary valencies are also referred to as ***co-ordination number***.
- Specific orientation of ligands in space around a central atom give rise to isomerism in co-ordination compounds.

Isomerism

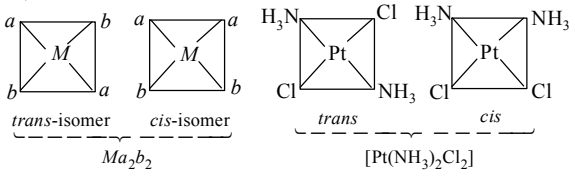
- Structural isomerism arises due to different positions of ligands around a metal atom. There are four types of structural isomerism *e.g.*
  - **(i)** ***Ionisation isomerism***: Complexes having the same empirical formula but giving different ions in solution state. This type of isomerism occurs when there is an interchange of groups between the co-ordination sphere of the metal ion or the ions outside the sphere.
    e.g. $[Co(NH_3)_5\ Br]SO_4$ and $[Co(NH_3)_5SO_4]Br$.
  - **(ii)** ***Hydration isomerism***: This type of isomerism occurs when a co-ordination group is replaced by water of hydration *e.g.* $CrCl_3\ .\ 6H_2O$ exists in the form of three isomers: $[Cr(H_2O)_6]Cl_3$; $[Cr(H_2O)_5Cl]Cl_2{\cdot}H_2O$ or $[Cr(H_2O)_4{\cdot}Cl_2]Cl{\cdot}2H_2O$
  - **(iii)** ***Salt and linkage isomerism*:** This type of isomerism occurs when the ligand possesses two possbilities in the mode of attachment in the metal atom or ion.
  - **(iv)** ***Co-ordination isomerism*:** This type of isomerism occurs when both complex cations and complex anions are present in the complex molecules.
- ***Stereoisomerism*** is the type of isomerism in which two substances of the same composition and even constitution differ only in the relative position in space assumed by certain of their constituent atoms or groups. It is of two types - ***geometrical*** and ***optical***.
- ***Geometrical isomers*** are possible for both square planar and octahedral complexes, but not tetrahedral.
- ***Optical isomers*** are possible for both tetrahedral and octahedral complexes, but not square planar.
- The earliest examples of stereoisomerism involve complexes of Co^{3+}. In 1889, Jorgensen observed purple and green salts of $[Co(en)_2Cl_2]^+$, which Werner later correctly identified as the *cis*- and *trans*- geometric isomers. In 1911, the first resolution of optical isomers was reported by Werner and King for the complexes *cis*-$[CoNH_3(en)_2X]^{2+}$, where X = Cl^- or Br^-.
- **Geometrical isomers :** This isomerism is due to ligands occupying different positions around the central metal atom or ion. The ligands occupying positions either adjacent or opposite to one another. This type of isomerism is also known as *trans* isomers. The number of geometric isomers expected for common stereochemistries are as follows.

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- **Square planar :**

General formula	*Number of isomers*
Ma_2b_2	2 (*cis*- and *trans*-)

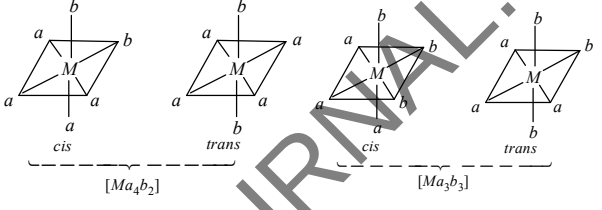
In this example, *a* and *b* are monodentate.



Octahedral :

General formula	*Number of isomers*
Ma_4b_2	2 (*cis*- and *trans*-)
Ma_3b_3	2 (*fac*- and *mer*-)

Here *a* and *b* are monodentate ligands.



- **Optical isomers** are related as non-superimposable mirror images and differ in the direction with which they rotate plane polarized light. These isomers are referred to as enantiomers or enantiomorphs of each other and their non-superimposable structures are described as being asymmetric.
- Various methods have been used to denote the absolute confi-guration of optical isomers such as D or L, R or S and Λ of Δ.
- The two isomers have identical chemical properties and just denoting their absolute configuration does not give any information regarding the direction in which they rotate plane-polarized light. This can only be determined from measurement and then the isomers are further distinguished by using the prefixes leavo (– or *l*) and dextro (+ or *d*) depending on whether they rotate left or right. The use of *l*- and *d*- is not recommended since it may appear to conflict with *L* and *D*.

Bonding in co-ordination compounds

Though Werner's theory was able to explain a number of properties of the co-ordination compounds, it could not answer the following basic questions.

(i) Why only certain elements form co-ordination compounds and not others?
(ii) Why the co-ordination sphere/entity has a definite geometry?
(iii) Why these compounds possess definite magnetic and optical properties?

To answer these questions a number of attempts have been made to extend the existing different theories of bonding to co-ordination compounds. These theories are

(a) Valence bond theory
(b) Crystal field theory
(c) Ligand field or molecular orbital theory

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- **Valence bond theory**

This theory was extended to the coordination compounds by Pauling in 1931. The following are the main postulates of this theory:

(i) In this approach, the basic assumption made is that the metal-ligand bond arises by the donation of pairs of electrons by ligands to the metal atom/ion.

(ii) In order to accommodate these electrons, the metal ions must possess requisite number of vacant orbitals of equal energy. These orbitals of metal atom undergo hybridisation to give a set of hybrid orbitals of equal energy.

(iii) Sometimes, the unpaired $(n-1)d$ electrons pair up as fully as possible prior to bond formation thus making some $(n-1)d$ orbitals vacant. The central metal atom then makes available a number of empty orbitals equal to its coordination number for the formation of coordinate bonds with suitable ligand orbitals.

(iv) With the approach of the ligands, metal-ligand bonds are formed by the overlap of these orbitals with those of the ligands, i.e. by donation of electron pairs by the ligands to the empty hybridised orbitals. Consequently, these bonds are of equal strength and directional in nature.

(v) Octahedral, square planar and tetrahedral complexes are formed as a result of d^2sp^3 (or sp^3d^2), dsp^2 and sp^3 hybridisation respectively.

- **Limitations of valence bond theory**

This theory is failed to explain the spectra of complex (why most of the complexes are coloured?). Valence bond theory is also unable to explain why at one time the electrons are rearranged against Hund's rule while at other times the electronic configuration is not disturbed. This theory was unable to explain why certain complexes are labile while others are inert.

- **Crystal field theory (CFT)**

This theory advanced by Brethe and Van Vleck was originally applied mainly to ionic crystals and is therefore, called crystal field theory.

- If the complex is formed by the use of inner *d*-orbitals for hybridisation (d^2sp^3) it is called inner orbital complex. Such a complex is also called a low spin complex *e.g.* $[Fe(CN)_6]^{3-}$ and $[Co(NH_3)_6]^{3+}$ etc.
- If the complex is formed by the use of outer *d*-orbitals for hybridisation (sp^3d^2), it is called an outer orbital complex. Such a complex is also called a **high spin complex.**
- **Crystal field theory** is based on the assumption that the metal ions and the ligands act as point charges and the interaction between them are purely electrostatic.
- The splitting of five degenerate *d*-orbitals of the metal into different sets of orbitals having different energies in the presence of electrical field of ligands is called **crystal field** splitting.

  Crystal field splitting will be different in different structures with different coordination numbers.
- **Crystal field splitting energy,** (Δ_o for octahedral structure or Δ_t for tetrahedral structure) is the difference between the various sets of energy levels formed by crystal field splitting.
- **Weak field ligands** are those ligands which cause a small degree of crystal field splitting *e.g.* I^-, Br^-, Cl^-, NO_3^-, F^-, OH^-, $C_2O_4^{2-}$ $(ox^{2-})^-$, H_2O etc.

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- **Strong field ligands** are those ligands which cause a high degree of splitting *e.g.* CO, CN^-, NO_2^- etc.
- **Spectrochemical series** is a series in which ligands are arranged in order of increasing magnitude of crystal field splitting.
  $I^- < Br^- < Cl^- < NO_3^- < F^- < OH^- < ox^{2-} < H_2O < p_y =$
  $NH_3 < en <$ dipy $<$ *o*-phen $< NO_2^- < CN^- <$ CO.
- **Limitations of crystal field theory**

(i) CFT considers only the metal ion *d*-orbitals and gives no consideration at all to other metal orbitals such as *s*, p_x, p_y and p_z orbitals and the ligand π-orbitals. Therefore, to explain all the properties of the complexes dependent on the π-ligand orbitals will be outside the scope of CFT. CFT does not consider the formation of π-bonding in complexes.

(ii) CFT is unable to account satisfactorily for the relative strengths of ligands, e.g. it gives no explanation as to why H_2O appears in the spectrochemical series as a stronger ligand then OH^-.

(iii) According to CFT, the bond between the metal and ligand is purely ionic. It gives no account of the partly covalent nature of the metal-ligand bonds. Thus the effects directly dependent on covalency cannot be explained by CFT.

- **Comparison between VBT and CFT**

The points showing the comparison between the two theories are given below.

1. Inner orbital octahedral complexes of VBT are the same as the spin-paired or low-spin octahedral complexes of CFT. Similarly outer-orbital complexes of VBT are the same as the spin-free or high spin octahedral complexes of CFT.
2. In the formation of some inner-orbital octahedral complexes of VBT, the promotion of an electron from *d*-orbital to *s*-orbital is required, while in the formation of spin-paired octahedral complexes of CFT no such promotion is required.
3. According to VBT, the metal-ligand bonding in complexes is only covalent, since VBT assumes that ligand electrons are donated to the vacant *d*-orbitals on the central cation. On the other hand, CFT considers the bonding to be entirely electrostatic. Thus, CFT does not allow the ligand to enter the metal *d*-orbitals.

- **Ligand field or Molecular orbital theory**

The valence bond theory is based on the assumption that the formation of a molecule involves an interaction between the electron waves of only those atomic orbitals of the participating atoms which are half filled. These atomic orbitals mix with one another to form a new orbital of greater stability while all other orbitals on the atoms remain undisturbed or maintain their individual identity. But this cannot be correct because the nucleus of one approaching atom is bound to affect the electron waves of nearly all the orbitals of the other atom. Besides this, the valence bond theory fails to explain the formation of coordinate bond, the paramagnetic character of O_2 molecule and the formation of odd electron molecules or ions such as H_2^+ ion where no pairing of electron occurs.

Molecular orbital theory of chemical bonding is more rational and more useful in comparison to valence bond theory. This theory was put forward by Hund and Mulliken. According to this theory, all the atomic orbitals of the atoms participating in molecule formation get disturbed when the concerned nuclei approach nearer. They all get mixed up to give rise to an equivalent number of new orbitals that belong to the molecule now. These are called molecular orbitals. The electrons belonging originally to the participating atoms are now considered to be moving along the molecular orbitals under the influence of all the nuclei.

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- **Organometallics**

Organometallic compounds are the compounds which contain atleast one carbon-metal bond. Zeisse, in 1830, prepared the first organometallic compound by the action of ethylene on a solution of potassium chloroplatinate(II). Grignard reagent RMgX, is a familiar example of organometallic compound where R is an alkyl group. Thus an organometallic is a compound which contains at least one of the following bond.

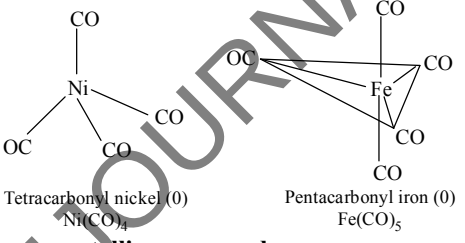
Metal – carbon

Metalloid (B, Si, As, Te) – carbon

- **σ-bonded compounds** are the ones in which organic group is bonded to a metal atom through a normal 2-electron covalent bond *e.g.* R-Mg-X, $(CH_3—CH_2)_2$ Zn etc.
- **π-bonded organometallic compounds** are generally formed by transition elements *e.g.* Zeise's salt, ferrocene, dibenzene chromium etc.
- **σ- and π-bonded organometallic compounds:**

Metal carbonyls, compounds formed between metal and carbon monoxide belong to this class. These compounds possess both σ- and π-bonding. The oxidation state of metal atoms in these compounds is zero. Carbonyls may be monomeric, bridged or polynuclear. Carbonyls are mainly formed by the transition metals of VIth, VIIth and VIIIth groups.

Some well known complexes are :



- **Applications of organometallic compounds**

The applications of organometallic compounds are numerous. Progress in this area introduced new reagents and catalysts for synthesis.

Some important ones are :

(i) Tetraethyl lead (TEL) is used as antiknock compound in gasoline.

(ii) Silicons are used as polymers of unique properties.

(iii) Organoalkali and Grignard reagents are used in many organic synthetic reactions.

(iv) The extraction and purification of nickel is based on the formation of organometallic compound, $Ni(CO)_4$.

(v) Organometallic compounds are used as homogeneous and heterogeneous catalysts. Wilkinson's catalyst, $[Rh(P.Ph_3)_3Cl]$ is used as homogeneous catalyst in the hydrogenation of alkenes. Zeigler Natta catalyst [$TiCl_4$ and triethyl aluminium] acts as a heterogeneous catalyst in the polymerization of ethylene into polyethylene.

- **Homogeneous catalysis** reactions are catalysed by soluble transition metal complexes. The reactant and the catalyst are in the same physical state *e.g.* hydrogenation of alkenes by the use of **Wilkinson's catalyst $(Ph_3P)_3RhCl$.**
- **Heterogeneous catalysis** reactions in which reactants and catalyst are in different physical states, organometallic compounds are used as catalyst in heterogeneous catalysis. *e.g.* In polymerisation of olefins, Zeigler Natta catalyst is used.

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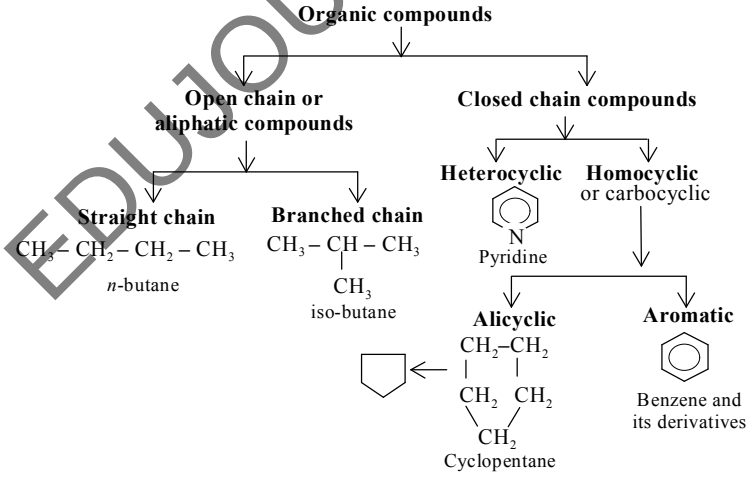
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CHAPTER

basic concepts of organic chemistry

- Berzelius (1808) defined organic chemistry as the chemistry of substances found in living matter and gave the Vital force theory.
- The discovery that shocked the vital force theory was Wohler's synthesis of urea from NH_4CNO.
- The first organic compound synthesised from its elements was acetic acid.
- Compounds of carbon having open chain of carbon atoms, branched or unbranched are called **acyclic compounds** or **aliphatic compounds**.
- Compounds of carbon having a closed chain of carbon as well as of other atoms are called **cyclic compounds**.
- **Carbocyclic compounds** are compounds of carbon having closed chain entirely made of carbon atoms.

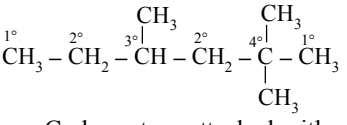


- **Aromatic compounds** are closed chain of only carbon atoms with alternate single and double bonds.
- **Alicyclic compounds** are closed carbon chains except characteristic benzene ring, resembling in properties with acyclic compounds.
- **Heterocyclic compounds** are compounds of carbon having closed chain made up of carbon and other atoms.
- **Trivial names :** Initially organic compounds were named after the source from which they were obtained or from their characteristic properties.

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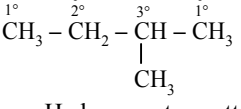
Compound	Name	Source
CH_3OH	wood spirit	obtained by destructive distillation of wood
NH_2CONH_2	urea	obtained from urine
CH_3COOH	acetic acid	obtained from acetum-vinegar
$H_2C_2O_4$	oxalic acid	obtained from oxalis plant
$CH(OH)COOH$ $\vert$ CH_2COOH	malic acid	obtained from malum (apple)

- **Primary, secondary and tertiary carbon atoms:**



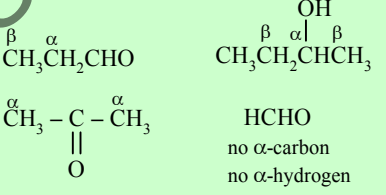
  - Carbon atom attached with one carbon atom : Primary or 1° carbon
  - Carbon atom attached with two carbon atoms : Secondary or 2° carbon
  - Carbon atom attached with three carbon atoms : Tertiary or 3° carbon
  - Carbon atom attached with four carbon atoms : Quarternary or 4° carbon

- **Primary, secondary and tertiary hydrogen atoms:**

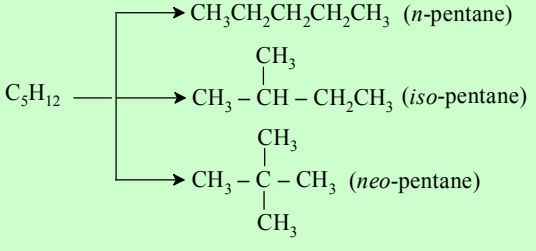


  - Hydrogen atom attached with 1° C-atom : Primary or 1° hydrogen
  - Hydrogen atom attached with 2° C-atom : Secondary or 2° hydrogen
  - Hydrogen atom attached with 3° C-atom : Tertiary or 3° hydrogen

(i) The carbon atom in the structure of an organic compound to which a functional group is attached is known as α-carbon and the corresponding hydrogen atom is referred to as α-hydrogen. The carbon atom adjacent to an α-carbon is known as β-carbon.



(ii)



n *i.e.* alkane is unbranched.
iso *i.e.* alkane contains $(CH_3)_2CH$ – and no other branches.
neo *i.e.* alkane contains $(CH_3)_3C$ – and no other branches.

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IUPAC (International Union of Pure and Applied Chemistry) names

- It is pronounced as eye-you-pack.
- IUPAC nomenclature involves the use of following terms:

(a) **Word root** : It represents the number of carbon atoms in the parent chain.

No. of carbons		Word root	No. of carbons		Word root
One	-	meth	Four	-	but
Two	-	eth	Five	-	pent
Three	-	prop	Six	-	hex

(b) **Primary suffix :** It is used to indicate saturation or unsaturation in the carbon chain. It is added to word root.

Carbon chain	Primary suffix
Saturated carbon chain C – C	ane
Unsaturated carbon chain	
one C = C	ene
two C = C	a diene
one C ≡ C	yne
two C ≡ C	a diyne

(c) **Secondary suffix :** It is used to indicate the functional group in the organic compound. It is added to primary suffix by removing its terminal 'e'.

(d) **Prefix :** The part of the name which appears before word root is called prefix.

(i) *Alkyl groups* - an alkyl group is formed by removing one hydrogen atom from an alkane. The symbol R – is often used to represent an alkyl group.

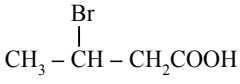
Alkane	Alkyl group
CH_4, Methane	$-CH_3$, methyl group
C_2H_6, Ethane	$-C_2H_5$, ethyl group
C_3H_8, Propane	$-C_3H_7$, propyl group

(ii) Some functional groups are always indicated by the prefixes.

e.g.				
	$-NO_2$	nitro	RNO_2	nitroalkane
	– Cl	chloro	RCl	chloroalkane
	– Br	bromo	RBr	bromoalkane

- **Functional groups :** A functional group is an atom or group of atoms in a molecule that gives the molecule its characterisitic chemical properties.

Prefix + Root word + Primary suffix + Secondary suffix



Prefix = bromo, root word = but

Primary suffix = –ane, secondary suffix = –oic acid

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	Functional group	Structure	Prefix	Suffix
1.	Carboxylic Acid	$-\overset{\overset{O}{\|\|}}{C}-OH$	Carboxy	- oic acid
2.	Acid amide	$-\overset{\overset{O}{\|\|}}{C}-NH_2$	Carbamoyl	- amide
3.	Acid chloride	$-\overset{\overset{O}{\|\|}}{C}-Cl$	Chloroformyl	- oyl chloride
4.	Alcohol	$-OH$	Hydroxy	- ol
5.	Aldehyde	$-\overset{\overset{O}{\|\|}}{C}-H$	Formyl or aldo	- al
6.	Ketone	$-\overset{\overset{O}{\|\|}}{C}-$	Keto or oxo	- one
7.	Ester	$-\overset{\overset{O}{\|\|}}{C}-O-R$	Alkyl oxocarbonyl	carboxy-late
8.	Ether	$-O-R$	Alkoxy	–
9.	Oxirane	$-C\underset{O}{\overset{\diagdown\ \diagup}{—}}C-$	Epoxy	–
10.	Halide	$-X$	Halo	_
11.	Amine	$-NH_2$	Amino	amine
12.	Carbonitrile	$-C\equiv N$	Cyano	nitrile
13.	Nitro derivative	$-NO_2$	Nitro	_
14.	Nitroso derivative	$-NO$	Nitroso	_
15.	Azo group	$-N=N-$	Azo	_
16.	Sulphide	$-S-R$	Alkyl thio	_
17.	Sulphonic derivative	$-SO_3H$	Sulpho	sulphonic
18.	Thio alcohol	$-SH$	Mercapto	thiol
19.	Double bond	$C=C$	–	ene
20.	Triple bond	$C\equiv C$	–	yne

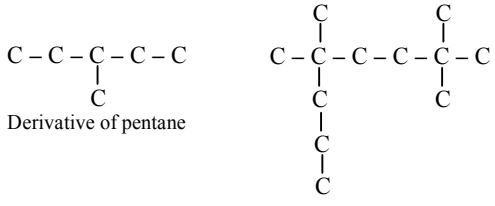
- **IUPAC Rules for Naming Complex Compounds**

  IUPAC name for an organic compound is given according to the following rules:

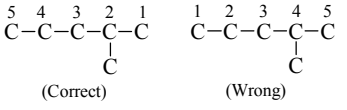
For Saturated Hydrocarbons

- *Longest chain rule*: The longest continuous chain of carbon atoms, which may or may not be straight, is selected.

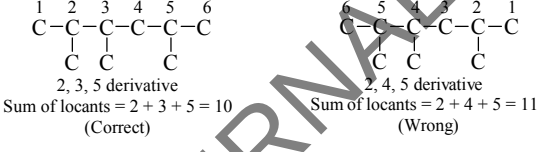
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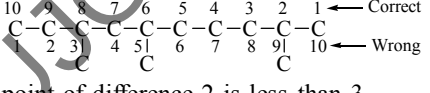
- *Lowest number rule*: The longest carbon chain is numbered as 1, 2, 3, 4 etc. starting from that end which gives the smallest possible number to the substituents.



- In case, the parent chain has two or more substituents, numbering must be done in such a way that the sum of locants on the parent chain is the lowest possible.

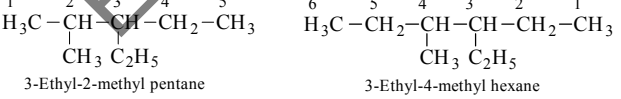


- When the length of the carbon chain is long, the lowest set of locants rule and lowest sum rule gives different results. Hence, the set of locants is preferred which has a lower number at the first point of difference even, if it violates the lowest sum rule.



Because at the first point of difference 2 is less than 3.

- If there are different alkyl substituents attached to the parent chain, their names are written in the alphabetical order.

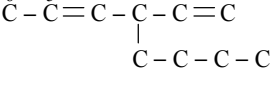


When different alkyl substituents at equivalent positions, then numbering of the parent chain is done in such a way that the alkyl group which comes first in the alphabetical order gets the lowest number.

- If a substituent is present two or more times, this is indicated by the prefix di-, tri-, tetra-, etc. added to the substituent.

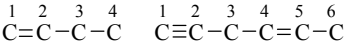
For Unsaturated Hydrocarbons

- *Longest chain*: The longest chain is so selected as to include maximum number of double or triple bonds, even if it is not the actual longest chain of carbon atoms.

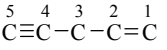


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- *Lowest number rule*: Lowest number is assigned to the first unsaturated carbon.



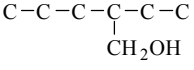
- If double and triple bonds are at the same position from either ends, lowest number is assigned to the double bond.



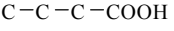
- If double and triple bonds are present in a compound, it is named as alkenyne.

For Compounds Containing Mono-Functional Group

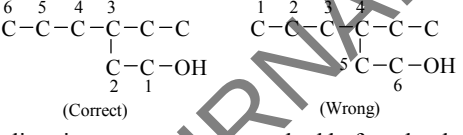
- *Longest chain*: Longest chain is so chosen as to include the functional group.



- The carbon atom of functional group is to be included in deciding the longest carbon chain.


4C atom chain

- *Lowest number rule*: Lowest number is assigned to functional group.

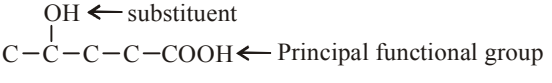

(Correct) (Wrong)

- Numerical prefixes di-, tri-, tetra-, etc. are attached before the designations of functional group if two or more identical groups are present.

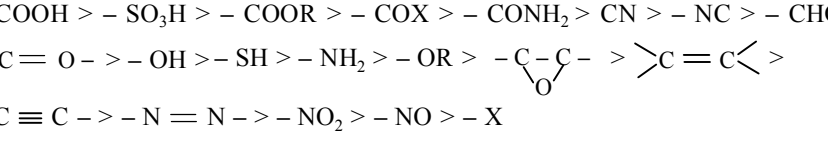

Ethane-1, 2-diol.

For Compounds Containing Poly-Functional Group

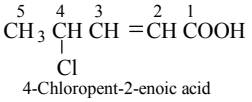
- In IUPAC system, one of the functional groups is chosen as the principal functional group and the remaining functional groups are treated as substituents and indicated by prefixes.



- **Trend of preference :**



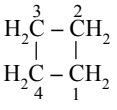
- The parent chain is so selected that it includes the maximum number of functional groups including the principal group.
- Principal functional group gets the lowest number. The following decreasing order of preference for giving the lowest numbers is followed.
  Principal functional group > Double bond or Triple bond > Substituents


4-Chloropent-2-enoic acid

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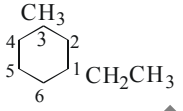
For Alicyclic Compounds

- These are named by adding the prefix cyclo to the name of alkane having the same number of carbon atoms as in the ring.



cyclobutane

- In substituted cycloalkanes, the numbering of the carbon atoms in the ring is done in such a way that the substituent which comes first in the alphabetical order is given the lowest possible number provided.



1-ethyl-3-methyl cyclohexane

- If ring contains greater number of carbon atoms than side chain, it is named as derivative of cycloalkane. If chain contains greater number of C atoms than ring, it is considered as the derivative of alkane.

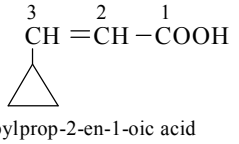


Propyl cyclobutane



1-Cyclobutyl pentane

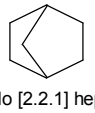
- If however, the side chain contains a multiple bond or a functional group, the alicyclic ring is treated as substituent irrespective of the size of the ring.



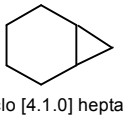
3-Cyclopropylprop-2-en-1-oic acid

For Bicyclo and Spiro Compounds

- Bicyclo compounds contain two fused rings with the help of a bridge. While naming the bicylcoalkane we write an expression between the word bicyclo and alkane (in square bracket), that denotes the number of carbon atoms in each bridge. The numerals are wirtten in descending order and the numbers are separated by full stops.



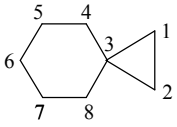
Bicyclo [2.2.1] heptane



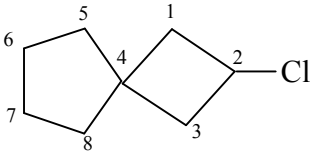
Bicyclo [4.1.0] heptane

- Bicyclic compounds in which the two rings have only common carbon atom are called spiro compounds. They are prefixed by the word spiro followed by brackets containing the number of carbon atoms in each ring in ascending order and then by the name of parent hydrocarbon containing total number of carbon atoms in the two rings.

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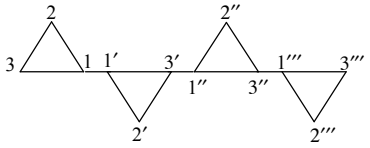
spiro [2.5] octane



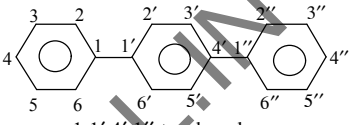
2-chlorospiro [3.4] octane

For unbranched identical hydrocarbon units joined by a single bond

These are named by placing a suitable numeral prefix as bi, ter, quater, quinque for two, three, four, five respectively before the name of hydrocarbon unit. Starting from either end, the carbon atoms of each repetitive hydrocarbon unit are numbered with unprimed and primed arabic numerals such as 1, 2, 3, ..., 1′, 2′, 3′, ..., 1″, 2″, 3″, ... etc.



1,1′,3′,1″,3″,1‴-quatercyclopropane



1,1′,4′,1″-terphenyl

- Organic compounds having same molecular formula but different physical and chemical properties are called isomers and phenomenon is known **as isomerism.**

ISOMERISM
(The phenomenon shown by two or more organic compounds having same molecular formula but different properties (physical & chemical) is known as isomerism.

Structural Isomerism
Shown by compounds having same molecular formula but different structural formula.

- **Chain Isomerism**
  Shown by compounds having same molecular formula but different carbon chains
- **Position Isomerism**
  In this, the position of substituent or functional groups are different
- **Functional Isomerism**
  It is due to the difference in the nature of functional groups present in the isomers.
- **Metamerism**
  Due to the different nature of alkyl groups around a poly-valent functional groups in position isomers
- **Tautomerism**
  The phenomenon in which a single compound exists in two readily interconvertible structures that differ in the relative position of atomic nucleus.

Stereo Isomerism
Shown by compounds having same molecular formula but different spatial arrangement

- **Geometrical Isomerism**
  Shown by compounds possesses same structural formula but differ in their spatial arrangement of the groups around a doubly bonded carbon atoms
  - **Cis-** Like atoms or groups at the same side of the double bond
  - **Trans-** Like atoms or groups across the double bond
- **Optical Isomerism**
  Arises from different arrangement of atoms or groups in three dimensional space resulting in two isomers which are mirror images of each other.

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- *Stereochemistry* involves the study of the relative spatial arrangement of atoms within molecules.
- ***Structural isomerism*** is a form of isomerism in which molecules with the same molecular formula have atoms bonded together in different orders.

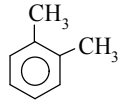
Various types of structural isomerism :

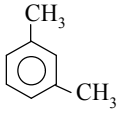
- *Chain isomerism* occurs when the way carbon atoms are linked together is different from compound to compound. It is an example of structural isomerism, and is also called nuclear isomerism.

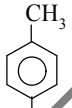
  e.g. C_5H_{12} : $CH_3-CH_2-CH_2-CH_2-CH_3$ (*n*-pentane), $H_3C-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-CH_3$ (isopentane), $H_3C-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-CH_3$ (neopentane)

- *Position isomerism* is an example of structural isomerism, it occurs when functional groups are in different positions on the same carbon chain.

  e.g. $C_6H_4(CH_3)_2$:

  
  o-xylene

  
  m-xylene

  
  p-xylene

- *Functional isomerism* is an example of structural isomerism, it occurs when substances have the same molecular formula but different functional groups. This means that functional isomers being to different homologous series.

  e.g. C_3H_9N : $CH_3-CH_2-CH_2-NH_2$ (propanamine), $CH_3-CH_2-NH-CH_3$ (N-methylethanamine), $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{N}}}}-CH_3$ (N,N-dimethylmethanamine)

- *Metamerism* : This type of isomerism occurs when the isomers differ with respect to the nature of alkyl groups around the same divalent functional group.

  e.g. $C_4H_{10}O$: $CH_3-O-CH_2-CH_2-CH_3$ (*n*-propyl methyl ether), $CH_3-CH_2-O-CH_2-CH_3$ (diethyl ether)

- *Ring-chain isomerism* : In this type of isomerism, one isomer is open chain but another is cyclic.

  e.g. C_3H_6 : $CH_3-CH=CH_2$ (propene), $H_2C\overset{\displaystyle CH_2}{\diagup\diagdown}CH_2$ (cyclopropane)

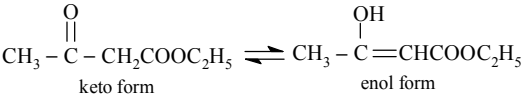
- *Tautomerism* : This type of isomerism is due to spontaneous interconversion of two isomeric forms into each other with different functional groups.

  Conditions : (i) Presence of a $-\overset{\displaystyle O}{\overset{||}{C}}-$ or $-\overset{\displaystyle O}{\overset{||}{N}}$ or $-C\equiv N-$ bond.

  (ii) Presence of at least one α-H atom which is attached to a saturated C-atom.

  e.g. Acetoacetic ester.

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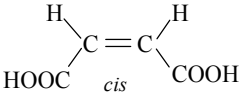


- ***Stereoisomerism*** **:** Compounds with the same molecular formula but having difference in the spatial arrangement of atoms or groups are called stereoisomers and the phenomenon is called stereoisomerism.

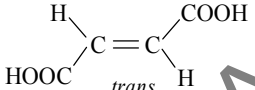
Various types of stereoisomerism

- *Geometrical isomerism* : Compounds with double bonds, or alicyclic rings can exhibit isomerism, due to the attached groups lying above or below the plane of the double bond or ring.

  (a) *cis-trans isomerism*: The *cis* compound is the one with the groups on the same side of the bond, and the *trans* has the groups on the opposite sides. The different isomers have different physical and chemical properties. e.g.



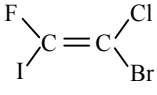
maleic acid



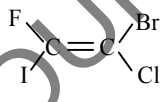
fumaric acid

  (b) *E-Z isomerism :* This isomerism arises when all the four groups linked to the doubly bonded C-atoms are different. It can also be used for the geometrical isomers which can be represented by *cis* and *trans* notations.

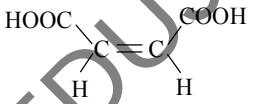
e.g.



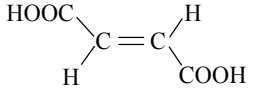
(*Z*) 1-bromo-1-chloro-2-fluoro-2-iodoethene



(*E*) 1-bromo-1-chloro-2-fluoro-2-iodoethene



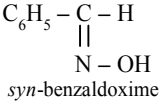
(*Z*) but-2-enoic acid



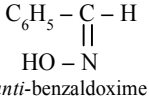
(*E*) but-2-enoic acid

  (c) *Syn-anti isomerism* : In aldoximes, if H and OH groups are on the same side of C = N bond, then the configuration is *syn* and if on opposite side of C = N bond, then the configuration is *anti*.

e.g.



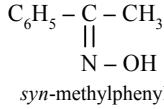
syn-benzaldoxime



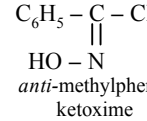
anti-benzaldoxime

  In ketoximes, prefixes *syn* and *anti* are related to the position of the first group named with respect to the OH group whether lying on the same side or opposite side of the double bond.

e.g.



syn-methylphenyl ketoxime

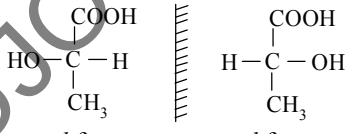


anti-methylphenyl ketoxime

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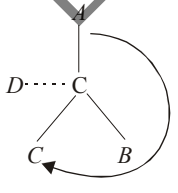
- ***Optical isomerism*** **:** Compounds having similar physical and chemical properties but differing only in the behaviour towards polarised light are called optical isomers and this phenomenon is called optical isomerism.
- The carbon atom linked to four different groups is called *chiral carbon*.
- Non-superimposable mirror images are called *enantiomers* which have same physical and chemical properties and also rotate the plane polarised light up to same extent but in opposite direction.
- *Fischer Projection:* An optically active compound can be represented by Fischer Projection which is planar representation of three dimensional structure. The points which must be followed while writing the Fischer projection are given as:
  (i) Asymmetrical or chiral carbon atom is kept at the intersection of two crossed lines.
  (ii) In vertical lines, groups are arranged in such a way that are seemed to go away from the observer i.e. longest possible chain is kept in vertical line with the most oxidised carbon atom at the top.
  (iii) In horizontal lines, groups are arranged in such a way as they are seemed to come out of the plane i.e. they are seemed to come towards the observer. It is possible only if preferred group is kept towards left side.

  Fischer projection representation of lactic acid (2-hydroxypropanoic acid), $\overset{3}{CH_3}-\overset{2}{CH}(OH)-\overset{1}{COOH}$:

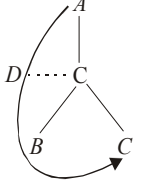


l-form *d*-form

- *Absolute configuration* : It is three directional representation of optically active compound. It is also said to be *R*-, *S*- system. (*R*-Rectus, *S*-Sinister).



R-Configuration

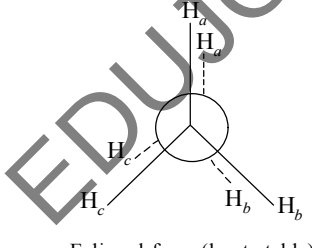


S-Configuration

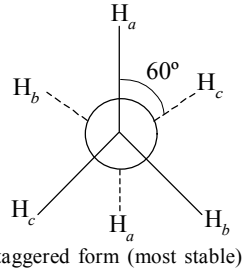
- *Determination of R, S configuration* : It involves the following steps:
  (i) Assignment of priority sequences of the group. Let the priority sequence among the given groups *A*, *B*, *C* and *D* are $A > B > C > D$.
  (ii) Rotation of eye from higher to lower priority sequence by keeping eye towards opposite side of lowest priority group i.e. rotating eye from 1 to 3 (*A* to *C*) via 2(*B*), while doing so if eye is rotating in clockwise then it is *R*-configuration and if in anticlockwise, then it is *S*-configuration.

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- *Sequence rule* : The following rules are followed for deciding the precedence order of the atoms or groups.
  (i) Highest priority is assigned to the atoms of higher atomic number attached to asymmetric carbon atom.
  (ii) If the first atom of a group attached to asymmetric carbon atom is same then we consider the atomic number of 2nd atom or subsequent atoms in group.
  (iii) If there is a double bond or triple bond, both atoms are considered to be duplicated or triplicated.
- *Diastereomers* are the stereoisomers which are not mirror images of each other. They have different physical properties but same chemical properties.
- *Meso compounds* are those compounds whose molecules are super imposable on their mirror images inspite of the presence of an asymmetric carbon atom.
- An equimolar mixture of the enantiomers (*d* & *l*) is called *racemic mixture*. The process of converting *d*- and *l*- form of an optically active compound into racemic form is called *racemisation*.
- The process by which *dl* mixture is separated into *d* and *l* forms with the help of chiral reagents or chiral catalysts is known as *resolution*.
- ***Conformational isomerism*** **:** The different arrangement of atoms in space that results from the carbon-carbon single bond free rotation by 360° are called conformations or conformational isomers or rotational isomers and this phenomenon is called conformational isomerism.
- *Newmann projection* : Here two carbon atoms forming the σ bond are represented by two circles, one behind the other so that only front carbon is seen. The C – H bonds of front carbon are depicted from the centre of the circle while C – H bonds of the back carbon are drawn from the circumference of the circle.

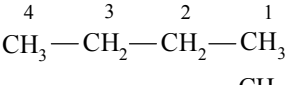


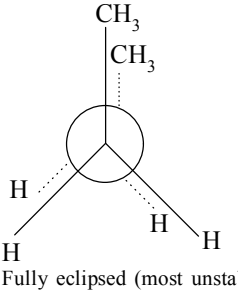
Eclipsed form (least stable)



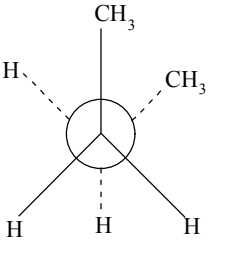
Staggered form (most stable)

➢ *Conformation of butane* :



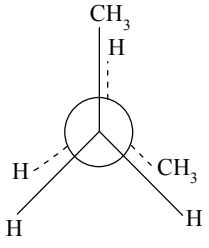


Fully eclipsed (most unstable)

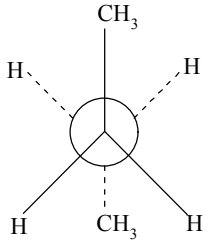


Gauche form

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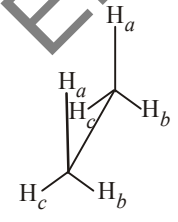
Eclipsed form



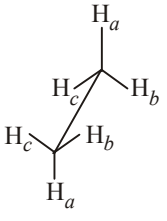
Anti-form (most stable)

The compound if contains	*Optically active forms*	*Optically inactive forms*
n different chiral carbon atoms and the molecule cannot be divided into two equal halves	2^n	0
an even number *n* of chiral carbons, but the molecule can be divided into two equal and similar halves via the central carbon	$2^{(n-1)}$	$2^{(n-2)/2}$
an odd number *n* of chiral carbons and the molecule can be divided into two equal and similar halves via the central carbon	$2^{(n-1)} - 2^{(n-1)/2}$	$2^{(n-1)/2}$

- Gauche conformations are also staggered but they have slightly (3.8 kJ/mol) more energy than anti-form because of methyl groups which are present at nearer position than the anti-form.
  The order of stability of these conformations is : Anti > Gauche > Eclipsed > Fully eclipsed.
- *Sawhorse representation* : Here C — C bonds are represented as oblique. Sawhorse representation of ethane structures are represented as



Eclipsed form (Least stable)



Staggered form (Most stable)

- The energy needed to break a bond in any compound is known as **bond energy.**
- Organic reactions proceed by cleavage of covalent bonds. A cleavage or fission of bond takes place in two ways :
  Homolytic fission - When bond between two atoms $A - B$ breaks in such a way that each fragment carries one unpaired electron (free radicals) is called homolytic fission.

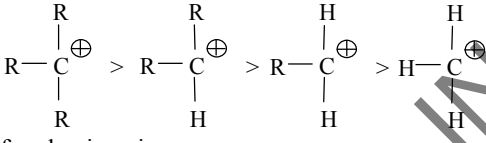
$$A - B \rightarrow \underset{\text{(free radicals)}}{\overset{\bullet}{A} + \overset{\bullet}{B}}$$

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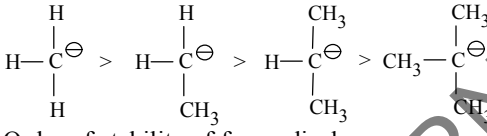
Heterolytic fission - When covalent bond between $A - B$ breaks in such a way that the shared pair of electron stays on one of the atoms, is called heterolytic fission.

$$A : B \rightarrow A^+ + : B^- \quad \text{or} \quad A^- : + B^+$$

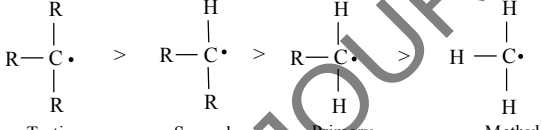
- The ionic species which carries positive charge on central carbon atom is called **carbonium ion** and species which carries negative charge on central carbon is called **carbanion.**
- Carbene (R_2C:) are reactive species containing a formally divalent carbon atom.
- Greater the number of alkyl groups attached to carbon carrying negative charge, the lesser is stability of carbonium ion. Order of stability of carbonium ion :



- Order of stability of carbanions is



- Order of stability of free radicals :



- Carbanion has unshared electron pair at central atom. Central carbon atom is in sp^3 hybridised state.
- In carbonium ion (carbocation) the carbon atom has 6 electrons and is deficient of electrons. The central atom in carbonium ion is in sp^2 hybridised state.
- Carbanions behave as nucleophiles, therefore initiate addition as well as substitution reactions.
- Decreasing order of –I effect is

  $\overset{\oplus}{N}R_3 > NO_2 > CN > F > Cl > Br > I > OH > OCH_3 > C_6H_6 > — CH = CH_2$
- Decreasing order of +I effect is

  $— O > — COO — > (CH_3)_3C— > (CH_3)_2CH— > CH_3CH_2 — > CH_3$
- Positively charged and electron deficient species are called **electrophiles** (electron seeking or electron loving). Electrophiles are always Lewis acids.

  e.g. NO_2^+, Br^+, Cl^+, H_3O^+, RN_2^+, Ag^+, $CH_3CH_2^+$, SO_3, $SOCl_2$, $AlCl_3$, BF_3, $ZnCl_2$, $FeCl_3$.
- **Nucleophiles or nucleophilic reagents** - Anions or molecules with unshared pairs of electrons having tendency of donating electron pairs are called nucleophiles (nucleus loving). Nucleophiles are always Lewis bases.

  e.g. CH_3O^-, OH^-, $C_2H_5O^-$, CN^-, H^-, CH_3CO^-, HSO_3^-, NH_3, NH_2^-, H_2O, RMgBr, $LiAlH_4$, $>O:$, $>S:$, –N, etc.

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Comparison of nucleophiles and electrophiles

Nucleophilic entities (Nucleophiles)	Electrophilic entities (Electrophiles)
(i) Electron rich	Electron deficient
(ii) Donate an electron pair	Accept an electron pair
(iii) Attack on electron-deficient atom	Attack on electron-rich atom
(iv) Lewis bases	Lewis acids
(v) Possesses an unshared pair of electrons which are not too strongly held to the atomic nucleus (usually atoms in groups 15 and 16 of the periodic table)	Possesses an empty orbital to receive the electron pair from the nucleophile.
(vi) Are able to increase their covalency by one unit	Are able to form an extra or alternative bond with the nucleophile
(vii) Are often anion	Are usually cations

- Process of electron shifting along a carbon chain due to the presence of a polar covalent bond in it is called **inductive effect** or **transmission effect.**
- When hetero atom is such that it attracts electron towards itself is said to be **–I effect.** When hetero atom pushes electron away from itself it exerts a **+I effect.**
- The phenomenon due to which a compound is said to be a hybrid of various canonical forms or resonating forms is termed as **resonance.**
- Greater the resonance energy, greater is the stability of a molecule and greater will be the reactivity.
- More the number of covalent bonds in a resonating structure more is its stability.
- **Mesomeric effect** is the effect of electron redistribution that can take place in unsaturated and specially in conjugated systems via their π orbitals.
- When the transfer of electrons takes place towards the attacking reagent it is said to be ***–E* effect** and when the transfer occurs away from attacking reagent the effect is called ***+E* effect.**
- The replacement of an atom or group from a molecule by a different atom or group is known as **substitution reaction.**

  e.g. $CH_3OH + HBr \rightarrow CH_3Br + H_2O$
- Reactions in which atoms or group of atoms are added to a molecule are known as addition reactions.

  $CH_2 = CH_2 + HBr \rightarrow CH_3 - CH_2Br$
- **Elimination reactions** are the reverse of addition reactions and involve loss of atoms or group of atoms from a molecule to form a multiple linkage.

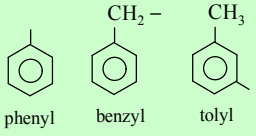
  e.g. $CH_3CH_2OH \xrightarrow{\text{Conc. } H_2SO_4} \underset{\text{(dehydration)}}{CH_2=CH_2} + H_2O$

  $CH_3 - CH_2Cl \xrightarrow{\text{Alc. KOH}} \underset{\text{(dehydrohalogenation)}}{CH_2 = CH_2} + HCl$

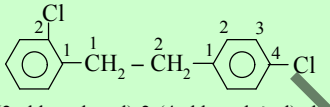
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Additional Information

- The aryl group obtained by removing one hydrogen atom from benzene ring is named as **phenyl** and not benzyl. Similarly in case of toluene it is called **tolyl**.
  The aryl group obtained by removing a hydrogen atom from side chain in case of toluene is named as **benzyl**.



- When two or more prefixes consist of identical words, the priority for citation is given to that group which contains the lowest locant at the first point of difference. For example,

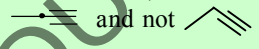


1-(2-chlorophenyl)-2-(4-chlorophenyl)ethane

- Bond line notations for alkynes:
  - bond line notation for but-1-yne is

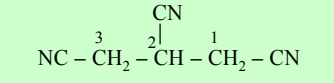


  - bond line notation for propyne is

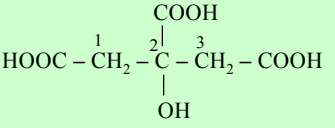


  This is because the bond angle involved in *sp*-hybridised carbon atom is 180° and not 120°.

- According to latest conversions as per 1993 recommendations of IUPAC nomenclature, if an unbranched carbon chain is directly linked to more than two like functional groups, the organic compound is named as a derivative of the parent alkane which does not include carbon atoms of the functional group. For example,



propane-1,2,3-tricarbonitrile
(formerly 3-cyanopentane-1,5-dinitrile)



2-hydroxypropane-1,2-3-tricarboxylic acid
(formerly 3-carboxy-3-hydroxypentane-1,5-dioic acid

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End

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CHAPTER

purification & analysis

- Purification means the removal of undesirable impurities associated with a particular organic compound, *i.e.*, to obtain the organic compound in pure state.
- The method applied to purify a definite compound depends on the nature of the organic compound and the impurities present in it.
- Some important methods of purification are as follows:

(i) Crystallisation < (a) Simple crystallisation
(b) Fractional crystallisation

(ii) Sublimation

(iii) Distillation
- (a) Simple distillation
- (b) Fractional distillation
- (c) Vacuum distillation
- (d) Steam distillation

(iv) Solvent extraction

(v) Chromatography
- (a) Column or Adsorption chromatography
- (b) Thin layer chromatography
- (c) Paper chromatography
- (d) Gas chromatography
- (e) Ion-exchange chromatography.

	Type of chromatogaphy	Mobile/ Stationary Phase	Uses
1.	Adsorption or column chromatography	Liquid/Solid	Large scale separations
2.	Thin-layer chromatography	Liquid/Solid	Qualitative analysis (identi-fication and characterisation of organic compounds)
3.	High performance liquid chromatography	Liquid/Solid	Qualitative and quantitative analysis
4.	Gas-liquid chroma-tography (GLC)	Gas/Liquid	Qualitative and quantitative analysis
5.	Paper or Partition chromatography	Liquid/Liquid	Qualitative and quantitative analysis of polar organic compounds (a-amino acids, sugars and inorganic compounds)

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- Simple crystallisation involves selection of a solvent in which the given substance is more soluble at higher temperature than at room temperature. In addition to water various other solvents like alcohol, ether, benzene, CCl_4 etc. are used for crystallisation.
- Fractional crystallisation is the process of separation of different components of a mixture by repeated crystallizations.
- Fractional crystallisation is carried out to separate organic solids having small difference in their solubilities in suitable solvent.
- Sublimation involves separation of volatile substance from non-volatile solids by heating.
- Simple distillation is used when the boiling point of the components differ widely (30°-50°C).
- If the boiling points of the two liquids of a mixture are very close *i.e.* differ by (10°-15°C) fractional distillation is done to separate the mixture.
- Petroleum and its components are separated by fractional distillation.
- Fractionating column is used to increase the cooling surface area and obstruct the path of ascending vapours and descending liquid in fractional distillation.
- Azeotropes are constant boiling mixtures so they are separated by azeotropic distillation.
- Normal boiling point of a liquid is that temperature at which its vapour pressure is equal to atmospheric pressure. If atmospheric pressure is reduced the liquid will boil earlier than its normal boiling point. This principle is used in the purification of those compounds which decompose at their normal boiling points, i.e., by carrying out distillation under reduced pressure.
- The process of separation of an organic compound from its aqueous solution by shaking with a suitable organic solvent is termed solvent extraction. The solvent should be immiscible with water and the organic compound to be separated should be highly soluble in it.
- A mixture of *o*-hydroxyacetophenone and *p*-hydroxy-acetophenone can be separated by steam distillation as *o*-hydroxyacetophenone due to chelation is steam volatile but due to intermolecular H-bonding *p*-hydroxyacetophenone is not.
- Tswett discovered chromatography in which the separation and purification is brought about by the differential movement of the individual components of a mixture through a stationary phase under the influence of a mobile phase.
- Column chromatography is based upon differential adsorption - desorption of different components of mixture.
- The technique of gas chromatography is suitable for compounds which vapourize without decomposition.
- In Lassaigne's test, the organic compound is fused with a piece of sodium metal to convert covalent compounds into ionic compounds (NaCN, Na_2S, NaX).
- In the Lassaigne's test for detection of nitrogen in an organic compound the blue colour is due to the formation of ferricferrocyanide, $Fe_4[Fe(CN)_6]_3$.
- A violet colour with sodium nitroprusside is the test for sulphur, is due to the formation of $Na_4[Fe(CN)_5NOS]$.

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- Copper wire test for halogens is known as Beilstein test.
- Beilstein's test is not given by fluorine since cupric fluoride is not volatile.
- A freshly prepared $FeSO_4$ solution is used in Lassaigne's test for nitrogen as on keeping $FeSO_4$ solution oxidises to basic ferric sulphate and cannot be used for detection.
- Siwolowoff"s method for determining the boiling point of liquid is used when the amount of liquid available is small.
- Sometimes crystallisation can be induced by adding a few crystals of the pure substance to the concentrated solution. This is called as seeding.
- Lithium is not used in Lassaigne's test since it reacts slowly and its compounds are generally covalent. Potassium can also not be used since it reacts evidently and cannot be handled.

Qualitative analysis of organic compounds

- ***Detection of carbon and hydrogen*** **:**
  A small amount of dry organic compound containing carbon and hydrogen are oxidised by cupric oxide to carbon dioxide and water respectively.
  $C + 2CuO \longrightarrow 2Cu + CO_2$ (turns lime water milky)
  $H_2 + CuO \longrightarrow Cu + H_2O$ (turns anhydrous $CuSO_4$ blue)
- ***Lassaigne's test :*** In Lassaigne's test, the organic compound (containing N, S, halogen) is fused with sodium metal as to convert these elements into ionisable inorganic substances.

Elements	Sodium salt of ions in Lassaigne's extract
N	NaCN
S	Na_2S
Both N and S	NaCNS
Halogen (*X*)	Na*X*

- ***Detection of nitrogen*** **:**
  $2NaCN + FeSO_4 \longrightarrow Fe(CN)_2 + Na_2SO_4$
  $Fe(CN)_2 + 4NaCN \longrightarrow Na_4[Fe(CN)_6]$
  $4Na_4[Fe(CN)_6] + 4FeCl_3 \longrightarrow \underset{\text{Prussian blue}}{Fe_4[Fe(CN)_6]_3} + 12NaCl$
- ***Detection of sulphur*** **:**
  (i) $Na_2S + Na_2[Fe(CN)_5NO] \longrightarrow \underset{\text{violet}}{Na_4[Fe(CN)_5NOS]}$
  (sodium nitroprusside)
  (ii) $Na_2S + (CH_3COO)_2Pb \longrightarrow \underset{\text{black}}{PbS\downarrow} + 2CH_3COONa$
- ***Detection of nitrogen and sulphur together*** **:**
  $3NaCNS + FeCl_3 \longrightarrow \underset{\text{blood red colour}}{Fe(CNS)_3} + 3NaCl$
  Note : This test fails in case of diazo compounds.

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- ***Detection of halogen*** **:** $AgNO_3$ test :

$$Na\mathit{X} + \text{conc. } HNO_3 + AgNO_3$$

white ppt. soluble in aq. NH_3	yellow ppt. partially soluble in aq. NH_3	yellow ppt. insoluble in aq. NH_3
$\rightarrow$ chlorine	$\rightarrow$ bromine	$\rightarrow$ iodine

Note: From colour of AgBr and AgI, it is difficult to judge whether one organic compound contains bromine or iodine. To confirm their presence, the filtrate is supplemented by chlorine water test (layer test). To acidified filtrate 1 ml of $CHCl_3$ or CCl_4 is added followed by addition of excess of chlorine water with constant shaking. If chloroform layer becomes yellow or brown, bromine is present and if violet, iodine is present.
Beilstein test or copper wire test is also used to detect halogens.

Quantitative analysis of organic compounds

- ***Estimation of carbon and hydrogen - Leibig's method***

$$C_xH_y + \underset{\text{excess}}{O_2} \longrightarrow x CO_2 + y/2\ O_2$$

$$\% \text{ of carbon} = \frac{12}{44} \times \frac{\text{wt. of } CO_2}{\text{wt. of organic compound}} \times 100$$

$$\% \text{ of hydrogen} = \frac{2}{18} \times \frac{\text{wt. of } H_2O}{\text{wt. of organic compound}} \times 100$$

- ***Estimation of nitrogen:***

(i) *Duma's method* : In this method a nitrogen containing compound is strongly heated with cupric oxide in the atmosphere of CO_2 to get free nitrogen along with CO_2 and water.

$$C_xH_yN_z + CuO \xrightarrow{\Delta} x CO_2 + \frac{y}{2} H_2O + \frac{z}{2} N_2 + Cu$$

$$\% \text{ of nitrogen} = \frac{28}{22400} \times \frac{\text{vol. of } N_2 \text{ collected at N.T.P. in c.c}}{\text{wt. of organic compound}} \times 100$$

(ii) *Kjeldahl's method* : In this method nitrogen containing compound is heated with conc. H_2SO_4 in presence of copper sulphate to convert nitrogen into ammonium sulphate which is decomposed with excess of alkali to liberate ammonia. The ammonia evolved is estimated volumetrically. The percentage of nitrogen is then calculated from the amount of ammonia evolved.

$$\% \text{ of nitrogen} = \frac{1.4 \times \text{vol. of acid used in ml} \times \text{normality of acid}}{\text{wt. of organic compound}}$$

- ***Estimation of halogens - Carius method:*** A weighed amount of the organic compound is heated with fuming HNO_3 in Carius tube containing few crystals of $AgNO_3$. Halogen present in the compound is converted into insoluble AgX which is separated and weighed.

$$\mathit{X} + AgNO_3 + \text{conc. } HNO_3 \longrightarrow Ag\mathit{X}$$

$$\% \text{ of } \mathit{X} = \frac{\text{At. mass of } \mathit{X}}{108 + \text{At. mass of } \mathit{X}} \times \frac{\text{wt. of } Ag\mathit{X}}{\text{wt. of org. compound}} \times 100$$

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Determination of molecular mass

- **Physical methods:**

(A) ***For volatile compound*** **:**

(i) *Victor Meyer's method*

$$\text{Molecular mass of substance} = \frac{\text{wt. of organic compound} \times 22400}{\text{volume of vapours obtained}}$$

(ii) *Duma's method*

$$\text{Molecular mass of substance} = \frac{\text{mass of vapours} \times 22400}{\text{volume of vapours at N.T.P.}}$$

(B) ***For non-volatile compounds*** **:**

(i) *Depression of freezing point*

$$\text{Molecular mass of substance} = \frac{1000 \times \text{molar depression constant of pure solvent} \times \text{wt. of solute}}{\text{wt. of solvent} \times \text{depression in freezing point}}$$

(ii) *Elevation in boiling point*

$$\text{Molecular mass of substance} = \frac{1000 \times \text{molal elevation constant of pure solvent} \times \text{wt. of solute}}{\text{wt. of solvent} \times \text{elevation in boiling point}}$$

- **Chemical methods:**

(i) *Silver salt method for organic acids* :

$$R - COOH \longrightarrow R - COOAg \xrightarrow{\Delta} Ag$$

$$\text{Molecular wt. of acids} = \left(\frac{\text{wt. of silver salt} \times 108}{\text{wt. of metallic silver}} - 107\right) \times \text{basicity of acid}$$

(ii) *Platinum chloride method for organic bases*

$$B + HCl \longrightarrow B \cdot HCl$$

$$2B \cdot HCl + PtCl_4 \longrightarrow B_2H_2PtCl_6 \xrightarrow{\Delta} Pt$$

$$\text{Molecular weight of base} = \frac{1}{2}\left(\frac{\text{wt. of platinum salt} \times 195}{\text{wt. of platinum}} - 410\right) \times \text{acidity of the base}$$

(iii) *Titration method for organic acids and organic bases*

$$\text{Equivalent weight of acid} = \frac{\text{wt.of acid} \times 1000}{\text{normality of alkali} \times \text{vol. of alkali used for end point}}$$

Molecular weight of acid = Eq. wt. of acid × basicity

Similarly,

$$\text{Equivalent weight of base} = \frac{\text{weight of base} \times 1000}{\text{normality of acid} \times \text{vol. of acid used for end point}}$$

Molecular wt. of base = Eq. wt. of base × acidity

- *Empirical formula* : The formula of a compound which gives the simplest whole number ratio of the atoms of various elements present in one molecule of the compound is called empirical formula of the compound.
- *Molecular formula* : The formula of a compound which gives the actual ratio of the atoms of various elements present in one molecule of the compound is called the molecular formula of the compound.

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- Molecular formula = $n \times$ empirical formula, where n = 1, 2, 3, ...
- Molecular weight = 2 × vapour density.
- All compounds containing an odd number of nitrogen atoms (*i.e.* 1, 3, 5, ...) have odd masses and those with even number of nitrogen atoms (*i.e.* 2, 4, 6, ...) have even masses. This is called **nitrogen rule**.
- Empirical formula of a compound represents simplest ratio of atoms.
- Molecular formula of a compound represents actual number of atoms of the various atoms present in one molecule.
  Molecular formula = $n \times$ empirical formula
- Eudiometry is a direct method for determination of molecular formula of gaseous hydrocarbons without determining the percentage composition of various elements in it and without knowing the molecular weight of the hydrocarbon.
- A mixture of benzene (b.p. 80°C) and chloroform (b.p. 61.5°C) is separated by distillation (fractional distillation) as their boiling points are close.
- Since alcohol and water form a constant boiling mixture (azeotrope) therefore absolute alcohol is prepared by azeotropic distillation.
- Human hair on heating strongly with soda-lime smells of ammonia as they contain amino acids.
- Sugar can be separated by using paper chromatography.
- On adding $FeCl_3$ solution to acidified Lassaigne's extract a blood red colouration is produced due to the formation of ferric thiocyanate or sulphocyanide, $Fe(CNS)_3$ indicates the presence of N and S.
- Aniline is purified by steam distillation as it is steam volatile.
- Equivalent weight of an acid is equal to molecular weight/basicity.
- Molecular mass of a volatile substance can be obtained by Victor-Meyer's method.
- Anhydrous copper sulphate is used to test the presence of water in a liquid as anhydrous $CuSO_4$ turns blue in presence of water.

$$\underset{\text{white}}{CuSO_4} + 5H_2O \rightarrow \underset{\text{blue}}{CuSO_4 \cdot 5H_2O}$$

- Thiophene can be removed from commercial benzene by shaking it with concentrated H_2SO_4.
- In Kjeldahl's method, during digestion, the nitrogen of the organic compound is converted into $(NH_4)_2SO_4$.
- Fusion of organic compound with fusion mixture ($Na_2CO_3 + K_2CO_3$) converts phosphorus into Na_3PO_4.
- For the detection of phosphorus, the organic compound after fusion with Na_2O_2 is extracted with water, boiled with HNO_3 and then ammonium molybdate is added to it we get the yellow precipitate of ammonium phosphomolybdate.
- In organic compounds phosphorus is estimated as magnesium pyrophosphate, $Mg_2P_2O_7$.
- The Lassaigne's extract is boiled with dilute HNO_3 before testing for halogen because Na_2S and NaCN are decomposed by HNO_3 otherwise Na_2S will give black ppt. of Ag_2S and NaCN will give white ppt. of AgCN which would interfere with the test of halogens.

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- In Kjeldahl's method for estimation of nitrogen, K_2SO_4 is added to raise the boiling point of H_2SO_4 to ensure complete conversion of nitrogen into $(NH_4)_2SO_4$ while $CuSO_4$ or mercury is added to catalyse the above conversion.
- Steam distillation can be regarded as analogous to distillation under reduced pressure.
- Messenger's method is used for estimation of sulphur. In this method, the organic compound is heated with alkaline $KMnO_4$ solution when sulphur of the organic compound is oxidised to K_2SO_4 which is estimated as $BaSO_4$.
- Alkaline solution of pyragallol *i.e.* 1,2,3-trihydroxy benzene is used to absorb oxygen.
- Soxhlet extractor is used for continuous extraction of organic compounds with a minimum amount of organic solvent.
- Lassaigne's test for the detection of nitrogen will fail in case of $\mathbf{H_2N \cdot NH_2 \cdot 2HCl}$ because for Lassiagne's test of N, compound must contain N in addition to carbon, so that NaCN can be formed in sodium extract.
- Azo compounds does not give a positive Lassaigne's test for N as azo compounds on moderate heating lose N_2, before the sodium melts in preparation of sodium extract.
- **Organic compounds** are studied separately from others, because of the special characteristics of carbon compounds like catenation, formation of compounds both with electropositive and electronegative elements and their tendency to show isomerism.
- The most suitable method of separation of 1 : 1 mixture of ortho and para nitrophenols is **distillation** as para nitrophenol has higher boiling point due to H-bonding.
- Phenol is soluble in **NaOH solution** because of their weakly acidic behaviour.
- Carbon shows maximum capacity of catenation because **C–C bond strength is very high.**
- A mixture of camphor and benzoic acid can be separated by **chemical methods** as both possess the sublimation nature; benzoic acid reacts with alkalies, whereas camphor does not.
- The function of boiling the sodium extract with conc. HNO_3 before testing for halogens is **to destroy CN^- and S^{2-} ions which will otherwise give ppt.**
- **Simple distillation** can be used to separate a mixture of ether (b. pt. 35ºC) and toluene (b.pt. 110ºC) as simple distillation is used when the boiling point of two components differ widely (30°-50°C).
- When petroleum is heated gradually, first batch of vapours evolved will be rich in **petroleum ether.**
- For detection of sulphur in an organic compound, sodium nitroprusside is added to the sodium extract. A violet colour is obtained due to the formation of $\mathbf{Na_4[Fe(CN)_5NOS]}$, (sodium thionitroprusside).
- Raw juice in sugar factories is generally concentrated by **vacuum distillation** as at low pressure boiling point is lowered and evaporation of water becomes more fast.
- Turpentine oil can be purified by **steam distillation** as it is steam volatile.
- To determine the weight of halogen in the organic compound, the compound is heated with fuming HNO_3 in presence of $\mathbf{AgNO_3}$ which gives precipitation of silver halides.

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- Salts can be obtained from a concentrated sea water by **crystallisation** as crystallisation of conc. solution separates out salts.
- Silica gel is used for keeping away the moisture because it **adsorbs H_2O.**
- Anhydrous $CaCl_2$ is used as drying agent because it **absorbs water molecules.**
- **Lavoisier** is called the father of Chemistry.
- If an organic compound contains C, H and S. When C and H are to be estimated the combustion tube at the exit should contain a **lead chromte.**
- If the compound contains C, H and halogen. When C and H are to be estimated the combustion tube at the exit should contain a **silver spiral.**
- The technique of gas chromatography is suitable for compounds which are **vaporised without decomposition.**
- Generally it is more difficult to purify organic compounds than inorganic compounds because **physical constants of organic compounds and the impurities associated with them are very close to each other.**
- **Sublimation** cannot be used for purification of liquids because sublimation is the process involving direct conversion of a solid species to gaseous phase.
- When sodium extract is prepared, generally the substance ignites H_2 as hydrogen of organic compound ignites.
- First systematic classification of naturally occurring compounds was given by **Lemery.**
- **Na metal** is most commonly used to dry organic liquids.
- An Azeotropic mixture of ethanol and water is first treated with **anhydrous lime and C_6H_6** before subjecting for fractional distillation to separate them as anhydrous lime or C_6H_6 disturbs the nature of azeotropic mixture of alcohol and water.
- There is no test (Direct) for the detection of **O** in an organic compound.
- The latest technique used for the purification of organic compounds containing minute quantities is **chromatography.**
- Boiling point of a liquid can be increased by **increasing the pressure** because liquids boil above boiling point if atmospheric pressure is higher than 1 atm.
- The substance used as an adsorbent in the column chromatography is **Al_2O_3.**
- The sodium extract of organic compound containing sulphur on acidification with acetic acid and them adding lead acetate solution gives a black precipitate.
- Organic compound + CaO + Na_2CO_3 $\xrightarrow[\text{a Pt. crucible}]{\text{heat in}}$ Cool the solution and add dil. HNO_3 and then $AgNO_3$. A precipitate of AgX is dried and weighed and the percentage of halogen is obtained as usual. This is Schiffs and Piria method used for the estimaion of **halogens.**

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End

22

CHAPTER

hydrocarbons

- Certain organic compounds contain only two elements, hydrogen and carbon and hence are known as **hydrocarbons**.
- Methane is the major constituent of natural gas (upto 97%).
- Methane is colourless and when liquefied is less dense than water.
- The quantity of heat evolved when one mole of a hydrocarbon is burned to carbon dioxide and water is called the **heat of combustion**, for methane its value is 213 kcal.
- The main sources of hydrocarbon are **natural gas**, **petroleum** and **coal**.
- **Petroleum** is the major source of aliphatic hydrocarbons while coal is a good source of aromatic hydrocarbons.
- The alkanes or saturated aliphatic hydrocarbons are also known as **paraffins** because of their less reactivity towards chemicals.
- Alkanes are represented by general formula $\mathbf{C_nH_{2n+2}}$ and in all alkanes each carbon atom is sp^3 hybridised.
- Relative reactivities of halogen towards alkanes is in the order $F_2 > Cl_2 > Br_2$.
- Methane and chlorine do not react in the dark at room temperature.
- Paraffins are generally insoluble in water but are soluble in organic solvents. The solubility decreases with increase in molecular weight.
- Except for the very small alkanes, the boiling point rises 20 to 30 degrees for each carbon that is added to the chain.
- A branched chain isomer has a lower boiling point than a straight chain isomer. Thus *n*-butane has a boiling point of 0°C and isobutane –12°C.
- The **specific gravity** of alkanes increases very slowly with increase in molecular weight, until it becomes constant at about 0.77.
- Pyrolysis of alkanes particularly when petroleum is concerned is known as **cracking**.
- In marshy places methane is produced by bacterial decomposition of organic matters, and hence known as **marsh gas**.
- A series of compounds in which each member differs from the next member by a constant amount is called a homologous series and the numbers of the series are called **homologues**.
- Since fluorination of alkanes is highly explosive reaction, the fluorine derivatives are prepared indirectly through bromo or iodo derivatives by action of HgF or HgF_2.
- The process of iodination is reversible and it is carried out in the presence of oxidising agents like iodic acid, nitric acid, mercuric oxide etc., which destroy the hydroiodic acid formed in the reaction.

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- A primary (1°) carbon atom is attached to only one other carbon atoms, a secondary (2°) is attached to two others, and a tertiary (3°) to three others.

ALKANES

The chemical inertness of alkanes (saturated hydrocarbons) is due to
(i) the non-polar nature of C – H bond and
(ii) presence of strong C – H and C – C bonds.

Methods of Preparation

- *By catalytic reduction of unsaturated hydrocarbons (Sabatier Senderen's reaction):*

$$R-CH=CH_2 + H_2 \xrightarrow[\Delta]{Ni} R-CH_2-CH_3$$

$$R-C\equiv C-R' + H_2 \xrightarrow[\Delta]{Ni} R-CH=CHR' \xrightarrow[\Delta]{Ni} RCH_2CH_2R'$$

- *By the reduction of alkyl halides*:

$$R-X \xrightarrow{LiAlH_4 \text{ or } NaBH_4} R-H$$

$$R-X \xrightarrow[Zn, HCl \text{ or } Zn\text{-}Cu, C_2H_5OH]{Zn, CH_3COOH \text{ or}} R-H$$

$$R-I \xrightarrow[150°C]{\text{red P, HI}} R-H$$

- *By decarboxylation of carboxylic acids*:

$$R-COONa + NaOH \xrightarrow[\Delta]{CaO} R-H + Na_2CO_3$$

- *By Wurtz reaction :*

$$R-X + 2Na + X-R \xrightarrow{\text{dry ether}} R-R + 2NaX$$

- *By Kolbe's electrolytic method:*

$$R-COONa \rightleftharpoons R-COO^- + Na^+$$

At anode : $R-COO^- \longrightarrow R-C\dot{O}O + e$ (unstable)

$$2R-C\dot{O}O \longrightarrow R-R\uparrow + 2CO_2\uparrow \quad \text{(alkane)}$$

$$2KOH + CO_2 \longrightarrow K_2CO_3 + H_2O$$

At cathode : $Na + e \longrightarrow Na$ (primary reaction)

$$Na + H_2O \longrightarrow NaOH + \frac{1}{2}H_2\uparrow \quad \text{(secondary reaction)}$$

Methane cannot be obtained in this method, while sodium acetate on electrolysis liberates ethane at anode along with CO_2 gas.

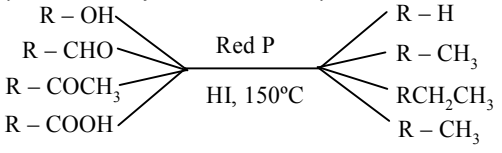
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- *By Clemmensen reduction* :

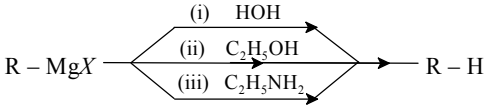
$$R - CHO + 4H \xrightarrow[\text{conc.HCl}]{Zn - Hg} R - CH_3 + H_2O$$

$$RCOCH_3 + 4H \xrightarrow[\text{conc.HCl}]{Zn - Hg} R - CH_2CH_3 + H_2O$$

- *By reduction of alcohols, aldehydes, ketones and carboxylic acids*



- *From Grignard reagent* :



- *Corey-House synthesis* :

$$R_2CuLi + R'X \xrightarrow{\text{dry ether}} R - R' + RCu + LiX$$

Chemical Properties

- *Nitration*

$$RH + HONO_2 \xrightarrow{\Delta} RNO_2 + H_2O$$

- *Isomerisation*

$$CH_3CH_2CH_2CH_3 \xrightarrow[\Delta]{\text{anhy. } AlCl_3} CH_3\underset{\displaystyle CH_3}{\underset{|}{C}}HCH_2CH_3 + CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-CH_3$$

- *Sulphonation*

$$R{-}H + H_2SO_4 \xrightarrow{\Delta} RSO_3H + H_2O$$

- *Oxidation*

$$CH_4 + O_2 \xrightarrow{\text{Cu tubes}} CH_3OH$$

$$CH_4 + O_2 \xrightarrow[550\ K]{MoO} HCHO + H_2O$$

$$CH_4 + O_2 \xrightarrow{(CH_3COO)_2\ Mn} HCOOH$$

- *Halogenation*

$$CH_4 + Cl_2 \xrightarrow{UV} CH_3Cl + CH_2Cl_2 + CHCl_3 + CCl_4$$

- *Cracking*

$$CH_3CH_2CH_3 \xrightarrow[\text{cracking}]{\Delta} CH_4 + CH_2 = CH_2$$

- *Aromatisation*

$$C_6H_{14} \xrightarrow[870\ K]{Al_2O_3/Cr_2O_3} C_6H_6$$

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ALKENES

- Ethylene is a flat molecule and the carbon-carbon "double bond" is made up of a strong σ-bond and weak π-bond.
- The C – C distance in ethylene molecule is 1.34 Å as compared with the C – C distance of 1.53 Å in ethane.
- **Electron diffraction** and **spectroscopic studies** show ethylene to be a flat molecule with bond angles very close to 120°.
- The particular kind of diastereomers that owe their existence to hindered rotation about double bonds are called **geometric isomers**.
- ***Cis-*** and ***trans-*** refers to groups that are on the same side or opposite side of the molecule.
- Boiling point of alkenes rises with increasing carbon number and the boiling point rise is 20-30 degrees for each added carbon and branching lowers the boiling point.
- **Saytzeff's rule** states that, in dehydrohalogenation the preferred product is the alkane that has the greater number of alkyl groups attached to the doubly bonded carbon atoms.
- The quantity of heat evolved when one mole of an unsaturated compound is hydrogenated is called the heat of hydrogenation.
- **Heats of hydrogenation** show that the stability of an alkene depends upon the position of the double bond. The greater the number of alkyl groups attached to the doubly bonded carbon atoms, the more stable is the alkene.
- **Markownikoff's rule** states that, in the addition of an acid to the carbon-carbon double bond of an alkene, the hydrogen of the acid attaches itself to the carbon that already holds the greater number of hydrogens.
- Alkenes react with cold concentrated sulphuric acid to form compounds of the general formula $ROSO_3H$ known as alkyl hydrogen sulphates.
- Water adds to the more reactive alkenes in the presence of acids to yield alcohols.
- Alkenes are readily converted by chlorine or bromine into saturated compounds that contain two atoms of halogens attached to adjacent carbons.
- Alkenes reacts with mercuric acetate in the presence of water to give hydroxy-mercurial compounds which on reduction yields alcohol. This reaction is called **oxymercuration-demercuration reaction**.
- Oxymercuration-demercuration is highly regioselective and gives alcohols corresponding to Markownikov addition of water to the carbon-carbon double bond.
- With the reagent diborane, $(BH_3)_2$, alkenes undergo hydroboration to yield alkylboranes R_3B, which on oxidation gives alcohols.
- **Hydroboration reaction** is carried out in an ether, commonly tetrahydrofuran or 'diglyme' (dimethylene glycol methyl ether, $CH_3OCH_2CH_2OCH_2CH_2OCH_3$).
- The hydroboration-oxidation process gives products corresponding to anti-Markownikov addition to water to the carbon-carbon double bond.
- Oxidising agents like cold alkaline potassium permanganate and peroxy acids such as peroxyformic acid (HCO_2OH) converts alkenes into 1,2-diols, dihydroxy alcohols containing the two – OH groups on adjacent carbons.

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- **Ozonolysis** (cleavage by ozones) is carried out in two stages, first addition of ozone to the double bond to form an ozonide and second, hydrolysis of the ozonide to yield the cleavage products.

Methods of preparation

- *Dehydration of alcohols*

$$RCH_2CH_2OH \xrightarrow[443\ K]{H_2SO_4} RCH = CH_2 + H_2O$$

- *Dehydrohalogenation of alkyl halides*

$$RCH_2CH_2X \xrightarrow{KOH(alc.)} RCH = CH_2 + KX + H_2O$$

- *Dehalogenation of vicinal dihalides and geminal dihalides*

$$\underset{\text{geminal dibromide}}{RCH_2 - CH(Br)_2} \xrightarrow[MeOH]{Zn\ dust} RCH = CH_2$$

$$RCH(Br) - CH_2(Br) \xrightarrow[MeOH]{Zn\ dust} RCH = CH_2$$

(vicinal dibromide)

- *Partial reduction of alkyne*

$$RC \equiv CH + H_2 \xrightarrow[catalyst]{Lindlar's} RCH = CH_2$$

- *Kolbe's Synthesis*

$$\begin{matrix} CH_2COONa \\ | \\ CH_2COONa \end{matrix} + 2H_2O \xrightarrow{electricity} CH_2 = CH_2 + 2CO_2 + H_2 + 2NaOH$$

- *Cracking of Natural gas and Petroleum*

$$C_2H_6 \xrightarrow{875\ K} C_2H_4 + H_2$$

$$CH_3CH_2CH_3 \xrightarrow{875\ K} CH_3CH = CH_2 + CH_2 = CH_2 + CH_4 + H_2$$

Chemical Properties

- *Combustion reaction* :

$$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$$

$$C_3H_6 + \frac{9}{2}O_2 \rightarrow 3CO_2 + 3H_2O$$

- Reaction with halogen :

$$R - CH_2 - CH = CH_2 \xrightarrow{Cl_2}$$

- low temp., CCl_4 soln. → $R - \underset{Cl}{CH} - \underset{Cl}{CH_2}$ (ionic addition)
- high temp., gas phase → $RCH(Cl)CH=CH_2$ (free radical substitution)

Addition of halogen is stereoselective (trans).

- *Hydrogenation* :

$$R - CH = CH_2 \xrightarrow[200-300°C]{Ni} \underset{\text{alkane}}{RCH_2 - CH_3}$$

- *Reaction with halogen acid* :

$$RCH = CHR + HX \rightarrow RCH_2CH_2X$$

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Order of reactivity of the halogen acid is

HI > HBr > HCl > HF.

- *Markownikoff's rule:* "When addition takes place across the multiple bond of unsymmetrical alkene or alkyne, then the negative part of the addendum goes to that carbon atom, which contains lesser number of hyrogen atom(s)".
- *The peroxide effect :* The presence of oxygen or peroxides that are formed when the alkene stands exposed to the air, or added peroxides such as benzoyl peroxide, causes the addition of HBr to take place in the direction opposite to that predicted by Markownikoff's rule. HCl, HI, HF do not exhibit this abnormal reaction.

  The mechanism of the peroxide effect is a free radical chain reaction.
- *Reaction with hypohalous acid* :

$$RCH_2{=}CHR + HOX \longrightarrow \underset{\text{Halohydrin}}{R-\underset{OH}{CH}-\underset{X}{CHR}}$$

- *Hydroxylation*:

  Alkaline $KMnO_4$ oxidises alkene to glycol and pink colour fades. This is a test for unsaturation, called Baeyer test and alkaline $KMnO_4$, ($MnO_4^- + OH^-$) is called Baeyer's reagent.

$$\begin{matrix}RCH\\ \|\\ RCH\end{matrix} + MnO_4^- \longrightarrow \begin{matrix}RCH-O\\ |\\ RCH-O\end{matrix}\!\!>\!Mn\!<\!\begin{matrix}O^-\\ O\end{matrix} \xrightarrow{OH^-} \begin{matrix}RCH-O^-\\ |\\ RCH-OMnO_3H^-\end{matrix} \xrightarrow{OH^-} \begin{matrix}RCH-OH\\ |\\ RCH-OH\end{matrix}$$

- *Ozonolysis* :

$$RCH{=}CHR \xrightarrow{O_3} \underset{\text{molozonide}}{\begin{matrix}RCH-O\\ |\\ O\\ \diagdown\ \diagup\\ O\end{matrix}\!\!-\!CHR} \longrightarrow \left[\begin{matrix}RCH & \overset{-}{O}-O\\ \|\ \ \ \ \ & \\ O & \end{matrix}\!\!-\!\underset{+}{CHR}\right]$$

$$\longrightarrow \underset{\text{ozonide}}{\begin{matrix}O-O\\ \diagup\ \ \diagdown\\ RCH\ \ \ CHR\\ \diagdown\ \ \diagup\\ O\end{matrix}} \xrightarrow{Zn/AcOH} RCHO + RCHO$$

$$\begin{matrix}O-O\\ \diagup\ \ \diagdown\\ R_2C\ \ \ CHR'\\ \diagdown\ \ \diagup\\ O\end{matrix} \xrightarrow[\text{Ni etc.}]{\text{Reduction with } Zn\text{-AcOH or } H_2\text{-Raney}} R_2CO + R'CHO$$

$$\xrightarrow[\text{or } NaBH_4]{\text{Reduction with } LiAlH_4} R_2CHOH + R'CH_2OH$$

$$\xrightarrow{\text{Oxidation with } Ag_2O,\ H_2O_2 \text{ or peroxides}} R_2CO + R'COOH$$

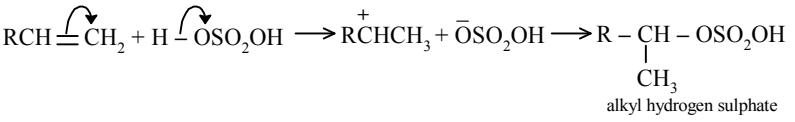
- *Hydroboration* :

$$RCH{=}CH_2 \xrightarrow{B_2H_6} (RCH_2CH_2)_3B \xrightarrow[NaOH]{H_2O_2} 3RCH_2CH_2OH + H_3BO_3$$

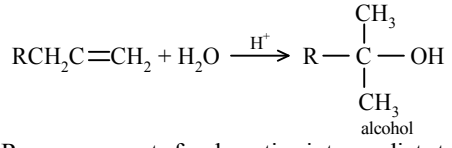
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The addition is *syn*-addition.

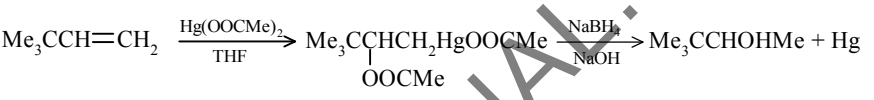
- *Reaction with conc.* H_2SO_4 :



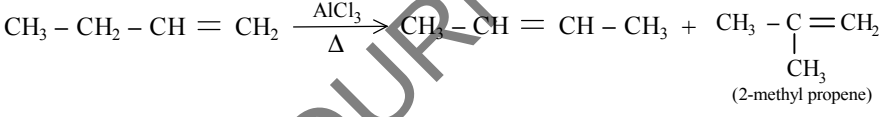
- *Hydration* :



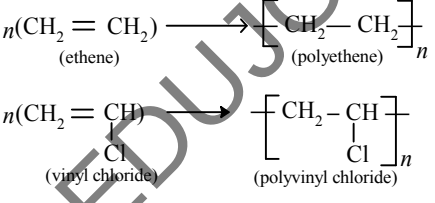
Rearrangement of carbocation intermediate to form more stable carbocation can be avoided by treatment of the alkene with mercuric acetate in THF, followed by reduction in aqueous NaOH with $NaBH_4$.



- *Isomerisation* :



- *Polymerisation* :



ALKYNES

- Alkynes have carbon-carbon triple bond.
- Alkynes have linear arrangement with *sp* hybridisation of carbon atoms, and the angle between the orbital is 180°.
- Alkynes are insoluble in water but quite soluble in the usual organic solvents of lone polarity i.e. ether, benzene, CCl_4 etc.
- Boiling point of alkynes increases with the increasing carbon number.
- Acetylene can be prepared by the controlled high-temperature partial oxidation of methane.
- Addition of acetylene to lithium amide dissolved in ether gives ammonia and lithium acetylide.
- When addition of HBr to an alkene take place in the presence of peroxide, addition occurs

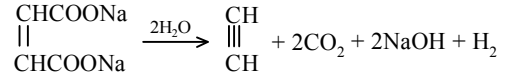
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in an anti-Markownikov manner i.e. Br is added to the carbon having large number of H-atom.

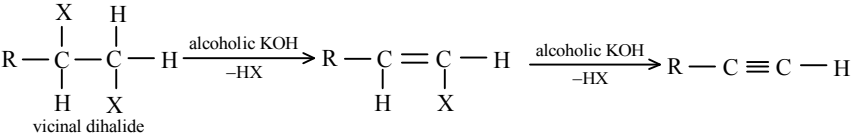
- Alkene decolourises Br_2 in CCl_4 following addition of Br_2 across double bond. This serves as a test of unsaturation. It follows anti-addition.
- The hydrogenation of alkenes is a *syn* addition carried out by many reagents such as Na in liq. NH_3 and alcohol, H_2, PtO_2/CH_3COOH, H_2/Ni at 573 K, H_2/Pd carbon in ethanol.
- Oxidation of alkenes by hot concentrated $KMnO_4$ gives acids or ketones depending upon the structure of alkenes. The terminal alkenes gives acids (ketones) and CO_2, whereas non-terminal alkenes gives mixture of acids and ketones.
- Oxidation of cycloalkenes leads to the ring opening and gives dicarboxylic acids or keto acids.
- Oxidation of alkynes with neutral $KMnO_4$ gives diketones, with acidic $KMnO_4$ and by ozone gives a mixture of acids.
- Addition of water to alkynes also follows Markownikov's rule, the hydrogen atom becomes attached to the carbon atom with the greater number of hydrogen atoms.
- Propene undergoes allylic bromination when it is treated with N-bromo succinimide (NBS) in CCl_4 in presence of peroxides or light.
- Addition of HX in cycloalkanes also follow Markownikov's addition.
- Dienes having an alternate system of single and double bonds are called **conjugated dienes**.
- Hydrocarbons containing cumulated double bonds are called allenes or **cumulated dienes**.
- Addition of Br_2 to 1,3-butadiene gives a mixture of 1,2- and 1,4-addition products.
- **Diels-Alder reaction** is an important reaction of conjugated dienes with double bonded compounds to form unsaturated cyclic compounds.
- Cycloalkanes are closed chain hydrocarbons having C_nH_{2n} as the general formula.

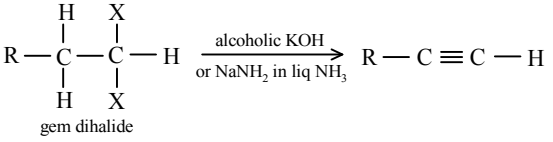
Methods of Preparation

- *From calcium carbide* : $CaC_2 + H_2O \rightarrow Ca(OH)_2 + HC \equiv CH$
- *By Kolbe's electrolysis method* :



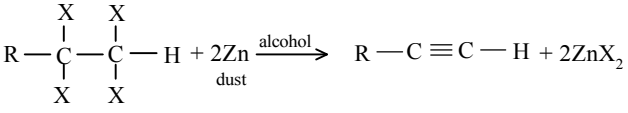
- *By dehydrohalogenation of vicinal and gem-dihalides*:



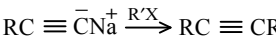


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- *By dehalogenation of tetrahalide or trihalide* :



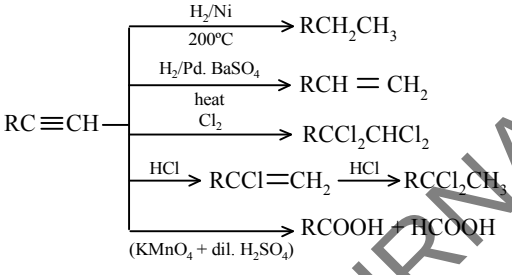
- By alkylation of acetylides :



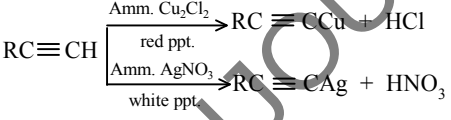
R′X should be 1° alkyl halide since higher 2° and 3° give mainly alkenes when they react with sodium salt of alkyne.

Chemical Properties

- *Addition reactions* :



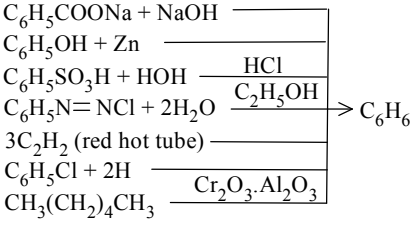
- *Substitution reactions* :



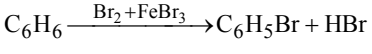
BENZENE

- Benzene is isolated from the middle oil fraction obtained from coal tar by fractional distillation.

Methods of Preparation



Chemical Properties



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$$C_6H_6 \xrightarrow{CH_3COCl + AlCl_3} HCl + \underset{\text{(Acetophenone)}}{C_6H_5COCH_3}$$

$$C_6H_6 \xrightarrow{CH_3Cl + AlCl_3} HCl + C_6H_5CH_3 \text{ (Toluene)}$$

$$C_6H_6 \xrightarrow{H_2SO_4} H_2O + C_6H_5SO_3H$$

$$C_6H_6 \xrightarrow{HNO_3 + \text{conc. } H_2SO_4} C_6H_5NO_2$$

$$C_6H_6 \xrightarrow[773\ K]{O_2 + V_2O_5} \begin{matrix} HC - CO \\ \| \\ HC - CO \end{matrix} \diagdown\!\diagup O$$

$$C_6H_6 \xrightarrow{H_2 + Ni \text{ or } Pt} C_6H_{12}$$

- γ-isomer of BHC is called gammaxene.
- *Gattermann aldehyde synthesis reaction* :
  In this reaction benzene is treated with a mixture of HCl and HCN in presence of anhydrous $AlCl_3$. The product obtained is benzaldehyde.

$$HCl + HCN \longrightarrow HN = CH - Cl$$

$$C_6H_6 + Cl - CH = NH \xrightarrow[\Delta]{AlCl_3} C_6H_5 - CH = NH + HCl$$

$$\underset{\text{(aryl imine)}}{C_6H_5 - CH = NH} \xrightarrow[\Delta]{H_2O} \underset{\text{(benzaldehyde)}}{C_6H_5 - CHO} + NH_3$$

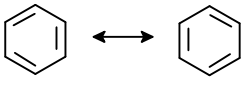
- *Chloromethylation reaction* :
  A chloromethyl group ($-CH_2Cl$) is introduced in the benzene ring by heating it with formaldehyde and HCl in presence of anhydrous $AlCl_3$ as catalyst, to form benzyl chloride. This process is called as chloromethylation.

$$\underset{\text{(benzene)}}{C_6H_6} + HCHO + HCl \xrightarrow[\Delta]{AlCl_3} \underset{\text{(benzyl chloride)}}{C_6H_5 - CH_2 - Cl} + H_2O$$

- *Ozonolysis* :

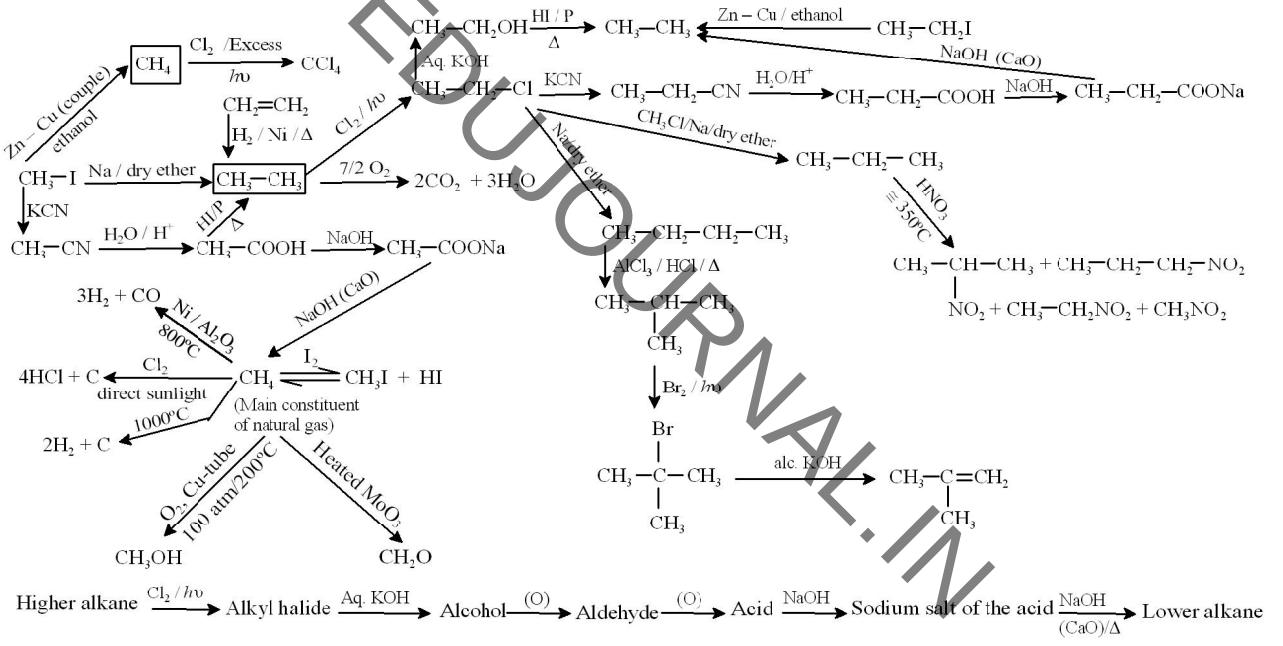
$$C_6H_6 + 3O_3 \xrightarrow{Zn/H_2O} \underset{\text{glyoxal}}{\begin{matrix} 3CHO \\ | \\ CHO \end{matrix}}$$

- When higher homologues of benzene are oxidised using $KMnO_4/OH^-$ or $Na_2Cr_2O_7/H^+$ or $KMnO_4/H^+$, the entire side chain (which contain at least one H at $\alpha - C$) is oxidised to –COOH.
- Benzene exists in the form of a resonance hybrid.



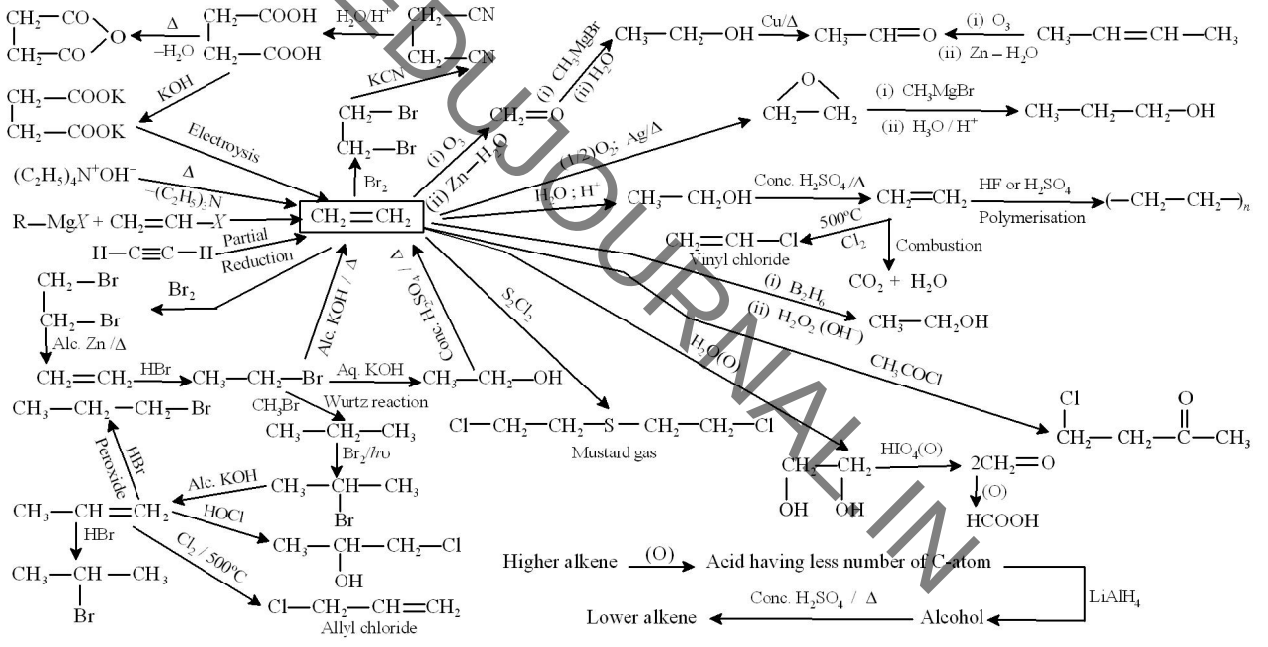
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Flow Chart for Methane and Ethane



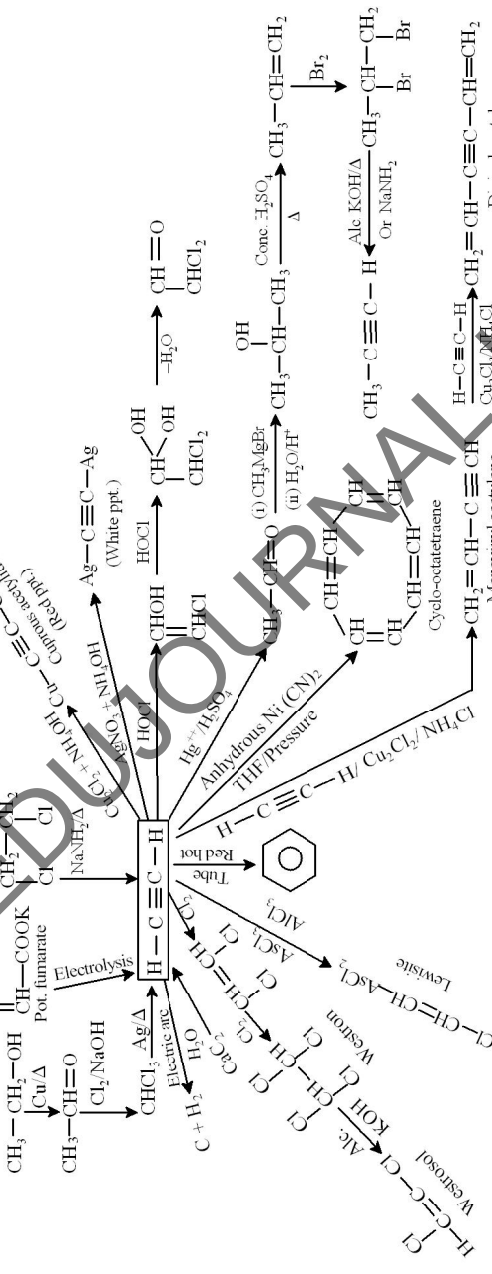
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Flow Chart for Ethylene



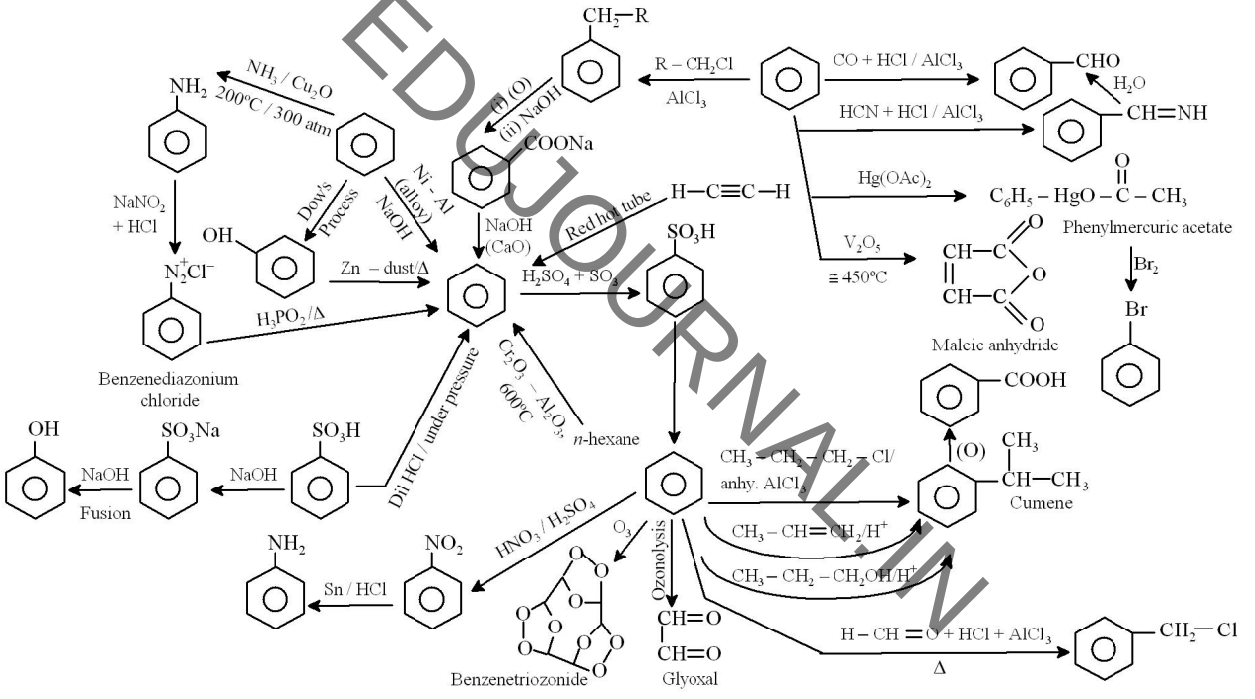
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Flow Chart for Acetylene



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Flow Chart for Benzene



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23

CHAPTER

alkyl and aryl halides

- **Halogen derivatives** are the compounds obtained by the replacement of one or more hydrogen atoms of the hydrocarbon by corresponding number of halogen atoms.
- Because of their greater molecular weight, haloalkanes have considerably higher boiling points than alkanes with the same number of carbons.
- **Haloalkanes** are insoluble in water due to the fact that they can neither form hydrogen bonds with water nor they can break the hydrogen bonds already existing between their molecules.
- Alkyl halides are quite soluble in organic solvents like alcohol, ether, acetone, chloroform, carbon tetrachloride.
- For a given halogen, the boiling point rises with increasing carbon number; as with alkanes, the boiling point rise is 20-30 degrees for each added carbon except for the very small homologs.
- Densities of alkyl halides are in the order
  $R - I > R - Br > R - Cl$.
- Density of alkyl halides goes on decreasing with the increase in the number of carbon atoms in their molecules.
- Relative reactivity of haloalkanes with respect to halogen atom is in the order :
  $RI > RBr > RCl > RF$.
- Haloalkanes undergo hydrolysis on boiling with aqueous alkali to form alcohols.
- Haloalkanes are converted to ethers with alcoholic sodium or potassium alkoxides. This reaction is called **Williamson's synthesis**.
- Haloalkanes reacts with sodium or potassium nitrate to form **alkyl nitrite**.
- Haloalkanes on being heated with an aqueous ethanolic solution of sodium or potassium hydrosulphide form thioalcohols.
- **Nitroalkanes** are formed when an aqueous ethanolic haloalkane is treated with silver nitrite.
- Haloalkanes on treatment with silver salt of a carboxylic acid in ethanol give **esters**.
- Alkyl chlorides or bromides when treated with sodium or potassium iodide in acetone undergo halogen exchange to form alkyl iodides. This is known as **Finkelstein reaction**.
- When a haloalkane is heated with concentrated alcoholic solution of KOH, a molecule of hydrogen halide is eliminated and alkene is formed. This is an example of **β-elimination reaction**.

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- For a given alkyl halide the ease of dehydrohalogenation is
  $$R - I > R - Br > R - Cl.$$
- For a given halogen, the ease of elimination is in the order
  tertiary (3°) > secondary (2°) > primary (1°) alkyl halide.
- **Saytzeff's rule** states that in case a haloalkane can eliminate hydrogen halide in two different ways, then the more substituted alkene is the major product of the dehydrohalogenation.
- Primary alkyl halides follows E2 mechanism (elimination bimolecular).
- For E2 and E1 mechanism, the order of reactivity for a given halogen follows the sequence
  E1 tertiary > secondary > primary alkyl halide
- Tertiary alkyl halides follow E1 mechanism.
- Haloalkanes react with sodium in the presence of dry ether to form alkanes. This reaction is called **Wurtz reaction**.
- E2 mechanism takes place in one step whereas E1 mechanism by two steps.
- Unsymmetrical alkenes can be obtained from alkyl halides by **Corey-House reaction** and the reaction is carried out by treating lithium dialkyl copper with an alkyl halide.
- Haloalkanes react with magnesium in presence of dry ether to form alkyl magnesium halides generally called **Grignard reagents**.
- Alkyl iodides can be easily reduced to alkanes by reducing it with HI in the presence of red phosphorus at 420 K.
- Haloalkanes can be reduced to alkanes by H_2 in presence of finely divided nickel, palladium or platinum (catalytic reduction).
- When a higher alkyl halide is heated to 570 K in the presence of a Lewis acid like anhydrous aluminium chloride which acts as a catalyst, haloalkanes undergo molecular rearrangement to give an isomeric haloalkane. This reaction is also called **isomerization** or **rearrangement reaction**.
- Alkyl halides react with benzene in the presence of a Lewis acid like anhydrous $AlCl_3$ to form homologues of benzene.
- Aryl halides are compounds containing halogen attached directly to an aromatic ring and they have the general formula ArX where Ar is phenyl, substituted phenyl or a group derived from some other aromatic system.
- Bromobenzene can be obtained from benzene diazonium chloride by treating it with CuBr dissolved in HBr.
- In **Balz-Schiemann reaction**, fluoroarenes are obtained by treating corresponding diazonium salt with fluoroboric acid (HBF_4) and it is filtered, dried and heated to get fluoroarenes.
- Aryl halides are generally colourless liquids or crystalline solids at room temperature.
- The melting and boiling points of haloarenes decrease in the following order.
  aryl iodides > bromides > chlorides > fluorides
- The melting point of *p*-isomer is generally 70-100 K higher than the melting points of *ortho* and *meta* isomer.

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- For the same halogen atom, the melting and boiling points increase as the size of the aryl group increases.
- Densities of aryl halides follow the order :
  aryl iodides > bromides > chlorides > fluorides.
- Haloarenes are soluble in organic solvents like benzene, acetone, chloroform, CCl_4 etc.
- Haloarenes are insoluble in water because they cannot form hydrogen bond with water molecules.
- **Benzonitrile** is formed when aryl bromide is heated with CuCN at 500 K in presence of pyridine or dimethyl formaldehyde (DMF).
- **Phenol** is formed when chlorobenzene is heated with an aqueous solution of NaOH at 625 K and under a pressure of 300 atmosphere.
- Electron-withdrawing groups like $-NO_2$, $-CN$, $>C=O$, $-COOH$ and $-SO_3H$, particularly when present in ortho or para position with respect to halogen activate aryl halide towards nucleophilic substitution.
- Electron-releasing groups like $-NH_2$, $-OH$, $-OR$, $-R$ etc. deactivate aryl halides towards nucleophilic aromatic substitution.
- Low reactivity of aryl halides is due to the fact that the C – X bond is far less polar in aryl halides than in alkyl halides.
- **Diaryls** are produced when aryl halides are treated with sodium in the presence of dry ether. This reaction is called Fittig reaction.
- In **Wurtz-Fittig reaction**, haloarenes are treated with an ethereal solution of an alkyl halide in the presence of sodium to form alkyl derivatives of benzene.
- When an iodoarene is heated with copper powder in a sealed tube, diaryl is formed. This is called **Ullmann reaction**.
- By the action of nickel-aluminium alloy, haloarenes can be reduced to the corresponding arenes.
- Aryl halides undergo electrophilic substitution reactions in the benzene ring such as halogenation, sulphonation, nitration and Friedel-Craft's reaction.
- Pure chloroform can be obtained by distilling chloral hydrate [$CCl_3CH(OH)_2$ or $CCl_3CHO{\cdot}2H_2O$] with concentrated aqueous NaOH solution.
- Inhalation of chloroform vapours produces loss of consciousness and is used as a general anaesthetic in surgery.
- **Chloroform** condenses with acetone in the presence of an alkali to give chloretone which is used as a hypnotic.
- Chloroform when treated with chlorine in the presence of sunlight is converted into carbon tetrachloride.
- Chloroform on warming with aniline (or any other 1° amine) and alcoholic potash gives phenyl isocyanide (phenyl carbyl amine) which has an extremely unpleasant odour and this reaction is used as a test for 1° amine. This reaction is known as **Hofmann's carbylamine reaction**.

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- Reduction of chloroform with Zn and hydrochloric acid gives dichloromethane (methylene chloride).
- **Chloroform** is a common ingredient of cough syrups.
- **Iodoform test** is used to test the presence of $CH_3-\underset{|}{CH}-OH$ or $CH_3-\overset{\overset{O}{||}}{C}-$ groups in alcohols, aldehydes or ketones.
- When a compound containing $CH_3-\underset{|}{\overset{OH}{CH}}-$ or $CH_3-\overset{\overset{O}{||}}{C}-$ is refluxed with I_2 and NaOH, a yellow precipitate of iodoform is obtained. The formation of yellow precipitate indicates the presence of either of these two groups.
- CCl_4 is insoluble in water but dissolves readily in organic solvents such as ether, alcohol etc.
- **CCl_4 vapours** are non-inflammable and is used as a fire extinguisher under the name pyrene.
- With antimony trifluoride in the presence of $SbCl_5$ as catalyst, carbon tetrachloride yields **dichlorofluoromethane (freon)**.
- CCl_4 is used as a medicine for the elimination of hook worms.
- Dichlorodifluoro methane, CCl_2F_2 is called **freon**.
- Freons are used as a **refrigerent** in domestic refrigerators and air conditioners.
- **Benzenehexachloride** is used as an insecticide and pesticide in agriculture, under the trade name **gammaxane** or **lindane** or **666**.

Methods of preparation of haloalkanes

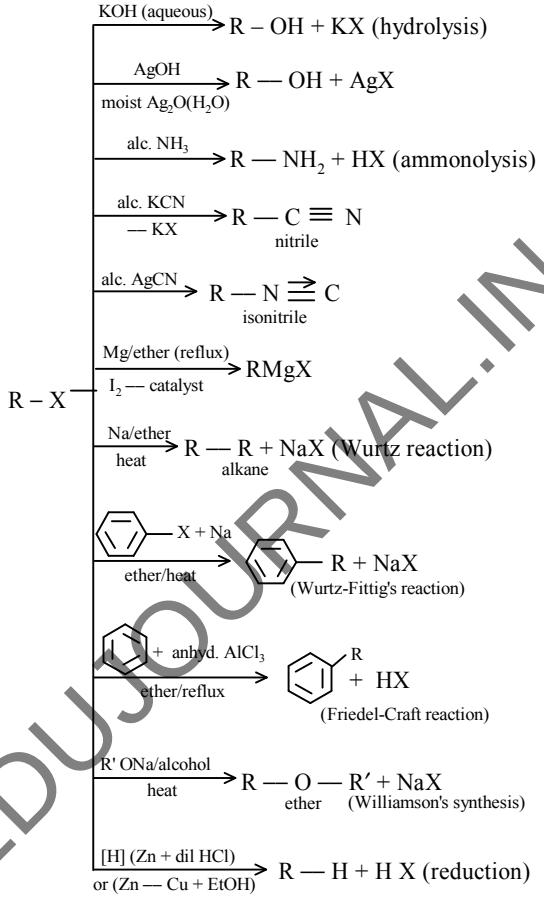
- Alkanes (RH) $\xrightarrow{\text{By halogenation}}$ $\xrightarrow{Br_2,\ AlBr_3}$ R–Br + HBr; $\xrightarrow[\text{Sunlight}]{Cl_2}$ R–Cl + HCl; $\xrightarrow[HIO_3 \text{ or } HNO_3 \text{ (oxi. agents)}]{I_2}$ R–I + HI
- Alkenes ($R-CH=CH_2$) $\xrightarrow{HX}$ $\xrightarrow[\text{Peroxide}]{}$ $R-CHX-CH_3$; $\xrightarrow{\text{Peroxide}}$ $R-CH_2-CH_2X$
- Alcohols (R — OH) $\xrightarrow{PCl_5}$ R–Cl; $\xrightarrow[PBr_3]{\text{Red P} + Br_2}$ R–Br; $\xrightarrow[PI_3]{\text{Red P} + I_2}$ R–I; $\xrightarrow{SOCl_2}$ R–Cl

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- **Properties**

Chemical Properties :

(i) *Nucleophilic substitution (S_N) reactions* :



(ii) *Dehydrohalogenation* :

$$RCH_2\underset{\displaystyle X}{\underset{|}{C}}HCH_3 + KOH \xrightarrow{\text{alcohol}} RCH{=}CHCH_3 + KX + H_2O$$

According to Saytzeff's rule, H-atom is eliminated preferentially from the adjacent C-atom which is joined to the least number of H-atoms.

Chloroform ($CHCl_3$) :

- **Preparation :** By the action of moist bleaching powder on ethanol or acetone.

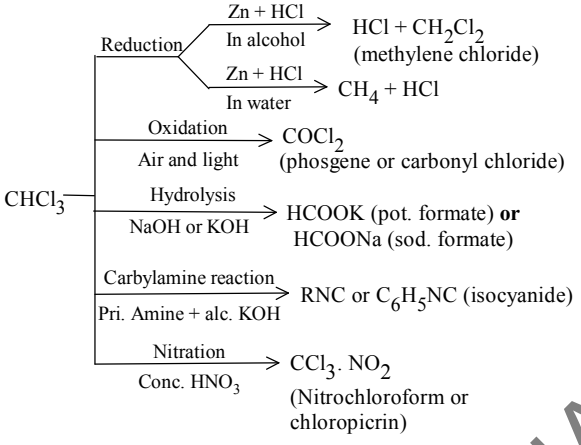
$$CH_3CH_2OH \xrightarrow{\text{Oxi.}} CH_3CHO \text{ (acetaldehyde)}$$

$$CH_3CHO \xrightarrow{\text{chlorination}} CCl_3 \cdot CHO \text{ (chloral)}$$

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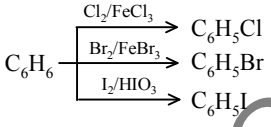
$$CCl_3CHO \xrightarrow[Ca(OH)_2]{\text{hydrolysis}} CHCl_3 \text{ (chloroform)}$$

- **Chemical Properties :**



Methods of Preparation of haloarenes

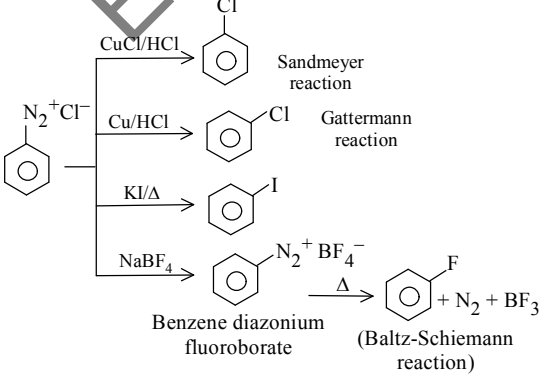
- *By direct halogenation of benzene* :



- It is an electrophilic substitution reaction.
- Low temperature and the presence of a halogen carrier favour nuclear substitution. The function of the halogen carrier is to generate the electrophile for the attack.

$$\underset{\text{Lewis acid}}{Cl_2 + FeCl_3} \rightarrow \underset{\text{electrophile}}{Cl^+ + FeCl_4^-}$$

- *From benzene diazonium salt* :



- *By Raschig process* :

$$2C_6H_6 + 2HCl + O_2 \xrightarrow[500\ K]{CuCl_2} 2C_6H_5Cl + 2H_2O$$

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- By Hunsdiecker reaction :

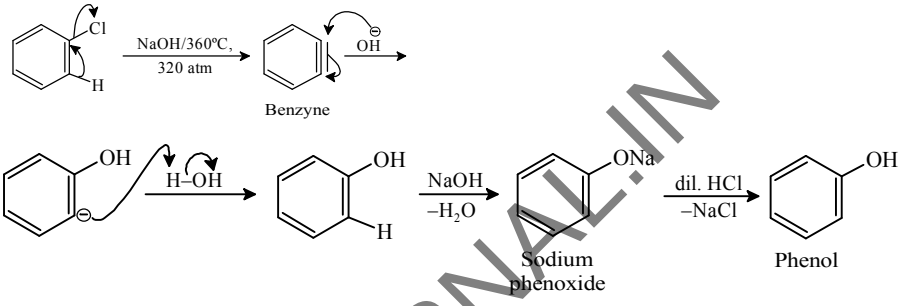
$$C_6H_5COOAg + Br_2 \xrightarrow[CCl_4,\ 350\ K]{\text{Distillation}} C_6H_5Br + AgBr + CO_2$$

Chemical Properties :

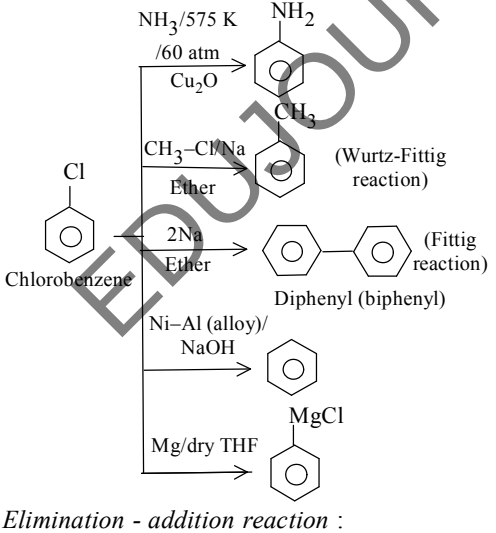
- *Nucleophilic substitution reaction of Chlorobenzene*:
  Reaction with NaOH : Dow's process

$$C_6H_5Cl + 2NaOH \xrightarrow[320\ atm]{360^oC} C_6H_5ONa \xrightarrow{\text{dil HCl}} C_6H_5OH$$

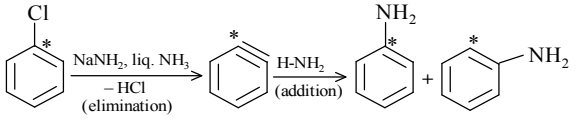
This reaction proceeds through benzyne intermediate.



- If both the *o*-position w.r.t. Cl atom is blocked, then benzyne intermediate is not obtained.



- *Elimination - addition reaction* :



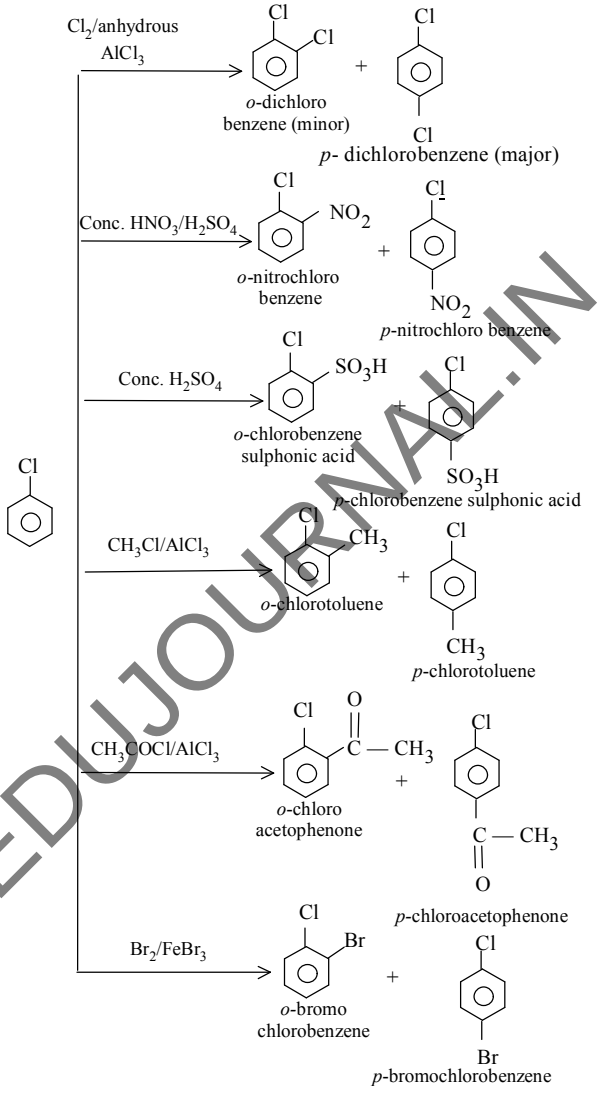
Since in this reaction, first elimination of HCl occurs and then addition of NH_3 takes place, it is called elimination-addition reaction.

Substitution at the C* that is attached to the leaving group is called direct substitution,

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substitution at the adjacent carbon is called *cine substitution*.

- *Electrophilic substitution reaction* :



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End

24
CHAPTER

alcohols, phenols and ethers

- **Alcohols** are compounds of the general formula ROH where R is any alkyl or substituted alkyl group.
- An alcohol is classified as primary, secondary or tertiary according to the kind of carbon that bears the **OH group**.
- Aliphatic hydroxy compounds in which the hydroxyl group is linked to an aliphatic carbon chain are called **aliphatic alcohols**.
- Aromatic hydroxy compounds in which the hydroxyl group is linked to the side chain of an aromatic hydrocarbon are called **aromatic alcohols**.
- An alcohol molecule is dipolar in nature with the oxygen carrying a partial negative charge (δ^-) and carbon and hydrogen each carrying a partial positive charge (δ^+).
- Alcohols are further classified as **monohydric**, **dihydric**, **trihydric** and **polyhydric** according as their molecules contain one, two or three or many hydroxyl groups respectively.
- Characteristic or functional groups of primary, secondary and tertiary alcohols are – CH_2OH, > CHOH and $\rangle C - OH$ respectively.
- The general formula of monohydric alcohols is $\mathbf{C_nH_{2n+1}OH}$ where n = 1, 2, 3 ... etc. or ROH where R is any alkyl group.
- According to common system of nomenclature, monohydric alcohols are called **alkyl alcohols**.
- According to IUPAC system, the parent structure with the longest continuous carbon chain that contains the OH group is selected.
- The carbon atom carrying the OH group gets the smallest number. The positions of other groups attached to the parent chain are indicated by suitable numbers.
- The general formula of dihydric alcohols is $(CH_2)_n(OH)_2$ where n = 2, 3, 4, ... etc.
- Dihydric alcohols are also known as **glycols** because of their sweet taste.
- In the IUPAC system, glycols are named as diols and their class name is **alkane diols**.
- In IUPAC system, trihydric alcohols are called **alkane triols**.
- Alcohols show increase in boiling point with increasing carbon number and decrease in boiling point with branching. [Reason: Alcohols like water are associated liquid and their abnormally high boiling points are due to the greater energy needed to break the hydrogen bonds that holds the molecules together].
- At ordinary temperature, lower members of alcohols are colourless liquids with distinct smell.

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- Higher members of alcohols are colourless, odourless, waxy solids.
- Amongst isomeric alcohols, the boiling points decrease with branching due to corresponding decrease in surface area. *i.e.* boiling points decrease in the order primary > secondary > tertiary.
- The lower alcohols are highly soluble in water due to the formation of hydrogen bonds between alcohol and water molecules.
- The solubility of alcohols decreases with the increase in molecular mass of the alcohol.
- Amongst isomeric alcohol, the solubility increases with branching. This is due to the reason that as the branching increases, the surface area of the non-polar hydrocarbon part decreases and the solubility increases.
- Lower alcohols form solid derivatives with metallic salts in which alcohol molecules show **solvation phenomenon**.
- Alcohols containing four or more carbon atoms exhibit chain isomerism due to difference in the nature of the carbon chain attached to the hydroxyl group.
- Alcohols containing three or more carbon atoms show position isomerism due to difference in the position of the hydroxyl group.
- Monohydric alcohols containing two or more carbon atoms show functional isomerism with ethers.
- Monohydric alcohols containing chiral carbon atoms exhibit enantiomers.
- Ethanol and methoxy methane are **functional isomers**.
- Alcohols are produced when haloalkanes (alkyl halides) are heated with aqueous sodium or potassium hydroxide or moist silver oxide.
- Reactive alkenes directly add to a molecule of water in the presence of mineral acid as a catalyst to form alcohols. The addition of water takes place in accordance with the Markownikoff's rule.
- In **hydroboration-oxidation** reaction, alkene is treated with diborane followed by treatment with water in the presence of H_2O_2 when alcohols is formed.
- Alkenes react with mercuric acetate, $(CH_3COO)_2Hg$ to form adducts which upon reduction with $NaBH_4$ in basic medium give alcohols. This two-step process is called oxymercuration-reduction or oxymercuration-demercuration and gives alcohols corresponding to Markownikoff's addition of water to alkenes.
- The reduction of aldehydes, ketones and esters with sodium and alcohol is commonly known as **Bouveault-Blanc reduction**.
- Grignard reagents react with aldehydes, ketones and esters to form addition products which upon decomposition with water or preferably with dilute HCl or dilute H_2SO_4 give alcohols.
- The process of breaking down large molecules into simpler ones in the presence of enzymes is called **fermentation**.
- Alcohols on heating with conc. H_2SO_4 at 435-445 K or phosphoric acid at 495-500 K are converted into alkenes on dehydration.
- The acidic character of alcohols is due to the electronegative oxygen atom which withdraws the electrons of the O – H bond towards itself.

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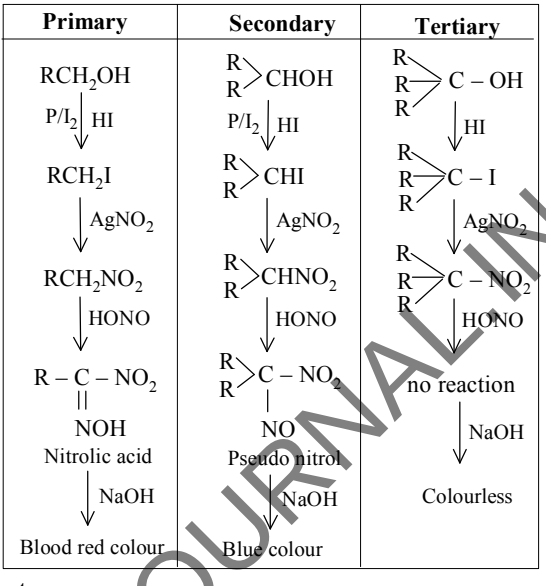
- Alcohols are weak acids (K_a = 1 × 10^{-16} - 10^{-18}) even weaker than water (K_w = 1 × 10^{-14}).
- Electron releasing inductive effect of the alkyl group makes the alcohols weaker acids than water.
- The acidic strength of alcohols follows the order : primary > secondary > tertiary.
- In esterification reaction, the order of reactivity of alcohols follows the order $CH_3OH > CH_3CH_2OH > (CH_3)_2CHOH > (CH_3)_3COH$ and that of carboxylic acid follows the order $HCOOH > CH_3COOH > (CH_3)_2CHCOOH > (CH_3)_3CCOOH$.
- Both alcohols and phenols react with Grignard reagents to form hydrocarbons. This reaction is called **Zerewitinoff's active hydrogen determination**.
- Strong bases like metal hydrides and metal amides react with alcohol to give H_2 and NH_3 respectively.
- The order of reactivity of alcohols towards HX is 3° > 2° > 1°.
- The order of reactivity of alcohols in the reactions involving the cleavage of C – OH bond follows the sequence : tertiary > secondary > primary.
- The order of reactivity of halogen acids with alcohols follows the sequence :

  HI > HBr > HCl.

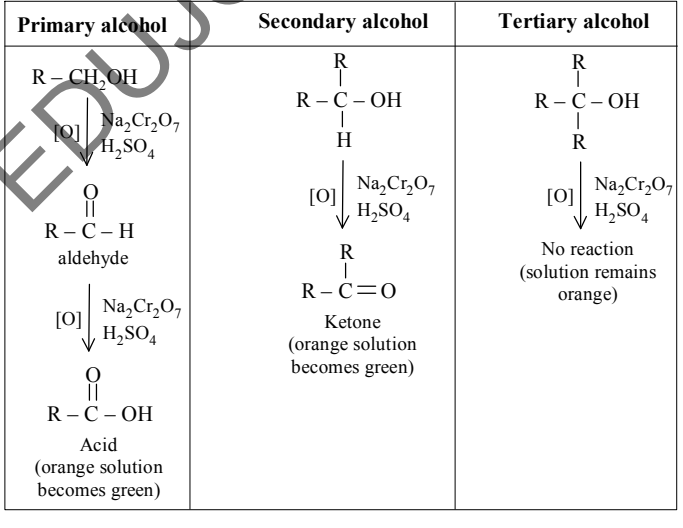
  [I^- is a better nucleophile than Br^- which in turn is better than Cl^- ion].
- The order of reactivity of alcohols differ widely in ease of dehydration. Ease of dehydration of alcohols - 3° > 2° > 1°.
- Medically ethanol is classified as a hypnotic (sleep producer), it is less toxic than other alcohols.
- Nearly all the ethanol used is a mixture of 95% alcohol and 5% water.
- The least reactive of the hydrogen halides, HCl requires the presence of anhydrous zinc chloride for reaction with primary and secondary alcohols.
- No catalyst is needed in the reactions of HCl with tertiary alcohols.
- The dehydration of 2° and 3° alcohols occur in accordance with the **Saytzeff rule** *i.e.* the more highly substituted alkene is always the major product.
- If the major product obtained due to the dehydration of alcohols in accordance with Saytzeff's rule is capable of showing *cis-trans* isomerism, then it is always the *trans*-product which predominates.
- The oxidation of an alcohol involves the loss of one or more hydrogens (α-hydrogens) from the carbon bearing the – OH group.
- A primary alcohol contains two α-hydrogens and can either lose one of them to form an aldehyde or both of them to form a carboxylic acid.
- A secondary alcohol can lose its only α-hydrogen to form a ketone.
- A tertiary alcohol contains no α-hydrogen and is not oxidised.
- One of the best and most convenient reagent used for the conversion of primary alcohols to aldehydes is **pyridinium chlorochromate ($C_5H_5NH^+CrO_3Cl^-$)**.

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- **Victor Meyer** test is based on the different behaviour of primary, secondary and tertiary nitroalkanes towards nitrous acid.
  (a) Primary alcohols produce a blood red colour.
  (b) Secondary alcohols produce blue colour.
  (c) Tertiary alcohols produce no colour.



- **Dichromate test :**



- Lucas reagent is a solution of HCl with $ZnCl_2$. With Lucas reagent
  Primary alcohol – no cloudiness
  Secondary alcohol – cloudiness in 5 minutes
  Tertiary alcohol – cloudiness immediately

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- **Methanol** is also called wood spirit since originally it was obtained by the destructive distillation of wood.
- Drinking of methyl alcohol causes blindness.
- Denatured alcohol is commonly known as **methylated spirit**.
- **Methanol** is used as an antifreeze for automobile radiators.
- **Ethanol** is used as power alcohol- a mixture of 20% absolute alcohol and 80% petrol (gasoline) with benzene or tetralin as cosolvent.
- **Methanol** is used for the manufacture of formaldehyde which is used in the manufacture of formaldehyde resins such as manufacture of bakelite, melamine-formaldehyde, urea-formaldehyde.
- Hydroxylation of a double bond can be achieved by the action of osmium tetroxide (OsO_4) and the cyclic osmate ester thus formed on decomposition with ethanolic Na_2SO_3 solution gives glycols in quantitative yield.
- Conversion of ethylene into ethylene glycol by the action of cold dilute alkaline $KMnO_4$ is called **hydroxylation**.
- When vapours of alcohols are passed over heated copper at 573 K, 1° alcohols give aldehydes, 2° alcohols give ketones and 3° alcohols give alkenes.
- Methylated spirit or denatured alcohol is obtained by adding methyl alcohol, acetone and pyridine to alcohol to make it unfit for drinking purposes.
- A cold dilute alkaline $KMnO_4$ solution is called **Baeyer's reagent**.
- **Rectified spirit** contains 96.5% alcohol and 4.4% water and is obtained by fermentation of carbohydrates.
- Ketones are reduced to secondary alcohols by aluminium isopropoxide in isopropyl alcohol. The reduction by this method is known as **Meerwein-Ponndorf-Verley (MPV) reduction** and is considered as an important method for fermentation of secondary alcohols.
- Cycloalkanols in presence of 50% HNO_3 at 55°C undergo cleavage forming dioic acids.
- Ethane-1,2-diol undergoes extensive intermolecular hydrogen bonding because of the presence of two – OH groups in its molecule.
- Due to extensive intermolecular hydrogen bonding, the boiling point of ethane –1,2-diol is quite high (470 K).
- With aldehydes and ketones in presence of *p*-toluenesulphonic acid (PTS) as catalyst, ethylene glycol gives cyclic acetals and cyclic ketals (1,3-dioxolanes) respectively.
- Ethylene glycol on oxidation with conc. HNO_3 mainly gives glycolic acid and oxalic acid.
- When ethylene glycol is treated with HIO_4 or lead tetra-acetate, carbon-carbon bond fission occurs to give formaldehyde.
- The per-iodic acid cleavage of 1,2-glycols is sometimes called as **Malaprade reaction**.
- **Ethylene glycol** is used for preparing 1,4-dioxane and polyethylene glycols which are used as industrial solvents.

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- **Glycerol** or glycerine occurs in almost all vegetable and animals oils and fats which are the triesters of glycerol with long chain fatty acids.
- Glycerol undergoes extensive intermolecular H-bonding because of the presence of three – OH groups.
- Due to extensive intermolecular hydrogen bonding, the boiling point of glycerol is quite high (563 K) even higher than that of ethylene glycol.
- Glycerol is miscible with water and alcohol in all proportions.
- A mixture of glyceryl trinitrate and glyceryl dinitrate absorbed on Kieselguhr is called **dynamite**.
- When glycerol is treated with a small amount of HI or PI_3, allyl iodide is formed.
- The smokeless powder cordite is a mixture of nitroglycerine, gun cotton and vaselin.
- Nitration of glycerol with a mixture of conc. HNO_3 + conc. H_2SO_4 gives **nitroglycerine**.
- When heated with acidified $KMnO_4$ solution, glycerol gets oxidised to oxalic acid, carbon dioxide and water.
- **Glycerol** is used as an antifreeze in automobile radiators.
- Ethanol is the only primary alcohol that gives iodoform test.
- **Isopropyl iodide** is obtained when glycerol is heated with excess PI_3.
- Haloform reaction does not take place with methanol.
- Acid catalysed dehydration of *t*-butanol is faster than *n*-butanol since *t*-butyl carbocation is more stable than *n*-butyl carbocation.
- **Phenols** are compounds of the general formula ArOH, where Ar is phenyl, substituted phenyl or some other aryl group (*e.g.* naphthyl).
- Phenol has a smaller dipole moment (1.54 D) than methanol because the C – O bond in phenol is less polar due to the electron withdrawing effect of the benzene ring while in methanol, C – O bond is more polar due to electron donating effect of the methyl group.
- The simplest phenols are liquids or low-melting solids.
- Phenols have quite high boiling points because of intermolecular hydrogen bonding.
- Phenol is soluble in water (9 g per 100 g) because of hydrogen bonding with water.
- *o*-Nitrophenol has a lower boiling point and it is steam volatile than the *m*- and *p*-isomers because *o*-nitrophenol exists as discrete molecules and cannot form H-bonds with water.
- Most phenols have K_a values in the neighbourhood of 10^{-10} and are thus considerably weaker acids than the carboxylic acids (K_a values about 10^{-5}).
- Phenols are produced when sodium salts of aromatic sulphonic acids are fused with NaOH at 300-350°C followed by acidification.
- In **Dow's process**, phenol is obtained when chlorobenzene is heated with 6-8% NaOH solution at 623 K under 300 atmospheric pressure.
- Phenols turn reddish brown due to atmospheric oxidation.
- Phenols are stronger acids than alcohol because the phenoxide ion left after the release of a proton is stabilised by resonance but the alkoxide ion is not.
- *o*-Nitrophenol is less acidic than *p*-nitrophenol due to intramolecular H-bonding which makes loss of a proton difficult.

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- Greater the number of electron withdrawing groups at the *o*- and *p*- positions more acidic is the phenol.
- Acidity of nitrophenols with respect to phenol decreases in the order : 2,4,6-trinitrophenol > 2,4-dinitrophenol > 4-nitrophenol or 2-nitrophenol > phenol.
- Electron donating group donates electrons, intensifies the negative charge, destabilizes the phenoxide ion with respect to phenol and thus decreases the acid strength.
- Acidic strength of cresols (alkyl phenols) decreases in the order : *m*-cresol > *p*-cresol > *o*-cresol.
- Due to –I effect of the halogen, all halophenols are more acidic than phenol.
- Acidity of all the *o*-halophenols decreases in the order: *o*-chlorophenol > *o*-bromophenol > *o*-iodophenol > *o*-fluorophenol.
- In *p*-fluorophenol, +R effect and –I effect of F almost balance each other and hence it is acidic as phenol itself.
- Phenols are soluble in aqueous solutions of NaOH or KOH since phenols react with alkalis (NaOH or KOH) to form salt and water.
- Electron-attracting substituents tend to disperse the negative charge of the phenoxide ion whereas electron-releasing substituents tend to intensify the charge.
- When esters of phenols are heated with aluminium chloride, the acyl group migrates from the phenolic oxygen to an *ortho* and *para* position of the ring thus yielding a ketone. This reaction is called **Fries rearrangement**.
- Treatment of phenol with chloroform and aqueous hydroxide introduces an aldehyde group – CHO, onto the aromatic ring generally *ortho* to the – OH. This reaction is known as the **Reimer-Tiemann reaction**.
- Phenol reacts with Grignard reagent to form hydrocarbons.
- Benzoylation of phenols in the presence of aq. NaOH is known as **Schotten Baumann reaction**.
- In **Kolbe's Schmidt reaction**, sodium phenoxide is heated with CO_2 at 390-410 K and at a pressure of 4-7 atmospheres sodium salicylate is formed as the major product.
- Sodium phenoxide when heated with CO_2 at 400 K under a pressure of 4-7 atmospheres followed by acidification gives salicylic acid. This reaction is known as **Kolbe's reaction**.
- Salicylic acid reacts with phenol in presence of $POCl_3$ to form **phenyl salicylate** (salol) which is used as an internal antiseptic.
- Phenol condenses with pthalic anhydride in presence of conc. H_2SO_4 to form phenolphthalein which is widely used as an indicator in acid-alkali titrations.
- Salicylic acid on acetylation with acetic anhydride in presence of CH_3COONa or a few drops of conc. H_2SO_4 gives **aspirin** which is used as an internal antiseptic.
- Phenol as such or its trichloroderivative i.e. **trichlorophenol** or TCP is used as a preservative for ink and other water-based colours.
- Phenol is used in the manufacture of drugs like salicylic acid, phenacetin, aspirin, salol etc.
- Ethers are compounds having general formula R – O – R′, Ar – O – R or Ar – O – Ar. [Ar is phenyl or some other aromatic group].

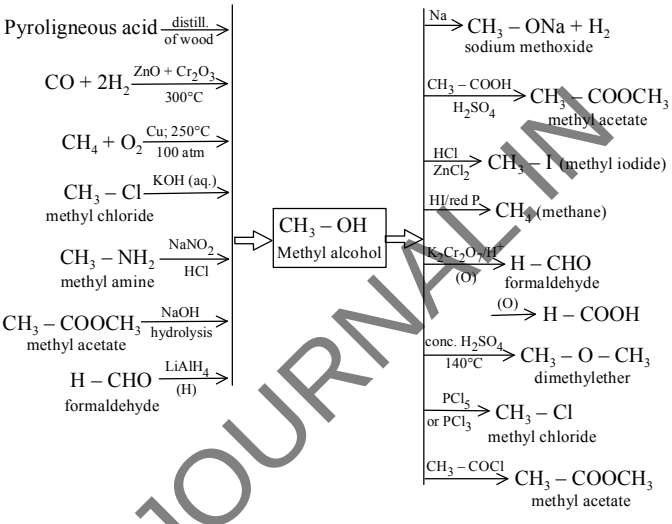
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- Ethers in which the groups R and R′ are same are called simple or symmetrical ethers while those in which the groups R and R′ are different are called **mixed** or **unsymmetrical ethers**.
- Ethers like water have a tetrahedral geometry *i.e.* oxygen is sp^3-hybridised.
- The C – O – C angle in ethers has been found to 110°.
- Methyl phenyl ether is called **anisole** and ethyl phenyl ether is called **phenetole**.
- Ethers having the same alkyl groups on either side of the oxygen atom but different arrangement of the carbon chain within the alkyl groups are called **chain isomers**.
- Ethers having the same molecular formulae but different alkyl groups on either side of the oxygen atom are called **metamers**.
- **Williamson's synthesis** of ethers involves the treatment of an alkyl halide with a suitable sodium alkoxide.
- Williamson's synthesis involves the nucleophilic displacement of the halide ion from the alkyl halide by the alkoxide ion by S_N2 mechanism.
- Ethers may be prepared by dehydration of alcohols either in the presence of acids or heated alumina.
- Dehydration of *tert*-butyl alcohol with conc. H_2SO_4 at 415 K yields only isobutylene.
- Due to the bent structure of ethers and polarity of C – O bond, ethers have a net dipole moment. i.e. ethers are polar in nature.
- Ethers have lower boiling points as compared to isomeric alcohols because of the fact that ethers do not form hydrogen bonds.
- Order of dehydration of alcohols leading to the formation of ethers is :
  primary > secondary > tertiary.
- The solubility of lower ethers in water is due to the formation of hydrogen bonds between water and ether molecules.
- All ethers are lighter than water.
- Ethers are inert compounds due to the reason that the functional group of ethers (– O –) does not contain any active site in their molecule.
- Ethers behave as Lewis bases on account of the presence of two lone pairs of electrons on the oxygen atom.
- Being Lewis base, ethers form coordinate complexes known as etherates with Lewis acids such as BF_3, $AlCl_3$, $FeCl_3$, Grignard reagent etc.
- Ethers dissolves in cold concentrated inorganic acids to form stable oxonium salts.
- Ethers are cleaved at C – O bond by hydroiodic acid or conc. hydrobromic acid when heated to 370 K. The reaction of hydroiodic acid with ethers forms the basis of **Zeisel's method** for the estimation of alkoxy groups such as methoxy, ethoxy etc.
- When exposed to air and light for a long time, ethers are oxidised to form peroxides.
- **Dimethyl ether** is used as a refrigerent and as a solvent at low temperature.
- Diethyl ether is used as an anaesthetic in surgery.
- A sample of diethyl ether free of all traces of water and alcohol is called absolute ether.
- Order of reactivity of halogen acids towards ether is HI > HBr > HCl

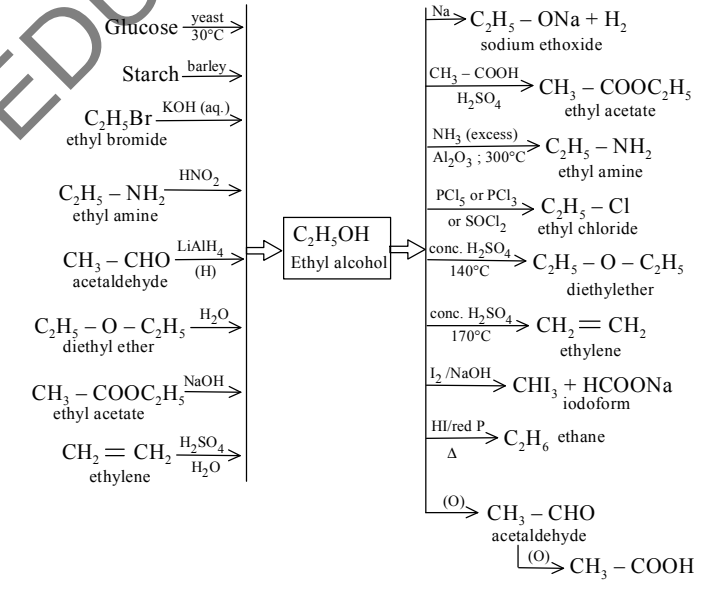
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- On heating with dilute H_2SO_4 under pressure ethers are hydrolysed to alcohols.
- When heated with conc. H_2SO_4, ethers form alcohols and alkyl hydrogen sulphates.
- Acid chlorides react with ethers when heated in the presence of anhydrous $ZnCl_2$ or $AlCl_3$ to form alkyl halides and esters.
- Ethers are used as a reaction medium for carrying out lithium aluminium hydride reduction and also for the preparation of Grignard and other organometallic reagents.

Flow chart for Methyl alcohol

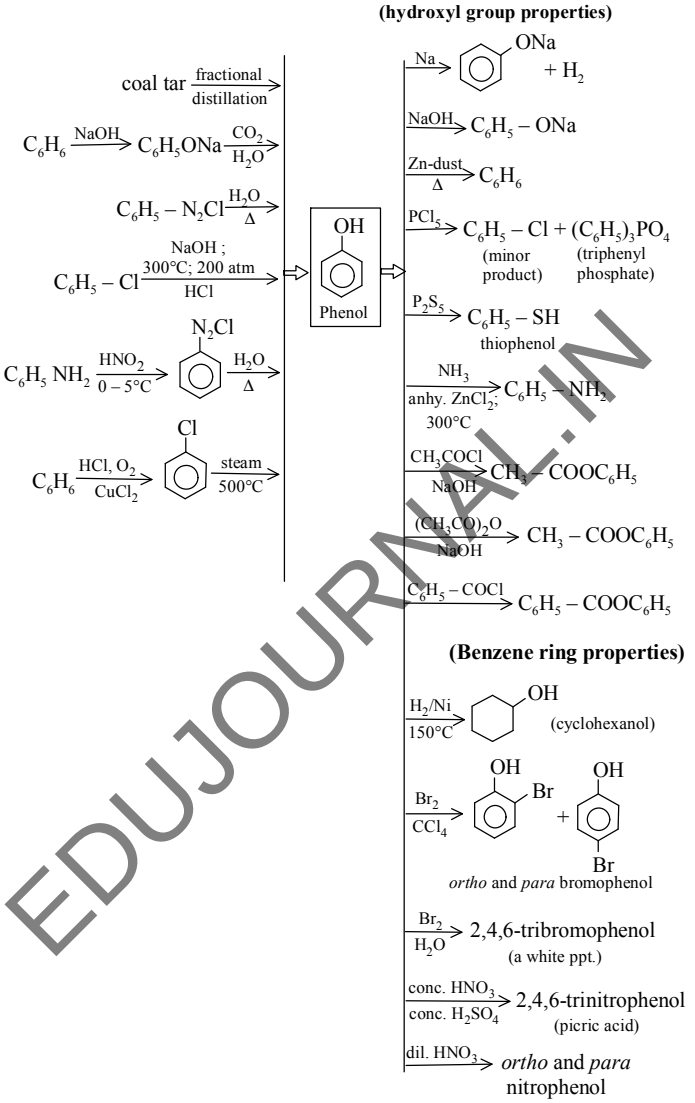


Flow chart for Ethyl alcohol



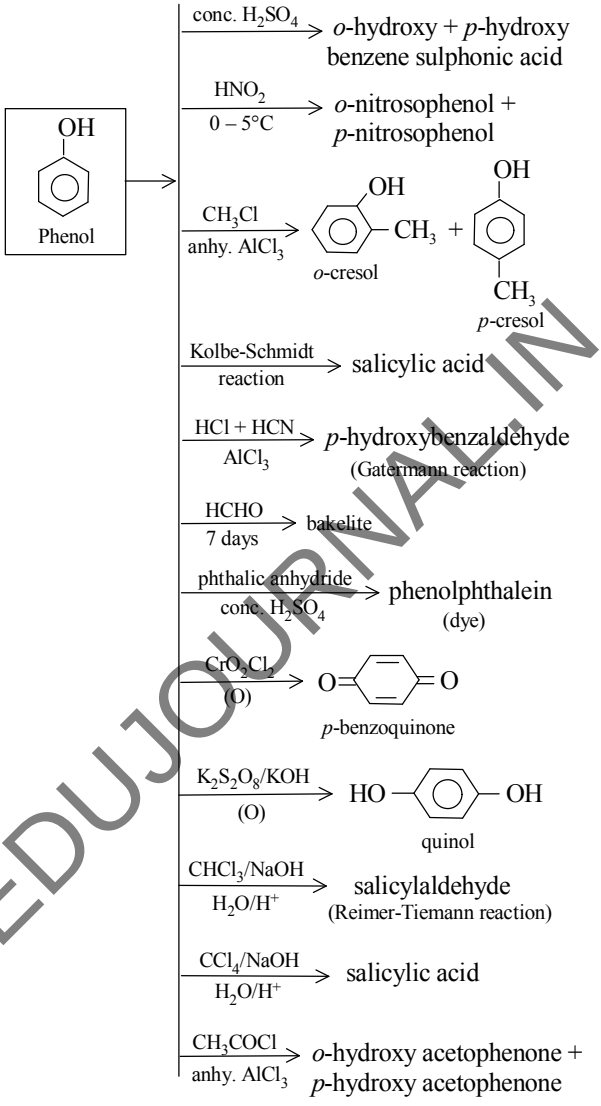
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Flow chart for Phenol



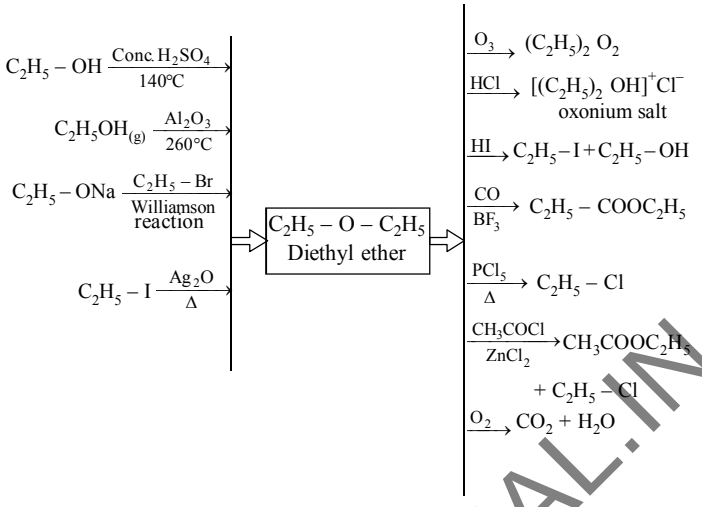
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Properties of Phenol



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Flow chart for Diethyl ether



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End

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CHAPTER

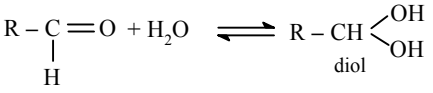
aldehydes and ketones

- Aldehydes are compounds of the general formula **RCHO** and ketones are compounds of the general formula **RR′CO**.
- Both aldehydes and ketones contain the carbonyl group $>C=O$ and are called **carbonyl compounds**.

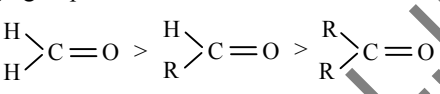
  The carbonyl group is polar in nature and carbonyl carbon is sp^2 hybridised.
- Aldehydes are prepared by the controlled oxidation of 1° alcohols using acidified potassium permanganate or acidified potassium dichromate.
- **Oppenauer oxidation** of 2° alcohols (saturated or unsaturated) with aluminium tert-butoxide in presence of excess of acetone gives ketones in good yields without the danger of being further oxidised to carboxylic acids.
- **Collin's reagent** ($CrO_3 \cdot 2C_5H_5N$) and pyridinium chlorochromate (PCC, $CrO_3 \cdot C_5H_5N$) are better oxidising agents than $K_2Cr_2O_7/H_2SO_4$ or $KMnO_4/KOH$ for converting 1° alcohols to aldehydes since these reagents oxidise 1° alcohols to the corresponding aldehydes and the aldehydes formed are not further oxidised to the carboxylic acids.
- Reduction of acid chlorides with H_2 in presence of Lindlar's catalyst (Pd deposited over $BaSO_4$ and partially poisoned by addition of S or quinoline) gives aldehydes - **Rosenmund reduction**.
- The reaction of acid chloride with dialkyl cadmium gives ketones.
- **Friedel-Craft's reaction** involves the treatment of an aromatic hydrocarbon with an acid chloride or acid anhydride in the presence of a Lewis acid such as anhydrous aluminium chloride in dry ether.
- **Etard's reaction** involves the oxidation of toluene with CrO_2Cl_2/CS_2 followed by decomposition of the complex thus formed with water.
- Friedel Craft's acylation of arenes with acid chlorides and anhydrides give ketones.
- **Reductive ozonolysis of alkenes** give aldehydes or ketones depending upon the structure of alkene.
- **Lithium organocuprates** react readily with acid chlorides to yield ketones.
- The boiling points of aldehydes and ketones are higher than those of hydrocarbons of comparable molecular masses.
- Among isomeric aldehydes and ketones, ketones have higher boiling points.
- **Benzaldehyde** has a smell of bitter almonds.
- Solubility of aldehydes and ketones in water decreases with the increase in molecular mass.

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- Diol formation of aldehydes with water also helps in solubility in water.



- Boiling points of ketones are slightly higher than those of isomeric aldehydes because ketones are relatively more polar than their corresponding isomeric aldehydes due to the presence of two electron repelling alkyl groups around the carbonyl carbon.
- Aldehydes and ketones can be reduced to hydrocarbons by the action of amalgamated zinc and concentrated hydrochloric acid - the Clemmensen reduction and of hydrazine NH_2NH_2, and a strong base like KOH or potassium tert-butoxide, the **Wolf Kishner reduction**.
- Aldehyde are more reactive than ketones and the order of reactivity is based upon the +I effect of the alkyl group as follows.



- As the size of the alkyl group increases, the reactivity decreases. The order of reactivity of various ketones is as follows.
  $CH_3COCH_3 > CH_3CH_2COCH_3 > CH_3CH_2COCH_2CH_3$
- Presence of electron attracting group on carbonyl compounds increases positive charge on carbonyl carbon due to –I effect and increases the reactivity. The order of reactivity of substituted aldehydes is as follows:
  $NO_2CH_2CHO > ClCH_2CHO > CH_3CHO$
- Aromatic aldehydes are more reactive than alkyl-aryl ketones which in turn are more reactive than diaryl ketones.
- The elements of HCN add to the carbonyl group of aldehydes and ketones to yield compounds known as **cyanohydrins**.
- Alcohols add to the carbonyl group of aldehydes in presence of anhydrous acids to yield **acetals**.
- In the presence of concentrated alkali, aldehydes containing no α-hydrogen undergo self-oxidation and reduction to yield a mixture of an alcohol and a salt of carboxylic acid.
- Aldehydes and ketones react with hydroxyl amine to form **oximes**.
- Aldehydes and ketones react with hydrazine forming **hydrazones**.
- Aldehydes and ketones react with semicarbazide to form **semicarbazones**.
- Aldehyde reduce Tollen's reagent and Fehling's solution.
- **Tollen's reagent** is an ammoniacal solution of silver nitrate and Fehling's solution is an alkaline solution of $CuSO_4$ containing some **Rochelle salt** *i.e.* sodium potassium tartarate.
- Ketones cannot be oxidised by weak oxidising agents such as Tollen's reagent, Fehling solution and these reagents are used for distinguishing aldehydes from ketones.
- Ketones are oxidised by strong oxidising agents like conc. HNO_3, $KMnO_4/H_2SO_4$, $K_2Cr_2O_7/H_2SO_4$.
- During oxidation of unsymmetrical ketones, the point of cleavage is such that keto group stays preferentially with the smaller alkyl group (**Popoff's rule**).

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- Two molecules of an aldehyde or a ketone having atleast one α-hydrogen atom condense in the presence of dilute alkali to form a β-hydroxy aldehyde or a β-hydroxy ketone. The reaction is called **aldol condensation**.
- Aldehydes and ketones containing α-hydrogen atoms undergo halogenation when treated with halogens in the presence of an acid or a base.
- Aldehydes and ketones like formaldehyde, benzaldehyde and benzophenone do not undergo aldol condensation since they do not have any α-hydrogen.
- In general a mixture of two aldehydes undergoes a Cannizzaro reaction to yield all possible products. If one of the aldehyde is formaldehyde, the reaction yields almost exclusively sodium formate and the alcohol corresponding to the other aldehyde such a reaction is called a crossed **Cannizzaro reaction**.
- The reaction between an aldehyde or a ketone with a phosphorus ylide to give a substituted alkene is called the **Wittig reaction** and the phosphorus ylide is commonly called the **Wittig reagent**.
- On heating with an ethanolic solution of KCN, two molecules of an aromatic aldehyde condense to form **benzoin**.
- Aromatic ketones do not form addition products with sodium bisulphite due to **steric hindrance**.
- Aldol condensation can also take place between two different aldehydes or ketones or between one aldehyde and one ketone.
- When aldehydes are treated with **Schiff's reagent**, its pink or magenta colour is restored and this reaction is used as a test for aldehydes because ketones do not restore the pink colour of Schiff's reagent.
- **Schiff's reagent** is an aqueous solution of magenta or pink coloured rosaniline hydrochloride which has been decolourised by passing SO_2.
- Formaldehyde reacts with ammonia to form **hexamethylene tetraamine**.
- Hexamethylene tetraamine is used as a urinary antiseptic under the name **urotropine**.
- Aldehydes and ketones react with primary amines in the presence of trace amount of an acid to form **azomethines** or **Schiff's bases**.
- **Perkin's reaction** involves heating of an aromatic aldehyde with an acid anhydride and its corresponding sodium salt.
- **Benzaldehyde** is prepared by the dry distillation of a mixture of calcium benzoate and calcium formate.
- Benzaldehyde on oxidation with alkaline potassium permanganate gives benzoic acid.
- Nitration of benzaldehyde with a mixture of conc. nitric acid and sulphuric acid gives *m*-nitrobenzaldehyde.
- **Paraldehyde** is used in medicine as hypnotic.
- When a few drops of conc. H_2SO_4 are added to acetaldehyde at room temperature, a rapid exothermic reaction occurs and a cyclic trimer called paraldehyde is formed.
- **Formaldehyde** is used in leather industry for tanning hides and as a reducing agent in silvering of mirrors and decolourising vat dyes.
- A 40% solution of formaldehyde in water is called **formalin** and is used for the preservation of biological or anatomical specimens.

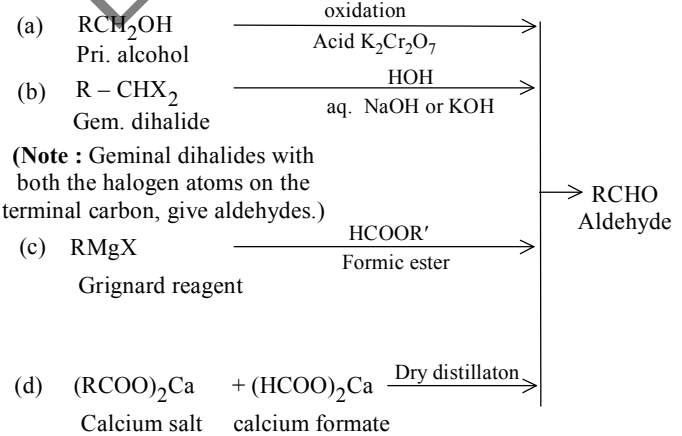
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- Formaldehyde is used in the manufacture of bakelite, resins and other polymers.
- When gaseous formaldehyde is allowed to stand it gives trioxane or **metaformaldehyde**.
- Nitration of hexamethylene tetraamine under controlled conditions gives the well known explosive **RDX (research and development explosive)**.
- Benzaldehyde reacts with ammonia to form a complex products called **hydrobenzamide**.
- A base-catalysed crossed aldol condensation between an aromatic aldehyde and an aliphatic aldehyde or a ketone is called **Claisen-Schmidt condensation** or simply **Claisen reaction**.
- Even aliphatic esters containing α-hydrogen atoms undergo Claisen-Schmidt condensation on treatment with an aromatic aldehyde in presence of a base.
- Ketones on reduction with magnesium amalgam and water form pinacole.
- All aldehydes and ketones having CH_3CO – group (attached either to H or to C) on treatment with an excess of halogen in presence of alkali (i.e. sodium hypohalite, NaOX) give haloform ($CHCl_3$, $CHBr_3$, CHI_3) and the salt of a carboxylic acid having one carbon atom less than the original aldehyde or ketone. This reaction is known as **haloform reaction**.
- When I_2 is used as the halogen in haloform reaction, yellow ppt. of iodoform are formed and the test is called **iodoform test**.
- **$LiAlH_4$** reduces aldehydes, ketones, acids, esters, amides and nitriles.
- Chain isomerism is exhibited by aldehydes containing four or more carbon atoms and ketones having five or more carbon atoms.
- Higher ketones and aromatic aldehydes show position isomerism.
- Aldehydes and ketones show functional isomerism among themselves but also with alcohols and ethers which may be either cyclic or acyclic.
- Aldehydes and ketones with a methyl or methylene group adjacent to $>C=O$ group on oxidation with selenium dioxide (SeO_2) at room temperature, give α-dicarbonyl compounds.
- Ketones can be reduced to corresponding secondary alcohols with aluminium isopropoxide in isopropyl alcohol.
- Formaldehyde does not give iodoform test.
- The acidity of α-hydrogens is partly due to the –I-effect of the carbonyl group which weakens the C^{α} – H bond and partly due to the resonance stabilisation of the resulting carbanion.
- The β^-, γ^-, δ^- ... etc. hydrogens are not acidic because the inductive effect decreases with distance and the resulting carbanions are not stabilised by resonance.
- Electrophilic substitution reactions in aromatic aldehydes and ketones occur at the *m*-position.
- Acetaldehyde readily dissolves in water, alcohol and ether in all proportions.
- **Acetaldehyde** is used in silvering of mirrors.
- When distilled with conc. H_2SO_4, acetone gives mesitylene *i.e.* 1,3,5-trimethyl benzene.
- **Acetone** is used as one of the constituents of liquid nail polish.
- **Mesityl oxide** (4-methylpent-3-en-2-one) is formed when two molecules of acetone in the presence of HCl combine with the elimination of one molecule of water.

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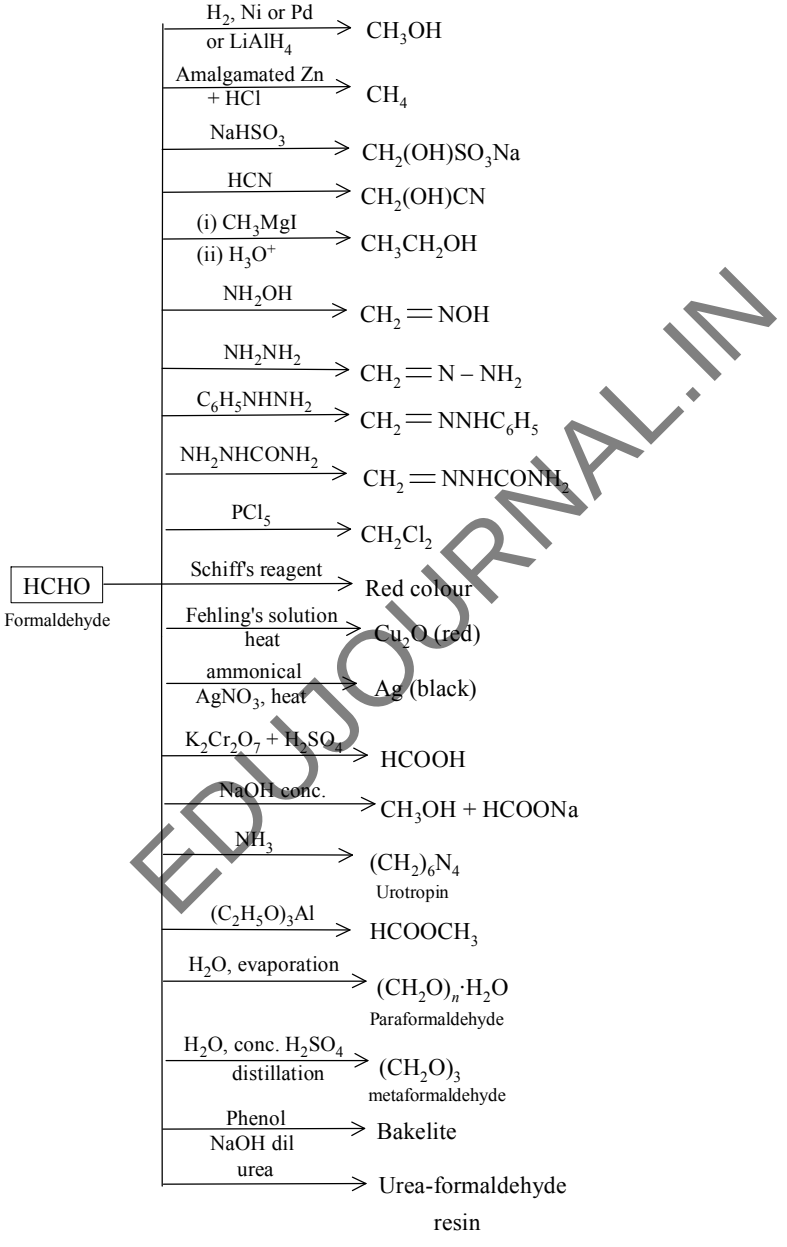
- Benzaldehyde is used as a flavouring agent in perfume industry.
- Benzaldehyde is used in the manufacture of dyes like malachite green.
- In **Gattermann-Koch aldehyde synthesis** a mixture of CO and HCl gas is passed through benzene at 323 K in presence of a catalyst consisting of anhydrous $AlCl_3$ and a small amount of CuCl to give benzaldehyde.
- Acetone is used as a solvent for acetylene, cellulose acetate, cellulose nitrate, celluloid, lacquers and varnishers.
- With phosphorus pentachloride, aldehydes and ketones gives **gem-dihalides**.
- Because of the small size of oxygen as compared to carbon, $>C=O$ bond length is shorter (1.23 Å) than that of C = C bond length (1.34 Å).
- Carbon-oxygen double bond is polar but carbon-carbon double bond is non-polar.
- Aliphatic aldehydes do not show position isomerism since aldehyde group being monovalent is always present at the end of the carbon chain.
- Formaldehyde cannot be prepared by **Rosenmund reduction** since formyl chloride HCOCl, is unstable at room temperature.
- In presence of hot dilute H_2SO_4 and $HgSO_4$ alkynes add up a molecule of water to form aldehydes and ketones.
- Aromatic aldehydes can be prepared by the oxidation of methyl benzenes with chromium trioxide in acetic anhydride.
- **Benzophenones** can also be prepared by Friedel-Craft's reaction of carbonyl chloride (phosgene) with excess of benzene.
- Acetone is a highly inflammable liquid and rapidly catches fire.
- Lower member of carbonyl compounds are miscible in water due to hydrogen bonding between oxygen of polar carbonyl group and hydrogen of water molecules.
- Cannizzaro reaction is a disproportionation reaction in which one molecule of an aldehyde is reduced while the other is oxidised.
- **Preparation of Aldehydes**

They can be summarized as follows.



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- **Properties of formaldehyde**



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- **Preparation of Ketones**

They can be summarized as follows.

(a) R_2CHOH (Sec. alcohol) $\xrightarrow[\text{Acid } K_2Cr_2O_7]{\text{oxidation}}$

(b) $R.CX_2.R$ (Gem. dihalide) $\xrightarrow[\text{aq. NaOH or KOH}]{\text{HOH}}$

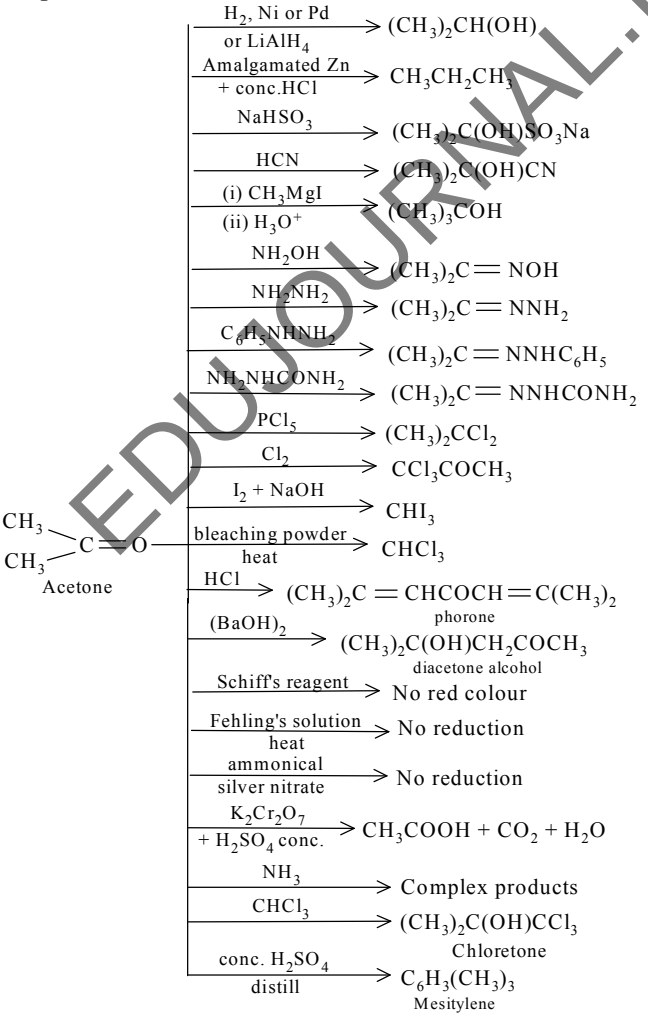
(Note : Geminal dihalides with both the halogen atoms on the middle carbon only gives ketones)

(c) $RMgX$ (Grignard reagent) $\xrightarrow[\text{Alkyl cyanide or alkyl nitrile}]{\text{RCN}}$

(d) $(RCOO)_2Ca$ (Calcium salt of fatty acid) $\xrightarrow{\text{Dry distillaton}}$

$\longrightarrow$ RCOR Ketone

- **Properties of acetone**



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Condensation reactions

(a) Condensation with hydroxylamine (NH_2OH)

$$CH_3CH = O \xrightarrow{NH_2OH} CH_3CH = NOH$$

Acetaldoxime

$$(CH_3)_2C = O \xrightarrow{NH_2OH} (CH_3)_2C = NOH$$

Acetoxime

(b) Condensation with phenylhydrazine ($C_6H_5NHNH_2$)

$$CH_3CH = O \xrightarrow{C_6H_5NHNH_2} CH_3CH = N \cdot NH \cdot C_6H_5$$

Acetaldehyde phenylhydrazone

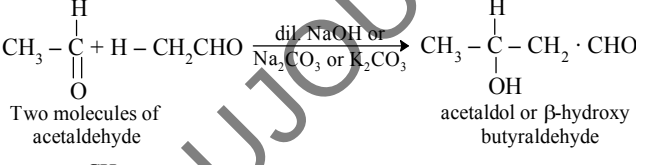
$$(CH_3)_2C = O \xrightarrow{C_6H_5NHNH_2} (CH_3)_2C = N \cdot NH \cdot C_6H_5$$

Acetone phenylhydrazone

Note : Oximes and phenylhydrazones, being crystalline solids, are used to identify aldehydes and ketones

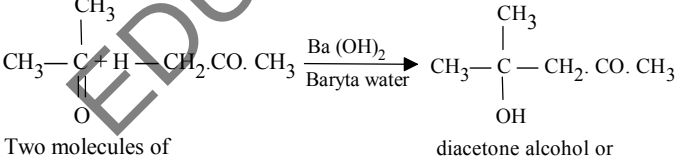
- **Aldol Condensation**

  This reaction is given only by aldehydes and ketones which contain α-hydrogen atom. When such aldehydes or ketones are treated with dil. alkali, two molecules undergo addition to give an aldol or a ketol respectively.



Two molecules of acetaldehyde

acetaldol or β-hydroxy butyraldehyde



Two molecules of acetone

diacetone alcohol or β – hydroxy ketone

Note : Aldol condensation may occur between

(i) Two aldehyde molecules (same or different)

(ii) Two ketone molecules (same or different)

(iii) An aldehyde and a ketone

- **Cannizzaro's Reaction**

  Aldehydes which do not contain α-hydrogen atom can only give this reaction. Other aldehydes as well as ketones do not give this reaction. *e.g.* formaldehyde (HCHO) and benzaldehyde (C_6H_5CHO) give this reaction.

  The reaction is brought about by 50% aqueous or alcoholic alkali (NaOH or KOH). The reaction involves two molecules of aldehyde, one of them is reduced to a primary alcohol

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while the other is oxidized to an acid. In this case, reduction and oxidation take place simultaneously and called **base catalysed auto-redox reaction.**

$$\underset{\text{Formaldehye (2 molecules)}}{2HCHO} \xrightarrow[50\%\ NaOH]{\text{Warm}} \underset{\text{Methyl alcohol}}{CH_3OH} + \underset{\text{Formic acid}}{HCOOH}$$

Formic acid actually comes out in the form of its salt, *i.e.* sodium formate (HCOONa) due to the presence of alkali in the reaction mixture.

- **Reduction**

Aldehydes and ketones on reduction give primary and secondary alcohols respectively. The common reducing agents used are

(1) Hydrogen in the presence of finely divided Ni, Pt or Pd as catalyst.
(2) Raney Nickel
(3) Amalgams of Na, Mg or Zn and water or dil. acid.

$$RCHO \xrightarrow{\text{Redn.}} RCH_2OH \text{ (primary alcohol)}$$

$$R_2CO \xrightarrow{\text{Redn.}} R_2CHOH \text{ (secondary alcohol)}$$

Clemmensen's Reduction : Ketones on reduction with Zn amalgam and conc. HCl give alkanes, when $>C=O$ group is reduced to a $—CH_2—$ group

$$\underset{\text{Acetone}}{CH_3COCH_3} \xrightarrow[\text{conc. HCl}]{\text{Zn (Hg)}} \underset{\text{Propane}}{CH_3CH_2CH_3}$$

Tests of aldehydes and ketones

Tests	Aldehydes	Ketones
1. With Schiff's reagent	Give pink colour	No colour
2. With Fehling's solution	Give red precipitate	No precipitate is formed
3. With Tollen's reagent	Black precipitate of silver or silver mirror is formed	No black precipitate or silver mirror is formed
4. With 2,4-dinitro phenyl hydrazine	Orange-yellow or red well defined crystals with melting points characteristic of individual aldehydes	Orange-yellow or red well defined crystals with melting points characteristic of individual ketones
5. With sodium hydroxide	Give brown resinous mass (formaldehyde does not give this test)	No reaction
6. With sodium nitroprusside and few drops of sodium hydroxide	A deep red colour (formaldehyde does not respond to this test)	Red colour which changes to orange

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Reducing properties of Aldehydes (Not given by ketones)

The H–atom on carbonyl group in aldehydes is readily oxidized to –OH group, hence aldehydes act as reducing agents.

$$\underset{\text{Aldehyde}}{R-\overset{\displaystyle H}{\overset{|}{C}}=O} \longrightarrow \underset{\text{Acid}}{R-\overset{\displaystyle OH}{\overset{|}{C}}=O}$$

In these reactions aldehyde is oxidized to a carboxylic acid. Aldehydes readily reduce (a) Fehling's solution (b) Tollen's reagent and (c) Schiff's reagent

(a) With Fehling's Solution : It is an alkaline $CuSO_4$ solution containing Rochelle salt (sodium potassium tartarate). Aldehyde reduces it to give a red ppt. of cuprous oxide, (Cu_2O)

(b) With Tollen's Reagent : It is an ammoniacal $AgNO_3$ solution. It contains the complex ion, $[Ag(NH_3)_2]^+$. Aldehyde reduces it to give silver mirror, hence called as silver mirror test.

$$CH_3CHO + 2[Ag(NH_3)_2]OH \xrightarrow{\text{warm}} CH_3COONH_4 + 2Ag\downarrow + 3NH_3 + H_2O$$

(c) With Schiff's Reagent : It is *p*–rosaniline hydrochloride (Magenta dye) solution decolourized by passing SO_2 gas. Aldehyde restores the original pink colour of the reagent.

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End

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CHAPTER

carboxylic acids and their derivatives

- Organic compounds containing – COOH as the functional group are called **carboxylic acids**.
- Carboxylic acids are soluble in less polar solvents like ether, alcohol, benzene etc.
- Aldehydes are easily oxidised to carboxylic acids with mild oxidising agents such as **Tollen's reagent**.
- Hydrolysis of esters either with mineral acids or alkalies yields carboxylic acids.
- Most of the aromatic acids exist as colourless solids with no distinct smell.
- Carboxylic acids have higher boiling points than alcohols due to the fact that a pair of carboxylic acid molecules are held together not by one but by two hydrogen bonds.
- The odours of lower aliphatic acids progress from the sharp irritating odours of formic and acetic acid to the distinctly unpleasant odours of butyric, valeric and caproic acids.
- Higher acids have little odour because of their low volatility.
- The alkali metal salts of carboxylic acids (sodium, potassium, ammonium) are soluble in water but insoluble in non-polar solvents.
- Most of the heavy metal salts (iron, silver, copper etc.) are insoluble in water.
- Benzoic acid, the simplest aromatic carboxylic acid, is nearly insoluble in cold water since the non-polar hydrocarbon part outweighs the effect of the polar – COOH part.
- The melting and boiling points of aromatic acids are usually higher than those of aliphatic acids of comparable molecular masses. [Reason : It is due to the reason that planar benzene rings in these acids can pack closely in the crystal lattice than zig-zag structure of aliphatic acids].
- Carboxylic acids exist as cyclic dimers in solid, liquid and even in vapour state due to intermolecular hydrogen bonding.
- The melting point of an acid with an even number of carbon atoms is higher than that of the acid with odd number of carbon atoms immediately below and above it in the series. This is known as **oscillation** or **alternation effect**.
- The first two members of the series have exceptionally high melting points due to the fact that they are associated as a polymer in liquid and solid states and not as a dimer due to small size of the non-polar part.
- Carbonation of Grignard reagents with dry ice in dry ether followed by acid hydrolysis gives carboxylic acids.

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- The greater the value of K_a, the dissociation constant of the acid, greater is the tendency of acid to ionise and hence stronger is the acid.
- The strength of an acid is expressed in terms of its pK_a value, which is the negative logarithm of the equilibrium constant K_a, *i.e.* $pK_a = -\log K_a$.
- Smaller the value of pK_a, stronger is the acid.
- Carboxylic acids fail to give the characteristic properties of carbonyl compounds because in resonance, the carbonyl group loses a part of its double bond character.
- Greater stability of carboxylate anion than carboxylic acid is responsible for the ionization and acidic character of the carboxylic acids.
- Carboxylic acids are stronger than alcohols due to the fact that both carboxylic acid and carboxylate anion are stabilized by resonance but neither the alcohols (ROH) nor their alkoxide ions (RO^-) are stabilized by resonance.
- Carboxylic acids are stronger than phenols.
- An electron withdrawing group (–I effect) on the negatively charged carboxylate ion will attract the electrons towards itself and thereby dispersing the negative charge. This stabilises the carboxylate ion and thus increases the acidic character of the acid.
- An electron releasing group on the negatively charged carboxylate ion will push the electrons towards the carboxylate ion and intensifying the negative charge on it and destabilises the carboxylate ion and thus decreases the acidic strength of the acid.
- **Formic acid** is a stronger acid than acetic acid since +I effect of the methyl group intensifies the negative charge on the carboxylate ion thereby acetate ion less stable than formate ion.
- **Chloroacetic acid** is stronger acid than CH_3COOH because chlorine atom due to its – I effect stabilises the chloroacetate ion relative to acetate ion by dispersing its negative charge.
- The electron withdrawing effect (–I effect) of the halogen atom decreases in the order: F > Cl > Br > I. Acidic strength of α-haloacids decreases in the order $FCH_2COOH > ClCH_2COOH > BrCH_2COOH > ICH_2COOH$.
- α-Chlorobutyric acid is a stronger acid than β-chlorobutyric acid which in turn is a stronger acid than γ-chlorobutyric acid since the distance between electronegative group and the COOH group increases the dispersal of the negative charge of the corresponding carboxylate ions and acidic strength decreases.
- Greater the number of electron-withdrawing substituents, greater would be the dispersal of the negative charge and hence stronger will be the acid.
- Trichloroacetic acid is a stronger acid than dichloroacetic acid which in turn is a stronger acid than monochloroacetic acid.
- In aromatic carboxylic acids, electron-donating substituents like CH_3 –, HO –, etc. tend to decrease the strength of the acid due to intensification of negative charge on the carboxylate anion.
- Electron-withdrawing groups like – NO_2, – Cl etc. tend to increase the strength of the acid due to the dispersal of the negative charge on the carboxylate anion.
- *o*-Substituted benzoic acids are generally stronger acids than benzoic acids regardless of the nature (+I or –I) of the substituent. This is called ortho effect and is probably due to a combination of steric and electronic factors.

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- Carboxylic acids neutralize alkalies forming salts.
- Among nitrobenzoic acids, the relative acid strength follows the sequence : *o*-nitrobenzoic acid > *p*-nitrobenzoic acid > *m*-nitrobenzoic acid > benzoic acid.
- The relative acid strength of *o*-, *m*- and *p*- toluic acids as compared to benzoic acid follows the sequence :
  o-toluic acid > benzoic acid > *m*-toluic acid > *p*-toluic acid
- The acid-weakening effect of the electron donating substituents and acid strengthening effect of the electron-withdrawing substituents is more pronounced at *p*- than at *m*-position.
- The relative strength of *o*-, *m*- and *p*- hydroxy benzoic acids relative to benzoic acids follows the sequence:
  salicylic acid > *m*-hydroxybenzoic acid > benzoic acid > *p*-hydroxybenzoic acid
- **Methanoic acid** is used for the dehydration of hides in leather industry.
- **Benzoic acid** is used in medicine as urinary antiseptic.
- **Sodium benzoate** being less toxic is used for preserving food products such as tomato sauce (ketch up) and fruit jams and juices.
- In the presence of a small amount of phosphorus, aliphatic carboxylic acids react smoothly with chlorine or bromine to yield a compound in which α-hydrogen has been replaced by halogen. This is the **Hell-Volhard-Zelinsky reaction**.
- When carboxylic acids are heated with alcohols in the presence of a few drops of conc. H_2SO_4 or dry HCl gas, esters are formed. The reaction is known as **Fischer esterification**.
- Dry distillation of calcium salts of carboxylic acids yields aldehydes or ketones.
- Benzoic acid does not undergo Friedel-Craft's alkylation or acylation due to the deactivation of the benzene ring by the electron withdrawing carboxyl group.
- Carboxylic acids give brisk effervescence with sodium bicarbonate due to the evolution of carbondioxide.
- The silver salts of carboxylic acids on treatment with Br_2 in refluxing CCl_4 give alkyl aryl bromides containing one carbon atom less than the parent acid.
- Formic acid reduces Fehling's solution to red ppt. of Cu_2O, Tollen's reagent to silver mirror, decolourises acidified $KMnO_4$ and $K_2Cr_2O_7$ solutions.
- Formic acid is used as a remedy for gout and neuritis.
- **Ethanoic acid** is used in the manufacture of plastics (polyvinyl acetate), rayon (cellulose acetate) and silk.
- The **Hell-Volhard Zelinsky** reaction is used for preparing an α-haloacid.
- **Vinegar** is a dilute aqueous solution of ethanoic acid.
- Heating a mixture of sodium benzoate and sodalime gives benzene.
- Formic acid reduces mercuric salts to mercurous salts.

FUNCTIONAL DERIVATIVES OF CARBOXYLIC ACIDS

- Acid chlorides, anhydrides, amides, esters are compounds in which the – OH of a carboxyl group has been replaced by – Cl, – OOCR, – NH_2 or – OR.
- Acid chlorides fume in air due to the formation of hydrochloric acid by hydrolysis.

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- The boiling points of acyl halides are lower than their corresponding carboxylic acids due to the absence of intermolecular hydrogen bonding in acyl halides.
- Acid chlorides are readily soluble in most of the organic solvents such as benzene, ether, chloroform etc.
- Both aliphatic and aromatic acid chlorides readily undergo nucleophilic acyl substitution even with weak nucleophiles such as alcohols, water etc. because the –I effect and +R effect of the chlorine atom tends to make the acyl carbon electron deficient.
- Aromatic acid chlorides are less reactive than their aliphatic counter parts due to the electron donating effect of the benzene ring which tends to reduce the electron deficiency of aromatic acids.
- Alcohols and phenols react with acid chlorides to form **esters**.
- Alkanoyl cyanides are formed when acyl chlorides react with potassium cyanide.
- The reaction of an aromatic acyl chloride with an alcohol or a phenol is usually carried out in presence of a base such as aq. NaOH or pyridine. This reaction is called **Schotten-Baumann reaction**.
- Acyl chlorides react with ammonia, primary and secondary amines to form amides. This reaction is called **ammonolysis**.
- Acyl chlorides are reduced to corresponding aldehydes with hydrogen in presence of palladium deposited over $BaSO_4$ and partially poisoned by the addition of sulphur or quinoline as catalyst. This reaction is called **Rosenmund's reduction**.
- Acyl chlorides react with organocadmium compounds to form ketones.
- Acetyl chloride and benzoyl chloride are chiefly used as acetylating and benzoylating reagents for the preparation of acetyl and benzoyl derivatives of alcoholic, phenols and amines.
- Friedel Craft's reaction of aromatic hydrocarbons with acid chlorides or anhydrides in presence of anhyd. $AlCl_3$ gives aromatic ketones.
- Acyl chloride is widely used for the detection and estimation of the number of – OH groups in an organic compound.
- Acid anhydrides are formed by the removal of a molecule of water either from two molecules of the same acid or one molecule each of the two different acids.
- Acid anhydrides are prepared by treating an acid chloride with a carboxylic acid in presence of a base such as pyridine.
- Acid anhydrides are generally soluble in common organic solvents such as ether, benzene etc.
- Boiling points of acid anhydrides are higher than those of the acids from which they are derived due to the reason that the molecular size of the anhydride molecule is larger than the parent and results in van der Waal's forces of attraction which account for higher boiling points.
- Anhydrides on reduction with $LiAlH_4$ give 1° alcohols.
- Acid anhydrides are slowly hydrolysed by water to form carboxylic acids.
- Acid anhydrides react with alcohols to form esters.
- Acetic anhydride is used as a dehydrating agent in Perkin's reaction.
- Acid anhydrides react with ammonia and amines to produce acid amides.

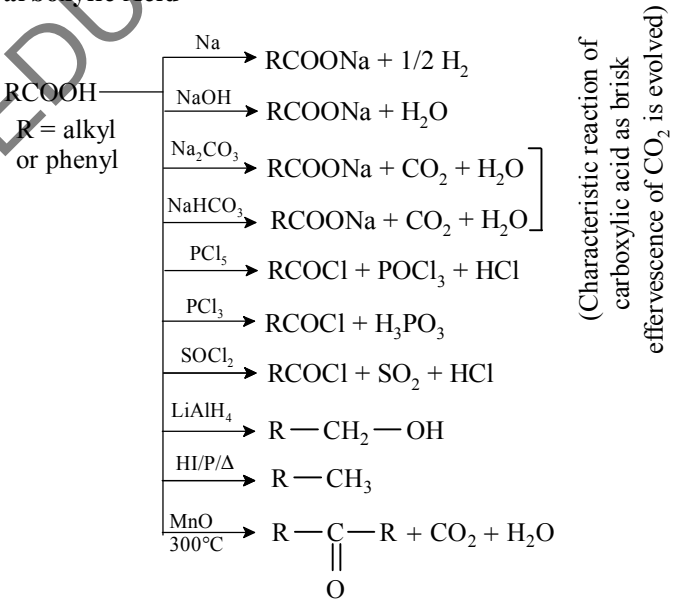
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- Oils and fats are esters of higher fatty acids (steric acid, palmitic acid, oleic acid etc.).
- General formula of esters is RCOOR′ where R may be H or any alkyl or aryl group while R′ is always either an alkyl or an aryl group.
- Methyl esters can be easily prepared by treating an acid with an ethereal solution of diazomethane.
- Aldehydes containing α-hydrogen atoms on treatment with aluminium ethoxide undergo condensation to produce esters. This reaction is called **Tischenko reaction**.
- The boiling points of esters are lower than those of the corresponding acids since they cannot associate by intermolecular H-bonding.
- Waxes are esters of higher fatty acids with higher monohydric alcohols such as myricyl and cetyl alcohols.
- The characteristic smell of bananas is due to isoamyl acetate.
- Esters are easily prepared by the action of alcohols on acid chlorides or anhydrides.
- Esters are insoluble in water but are soluble in organic solvents such as alcohol, ether, benzene etc.
- Esters react with ammonia, 1° amines, and 2° amines to form amides and substituted amides.
- Chemical reduction of esters is carried out by use of sodium metal and alcohol or by use of lithium aluminium hydride.
- Catalytic reduction of esters is carried out by treating an ester with H_2 in presence of copper chromite as catalyst at 525 K and under 200-300 atmospheric pressure.
- Esters containing α-hydrogen atoms undergo self condensation in presence of a strong base such as sodium ethoxide to form β-ketoesters. This reaction is called **Claisen condensation**.
- Esters are used for making artificial scent.
- Esters are widely used as industrial solvents for lacquers, oils, fats, varnishes and gums.
- The reaction which involves the replacement of the alkoxy part of the ester by the alkoxy part of the alcohol taken in excess is called **trans-esterification**.
- Trans-esterification is catalysed by acid (H_2SO_4 or dry HCl) or base (usually alkoxide ion).
- Esters are hydrolysed rapidly in acidic or alkaline solution.
- **Acidic hydrolysis** of esters yields a carboxylic acid and an alcohol, whereas alkaline hydrolysis of ester gives an alcohol and the salt of a carboxylic acid.
- **Alkaline hydrolysis** of esters is commonly called saponification.
- Acidic hydrolysis is reversible but alkaline hydrolysis is irreversible.
- Amides are derivatives of acids in which – OH part of the – COOH group is replaced by – NH_2, – NHR or – NR_2 groups.
- Amides have high melting and boiling points due to strong intermolecular hydrogen bonding.
- The boiling points of amides are even higher than the acids from which they are derived even though their molecular masses are almost identical.

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- Due to resonance, acid amides are much weaker bases than amines. Infact, they are amphoteric in nature and hence react with strong acids and bases forming their corresponding salt.
- Lower amides (C_1 - C_6) are soluble in water due to the formation of hydrogen bonds with water.
- Acid amides are least reactive of all the acid derivatives towards nucleophilic acyl substitution reactions. This is because the electron deficiency of the acyl carbon due to –I effect of the NH_2 group is compensated to a large extent by its +R effect.
- Primary amides react with nitrous acid in cold to produce carboxylic acid and nitrogen gas.
- Acid amides on reduction with sodium and ethyl alcohol or lithium aluminium hydride in dry ether yield amines.
- **Dimethyl formamide** (DMF) $HCON(CH_3)_2$ is a very good solvent for polar and non-polar compounds.
- Primary amides on heating with Cl_2 or Br_2 in presence of an alkali give the corresponding 1° amines with one carbon atom less than the parent amide. This is Hofmann bromamide reaction and is used for descending the homologous series.
- Amides undergo dehydration with dehydrating agents such as P_2O_5, $POCl_3$, $SOCl_2$ etc. to form alkyl cyanides.
- Amides are hydrolysed by acids to form a carboxylic acid and ammonium salt.
- With alkalies amides are hydrolysed to give a salt of a carboxylic acid and free ammonia.
- The reduction of esters with sodium and alcohol to form alcohols is called **Bouveault-Blanc reduction**.
- Trans-esterification involves a reaction between esters and alcohols.

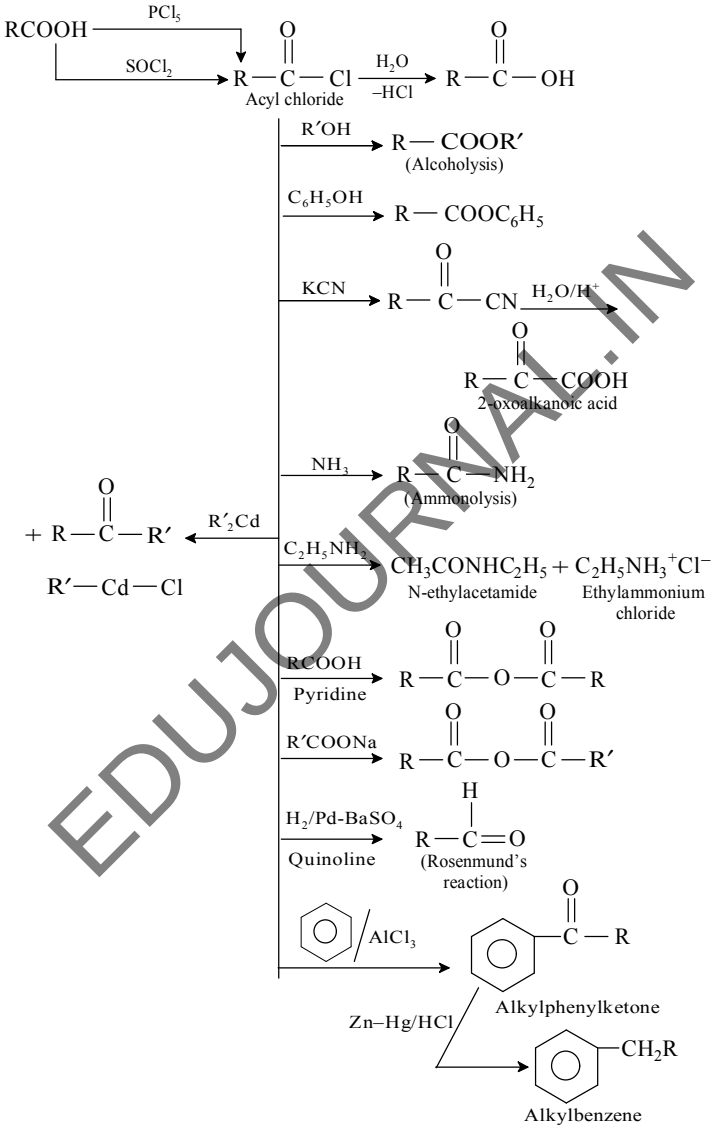
Reactions of Carboxylic Acid



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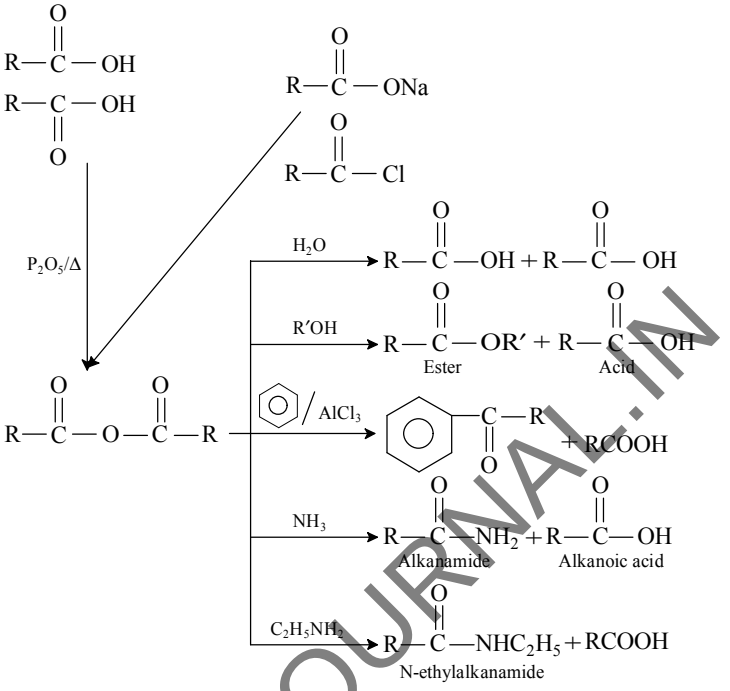
Functional Derivatives of Carboxylic Acid

- **Acid halide**

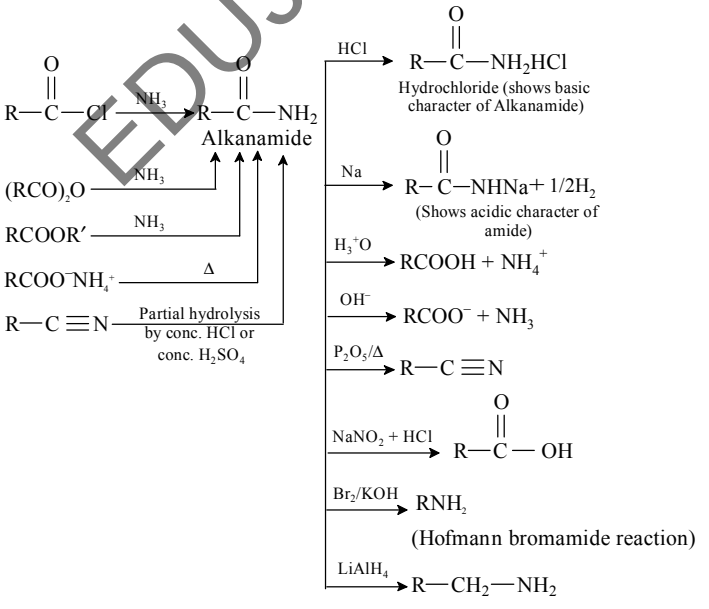


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- **Acid anhydrides**

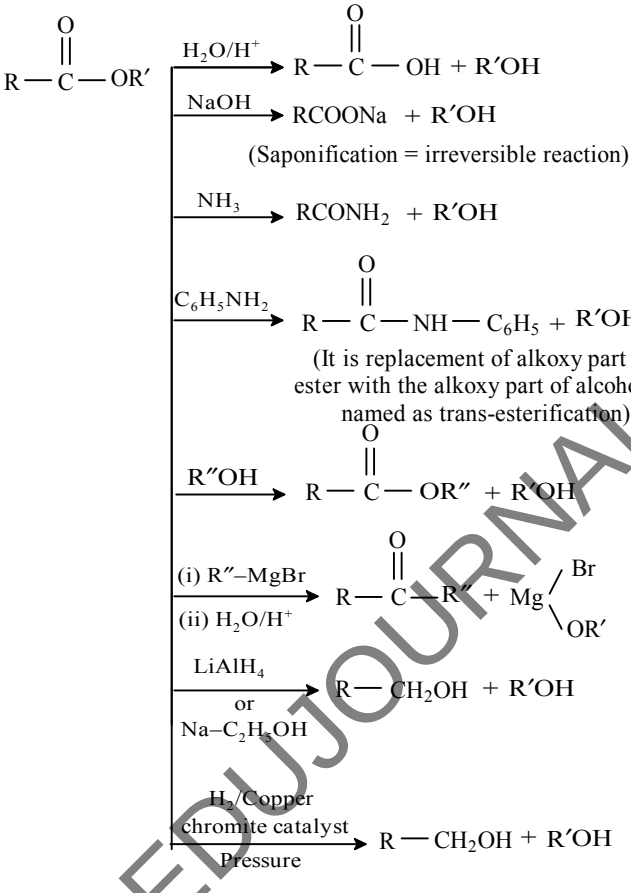


- **Acid amides**



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- **Esters**



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CHAPTER

nitrogen containing compounds

NITRO COMPOUNDS

- Organic compounds containing nitro as the functional group are called **nitro compounds**.
- Nitrogen in nitroalkanes or nitroarenes is sp^2 **hybridised**.
- The reaction which involves the replacement of one or more hydrogen atoms of a hydrocarbon by an equal number of nitro group is called **nitration**.
- **Nitrobenzene** is a pale-yellow liquid with a strong smell of bitter almonds.
- Nitroalkanes are colourless (when pure) liquid with a pleasant smell.
- Nitroalkanes and nitroarenes have higher boiling point than hydrocarbons of comparable molecular masses. This is because they are highly polar compounds with strong dipole-dipole interactions.
- Nitroalkanes are sparingly soluble in water while nitroarenes are insoluble.
- With Zn/HCl, Fe/HCl or Sn/HCl, both aliphatic and aromatic nitrocompounds can be reduced to the corresponding1° amines.
- Both aliphatic and aromatic nitro compounds are reduced to hydroxyl amines when reduced with Zn dust and NH_4Cl.
- Nitroalkanes are reduced to the corresponding primary amines with lithium aluminium hydride ($LiAlH_4$).
- Aromatic nitro compounds on reduction with lithium aluminium hydride gives **azo compounds**.
- Nitrite ion is an ambidient nucleophile since it has two sites (oxygen and nitrogen) through which it can attack an alkyl halide.
- Reduction of *m*-dinitrobenzene with sodium or ammonium sulphide gives *m*-nitroaniline.
- Reduction of nitro compounds with sulphides and polysulphides is called **zinin reduction**.
- Nitroalkanes containing α-hydrogen atoms show tautomerism.
- 1° and 2° nitroalkanes dissolve in aq. NaOH to form salts.
- Tertiary nitroalkanes do not react with nitrous acid since they do not contain α-hydrogen atoms.
- Electrolytic reduction of nitrobenzene in weakly acidic medium gives aniline while in strongly acidic medium gives *p*-aminophenol.
- Nitro group, due to its electron-withdrawing nature reduces the electron density at the *o*- and *p*- positions. Electron density is comparatively more at the *m*-position, *i.e.*, the nitro group is *m*-directing.

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- Nitrobenzene does not undergo Friedel Craft's reaction. Therefore nitrobenzene is used as a solvent during Friedel-Craft's reaction.
- Nitroalkanes are also widely used as a propellent in rockets *i.e.* nitromethane is a liquid propellant and nitrocellulose gel in nitroglycerine are used as solid propellants.
- 1° nitroalkanes are used for the manufacture of carboxylic acids and hydroxyl amine.
- The main reason for the acidic character of 1° and 2° nitroalkanes are (i) strong electron withdrawing nature of the nitro group and (ii) the resonance stabilisation of the carbanion of the salt so produced.
- A mixture of TNT (20%) and NH_4NO_3 (80%) called **amatol** is used in coal mining.
- A mixture of TNT (15%), NH_4NO_3 (65%), aluminium (17%) and charcoal (3%) is called **ammonal** and is used for blasting purposes.
- RDX (Research and Development Explosive) is also called **cyclonite**.
- **RDX** is prepared by controlled nitration of hexamethylene tetramine (obtained from formaldehyde) with fuming nitric acid at 293 K.
- Most of the nitroalkanes are quite stable and hence can be distilled without decomposition under atmospheric pressure.
- **Nitration** is a typical example of an aromatic electrophilic substitution reaction in which the nitronium ion (NO_2^+) acts as the electrophile.
- Because of acidic nature, 1° and 2° nitroalkanes react with halogens in presence of alkali to form the corresponding halonitroalkanes.
- Nitroalkanes are formed in excellent yields when 1° or 2° alkyl bromides or iodides are treated with alcoholic silver nitrite solution.

AMINES

- Amino derivatives of alkanes are referred to as amines and hence they are called **alkyl amines**.
- Amines are classified as primary amines (1°) RNH_2, secondary amines (2°) $R-\overset{R'}{\underset{|}{N}}H$ and tertiary amines (3°) $R-\overset{R'}{\underset{|}{N}}-R''$.
- Both aliphatic and aromatic amines can be obtained from corresponding nitrocompounds by reduction.
- Amines are polar compounds and except for tertiary amines can form intermolecular hydrogen bonds.
- Amines have higher boiling points than non-polar compounds of the same molecular weight but lower boiling point than alcohols or carboxylic acids.
- The lower boiling point of amines than alcohols or carboxylic acids are due to the fact that O – H bond is more polar than N – H bond and hydrogen bonding in alcohols and carboxylic acids is stronger than in amines.
- Amines of all three classes form hydrogen bonds with water. As a result, smaller amines are quite soluble in water.

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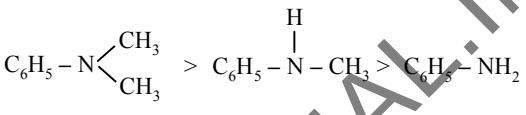
- Amines are soluble in less polar solvents like ether, alcohol, benzene, etc.
- Methyl and ethyl amines smell very **much like ammonia**, the higher alkyl amines have decidely "fishy" odours.
- Aromatic amines are generally very toxic, they are readily absorbed through the skin, often with fatal results.
- Aromatic amines are insoluble in water due to the larger hydrocarbon part which tends to retard the formation of H-bonds.
- Aromatic amines are readily oxidised in air to form coloured oxidation products.
- 1°, 2° and 3° amines because of the presence of a lone pair of electrons on the nitrogen atom behave as bases.
- Aqueous solution of amines behaves like NH_4OH and precipitate out iron, aluminium and chromium hydroxide from their salts.
- Amines react with water to form alkyl or aryl ammonium hydroxides which ionize to furnish hydroxyl ions.
- Amine salts are typical ionic compounds.
- Amine salts are non-volatile solids and when heated generally decompose before the high temperature required for the melting is reached.
- Amine salts are soluble in water but are insoluble in non-polar solvents such as benzene, chloroform, ether etc.
- Amines react with chloroplatinic acid (H_2PtCl_6) to form insoluble salts called **chloroplatinates**.
- All the amines behave as bases since they contain a lone pair of electrons on the nitrogen atom.
- All aliphatic amines are more basic than ammonia.
- In aqueous solution, the order of basicity is :

  2° amine > 1° amine > 3° amine
- In non-aqueous solution, (e.g. chlorobenzene) the order of basicity is

  3° amine > 2° amine > 1° amine
- The basic strength of an amine is determined by its basicity constant, K_b.
- Greater the value of K_b, stronger is the base.
- Basicity of an amine can be expressed in terms of its pK_b value which is the negative logarithm of the basicity constant, K_b, *i.e.* p$K_b = -\log K_b$.
- Smaller the value of pK_b, stronger is the base.
- All aromatic amines are weaker bases than ammonia.
- In general, electron-donating groups such as $-CH_3$, $-OCH_3$, $-NH_2$, etc. increase the basicity while electron-withdrawing substituents such as $-NO_2$, $-CN$, $-X$ (halogen), etc. decrease the basicity of amines.
- Electron-withdrawing group withdraws electrons destabilises the conjugate acid (cation) and thus decreases the basic strength.

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- *o*-substituted anilines are weaker bases than aniline regardless of the nature of the substituent whether electron-donating or electron withdrawing. This is called ortho-effect.
- The basic character of *o*-, *m*- and *p*-toluidine relative to aniline in the following order (effect of electron releasing substituents).
  p-toluidine ($K_b = 12 \times 10^{-10}$) > *m*-toluidine ($K_b = 5 \times 10^{-10}$) > aniline ($K_b = 4.2 \times 10^{-10}$) > *o*-toluidine ($K_b = 2.6 \times 10^{-10}$)
- Basic character of *o*-, *m*- and *p*-nitroanilines relative to aniline is in the following order (effect of electron-withdrawing substituents).
  Aniline ($K_b = 4.2 \times 10^{-10}$) > *m*-nitroaniline ($K_b = 2.9 \times 10^{-12}$) > *p*-nitroaniline ($K_b = 1.0 \times 10^{-13}$) > *o*-nitroaniline ($K_b = 6 \times 10^{-15}$).
- N-methyl amine is a stronger base than aniline and N,N-dimethyl aniline is even stronger than N-methylaniline.



- Aniline is a stronger base than diphenylamine which in turn is a much stronger base than triphenyl amine. Thus,
  $C_6H_5NH_2 > (C_6H_5)_2NH > (C_6H_5)_3N$
- Like NH_3, amines form coordinate complexes with metal ions such as Ag^+, Cu^{2+} etc. Thus AgCl dissolves in methyl amine and $CuCl_2$ forms a blue solution with ethyl amine.
- Primary and secondary amines react with Grignard reagent to give alkanes.
- Benzoylation of compounds containing an active hydrogen such as amines, alcohols and phenols with benzoyl chloride in the presence of dilute aqueous sodium hydroxide solution is known as **Schotten Baumann reaction**.
- Pure primary amines can be obtained from alkyl halides by **Gabriel phthalimide synthesis**.
- Aliphatic and aromatic nitro compounds can be reduced with hydrogen in the presence of Raney Ni, Pt or Pd at room temperature to obtain the corresponding 1° amine.
- Reduction of nitrocompounds can also be carried out with Sn/Fe in HCl or with Na and alcohol.
- Cyanides on reduction with H_2 and Raney Ni or with $LiAlH_4$ or with sodium and alcohol (Mendius reaction) give aliphatic or aryl primary amines. While isocyanides on reduction with H_2/Ni or $LiAlH_4$ give secondary amines.
- Reduction of oximes with $LiAlH_4$ or with Na/ethanol give 1° amines.
- **Hoffmann degradation** of primary amides with Br_2 and aqueous solution of KOH gives 1° amines with one carbon less than the original amide.
- **Reductive amination** of aldehydes and ketones in the presence of reducing agents can be carried out catalytically with H_2 and Raney Ni or by use of sodium cyanohydridoborate $NaBH_3CN$ gives 1° amines.

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- On heating with sodium, 1° and 2° amines gives effervescence of hydrogen while tertiary amines do not react.
- 1° and 2° amines react with acid chlorides or acid anhydrides to form substituted amides.
- Aromatic amines react with alkyl halides to give 2°, 3° and quarternary ammonium salts.
- 1° amines (aliphatic and aromatic) on heating with chloroform and an alcoholic solution of potassium hydroxide produce isocyanides or carbylamines which have a very unpleasant smell (carbylamine reaction) used as a test to distinguish 1° amines from 2° and 3° amines.
- Only 1° amines react with aldehydes and ketones in the presence of a trace of an acid which acts as a catalyst to produce **azomethenes** or also called **Schiff's base** or **anils**.
- Aliphatic 1° amines on warming with CS_2 form dithioalkyl carbamic acid which decompose on heating with $HgCl_2$ to give alkyl isothiocyanates having a characteristic smell like that of mustard oil. This reaction is called **Hofmann mustard oil reaction** and is used as a test for 1° amines.
- Direct nitration of aniline under controlled conditions gives a mixture of *m*-nitroaniline (60%) and *p*-nitroaniline (30%) along with some *o*-nitroaniline.
- Oxidation of aliphatic 1° amines with acidified $KMnO_4$ gives aldemine or ketemine which on hydrolysis gives aldehydes or ketones respectively.
- 2° aliphatic amines on oxidation with acidified $KMnO_4$ give tetra-alkyl hydrazine.
- Tertiary aliphatic amines are easily oxidised with hydrogen peroxide to amine oxide.
- Aromatic amines on oxidation with a mixture of sulphuric acid and $K_2Cr_2O_7$ forms a black dye of complex known as **aniline black**.
- Aromatic amines on oxidation with arsenic acid yields **violaniline** - a violet coloured substance.
- The conversion of aniline or other 1° amines into diazonium salts by the action of nitrous acid and dilute HCl is called **diazotisation**.
- **Hinsberg test** is used to distinguish 1°, 2° and 3° amines and it involves shaking the given amine with benzene sulphonyl chloride (Hinsberg reagent) in the presence of an excess of aqueous KOH solution.
- The reaction between an arene diazonium salt and another aromatic compound which has a strong electron-donating group (e.g. amines and phenols) attached to benzene nucleus is called **coupling reaction**.
- 1° and 2° aliphatic amines react with carbonyl chloride to form substituted ureas.
- Tertiary aliphatic amines are oxidised to the corresponding amine N-oxides by **Caro's acid**, ozone or H_2O_2.

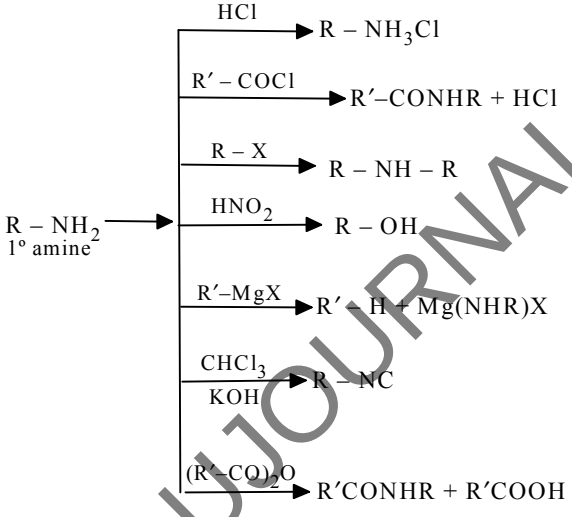
Preparation of alkyl amine

- $RCN \xrightarrow{H_2 \text{ or } LiAlH_4} RCH_2NH_2$
- $R-\overset{\overset{O}{\|}}{C}-Cl \xrightarrow{NaN_3} R-\overset{\overset{O}{\|}}{C}-N_3 \xrightarrow{C_2H_5OH/\Delta} R-N=C=O \xrightarrow{HOH} RNH_2 + CO_2$

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- $ROH + NH_3 \longrightarrow RNH_2$
- $RNC + H_2/Pt \longrightarrow RNHCH_3$
- $RCONH_2 + Br_2 + 4KOH \longrightarrow RNH_2 + 2KBr + K_2CO_3 + 2H_2O$
- $RCH = NOH + 4H \xrightarrow[H_2/Pt \text{ or } LiAlH_4]{Na(alc)} RCH_2NH_2 + H_2O$
- $RNC + HOH \xrightarrow{HCl} RNH_2 + HCOOH$
- $RCOOH + HN_3 \xrightarrow{conc.\ H_2SO_4} RNH_2 + CO_2 + N_2$

Properties:



CYANIDES & ISOCYANIDES

- **Cyanides (RCN)**

  These are compounds containing — C ≡ N functional group. They have general formula R — C ≡ N.

- **Preparation of Cyanides (RCN)**

  (i) $RCONH_2 + P_2O_5 \xrightarrow{\Delta} RCN + H_2O$

  (ii) $RCH = NOH \xrightarrow{(CH_3CO)_2O} RC \equiv N + H_2O$

  (iii) $RX + KCN\ (alc) \longrightarrow RCN + KX$

- **Reactions of Cyanides**

  (i) $R{-}C \equiv N \xrightarrow{2H_2O/H^+} RCOOH + NH_4^+$

  (ii) $R - C \equiv N \xrightarrow{LiAlH_4 \text{ or } H_2/Pt} RCH_2NH_2$

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(iii) $R-C\equiv N \xrightarrow{R'OH+H_2O} R-\underset{\text{(ester)}}{COOR'}$

(iv) $R-C\equiv N \xrightarrow{R'MgX,\ H_2O/H^+} R-\underset{\text{(ketones)}}{COR'}$

(v) $R-C\equiv N \xrightarrow{H_2O/OH^-} RCONH_2$

(vi) $R-C\equiv N \xrightarrow{SnCl_2/HCl} RCH=NHHCl \xrightarrow[\text{Boil}]{SnCl_2/HCl} RCHO$ (aldehyde)

- **Isocyanides or Carbylamines (RNC)**

  These compounds have general formula $R-N\overset{+}{\equiv}C^-$ with functional group —NC.

- **Preparation of Isocyanides**

  (i) $RX + AgCN \longrightarrow RNC$

  (ii) $RNH_2 + CHCl_3 + 3KOH \longrightarrow \underset{\text{(Carbylamine reaction)}}{RNC} + 3KCl + 3H_2O$

- **Reactions of Isocyanides**

  (a) $R-N\overset{+}{\equiv}C^- \xrightarrow{2HgO} R-N=C=O+Hg_2O$

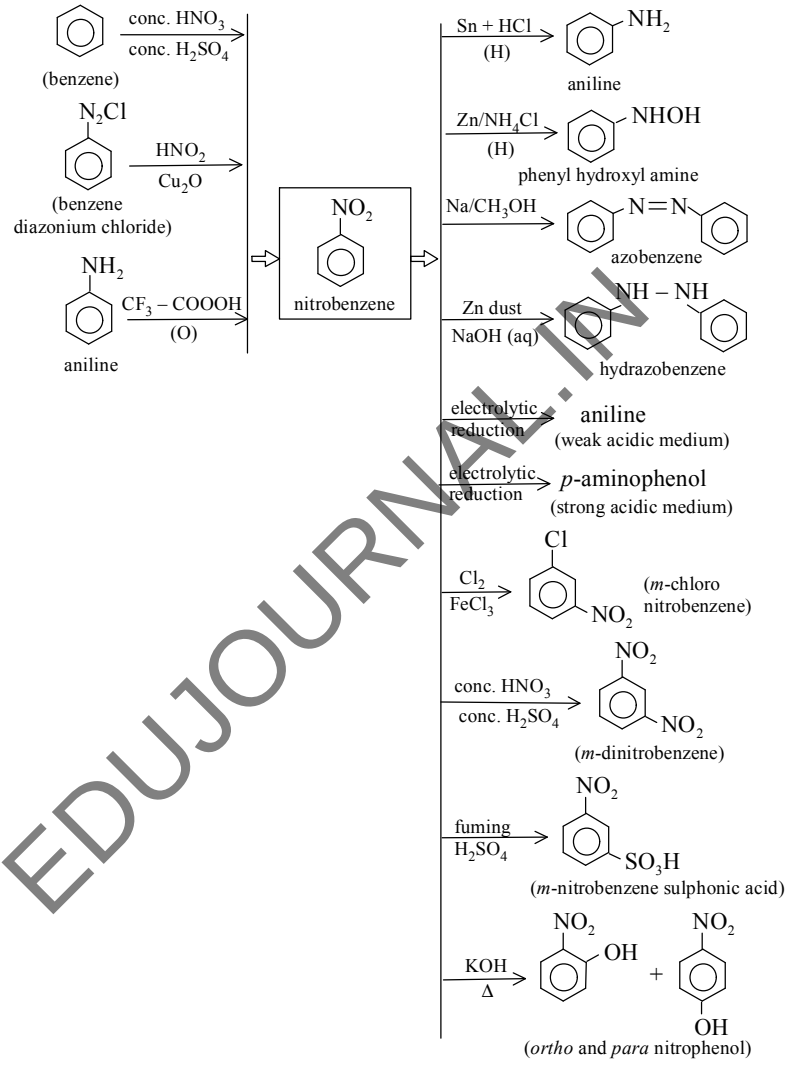
  (b) $R-N\overset{+}{\equiv}C^- \xrightarrow{Cl_2} R-N=CCl_2$

  (c) $R-N\overset{+}{\equiv}C^- \xrightarrow{1/8\,S_8} R-N=C=S$

  (d) $R-N\overset{+}{\equiv}C^- \xrightarrow[\text{Reduction}]{Na/C_2H_5OH} RNH-CH_3$

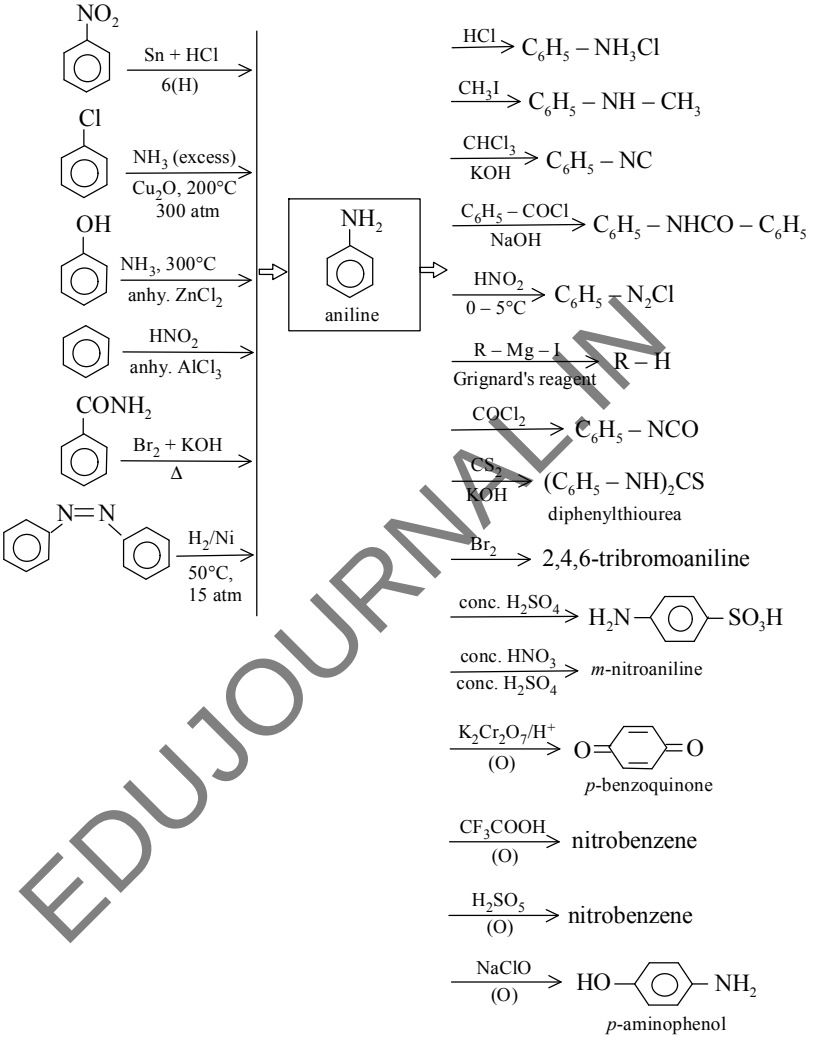
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Flow chart for nitrobenzene



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Flow chart for aniline



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CHAPTER

polymers

- **Polymers** are compounds of high molecular mass formed by the combination of a large number of small molecules called *monomers*.
- The small molecules which constitute the repeating units in a polymer are called monomer units.
- The process by which the monomers are transformed into polymers is called polymerisation.
- Polymers formed from one kind of monomer units are called homopolymers *e.g.* polyethylene

$$n\mathrm{CH_2{=}CH_2} \longrightarrow -(\mathrm{CH_2-CH_2}-)_n-$$

- Polymer formed from more than one kind of monomer units is called *copolymer* or *mixed polymers*. For example, nylon-66 is a polymer of two types of monomers : hexamethylene diamine and adipic acid.

$$\underset{\text{Hexamethylene diamine}}{n\mathrm{H_2N-(CH_2)_6-NH_2}} + \underset{\text{Adipic acid}}{n\mathrm{HOOC-(CH_2)_4-COOH}}$$

$$\xrightarrow{\text{Polymerisation}} \underset{\text{Nylon-66 (Co polymer)}}{(-\mathrm{NH-(CH_2)_6-NH-CO-(CH_2)_4-CO})_{\overline{n}}} + n\mathrm{H_2O}$$

- Various biomolecules (*e.g.* carbohydrates, proteins etc.) which are polymers are called *biopolymers.*
- A large number of synthetic polymers are long chain organic molecules. Such molecules are called *macro-molecules.*
- Polymers found in nature, mostly from plant and animal sources, are called natural polymers *e.g.* starch, cellulose, protein, silk, wool, natural rubber etc.
- The polymers which are prepared in the laboratory are known as *synthetic polymers or man made-polymers e.g.* polyethylene, synthetic rubber, polystyrene, nylon, P.V.C., teflon, orlon etc.
- **Linear polymers** are the polymers in which monomer units are linked together to form long straight chains *e.g.* polyethylene.
- **Branched chain polymers** are the polymers in which monomeric units are linked to constitute long chains (called *main chains*). These are *side chains* of different lengths which constitute branches. Such polymers have lower tensile strength and lower melting points as compared to linear polymers. *e.g.* low density polythene, glycogen, starch *etc.*
- **In cross-linked polymers** the monomeric units are linked together to constitute a three-dimensional network. Such polymers are hard, rigid and brittle *e.g.* bakelite, melamine.

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- When monomer units are separately added to form long chains without the elimination of any byproduct molecules, the product obtained is known as addition polymer *e.g.*

$$n CH_2{=}CH_2 \longrightarrow \underset{\text{Addition polymer}}{-(-CH_2-CH_2-)_n-}$$

The empirical formula for monomer and its addition polymer is same.

- Addition polymerisation proceeds by chain reaction which may be initiated by active species such as **free radical, cation or anion.**

(*i*) **Free radical polymerisation** generally occurs at high temperature and under pressure in presence of small amounts of organic peroxides. It proceeds in three steps : (*a*) initiation (*b*) propagation and (*c*) termination.

Chain initiation

$$\underset{\text{Organic peroxide}}{R-O-O-R} \xrightarrow{\text{Homolysis}} \underset{\text{Alkoxy free radical}}{R-\dot{O}+\dot{O}-R}$$

Chain propagation

$$R\dot{O}+CH_2=CH_2 \longrightarrow RO-CH_2-\underset{\text{free radical}}{\dot{C}H_2}$$

$$RO-CH_2-\dot{C}H_2 + n(CH_2=CH_2) \longrightarrow \underset{\text{Free radical}}{RO-(CH_2-CH_2)_n-CH_2-\dot{C}H_2}$$

In chain termination the long chain free radicals may either combine by coupling or by disproportionation leading to the final product. **Coupling** is the result of collisions in the growing chains either as such or in presence of a catalyst.

$$RO-(CH_2CH_2)_n-\dot{C}H_2 + \dot{C}H_2CH_2-(CH_2CH_2)_n-OR \xrightarrow{\text{coupling}} RO-(CH_2CH_2CH_2CH_2)_n-OR$$

Disproportionation is caused by the acceptance of one hydrogen atom by one free radical from the other which is converted to an alkene.

$$RO(CH_2CH_2)_n CH_2\dot{C}H_2 + H_2\dot{C}CH_2(CH_2CH_2)_n OR \xrightarrow[\text{Disproportionation}]{} RO(CH_2CH_2)_nCH_2CH_3 + \underset{\text{Alkene}}{CH_2{=}CH(CH_2CH_2)_nOR}$$

(*ii*) **Cationic polymerisation** normally occurs in the acidic medium in the presence of protonic acids (*e.g.* H_2SO_4) or Lewis acids (*e.g.*$AlCl_3$) *e.g.* polyisobutylene (a polymer used in manufacture of truck tyre, inner tubes) is formed as a result of cationic polymerisation of isobutylene in presence of BF_3 (Lewis acid) catalyst at 200 K.

(*iii*) **Anionic polymerisation** is noticed in alkenes having electron withdrawing groups present in them *e.g.* vinyl chloride etc. It is carried out in presence of a suitable base like sodamide ($NaNH_2$), *n*-butyl lithium etc.

- When monomers contain active functional groups (generally two) which react together with the elimination of a simple molecule such as H_2O, then the product formed is known as **condensation polymer** *e.g.* nylon-66, polyester, bakelite etc.

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- **Addition polymerisation** generally occurs between molecules containing double or triple bonds. In this type of polymerisation, the monomers simply add together forming polymeric chains *e.g.* polypropylene, polythene etc.
- **Chain growth polymerisation** is a result of addition polymerisation. For this type of polymerisation some initiator, like some organic peroxide is always needed for production of *free radical,* the monomers are then added to this free radical produced and addition takes place in a chain fashion *e.g.* ethylene, propylene, butadiene, tetrafluoroethylene, vinyl chloride etc. undergo chain growth polymerisation.
- **Step growth polymerisation** is a result of condensation polymerisation. The condensation takes place step by step. The condensation may take place with or without elimination of smaller molecules such as water, NH_3 etc. For example, polymerisation of *adipic acid* and *hexamethylene diamine* to form **Nylon-66**, *phenol* and *formaldehyde* to form **bakelite**, *terephthalic acid* and *ethylene glycol* to form **polyester**.
- **Classification** of polymers on the basis of magnitude of intermolecular forces present in them:
  (*i*) **Elastomers** are the polymers in which the polymer chains are held together by weakest intermolecular forces *e.g.* natural rubber.
  (*ii*) **Fibres** are the polymers which have a quite strong interparticle forces such as hydrogen bonds *e.g.* nylon, dacron, silk etc.
  (*iii*) **Thermoplastics :** the intermolecular forces are intermediate to those of elastomers and fibres *e.g.* polyethylene, polystyrene etc.
  (*iv*) **Thermo-setting polymers** are generally obtained fromsemi-fluid polymers with low molecular masses by heating in a mould when a hard, infusible and insoluble mass is formed. This is due to excessive cross linking between the chains forming a three dimensional network of bonds *e.g.* bakelite.
- **Difference Between Thermoplastics and Thermosetting Polymers**

	Thermoplastic	**Thermosetting**
(a)	Soft on heating.	Do not soften on heating.
(b)	Formed by the addition polymerization	Formed by condensation
(c)	Have linear structure *e.g.,* Teflon, PVC	Three dimensional structure, *e.g.,* Bakelite

- **Number average molecular mass** is obtained when the total molecular mass of a sample is divided by total number of molecules.

$$\bar{M}_n = \frac{\Sigma\ N_i\ \mathrm{M}_i}{\Sigma\ N_i}$$

- **Mass average molecular mass** is obtained when the total molecular mass of groups of molecules having particular molecular masses are multiplied with their respective molecular masses, the products are added and the sum is divided with the total mass of all the molecules.

$$\bar{M}_w = \frac{\Sigma\ N_i\ M_i^2}{\Sigma\ N_i\ M_i}$$

- **Poly dispersity index (P.D.I.)** is the ratio of weight average molecular mass to number average molecular mass.

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$$\text{P.D.I.} = \frac{\text{weight average molecular mass}}{\text{Number average molecular mass}} = \frac{\bar{M}_w}{\bar{M}_n}$$

In *natural polymers* it is lesser than unity because such molecules are generally *monodispersed.*

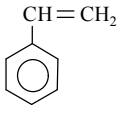
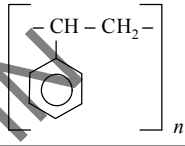
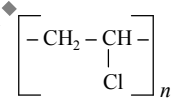
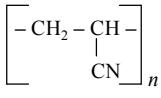
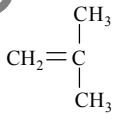
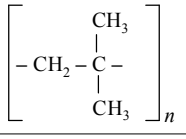
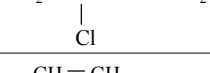
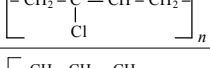
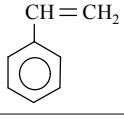
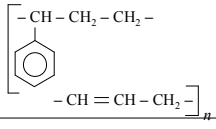
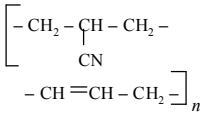
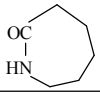
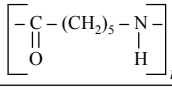
In *synthetic polymers* it is greater than unity because in such **polymers** $\bar{M}_w$ is greater than $\bar{M}_n$.

- **Polydienes** are the polymers of dienes *e.g.* synthetic rubber. In such polymers there are two double bonds.
- **Polyolefins** are the polymers derived from unsaturated hydrocarbons *e.g.* polyethylene, polypropylene, polystyrene etc.
- **Polyacrylates** can be obtained from a variety of acrylic monomers *e.g.* polymethylmethacrylate, polyethylacrylate etc.
- **Polyhalo-olefins** are the polymers obtained by polymerisation of halogenated unsaturated hydrocarbons *e.g.* PVC, PTFE (or **Teflon).**
- **Glycogen** is a polymer *of glucose* and can be represented as $(C_6H_{10}O_5)_n$. It is also known as animal starch. It is found in all animal cells, mainly liver. On combustion it provides energy needed for life processes.
- **Neoprene** is a polymer of chloroprene. *(i.e.* 2-chloro-1, 3-butadiene). It is synthetic rubber. It is fire resistant and is used in the manufacture of conveyor belts used in coal mines.
- **Nylon-6** is a polymer of *caprolactum*. It is used for moulding frictionless bearings, gears etc.
- **Nylon-66** is a polyamide fibre and is used for making textile fibres.
- **Rayon** is an *artificial silk*. It is obtained from cellulose and its derivatives. It does not absorb water.
- **Resilience** is the property of returning to original shape and size after the distortion forces are removed.
- **Silk** is a thread like natural polymer which is obtained from cocoons of silk worms. It is a natural *polyamide fibre.*
- **Thiokol** is synthetic rubber. It is obtained by polymerisation of ethylene dichloride and sodium poly-sulphide by condensation polymerisation. It is used as a rocket fuel.
- **Urea formaldehyde resin** is used for making china ware.
- **Melamine-formaldehyde resin** is used for making crockery.
- **Wool** is a natural polymer obtained from hair of sheep, goat etc.
- **Polyurethanes** are obtained by the action of di-isocyanate with a polyester having hydroxy groups on ends. These are used as leather substitutes.
- **Natural rubber** is a poly *cis*-isoprene. It is prepared from latex (obtained from rubber tree by coagulations with acetic acid). It is soft and tacky material. **Gutta-percha,** on the other hand, is poly *trans* isoprene. It is a hard and horny material.

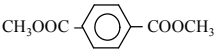
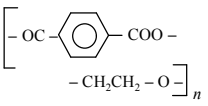
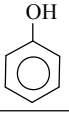
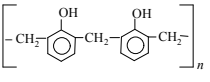
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- **Vulcanization** is a process in which natural rubber is treated with 3-5% sulphur. It introduces sulphur bridges between polymer chains thereby increasing its tensile strength, elasticity and resistance to abrasion. The rigidity of any vulcanized rubber depends upon the extent of sulphur cross-linking. *The process of vulcanization was developed by Charles Goodyear in 1839.*

Some industrially important polymers

No.	Polymer	Monomer	Structure
1	Polythene	$CH_2 = CH_2$	$[-CH_2-CH_2-]_n$
2	Polystyrene		
3	Polyvinylchloride (PVC)	$CH_2 = CHCl$	
4	Polytetrafluoro ethylene (PTFE) or Teflon	$CF_2 = CF_2$	$[-CF_2-CF_2-]_n$
5	Polyacrylo nitrile (PAN) or orlon	$CH_2 = CHCN$	
6	Butyl rubber		
7	Neoprene		
8	Styrene butadiene rubber (SBR) or (BuNa-S)	 and	
9	Nitrile rubber (BuNa-N)	$CH_2=CH-CH=CH_2$ $CH_2 = CHCN$ and $CH_2=CH-CH=CH_2$	
10	Nylon-6		

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No.	Polymer	Monomer	Structure
11	Nylon 66	$HOOC–(CH_2)_4–COOH$ and $H_2N – (CH_2)_6 – NH_2$	$[–CO–(CH_2)_4–CONH – (CH_2)_6 – NH –]_n$
12	Terylene (Dacron)	 and $HO – CH_2CH_2 – OH$	
13	Bakelite	 and HCHO	

- **Uses of some important polymers**

– **Polythene :** As insulator, packing material, household and laboratory ware, anticorrosive.

– **Polystyrene :** As insulator, wrapping material, house-hold articles and toys maker.

– **Polyvinyl chloride (PVC) :** In manufacture of raincoats, hand bags, leather clothes and vinyl flooring.

– **Teflon (PTFE) :** As lubricant, insulator and making cooking wares.

– **Polyacrylo nitrile (Orlon) :** In making synthetic fibres and synthetic wool.

– **Butyl rubber :** Used in place of natural rubber in industry

– **Neoprene :** As insulator, making conveyor belts and printing rollers

– **Buna-S :** In making automobile tyres and footwear

– **Buna-N :** In making oil seals, hose-pipes and tank linings

– **Nylon-6 :** In making carpets, ropes and tyre codes

– **Nylon-66 :** Synthetic fibres, fishing nets, ropes and tyre industries

– **Terylene :** Synthetic fibres, safety belts, tyre cords and tents

– **Bakelite :** In making gears, protective coatings and electric fittings

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End

29

CHAPTER

biochemistry

- **Biochemistry** is the branch of chemistry dealing with chemical changes occurring in living systems.
- Living systems are composed of biomolecules which are in a position of performing independent functions of life.
- **Biomolecules** are complex, lifeless organic molecules (compounds) which combine in a specific manner to produce life *e.g.* carbohydrates, proteins, amino acids, fats etc.
- **Energy and chemical change**
  Radiant energy from sun → heat and light → photosynthesis in plants → starches and oxygen → chemical energy for animals

$$6CO_2 + 6H_2O \xrightarrow[\text{chlorophyll}]{\text{sunlight}} \underset{\text{glucose}}{C_6H_{12}O_6} + 6O_2\uparrow$$

The living plants may convert glucose, thus produced into disaccharides, polysaccharides, starches, cellulose, proteins, or oils. Plants are thus primary source of energy for animals.

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O \; ; \; \Delta G = -2880 \text{ kJ mol}^{-1}$$

CARBOHYDRATES

- **Carbohydrates** are polyhydroxy aldehydes and ketones and contain at least one chiral carbon atom in their molecules *e.g.* sugar, starches, cellulose, glycogen, dextrins, gums etc. They are generally derived from plants.
- **Monosaccharides** are simple carbohydrates which cannot be hydrolysed into smaller units *e.g.* glucose, fructose etc.
- **Disaccharides** are sugars like cane sugar, maltose, lactose, which on hydrolysis produce two units of monosaccharides.
- **Polysaccharides** are the carbohydrates which are polymeric molecules and can be hydrolysed to give large number of monosaccharide units. The commonly occuring polysaccharides have the general formula $(C_6H_{10}O_5)_n$, example: starch, cellulose, etc.

$$\underset{\text{Starch or Cellulose}}{(C_6H_{10}O_5)_n} + nH_2O \xrightarrow{H^+} \underset{\text{Glucose}}{nC_6H_{12}O_6}$$

- **Oligosaccharides** are the carbohydrates which on hydrolysis give two to nine units of monosaccharides. They are further classified as di-, tri-, tetra-saccharides depending

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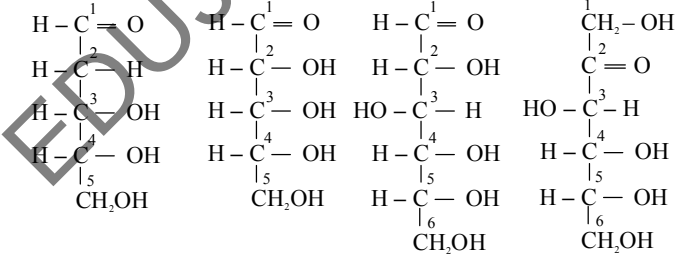
on the actual number of mono-saccharide units formed by the hydrolysis of a particular oligosaccharide e.g.

Disaccharides—Sucrose, maltose ($C_{12}H_{22}O_{11}$)

Trisaccharides—Raffinose ($C_{18}H_{32}O_{16}$)

Tetra-saccharides—Stachyose ($C_{24}H_{42}O_{21}$)

- Monosaccharides which contain an aldehyde (—CHO) group are called aldoses.
- Monosaccharides which contain a keto (> C = O) group are called ketoses.
- **Triose** is a monosaccharide that contain three carbon atoms in its molecule.
- **Tetrose** is a monosaccharide that contains four carbon atoms in its molecule.
- The oxygen bridge which links the monosaccharide units together to form oligosaccharides or polysaccharides is called glycosidic linkage. This linkage is called alpha, if the oxygen atom is below the plane of the linked monosaccharide units and beta if it is above.
- Carbohydrates with sweet taste are called sugars. All mono and oligo-saccharides are sugars.
- Carbohydrates without a sugar taste are called non-sugars. Polysaccharides are non-sugars.
- Carbohydrates which reduce Tollen's reagent and Fehling's solution are called reducing sugars. e.g. glucose, fructose, etc.
- Carbohyrates which do not reduce Fehling's solution and Tollen's reagent are called non-reducing sugars, e.g. sucrose.
- **Structures of a few monosaccharides**

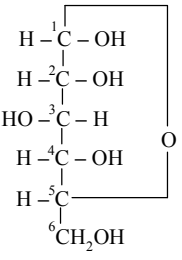


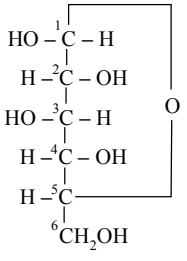
Ring structure of monosaccharides



- **Anomers** refer to a pair of stereoisomers which differ in configuration only around C_1 and C_1 carbon is called **anomeric carbon or glycosidic carbon.** Cyclic structures for two anomeric forms

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α-D-(+)-glucopyranose


β-D-(+)-glucopyranose

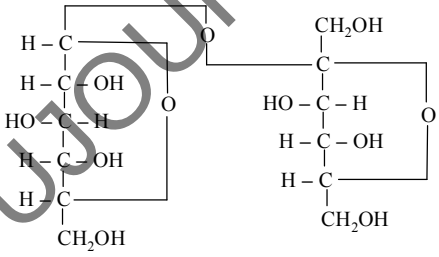
Cyclic structures are also known as **hemi-acetal** structures.

- The five membered ring containing an oxygen is called the furanose form.
- The six membered ring containing an oxygen atom is called the pyranose form.
- D(+) glucose exists in two anomeric forms *i.e.* α–D(+) glucopyranose and β-D(+) glucopyranose shown above. When these are separately dissolved in water, they undergo a change in specific rotation till it becomes constant after some time. **The change in specific rotation of isomers in aqueous solution** is called **Mutarotation.**

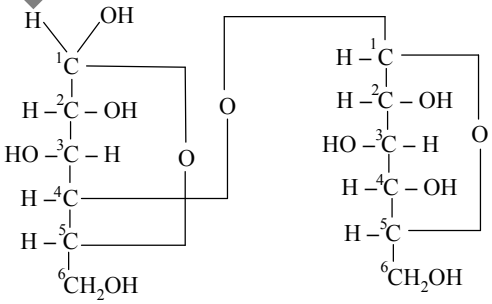
  α-D(+) glucose ⇌ Equilibrium ⇌ β-D(+) glucose

  $[\alpha]_D^t = +112°$ $[\alpha]_D^t = +52.7°$ $[\alpha]_D^t = +19°$

- **(+) Sucrose, $C_{12}H_{22}O_{11}$ (Cane sugar)**

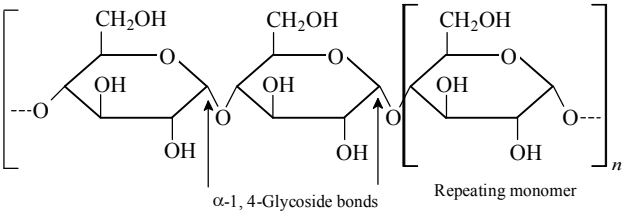


- **Maltose, $C_{12}H_{22}O_{11}$ (Malt sugar)**

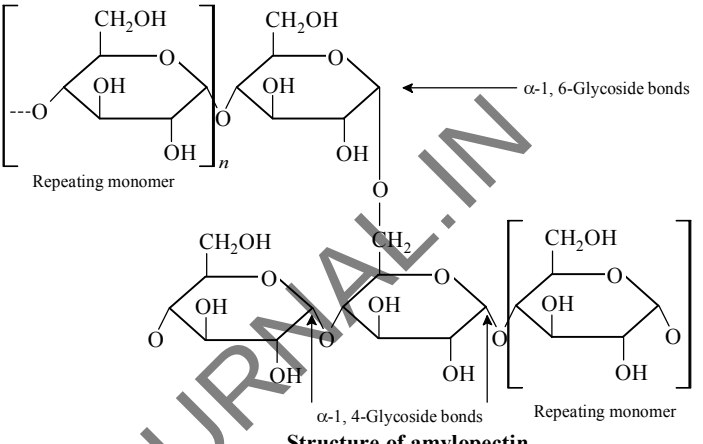

(reducing unit) (non-reducing unit)

- **Starch** $(C_6H_{10}O_5)_n$ consists of two compounds *i.e. amylose* (which is soluble in water) and amylopectin (which is insoluble in water). It contains 20% amylose and 80% amylopectin. The structures of *amylose* and *amylopectin* are given below:

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Structure of amylose



Structure of amylopectin

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- On boiling with dilute acid, starch ultimately yields glucose.

$$(C_6H_{10}O_5)_n \rightarrow \underset{\text{glucose}}{C_6H_{12}O_6}$$

- When treated with enzyme diastase starch yields maltose.

$$2(C_6H_{10}O_5)_n + nH_2O \rightarrow \underset{\text{maltose}}{nC_{12}H_{22}O_{11}}$$

Glucose ($C_6H_{12}O_6$)

It occurs in ripe grapes, honey and in most of the sweet fruits. Glucose is known as dextrose because it occurs in nature as the optically active dextrorotatory isomer.

- **Preparation :**

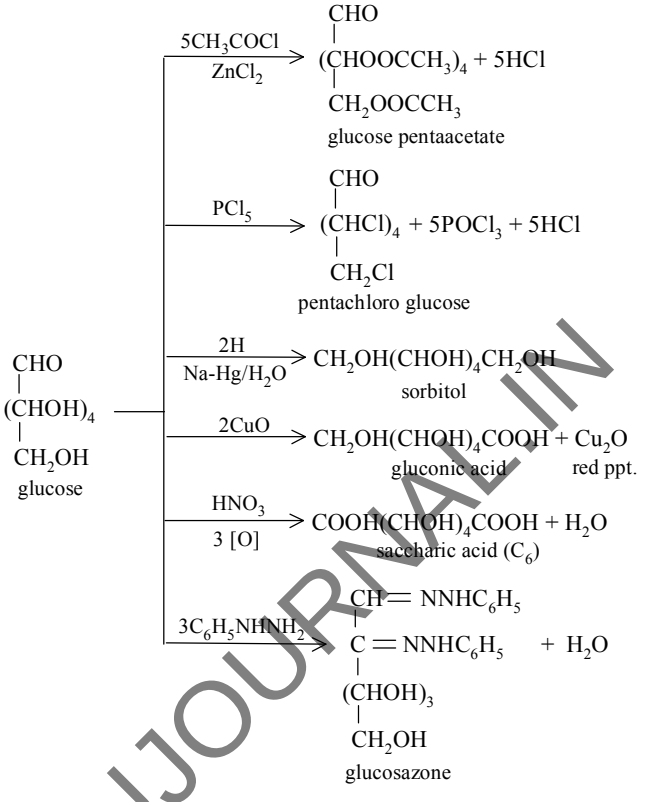
$$\underset{\substack{\text{cane sugar}\\\text{(sucrose)}}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6}$$

$$\underset{\text{starch}}{(C_6H_{10}O_5)_n} + nH_2O \xrightarrow{H^+} \underset{\text{glucose}}{nC_6H_{12}O_6}$$

Physical properties :

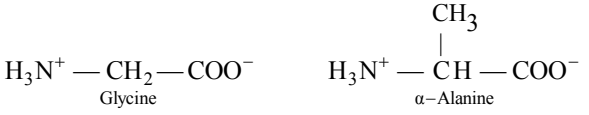
- The melting point of this colourless crystalline solid is 146°C.
- It is soluble in water but is sparingly soluble in alcohol and is insoluble in ether.
- It is optically active and the ordinary naturally occurring form is (+) glucose or dextro form.

Chemical reactions of glucose :



AMINO ACIDS

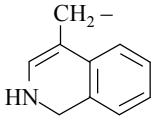
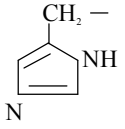
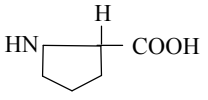
- Compounds containing carboxyl and amino groups. Amino acids obtained from proteins are α-amino acids, amino group is linked to the carbon atom having carboxyl group attached to it, *e.g.,* glycine, α-alanine, valine etc.
  Zwitter ion structure of glycine and α-alanine



- **Physical properties of α-amino acids**
  Since amino acids contain both a basic amino and an acidic carboxyl group, they exhibit some unusual properties, owing to their amphoteric character. They are all soluble in water and have high melting points indicating that they are actually salts and should be written with formula $RCH(NH_3^+)COO^-$. This zwitter ion or dipolar formula accounts for the behaviour of amino acids in proteins in solution. In acidic solution amino acid exists as a positive ion while in alkaline solution it exists as a negative ion.
  In electric field these ions will migrate towards the electrodes of opposite charge (+ve ions towards cathode and –ve ions towards anode). At a certain pH the dipolar ion

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exists as neutral ion and does not migrate to either electrodes. This pH is known as isoelectric point of amino acids. At this point usually pH is 5.5 to 6.3.

	Name of amino acid	Structure of R	Three letter symbol
1	Glycine	– H	Gly
2	Alanine	$-CH_3$	Ala
* 3	Valine	$-CH(CH_3)_2$	Val
* 4	Leucine	$-CH_2CH(CH_3)_2$	Leu
* 5	Isoleucine	$-CH(CH_3)-CH_2CH_3$	Ile
* 6	Arginine	$-(CH_2)_3NH-C(=NH)-NH_2$	Arg
* 7	Lysine	$-(CH_2)_4NH_2$	Lys
8	Glutamic acid	$-CH_2CH_2COOH$	Glu
9	Aspartic acid	$-CH_2COOH$	Asp
10	Glutamine	$-CH_2CH_2CONH_2$	Gln
11	Asparagine	$-CH_2CONH_2$	Asn
* 12	Threonine	$-CHOH{\cdot}CH_3$	Thr
13	Serine	$-CH_2OH$	Ser
14	Cysteine	$-CH_2SH$	Cys
* 15	Methionine	$-CH_2CH_2SCH_3$	Met
* 16	Phenylalanine	$-CH_2C_6H_5$	Phe
17	Tyrosine	$-CH_2C_6H_4OH$ (p)	Tyr
* 18	Tryptophan		Trp
* 19	Histidine		His
* 20	Proline		Pro

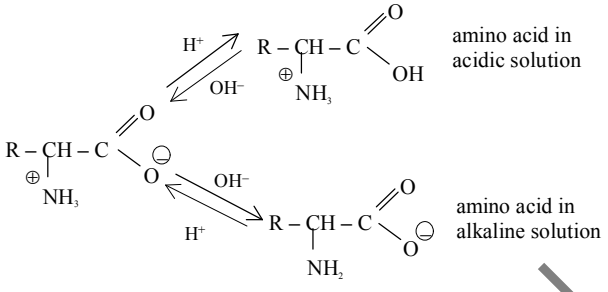
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- **Chemical properties of α-amino acids**

  Since these form salt with acids as well as with bases, their chemical reactions are similar to primary amines and carboxylic acids. Compounds which exhibit acidic and

basic property are called amphoteric substances and the phenomenon is known as amphoterism.

The equilibria are expressed as follows:



PEPTIDES

Condensation products of two or more molecules of α-amino acids is called peptides.

- **Peptide Linkage**

  Linkage which unites the α-amino acid molecules together is called peptide linkage. It is — CO — NH — linkage.

Proteins

- They are naturally occurring, nitrogeneous polymeric, complex organic compounds of very high molecular weights. They are important constituent of food. They occur in the protoplasm of all plants and animals. Plants can synthesize proteins from CO_2, H_2O and inorganic nitrates present in soil. Animals are not able to synthesize proteins, so they depend on plants for their protein supply.
- **Classification**

  (A) Simple Proteins : These proteins on hydrolysis, by acids or enzymes, give α-amino acids or their derivatives. For example,

  (a) Albumin : Serum albumin, egg albumin, lactalbumin.

  (b) Globulins : Serum globulin, tissue globulin and vegetable globulin (in seeds and nuts). Simple proteins are further subdivided into two groups.

  (i) Globular proteins : Their molecules have spherical, oval or elliptical shapes, *e.g.* egg albumin, caesin of milk, haemoglobin.

  (ii) Fibrous proteins : Their molecules have fibre-like structures, *e.g.* keratin (in hair), fibrin (in silk), collagen (in tendons), myosin (in muscles).

  (B) Conjugated proteins : They contain a simple protein united with a non-protein group. This non-protein group is called prosthetic group. For example,

  (i) Nucleo proteins : The prosthetic group is nucleic acid and they are found in the nucleus of cell.

  (ii) Glycoproteins : The non-protein part is a carbohydrae. It is found in egg-white.

  (iii) Phosphoproteins : These are compounds of protein with phosphoric acid, *e.g.* caesin of milk and ovovitellin of egg yolk.

  (C) Derived proteins : Proteins obtained by chemical decomposition of natural proteins are called derived proteins. The final product is α-amino acids.

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Structure of proteins

- **Primary structure :** The sequence in which various amino acids are arranged in a protein is known as the primary structure of a protein. The number, sequence and identity of amino acids in a protein constitute primary structure of a protein.
- **Secondary structure :** The coiling of the long strings of amino acids in a protein is its secondary structure. The α-*helix* is a common secondary structure. In α-*helix,* the peptide chain coils and the turns of the coil are held together by hydrogen bonds. Another type of secondary structure is possible in which the protein chains are stretched out. It is a β-*pleated* sheet structure.
- **Tertiary structure :** The folding and binding of α-helix into more complex shapes illustrates the tertiary structure of proteins. At normal pH and temperature, each protein will take the energetically most stable shape. This shape is specific to a given amino acids which form proteins.
- **Quaternary protein structure** results when several protein molecules are bonded together to form a still larger units.

Hydrolysis of Proteins and Peptide linkage

- Proteins are hydrolysed by acids, alkalies or enzymes.
  Proteins → Proteoses → Peptones → Peptides → α-amino acids
- The α-amino acid molecules are the building units or proteins joined with peptide linkage (— CO — NH). A large number of some of different amino acid molecules are joined by peptide linkages and form polypeptides which are macromolecules. A polypeptide with molar mass greater than 10,000 is termed as protein by convention. The number and type of amino acids and also the sequence in which they are arranged in the chain, decide the properties of proteins.

Colour Tests

- **Biuret Test :** Proteins give a violet or blue colour with 10% NaOH solution and a drop of very dilute copper sulphate. The test is due to [— CO — NH —] group and is given by all compounds containing this group.
- **Millon's Test :** Millon's reagent is a solution of mercuric and mercurous nitrate in nitric acid. Protein, when warmed with Millon's reagent, gives a white precipitate which changes to red.
- **Denaturation of proteins** involves irreversible precipitation of proteins. The complex three dimensional structure of proteins changes by change in pH, temperature, presence of salts or certain chemical compounds. Denaturation does not change primary structure but changes secondary and tertiary structures of proteins *e.g.* coagulation of albumin present in white part of egg when egg is boiled.

ENZYMES

- Most of the reactions occuring in living beings are too slow, to be effective, unless a catalyst is present. A catalyst which permits such a reaction to occur at useful rate is called an *enzyme.* Thus *enzymes* are essential biological catalysts which are required to catalyse biological reactions *e.g. maltase, amylase, lactase, invertase* etc. All enzymes are protein molecules.

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- **Enzyme inhibitors** are the substances which tend to reduce the activity of a particular enzyme. These are mostly the inorganic ions or the complex organic molecule *e.g.* salts of heavy metals. CN^-, DNP (dinitrophenol), high energy radiations etc.
- **Rennin** is an enzyme. It is used in cheese making. It coagulates over a million of times of its own weight of milk protein.
- Enzymes molecules have regions on to which reactant molecules (substrates) can temporarily bind. After this happens the reactions can occur readly to form new products. These highly specific regions are called *active sites.*
- The non-proteinous part present in a **protein** is called prosthetic group.
- The prosthetic group is necessary for protein to act as enzyme and is known as coenzyme. They may be defined as **a substance necessary for the activity of enzymes.** These are generally metal ions or small organic molecules. The common metal ions are Zn, Mg, Mn, Fe, Cu, Co, Mo, K and Na.
- The protein part of a **conjugated** protein is called **apoenzyme** and the molecule as a whole is called **holoenzyme.**

$$\underset{\text{(Conjugated protein)}}{\text{Holoenzyme}} \rightleftharpoons \underset{\text{(Protein)}}{\text{Apoenzyme}} + \underset{\text{(Prosthetic group)}}{\text{Co-enzyme}}$$

NUCLEIC ACIDS

- It is a polymer molecule found in living cells. The small constituent units of the molecule known as *nucleotides* are not identical, as are the monomers of a polymer such as polyethylene.
- The sub units of a nucleic acid molecule are called nucleotides. Each nucleotide consists of a pentose sugar (either ribose or deoxyribose) and a base (usually adenine, guanine, uracil, cytosine, thymine) plus a phosphate group which forms a bridge between the pentose parts of adjacent nucelotides in the nucleic acid chain. There are two classes of nucleic acids.
- **(i) DNA.** (Deoxyribonucleic acid) is any nucleic acid in which the pentose part of nucleotides are deoxyribose units.
- **(ii) RNA.** (Ribonucleic acid) is a nucleic acid in which the pentose part of the nucleotides are ribose units.
- A base joined to a sugar molecule is called nucleoside, e.g. adenosine which contains adenine and ribose; guanosine which contains ribose and guanine ; cytidine which contains ribose and cytosine.
- The genetic information for the cell is contained in the sequence of bases **A, T, G** and **C** in DNA molecule. When a cell divides, DNA molecules replicate and make exact copies of themselves so that each daughter cell will have DNA identical to that of the parent cell. The specificity of base pairing ensures the exact duplication of the sequence of bases in the newly synthesized strand of DNA.
- **Double helix structure of DNA**
  Nucleic acids control heredity at molecular level. The double helix of DNA is the storehouse of hereditary information of organism. This information is in coded form as sequence of bases along the polynucleotide chain.

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In 1953, J.D. Watson and H.C. Crick from X-ray diffraction studies of DNA proposed a double helical structure for DNA. The actual structure looks like a spiral - staircase (or twisted ladder) whose backbone is of phosphate and sugar, while the rungs of the ladder are the bases. These bases have also very specific connection with way that A (adenine) can H-bond only with T (thymine) while G (guanine) can H-bond only with C (cytosine). A and T are joined by two H-bonds while C and G are joined by three H-bonds.

- **Transcription** process involves copying of DNA molecules into a *complimentary* RNA molecule called *Messenger RNA* (*m*-RNA). This copying proceeds in accordance with the same base pairing as in replication but with a difference that the base **A** pairs with **U** in RNA. The pairing is as shown.

DNA	*m*-RNA
A	U
G	C
C	G
T	A

- **Translation** is the process where in *m*-RNA directs proteins synthesis in the cytoplasm of the cell with the involvement of another type of RNA molecule namely transfer RNA (*t*-RNA) and the ribosomal particles (RNA-protein complexes). The synthesis of protein may be represented as :

$$\text{DNA} \xrightarrow{\text{Replication}} m\text{RNA} \longrightarrow \underset{\uparrow\, t-\text{RNA}}{\text{Ribosomes}} \xrightarrow{\text{Translation}} \text{Polypeptide or Protein}$$

- The DNA sequence that acts as a code for specific protiens is called gene. Every protein in the cell has a corresponding gene.
- The relationship between the nucleotide triplets and amino acids is called genetic code. A sequence of three nucleotides which are adjacent on the nucleic acid chain of *m*-RNA is called a **codon.** The triplet serves to direct the addition of particular amino acid to the chain of amino acids being synthesized into a protein molecule by cell.
- **Mutation** is a chemical change in DNA molecule that lead to synthesis of proteins with an altered amino acid sequence.

LIPIDS

- Fats, oils and waxes belong to the group of naturally occurring compounds called Lipids. Lipids are those constituents of animals and plants which are soluble in organic solvents but insoluble in water. They can be divided into two categories:
  (a) Fats and oils, which yield long chain fatty acids and glycerol on hydrolysis.
  (b) Waxes, which yield long chain fatty acids and long chain alcohols on hydrolysis.
- Simplest and most abundant lipids are triglycerides. These are esters of glycerol with three fatty acids. Triglycerides are widely used in soaps, paints, varnishes printing inks and ointments.

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- Phospholipids are class of compounds that form structures of cell membranes. The classical example is the phosphoglycerides which contain glycerol, phosphate, two fatty acids and an alcoholic compound that may be choline, serine, ethanolamine or inositol. Phospholipids have good emulsifying and membrane-forming properties.
- Fats and oils are esters of glycerol with higher fatty acids. Glycerol reacts with these higher acids to give triesters of glycerol or glycerides of fatty acids, *e.g.*

$$\begin{array}{l} CH_2O\cdot OC\cdot R_1 \\ | \\ CHO\cdot OC\cdot R_2 \\ | \\ CH_2O\cdot OC\cdot R_3 \end{array}$$

R_1, R_2 and R_3 are higher alkyl groups which may be same or different. If they are same then triester is known as simple glyceride and if they are different then it is called mixed glyceride. The hydrocarbon part in the acid may be saturated or unsaturated, *e.g.* palmitic acid ($C_{15}H_{31}COOH$), stearic acid ($C_{17}H_{35}COOH$) are saturated acids while oleic acid ($C_{17}H_{33}COOH$) amd linoleic acid ($C_{17}H_{31}COOH$) are unsaturated acids.

- The glycerides which are liquid at ordinary temperatures are called oils, while those which are solids are called fats. Oils contain a large proportion of unsaturated acids as compared to fats.
  Oils when hydrogenated, unsaturated part is saturated and liquid oil is solidified to solid fats to give vegetable or artificial ghee.

Digestion of carbohydrates, proteins and lipids can be summarized as

(a) Polysaccharides (Starch) $\xrightarrow{\text{Amylases}}$ Disaccharides (Sucrose, lactose) $\xrightarrow{\text{Amylases}}$ Monosaccharides (Glucose, fructose, galactose)

(b) Proteins $\xrightarrow{\text{Pepsin}}$ Polypeptides $\xrightarrow{\text{Trypsin chymotrypsin}}$ α-Amino acids

(c) Lipids $\xrightarrow{\text{Bile acids}}$ Emulsified lipids $\xrightarrow{\text{Lipases}}$ glycerol + fatty acids

HORMONES

- A hormone is a secretion of ductless gland. They are a group of biomolecules which are produced in the ductless **(endocrine)** glands and are carried to the different parts of the body by the blood stream where they control various metabolic processes or show physiological activity which may be inhibitory or stimulatory. They are needed only in very small quantities and are not stored in the body.

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Classification of hormones

- **Steroid hormones** are a group of naturally occurring compounds having a structure that is based on four rings network. Out of these four, three are cyclohexane rings and one pentane ring *e.g.* steroids of sex hormones, bile acids etc. Steroid alcohols are called **sterols** *e.g.* cholesterol.
  - **(i)** One of the most important group of steroid hormones is sex hormones *e.g.* **Male sex hormones:**Testosterone, dihydrotestosterone, endrogens etc. **Female sex hormones:** Estrogens.
  - **(ii)** **Adrenal cortex hormones** regulate metabolic processes, control mineral and water balance. Examples of corticoids are cortisone, corticosterone and aldosterone.
- **Peptide hormones** are the hormones formed from nonapeptides made from nine amino acid residues e.g. oxytosin, vasopressin, etc.
  - **(i)** **Oxytosin** is a peptide hormone secreted by posterior lobe of pituitary gland. It causes contraction of uterus during child birth.
  - **(ii)** **Vasopressin** is peptide hormone secreted by posterior lobe of pituitary gland. It controls the reabsorption of water in the kidneys.
  - **(iii)** **Insulin** is a polypeptide hormone secreted by pancreas. It lowers blood glucose level by increasing the rate of conversion of glucose into glycogen.
  - **(iv)** **Angiotensin** is present in the blood plasma of the person with high blood pressure (hypertension). It is a potent vasoconstrictor *i.e.* it contains 8 amino acid residues.
- **Amine hormone** are water soluble amine compounds *e.g.* adrenaline (epinephrine) and thyroid hormones.
  - **(i)** **Adrenaline** increases the heart beat, the heart output and the blood pressure. It prepares the cardiovascular system for emergency action. It stimulates the break down of liver glycogen into blood glucose and is used as a fuel for anaerobic muscular work.
  - **(ii)** **Thyroid hormones** such as thyroxin are secreted by thyroid glands. It controls the metabolism of carbohydrates proteins and lipids.
- **Endocrine glands** refer to ductless glands.
- The secretion of hormone is under the control of the anterior lobe of a gland which is located at the base of the brain and is known as pituitary gland.
- Steroid alcohols are called **sterols** e.g. cholesterol.
- **Digitoxigenin** is a steroid and is extracted from a plant digitalis. It is used as a drug for regulating the functions of heart. It is also the raw material for the manufacture of a number of steroid drugs.

VITAMINS

The organic compounds (other than carbohydrates, proteins, fats or a group of biomolecules) which are essential for maintaining a normal health, growth and nutrition are called vitamins.

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- **Types of vitamins**
  (i) The vitamins A, D, E and K are **fat soluble vitamins.**
  (ii) Water **soluble vitamins** include vitamin B-complex (B_1, B_2, B_5, B_6, B_{12}, pentothenic acid, biotic or vitamin H and folic acid) and vitamin C.
- **Antiferments** are the substances which act as poison for enzymes *e.g.* $CHCl_3$, Hg, etc.
- **Pro-vitamins** are biologically inactive compounds that can be converted into active vitamins easily *e.g.* β-carotens (a pro-vitamin of vitamin A).
- **British gum** is the trade name of dextrine which is prepared by heating starch to about 2000°C. It is used as an adhesive.

	Name	Sources	Effects of deficiency
	Water-soluble vitamins		
1.	B_1 (Thiamine or Aneurin) ($C_{12}H_{18}N_4SOCl_2$)	Rice polishings, wheat flour, oat meal, eggs, yeast, meat, liver etc.	**Beri-beri**, loss of appetite and vigour, constipation, weak heart beat, muscle atrophy, even paralysis.
2.	B_2 or G (Riboflavin or Lactoflavin) ($C_{17}H_{20}N_4O_6$)	Cheese, eggs, yeast, tomatoes, green vegetables, liver, meat, cereals, etc.	**Cheilosis**, digestive disorders, burning sensations in skin and eyes, headache, mental depression, scaly dermatitis at angles of nares, corneal opacity, etc.
3.	B_3 (Pantothenic acid) ($C_9H_{17}O_5N$)	All food, more in yeast, liver, kidneys, eggs, meat, milk, sugarcane, groundnut, tomatoes.	**Dermatitis**, in cocks, greying of hairs, retarded body and mental growth, reproductive debility
4.	B_5 or P-P (Nicotinic acid or Niacin) $C_6H_5NO_2$ ($C_5H_4N - COOH$)	Fresh meat, liver, yeast, fish, cereals, milk, pulses, etc.	**Pellagra**, dermatitis, diarrhoea, dermentia, muscle atrophy, inflammation of mucous membrane of gut
5.	B_6 (Pyridoxine or Adermin) ($C_8H_{11}O_3N$)	Milk, cereals, fish, meat, liver, yeast synthesised by intestinal bacteria.	**Dermatitis**, **anaemia**, convulsions, nausea, insomnia, vomiting, mental disorders, depressed appetite
6.	Vit. H (biotin) ($C_{10}H_{16}N_2O_3S$)	Yeast, vegetables, fruits, wheat, chocolate, eggs, groundnut synthesised by intestinal bacteria.	**Skin lesions**, loss of appetite, weakness, hairfall, paralysis
7.	Folic acid group	Green vegetables, soyabean, yeast, kidneys, liver, synthesised by intestinal bacteria	Retarded growth, **anaemia**

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	Name	Sources	Effects of deficiency
8.	B_{12} (Cyanocobalamine) $C_{63}H_{88}O_{14}N_{14}PCo$)	Meat, fish, liver, eggs, milk, synthesised by intestinal bacteria	Retarded growth, **pernicious anaemia**
9.	Vit. C (Ascorbic acid) ($C_6H_8O_6$)	Lemon, orange and other citrus fruits, tomatoes, green vegetables, potatoes, carrots, pepper, etc.	Wound-healing and growth retarded, scurvy, breakdown of immune defence system, spongy and bleeding gums, fragile blood vessels and bones, exhaustion, nervous breakdown, high fever etc.
	Fat-soluble vitamins		
10.	Vit. A (Retinol or Axerophthol) ($C_{20}H_{30}O$)	Synthesized in cells of liver and intestinal mucous membrane from carotenoid pigments found in milk, butter, kidneys, egg yolk, liver, fish oil, etc.	Xerophthalmia-keratinized conjunctive and opaque and soft cornea. Stratification and keratinization in epithelia of skin, respiratory passages, urinary bladder, uterus and intestinal mucosa, night-blindness, impaired growth, glandular secretion and reproduction
11.	Vit. D (Ergocalciferol), (Sun shine vitamin) ($C_{28}H_{44}O$) and Cholecalciferol	Synthesized in skin cells in sunlight from 7-dehydrocholesterol also found in butter, liver, kidneys, egg yolk, fish oil, etc.	**Rickets** with osteomalacia; soft and fragile teeth
12.	Vit. E group Tocopherols (a, b, g) ($C_{29}H_{50}O_2$)	Green vegetables, oils, eggs yolk, wheat, animal tissues.	Sterility (impotency) and muscular atrophy
13.	Vit. K (Phylloquinone) ($C_{31}H_{46}O_2$)	Carrots, lettuce, cabbage, tomatoes, liver, egg yolk, cheese; synthesised by colon bacteria	**Haemorrhages**, excessive bleeding in injury, poor coagulation of blood

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End

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CHAPTER

chemistry in action

DRUGS

- **Drugs** are the chemicals used for treatment of diseases and for reducing pain. They are known as medicines.
- An ideal drug should satisfy the following requirements:
  - When administrated to the ailing individual or host, its action should be localised at the site where it is desired to act.
  - Should act on a system with efficiency and safety.
  - Should have minimum side effects.
  - Should not injure host tissues or physiological processes.
  - Should be non-resistant.
- The treatment of disease by means of chemicals that have a specific toxic effect upon the disease producing micro-organisms or that selectively destroy neoplastic tissues is called **chemotherapy**.
- Those chemical substances which prevent the growth of micro-organism or kill them but are not harmful to the living human tissues are called **antiseptics**. *e.g.* dettol, bithional, iodine, genatian violet, methylene blue, salicylic acid, picric acid, resorcinol, phenol etc.
- Those chemical substances that are used for bringing down the temperature of human body, in high fevers are known as **antipyretics**. e.g. Aspirin, paracetamol, analgin and phenacetin etc.
- The chemical substances which are used for the cure of mental diseases are known as **tranquilizers**. They reduce anxiety and tension and act on higher centres of nervous system. They are constituents of sleeping pills. They are called psycho-therapeutic drugs, *e.g.* barbituric acid and its derivatives such as seconal, luminal, equanil etc.
- **Analgesics** are the chemical substances which are used for relieving pain. *e.g.* aspirin, analgin etc.
- Analgesics are of two types:
  (i) **Narcotics** are drugs which produce sleep and unconciousness. *e.g.* morphine, codeine, heroine etc. Used in severe pain. They are very potent drugs and their chronic use leads to addiction.
  (ii) **Non-narcotics** are the drugs which are not potent and do not cause addiction. *e.g.* aspirin and analgin.

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- Those chemical substances which are used for killing micro-organisms or to stop their growth but are harmful to human tissues are called **disinfectants**. They are used to disinfect floors, toilets etc. but cannot be used directly to clean wounds. *e.g.* phenol (1%), sulphur dioxide etc.
- **Anaesthetics** are the drugs which produce loss of sensation. *e.g.* cyclopropane, nitrous oxide, xylocains etc.
- Those drugs which produce sleep and are habits forming are called hypnotics. *e.g.* luminal, seconal, etc.
- **Sulpha drugs** can be used in place of antibiotics. They inhibit the growth of micro-organisms. *e.g.* sulphanilamide, sulphadiazine, sulphaguanidine, etc.
- **Anti-malarials** are chemical substances used for the treatment of malaria. *e.g.* chloroquine, paraquine, primaquine, etc.
- **Sedatives** act as depressant and suppress the activities of central nervous system. Their high doses induce sleep. *e.g.*valium, barbiturates, etc.
- **Antidepressants** produce a feeling of well being and confidence in the person of depressed mood. These are also called mood booster drugs. *e.g.* vitalin, cocain, etc.
- **Antimicrobials** are the chemical substances used to cure infections due to microorganisms. They may be synthetic chemicals such as sulphonamides, para aminosalicylic acids or they may be antibiotics such as tetracycline, penicillin, chloramphenicol, etc.
- **Anti-fertility** drugs are the chemical substances used to control the pregnancy. These are also called oral contraceptives. Progestrogens either alone or in combination with oestrogens steroids are commonly used as antifertility drugs. *e.g.* norgestrel, ethylnodial, noresthistereone, lynestrenol etc. (all these are progestrogens).
- **Anti-histamines** are the chemical substances which diminish or abolish the main action of histamine released in the body and hence prevent allergic reactions such as hay fever, mild asthama, nasal discharge etc. Commonly used histamines are pheniramine maleate (avil), chlorpheniramine maleate (zeet), triprolidine (actidil), antazoline (antistine), dimethindene (forsital).
- Substances which neutralise the excess acid and raise the pH to appropriate level in stomach are called **antacids**.
  One of the most common ailment associated with digestion is acid gastritis. It is caused by excess of hydrochloric acid in the gastric juice.
- Aspirin is used to prevent heart attacks besides being antipyretic and analgesic agents.
- Magnesium hydroxide, magnesium carbonate, magnesium trisilicate, aluminium hydroxide gel, sodium bicarbonate are antacids which neutralise the HCl in the stomach but omeprazole and lansoprazole prevent the formation of acid in the stomach.
- The preservative C_6H_5COONa is metabolized in the body and is converted into hippuric acid (benzoyl glycine) which is secreted in urine.

DYES

- **Dyes** are coloured substances which can be applied in solution or dispersion to a substrate such as textile fibres (cotton, wool, silk, polyester, nylon), paper, leather, hair, fur, plastic material, wax, a cosmetic base or a dyestuff, giving it a coloured appearance.

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- If a compound absorbs light in the visible region, its colour will be that of the reflected light, *i.e.* complementary to that absorbed. For example, if a dye absorbs blue colour, it will appear yellow which is the complementary colour of blue.
- **Chromophores** are groups which impart colour to a compound, *i.e.* NO_2, NO, N = N, quinonoid structures.
- **Auxochromes** are groups which themselves do not absorb light (*i.e.* are not chromophores) but deepen the colour when introduced into the coloured compounds, *i.e.* – OH, $-NH_2$, – Cl, $-CO_2H$ etc.

Classification based upon source

- **Natural dyes** are obtained from plants. For example, alizarin, indigo, etc.
- **Synthetic dyes** are prepared in the laboratory. For example, martius yellow, malachite green, orange-I, orange-II, congo red, aniline yellow, etc.

Classification based upon constitution

(i) ***Nitro dyes*** - martius yellow,
(ii) ***Azo-dyes*** - aniline yellow, methyl orange, orange-I etc.
(iii) ***Triphenylmethane dyes*** - malachite green, magneta
(iv) ***Indigoid dyes*** - indigo, indigosol
(v) ***Anthraquinone dyes*** - alizarin and
(vi) ***Phthalein dyes*** - phenolphthalein.

Classification based upon their application

- **Acid dyes** are sodium salts of azo dyes containing sulphonic acid or carboxylic acid groups, *e.g.* orange-I, orange-II, congo red, methyl orange and methyl red. These dyes do not have any affinity for cotton but are used to dye, silk, polyurethane fibres. The affinity of acid dyes for nylon is higher than that for other types because polycaprolactam fibres contain a higher proportion of free amino groups.
- **Basic dyes** are the salts of azo and triphenylmethane dyes containing amino groups as auxochromes, e.g. aniline yellow, butter yellow, malachite green etc. These dyes are applied in their soluble acid solutions and get attached to the anionic sites present on the fabrics. Such dyes are used to dye polyesters and reinforced nylons.
- **Direct dyes** are water soluble dyes which are directly applied to the fabric from an aqueous solution. These dyes are most useful for fabrics which can form H-bonds such as cotton, wool, silk, rayon and nylon. Some example of direct dyes are congo red and martius yellow.
- **Disperse dyes** are applied to the fabric in the form of their dispersion in a soap solution in presence of some stabilizing agent such as phenol, cresol or benzoic acid. These dyes are used to dye synthetic fibres like polyesters, nylon and polyacrylonitrile. Many anthraquinone disperse dyes such as *celliton fast pink B* and *celliton fast blue B* are suitable for synthetic polyamide fibres.
- **Fibre reactive dyes** attach themselves to the fibre by an irreversible chemical reaction. Therefore, their dyeing is fast and colour is retained for a longer time. These dyes contain a reactive group which combines directly with the hydroxyl or the amino group of the fibre (cotton, wool, silk). Dyes which are derivatives of 2,4-dichloro-1,3,5-triazine are important examples of fibre reactive dyes.

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- **Insoluble azo dyes** constitute about 60% of the total dyes used. These are obtained by coupling of phenols, naphthols, arylamines, aminophenols adsorbed on the surface of a fabric with a diazonium salt. These dyes can be used to dye cellulose, silk, polyester, nylon, polypropylene, polyurethanes, polyacrylonitriles and leather. Azo dyes can also be used to dye foodstuffs, cosmetics, drugs, biological strains such as indicators etc. However, because of their toxic nature, these dyes are no longer permitted to dye food stuffs.
- **Ingrain dyes** are water insoluble azo dyes which are produced *in situ* on the surface of the fabric by means of coupling reactions e.g. para red.
- **Vat dyes** are insoluble dyes which are first reduced to a colourless soluble form (leuco compound) in large vats with a reducing agent such as alkaline sodium hydrosulphite and applied to the fabric and then oxidised to the insoluble coloured form by exposure to air or some oxidising agent such as chromic acid or perboric acid, *e.g.* indigo, Indigosol O, on the other hand, is readily soluble in water. It has affinity for cellulose and can be rapidly and quantitatively oxidised on the fibre with the formation of indigo. It is especially suitable for wool.
- **Mordant dyes** are applied to the fabric after treating them with a metal ion (mordant) which acts as a binding agent between the dye and the fabric. Depending upon the metal ion used, the same dye can give different colours. Thus, alizarin gives a rose red (turkey red) colour with Al^{3+} ions and blue colour with Ba^{2+} ions. These dyes are especially used for dyeing wool.

COSMETICS

- **Cosmetics** are chemical preparations that are used to cleanse, beautify and improve appearances. *e.g.* creams, perfumes, deodorants and talcum powder etc.
- Creams are stable emulsions of oils or fats in water. The two fundamental components of creams are
  (i) ***emmollients*** - prevent water loss from skin by forming water proofing coating and
  (ii) ***humectants*** - chemicals that attract water.
- **Perfumes** are solutions having pleasant odour. They are prepared by mixing various essential oils with alcohol and glycerine. They contain 70-95% alcohol which acts as solvent.
  The odoriferous contents are in the range of 2 - 10%.
- **Talcum powder** make the skin to look good, smell good and feel good. It is chiefly ground rock talc ($Mg_3Si_4O_{11}\cdot H_2O$) with a suitable scent added.
- **Deodorants** are the chemical substances which are used to mask, remove or control the perspiration odours and prevent their development. Some of the deodorants are antiperspirants also.
- **Deodorants** are perfumed preparation which do not affect perspiration whereas **antiperspirants** use astringent chemicals which inhibit the flow of perspiration. The most commonly used antiperspirant in deodorants is aluminium chlorohydrates, $Al_2(OH)_4Cl_2$ and $Al_2(OH)_5Cl$.

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DETERGENTS

- **Detergents** are the materials which are used for cleaning purposes. They are also called soapless soap.

Classification of detergents

On the basis of charge on the polar head, detergents are classified as :

- **Anionic detergents** - Their polar head is negatively charged

e.g. $CH_3-(CH_2)_{10}-CH_2-O-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-\bar{O}Na^+$

sodium lauryl sulphate

$CH_3-(CH_2)_{16}-CH_2-O-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-\bar{O}Na^+$

sodium stearyl sulphate

$CH_3-(CH_2)_9-\underset{|}{\overset{CH_3}{CH}}-C_6H_4-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-\bar{O}Na^+$

sodium dodecyl-benzenesulphonate

Such detergents are used to wash clothes.

- **Cationic detergents** - Their polar head is positively charged *e.g.* sapamine,

$[C_{17}H_{33}CONHCH_2NHCH_2CH_2-\overset{+}{N}(CH_3)_2]_2SO_4^{2-}$.

These are used as fabric softner and hair conditioner.

- **Non-ionic detergents** - Their polar head is neutral. *e.g.* Ethoxylate nonylphenol,

$C_9H_{19}-C_6H_4-(OCH_2CH_2)_6OH$.

Such detergents are used in dish washers.

PROPELLANTS

- **Propellant** is a combination of oxidiser and a fuel. A propellant on ignition undergoes combustion to release great quantities of hot gases.
- **Specific impulse** - The rocket propellant performance is measured in terms of specific impulse (I_s) which is given by the relation

$$I_s = \frac{1}{g}\sqrt{\left(\frac{2\gamma}{\gamma-1}\right)\left(\frac{gRT_C}{M}\right)\left(1-\frac{P_C}{P_e}\right)^{\frac{\gamma-1}{\gamma}}}$$

where γ = ratio of specific heat at constant pressure to specific heat at constant volume,
R = gas constant
T_C = combustion chamber temperature
M = average molecular mass of the exhaust products

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P_C = chamber pressure and P_e = external pressure.

From the above equation, it follows that conditions favouring high specific impulse are high chamber temperature and pressure, low molecular mass of the exhaust gases, and low external pressure. Thus, higher the temperature and pressure achieved in the chamber, the higher the kinetic energy of the gases escaping through the nozzle.

Types of propellants

Type	**Fuels**	**Oxidiser**
Solid propellants	Synthetic rubbers, synthetic resins, cellulose or its derivatives	Ammonium perchlorate, potassium perchlorate, ammonium nitrate, nitroglycerine
Liquid propellants	(i) Synthetic rubber, cellulose (ii) Liquid hydrogen (iii) Kerosene oil (iv) Hydrazine (v) Methyl hydrazine	Liquid oxygen Liquid oxygen Liquid oxygen HNO_3 N_2O_4
Hybrid propellants	Solid acrylic rubber	Liquid N_2O_4



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CHAPTER

practical chemistry

- Qualitative analysis deals with the identification of various constituents present in a given material. This analysis involves preliminary tests, wet tests for anions and cations, test for functional groups, etc.

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Preliminary tests:

- Colour : Blue (Cu^{2+}), green (Ni^{2+} or Cu^{2+}), deep green (Cr^{3+}), yellow or brown (Fe^{3+}), light pink (Mn^{2+}), pinkish violet (Co^{2+}).
- Smell : A pinch of mixture on rubbing with water gives characteristic smell. *e.g.* rotten eggs smell (sulphide), burning sulphur smell (some sulphites).

Analysis for acid radicals or anions

- ***Sulphite :*** Sulphite reacts with dilute H_2SO_4 producing SO_2 gas which turns acidified potassium dichromate solution green due to the reduction of dichromate to chromium sulphate which is green.

  $K_2Cr_2O_7 + 3SO_2 + H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$

  Sulphite reacts with barium chloride solution to produce barium sulphite (white ppt.) which is soluble in dilute HCl.
- ***Sulphide*** : Sulphide reacts with dilute H_2SO_4 liberating H_2S gas which turns lead acetate paper black due to the formation of black lead sulphide.

  $Pb(CH_3COO)_2 + H_2S \rightarrow \underset{\text{Black ppt.}}{PbS} + 2CH_3COOH$
- Soluble sulphide reacts with sodium nitroprusside to produce pink violet colour.

  $Na_2S + \underset{\text{Sodium nitroprusside}}{Na_2[Fe(CN)_5NO]} \rightarrow \underset{\text{Pink violet colour}}{Na_4[Fe(CN)_5NOS]}$
- ***Chloride*** : Chloride on heating with concentrated H_2SO_4 produces HCl gas which gives white precipitate of AgCl with $AgNO_3$ solution.

  $2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$

  $HCl + AgNO_3 \rightarrow AgCl\downarrow + HNO_3$
- The gas evolved by heating chloride with H_2SO_4 forms white fumes of ammonium chloride with NH_4OH.

  $NH_4OH + HCl \rightarrow \underset{\text{white fumes}}{NH_4Cl} + H_2O$
- On heating chloride with $K_2Cr_2O_7$ and concentrated H_2SO_4 a reddish chromyl chloride (CrO_2Cl_2) gas is produced which gives yellow solution with NaOH due to sodium chromate and on adding acetic acid, lead acetate solution produces a yellow precipitate of $PbCrO_4$. This test is known as *chromyl chloride test.*

  $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$

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$$K_2Cr_2O_7 + 2H_2SO_4 \rightarrow 2KHSO_4 + 2CrO_3 + H_2O$$

$$CrO_3 + 2HCl \rightarrow CrO_2Cl_2 + H_2O$$

$$CrO_2Cl_2 + 4NaOH \rightarrow \underset{\text{yellow colour}}{Na_2CrO_4} + 2NaCl + 2H_2O$$

$$Na_2CrO_4 + Pb(CH_3COO)_2 \rightarrow \underset{\text{yellow ppt.}}{PbCrO_4} + 2NaCl + 2H_2O$$

- ***Bromide*** : Bromide on heating with concentrated H_2SO_4 produces a reddish brown bromine gas which turns starch iodide paper blue due to the liberation of iodine from starch iodide. This iodine gives blue complex with starch.
- On adding $AgNO_3$ solution, a pale yellow precipitate of AgBr is formed which is hardly soluble in NH_4OH.

$$AgNO_3 + HBr \rightarrow AgBr\downarrow + HNO_3$$

$$AgBr + NH_4OH \rightarrow \underset{\substack{\text{Soluble silver ammonium}\\ \text{bromide complex}}}{[Ag(NH_3)_2]Br} + 2H_2O$$

- ***Iodide*** : Iodide on heating with concentrated H_2SO_4 gives a violet iodine gas which turns starch paper blue.
- On adding $AgNO_3$ solution, a yellow ppt. of AgI is formed which is insoluble in NH_4OH.

$$AgNO_3 + HI \rightarrow AgI\downarrow + HNO_3$$

Identification and separation of acidic radicals

- **Group I** : This group consists of radicals which are detected by dilute H_2SO_4 or dil. HCl. These are
  (i) carbonate (ii) sulphite (iii) sulphide (iv) nitrite (v) acetate.
- *Carbonate*

$$Na_2CO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + CO_2$$

$$\underset{\text{lime water}}{Ca(OH)_2} + CO_2 \longrightarrow \underset{\text{white ppt.}}{CaCO_3} + H_2O$$

$$\underset{\text{white ppt.}}{CaCO_3} + H_2O + CO_2 \longrightarrow \underset{\text{soluble}}{Ca(HCO_3)_2}$$

- *Sulphite*

$$Na_2SO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + SO_2$$

$$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + \underset{\text{green}}{Cr_2(SO_4)_3} + H_2O$$

- *Sulphide*

$$Na_2S + H_2SO_4 \longrightarrow Na_2SO_4 + H_2S$$

$$Pb(CH_3COO)_2 + H_2S \longrightarrow \underset{\text{black ppt.}}{PbS} + 2CH_3COOH$$

- *Nitrite*

$$2NaNO_2 + H_2SO_4 \longrightarrow Na_2SO_4 + \underset{\text{nitrous acid}}{2HNO_2}$$

$$3HNO_2 \longrightarrow H_2O + 2NO + HNO_3$$

$$2NO + O_2 \longrightarrow \underset{\text{(brown coloured)}}{2NO_2}$$

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- **Group II :** This group consists of radicals which are detected by concentrated H_2SO_4. These are
  (i) chloride (ii) bromide (iii) iodide (iv) nitrate (v) oxalate
- *Chloride*

$$NH_4OH + HCl \longrightarrow \underset{\text{white fumes}}{NH_4Cl} + H_2O$$

$$AgNO_3 + HCl \longrightarrow \underset{\text{ppt.}}{AgCl} + HNO_3$$

$$AgCl + 2NH_4OH \longrightarrow \underset{\text{(soluble)}}{Ag(NH_3)_2Cl} + 2H_2O$$

$$NaCl + H_2SO_4 \longrightarrow NaHSO_4 + HCl$$

$$MnO_2 + 4HCl \longrightarrow \underset{\text{(yellowish green gas)}}{MnCl_2 + 2H_2O + Cl_2\uparrow}$$

Confirmatory test

$$\text{Chloride} + K_2Cr_2O_7 \text{ (solid)} + \text{conc. } H_2SO_4 \xrightarrow{\text{heat}} \underset{\text{(reddish brown vapours)}}{CrO_2Cl_2}$$

$$CrO_2Cl_2 \xrightarrow{NaOH} \underset{\text{yellow solution}}{Na_2CrO_4} \xrightarrow[+ (CH_3COO)_2Pb]{CH_3COOH} PbCrO_4 \text{ (yellow ppt.)}$$

- *Bromide*

$$NaBr + H_2SO_4 \longrightarrow NaHSO_4 + HBr$$

$$2HBr + H_2SO_4 \longrightarrow \underset{\text{(brown gas)}}{Br_2\uparrow} + 2H_2O + SO_2$$

Confirmatory test

$$NaBr + AgNO_3 \longrightarrow AgBr + NaNO_3$$

$$AgBr + 2NH_4OH \longrightarrow Ag(NH_3)_2Br + 2H_2O$$

AgBr is sparingly soluble in NH_4OH solution.

- *Iodide*

$$2KI + 2H_2SO_4 \longrightarrow 2KHSO_4 + 2HI$$

$$2HI + H_2SO_4 \longrightarrow \underset{\text{(violet vapours)}}{I_2\uparrow} + SO_2 + 2H_2O$$

Violet vapours with starch produce blue colour.

$$I_2 + \text{starch} \longrightarrow \text{blue colour}$$

$$NaI + AgNO_3 \longrightarrow \underset{\text{yellow ppt.}}{AgI} + NaNO_3$$

$$AgI + NH_4OH \longrightarrow \text{not soluble}$$

- *Nitrate*

$$NaNO_3 + H_2SO_4 \longrightarrow NaHSO_4 + HNO_3$$

$$4HNO_3 \longrightarrow 2H_2O + \underset{\text{(light brown fumes)}}{4NO_2} + O_2$$

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$Cu + 4HNO_3 \longrightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$

Confirmatory test

$NaNO_3 + H_2SO_4 \longrightarrow NaHSO_4 + HNO_3$

$6FeSO_4 + 2HNO_3 + 3H_2SO_4 \longrightarrow 3Fe_2(SO_4)_3 + 4H_2O + 2NO$

$[Fe(H_2O)_6]SO_4 \cdot H_2O + NO \longrightarrow [Fe(H_2O)_5NO]SO_4 + 2H_2O$

(brown ring)

Ring test is not reliable in presence of nitrite, bromide and iodide.

- **Group III :** The radicals which do not give any characteristic gas with dilute acid and concentrated H_2SO_4. These are
  (i) sulphate (ii) phosphate (iii) borate (iv) fluoride
- *Sulphate*

$Na_2SO_4 + BaCl_2 \xrightarrow{H^+} BaSO_4 \downarrow + 2NaCl$

white ppt.

Identification of basic radicals

- Analysis of basic radicals includes the following steps.
  - Preparation of the original solution of the salt or mixture.
  - Separation of basic radicals into different groups.
  - Analysis of the precipitates obtained in different groups.
- *Separation of basic radicals into groups*

Group	Group reagent	Basic radical	Composition and colour of the precipitate	
I	Dilute HCl	Ag^+ Pb^{2+} Hg_2^{2+}	AgCl: white $PbCl_2$: white Hg_2Cl_2: white	Chloride insoluble in cold dilute HCl
II	H_2S in presence of dilute HCl	Hg^{2+} Pb^{2+} Bi^{3+} Cu^{2+} Cd^{2+} As^{3+} Sb^{3+} Sn^{2+} Sn^{4+}	HgS : black PbS: black Bi_2S_3: black CuS: black CdS: yellow As_2S_3: yellow Sb_2S_3: orange SnS: brown SnS_2: yellow	Sulphides insoluble in dilute HCl
III	NH_4OH in presence of NH_4Cl	Fe^{3+} Cr^{3+} Al^{3+}	$Fe(OH)_3$: reddish brown $Cr(OH)_3$:green $Al(OH)_3$: white	Hydroxides are insoluble in NH_4OH
IV	H_2S in presence of NH_4OH	Zn^{2+} Mn^{2+} Co^{2+} Ni^{2+}	ZnS: greenish white MnS: buff CoS: black NiS: black	Sulphides are insoluble in NH_4OH

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V	$(NH_4)_2CO_3$ in presence of NH_4OH	Ba^{2+} Sr^{2+} Ca^{2+}	$BaCO_3$: white $SrCO_3$: white $CaCO_3$: white	Carbonates are insoluble
VI	Na_2HPO_4	Mg^{2+}	$Mg(NH_4)PO_4$: white	
Zero	NaOH	NH_4^+	Ammonia gas is evolved	

- ***Pb^{2+} (lead)*** : The sulphide is dissolved in dilute HNO_3, solution with dilute H_2SO_4 gives a white precipitate.

  $Pb(NO_3)_2 + H_2SO_4 \rightarrow PbSO_4\downarrow + 2HNO_3$

- Lead sulphate is dissolved in concentrated ammonium acetate solution which gives a yellow precipitate of $PbCrO_4$ with K_2CrO_4 solution.
- ***Cu^{2+} (copper)*** : Sulphide on treatment with dilute HNO_3 and excess of NH_4OH, forms a deep blue coloured solution.

  $3CuS + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 2NO + 3S + 4H_2O$

  $Cu(NO_3)_2 + 4NH_4OH \rightarrow \underset{\text{deep blue solution}}{[Cu(NH_3)_4](NO_3)_2} + 4H_2O$

- On acidifying with acetic acid and adding potassium ferrocyanide, blue solution gives a chocolate coloured precipitate of $Cu_2[Fe(CN)_6]$.
- Fe^{3+}, Cr^{3+} and Al^{3+} comprise III group and the reagent is NH_4OH in presence of NH_4Cl.
- These radicals are precipitated as their hydroxides.
- ***Fe^{3+} (iron)*** : The brownish red precipitate of $Fe(OH)_3$ on treatment with dilute HCl and $K_4[Fe(CN)_6]$ solution, gives deep blue solution or precipitate.

  $Fe(OH)_3 + 3HCl \rightarrow FeCl_3 + 3H_2O$

  $4FeCl_3 + 3K_4[Fe(CN)_6] \rightarrow \underset{\text{Prussian blue}}{Fe_4[Fe(CN)_6]_3} + 12KCl$

- Addition of potassium thiocyanate solution gives a blood red colouration.

  $FeCl_3 + 3KCNS \rightarrow \underset{\text{blood red colour}}{Fe(CNS)_3} + 3KCl$

- ***Al^{3+} (aluminium)*** : The gelatinous precipitate of $Al(OH)_3$ on treatment with NaOH forms soluble $NaAlO_2$.

  $Al(OH)_3 + NaOH \rightarrow \underset{\substack{\text{sodium}\\\text{meta-aluminate}}}{NaAlO_2} + 2H_2O$

- Sodium meta aluminate on boiling with ammonium chloride gives $Al(OH)_3$ ppt.
- Zn^{2+} and Mn^{2+} are present in group IV and the reagent is H_2S in presence of NH_4OH.
- The radicals are obtained as their sulphides.
- ***Zn^{2+} (zinc) :*** The sulphide on treatment with HCl gives chloride, which gives a white precipitate with NaOH, which dissolves in excess of NaOH.

  $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$

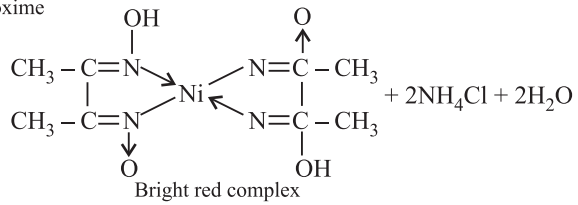
  $ZnCl_2 + 2NaOH \rightarrow Zn(OH)_2 + 2NaCl$

  $Zn(OH)_2 + 2NaOH \rightarrow \underset{\text{(soluble)}}{Na_2ZnO_2} + 2H_2O$

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- ***Ni^{2+} (Nickel)*** : Dimethyl glyoxime test

$$NiCl_2 + 2NH_4OH + 2 \text{ (Dimethyl glyoxime)} \longrightarrow \text{Bright red complex (ppt)} + 2NH_4Cl + 2H_2O$$



- Sodium hydroxide-bromine water test

$$NiCl_2 + 2NaOH \rightarrow 2NaCl + Ni(OH)_2\downarrow$$
(green ppt.)

$$Br_2 + H_2O \rightarrow HBr + O$$
$$2Ni(OH)_2 + H_2O + O \rightarrow 2Ni(OH)_3\downarrow$$
(black ppt)

- ***Ba^{2+} (barium)*** : The acetate on treatment with potassium chromate solution gives yellow precipitate of barium chromate.

$$Ba(CH_3COO)_2 + K_2CrO_4 \rightarrow BaCrO_4\downarrow + 2CH_3COOK$$

- The yellow ppt. of $BaCrO_4$ is dissolved in concentrated HCl.
- ***Ca^{2+} (calcium)*** : The acetate on treatment with ammonium oxalate gives a white ppt. of calcium oxalate.

$$Ca(CH_3COO)_2 + (NH_4)_2C_2O_4 \rightarrow CaC_2O_4\downarrow + 2CH_3COONH_4$$

- The white ppt. is dissolved in dilute H_2SO_4 and a drop of $KMnO_4$ solution is added which immediately decolorises.
- ***Mg^{2+} (magnesium)*** : This is a member of group VI and the reagent is disodium hydrogen phosphate.
- The salts give a white precipitate of magnesium ammonium phosphate when disodium hydrogen phosphate is added to ammoniacal solution of Mg^{2+}.

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Colour		**ODOUR**	
(A) Solids	Compounds	Mousy	acetamide, acetonitrile
Yellow	iodoform, nitro compounds and quinones	Fruity	esters
		Penetrating smell	HCHO, CH_3CHO and HCOOH
Orange	*o*-nitroaniline		
Brown-red	azo compounds, diamines, aromatic amines, amino-phenol	Pleasant	ketones (aliphatic and aromatic)
		Smell of bitter almonds	C_6H_5CHO, nitrobenzene, nitrotoluene
Pink	naphthols		
Colourless	simple phenols, carbohydrates	Vinegar smell	CH_3COOH
		Garlic smell	thiophenol, thioalcohol
		Wine like	alcohol

(B) Liquids		Fishy smell	aliphatic and aromatic amines
Brown-red	amines	Carbolic smell	phenols, cresols, naphthols
Yellow-orange	nitrocompounds, diketones	Ammonical smell	tertiary amines
Colourless	alcohols, aldehydes, ketones, lower aliphatic acid and their anhydrides	Sweet smell	chloroform
		Oil of winter green	methyl salicylate
		Characteristic aromatic smell	benzene, toluene

- ***Detection of nitrogen, sulphur and halogens***
- Nitrogen, sulphur and halogens in any organic compounds are detected by 'Lassaigne's test'.
- ***Preparation of Lassaigne's Extract (or Sodium Extract)***
- A small piece of sodium is heated gently in an ignition tube till the sodium melts. About 50 - 60 mg of the organic compound is added to this and the tube heated strongly for 2-3 minutes to fuse the material inside it. After cooling, the tube is carefully broken in a china dish containing about 20 to 30 mL of distilled water. The fused material along with the pieces of ignition tube is crushed with the help of a glass rod and the contents of the china dish are boiled for a few minutes. The sodium salts formed in the above reactions (*i.e.* NaCN, Na_2S, NaX or NaSCN) dissolve. Excess of sodium reacts with water to give sodium hydroxide. This alkaline solution is called Lassaigne's extract or sodium extract. The solution is then filtered to remove the insoluble materials and the filtrate is used for making the tests for nitrogen, sulphur and halogens.
- ***Reactions***
- An organic compound containing C, H, N, S and halogens when fused with sodium metal gives the following reactions.

$$\underset{\text{present in organic compound}}{C + N + Na} \xrightarrow{\text{Fusion}} \underset{\text{sodium cyanide}}{NaCN}$$

$$X(\text{Cl, Br, I}) + Na \xrightarrow{\text{Fusion}} NaX\ (X = \text{Cl, Br, I})$$

$$S + 2Na \xrightarrow{\text{Fusion}} Na_2S$$

(NaSCN) is formed during fusion, which in the presence of excess sodium forms sodium cyanide and sodium sulphide.

$$Na + C + N + S \xrightarrow{\text{Fusion}} NaCNS$$

$$NaCNS + 2Na \xrightarrow{\text{Fusion}} NaCN + Na_2S$$

Element	Sodium extract	Confirmatory test
Nitrogen	$Na + C + N \xrightarrow{\Delta} NaCN$	$(NaCN + FeSO_4 + NaOH) + FeCl_3 + \text{conc.HCl} \xrightarrow[\text{boil and cool}]{} \text{Blue or green colour.}$
Sulphur	$2Na + S \xrightarrow{\Delta} Na_2S$	(i) Na_2S + sodium nitroprusside $\longrightarrow$ A deep violet colour. (ii) $Na_2S + CH_3COOH + (CH_3COO)_2Pb \longrightarrow$ A black ppt.

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Halogen	$Na + X \xrightarrow{\Delta} NaX$	$NaX + HNO_3 + AgNO_3$ (i) White ppt. soluble in aq. NH_3 confirms Cl. (ii) Yellow ppt. partially soluble in aq. NH_3 confirms Br. (iii) Yellow ppt. insoluble in aq. NH_3 confirms I.
Nitrogen and sulphur together	$Na + C + N + S \xrightarrow{\Delta} NaCNS$	As in test for nitrogen ; instead of green or blue colour, blood red colouration confirms presence of N and S both.

Detection of organic functional groups

- **Alcoholic group** (– OH) (linked to aliphatic carbon chain)
  - *Sodium metal test*

$$2R - OH + 2Na \longrightarrow 2R - ONa + H_2 \uparrow$$

  - *Ester test*

$$R - OH + R' - COOH \overset{H^+}{\rightleftharpoons} R' - COOR + H_2O$$
ester

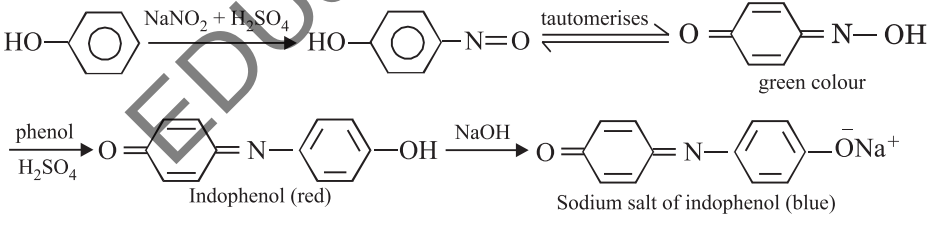
  - $2R - OH + (NH_4)_2Ce(NO_3)_6 \longrightarrow (ROH)_2Ce(NO_3)_4 + 2NH_4NO_3$
ceric ammonium nitrate — pink or red
  - *Acetyl chloride test*

$$R - OH + CH_3COCl \longrightarrow CH_3COOR + HCl_{(g)}$$

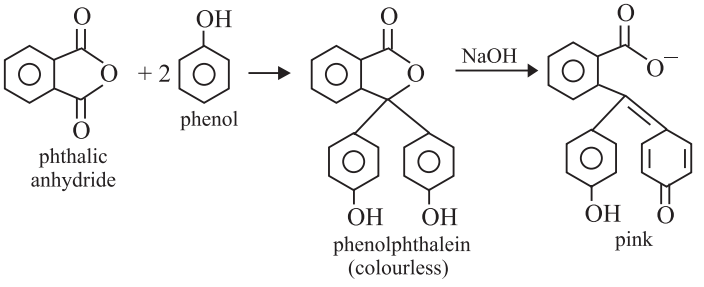
$$HCl + NH_3 \longrightarrow NH_4Cl$$
(white fumes)

Test for phenolic (– OH) group

- **Liebermann's test**



- **Phthalein test**

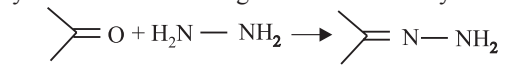


This test is also called fluorescein test.

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- **Tests for aldehyde group**

  The presence of a carbonyl group can be confirmed by treating the organic compound with hydrazine and observing the formation of hydrazones.



  Aldehyde/Ketone Hydrazone

  To identify aldehydic group, the following tests are performed:

  - *Tollen's test :* To about 5-10 ml of Tollen's reagent (ammoniacal $AgNO_3$), a small quantity of organic compound is added and it is heated on a water bath. A shining silver mirror or grey deposit on the inner wall of the test tube indicates the presence of – CHO group.

$$R - CHO + 2[Ag(NH_3)_2]OH + H_2O \longrightarrow RCOONH_4 + 3NH_3 + H_2O + 2Ag\downarrow$$

(silver mirror)

    Formic acid and α-hydroxy ketones also give the test.
  - *Fehling's test* : A small amount of the organic compound is boiled with some Fehling solution (alkaline solution of cupric ions complexed with sodium potassium tartarate), it gives red precipitate of Cu_2O.

    Salicylaldehyde does not reduce Fehling's solution.
  - *Benedict's test* : To 4-5 ml of Benedict's reagent (cupric ion complexed with citrate ion) a small quantity of the organic compound is added and the solution is heated to boiling. Formation of red precipitate indicates the presence of –CHO group.
  - *Schiff's test* : 5 ml of Schiff's reagent is taken in a test tube and shaken with organic compound (without heating). A pink colour is formed within two minutes.

- **Tests for ketones**

  Ketones do not respond to Fehling's, Tollen's and Benedict's tests. However, the following tests can be used to confirm the presence of a keto group:

  - *Iodoform test* : Ketones with $CH_3CO–$ group react with I_2 in alkali to give yellow precipitate of CHI_3. Carboxylic acid, its derivatives and active methylene compounds (except β-keto acids) do not respond to this test.

$$CH_3-\overset{\overset{O}{\|}}{C}-CH_3 + I_2 \xrightarrow{OH^-} \underset{\text{yellow}}{CHI_3\downarrow} + CH_3COO^-$$

  - *Nitroprusside test* : 1 ml of the organic compound is treated with 1 ml of freshly prepared solution of sodium nitroprusside followed by addition of excess of NaOH solution. A wine-red colour is obtained.

- **Tests for carboxylic group**

  Aliphatic acids are soluble in cold water and aromatic acids are soluble in hot water. Dicarboxylic acids, phenolic acids are more soluble than simple carboxylic acids.

  - *Litmus test* : A small amount of organic compound or its aqueous solution is added to a blue litmus paper. If the paper turns red, the acidic carboxylic group may present.
  - *Sodium bicarbonate test* : A small quantity of the organic compound is added to an aqueous solution of sodium bicarbonate solution. CO_2 effervescence confirms the presence of –COOH (picric acid, 2,4,6-trinitrophenol also gives a positive test).

- **Tests for primary amines**

  Amines are basic in nature, soluble in water and dilute HCl but insoluble in NaOH or Na_2CO_3.

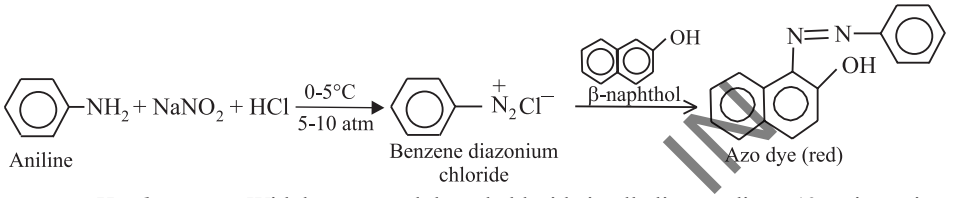
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- *Carbylamine test* : The organic compound is heated with alc. KOH and $CHCl_3$ in a test tube. A highly offensive smell is evolved due to the formation of isocyanides.

$$RNH_2 + CHCl_3 + 3KOH \longrightarrow \underset{\text{foul smell}}{R-N\equiv C} + 3KCl + 3H_2O$$

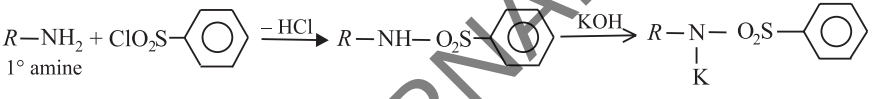
Aromatic amines like $C_6H_5 - NH_2$ also give this test.

- *Azo dye test* : This test is applicable for aromatic amines. The test involves the additioin of a small amount of the organic compound in dil. HCl and $NaNO_2$ (at 0-5°C) and alkaline β-naphthol (at 0-5°C) with constant shaking, a red dye is obtained.



Aniline — Benzene diazonium chloride — Azo dye (red)

- *Hinsberg test* : With benzene sulphonyl chloride in alkaline medium, 1° amines give an alkali soluble product.



1° amine

Test for Functional Groups

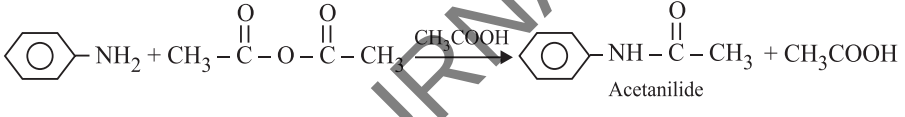
No.	Experiment	Observation	Inference
1.	O.C. + 3 cc saturated soln. of $NaHCO_3$	effervescences of CO_2 which changes lime water milky	–COOH (carboxylic)
2.	5 cc O.C. + 2 – 3 drops of ceric ammonium nitrate	a red colour	–OH (alcoholic)
3.	2 cc aq. or alc. soln. of O.C. + 1 – 2 drops neutral $FeCl_3$ soln	blue violet, red or deep green colour	–OH (phenolic)
4.	1 cc Schiff's reagent + 2 – 3 drops O.C. and shake	violet or red colour	–CHO (aldehydic)
5.	1 – 2 cc of sodium nitroprusside + 1 – 2 drops O.C. + NaOH	red or violet colour	> C = O (ketonic)
6.	2 cc aq. soln. of O.C. + 2 drops Molisch reagent + pour it in another test tube containing 1 – 2 c.c. conc. H_2SO_4	formation of red ring at the junction	Carbohydrate
7.	O.C. + 2 cc conc. H_2SO_4 & shake	insoluble or immiscible	Hydrocarbon
8.	0.5 g O.C. in 2 cc alcohol + 1 drop NaOH + 1 drop phenolphthalein	disappearance of pink colour	–COOR (ester)

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9.	0.3 g O.C. + 5 cc H_2O + 1 cc acetone + few drops sodium nitroprusside	formation of violet red colour	$-NH_2$ (amino)
10.	0.2 g O.C. + 1 c.c. NaOH + heat	smell of NH_3, red litmus changes to blue	$-CONH_2$ (amide)
11.	O.C. + 2 c.c alcoholic $AgNO_3$ + heat	(a) ppt. formed (b) no ppt.	aliphatic halogen aromatic halogen (means halogen attached to benzene nucleus)
12.	O.C. + caustic alkali (1 : 1) + dilute HCl	penetrating smell of SO_2 which on passing into acidic $K_2Cr_2O_7$ soln. produces green colour	$-SO_3H$ (sulphonic acid)

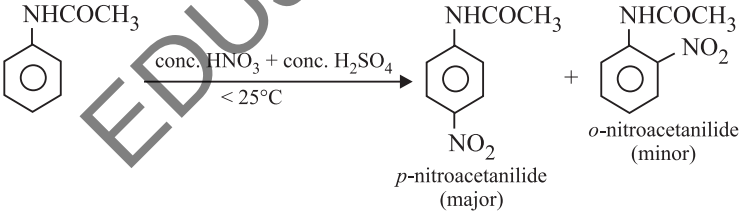
Chemistry involved in the preparation of some organic compounds

- **Acetanilide :**





- ***p*-Nitroacetanilide :**



- **Iodoform :**

Compounds containing $CH_3 - \underset{\displaystyle OH}{\underset{|}{CH}}$ group or CH_3CO- group can form iodoform on reaction with sodium hypoiodide. *e.g.* ethanol, acetaldehyde, acetone, etc.

$$CH_3CH_2OH \xrightarrow[\text{or NaOI}]{\text{KOI}} CH_3CHO \quad \text{(oxidation)}$$

$$CH_3CHO \xrightarrow[\text{or NaOI}]{\text{KOI}} CI_3CHO \quad \text{(iodination)}$$

$$CI_3CHO + NaOH \longrightarrow \underset{\text{Iodoform}}{CHI_3} + HCOONa \text{ (hydrolysis)}$$

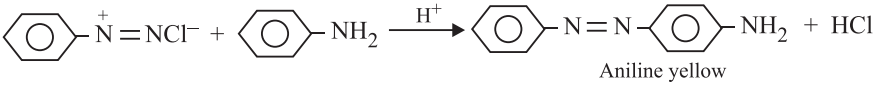
With acetone no initial oxidation takes place.

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$$CH_3COCH_3 \xrightarrow[\text{(iodination)}]{NaOI} CI_3COCH_3$$

$$CI_3COCH_3 + NaOH \xrightarrow[\text{(hydrolysis)}]{} CHI_3 + CH_3COONa$$

- **Aniline yellow :**



Aniline yellow

PHYSICAL

Volumetric analysis

- Volumetric analysis is a process by which the concentration or strength of a chemical substance is measured by measuring the volume of its solution taking part in a given chemical reaction. The main process of this analysis is called titration.

Titration

- Determination of strength of one solution using another solution of known strength under volumetric conditions is known as titration.

Some important terms

(i) Analyte : The substance being analyzed is known as analyte or titre.

(ii) Titrant : The substance added to the analyte in a titration is known as titrant.

(iii) Equivalence point : It is the point where reaction between two solutions is just complete or the point in a titration at which the quantity of titrant is exactly sufficient for stoichiometric reaction to be complete with the analyte.
At this point there is a sudden change in a physical property, such as indicator colour, pH, conductivity, or absorbance. It is also known as end point.

(iv) Indicator : A compound having a physical property (usually colour) that changes abruptly near the equivalence point of a chemical reaction is known as indicator. It indicates the attainment of end point.

(v) Standard solution : A solution whose concentration is known is called standard solution.

(vi) Standardization : It is the process in which concentration of a reagent is determined by reaction with a known quantity of second reagent whose concentration is known.

(vii) Primary standard substance : A reagent that is pure enough so that its standard solution can be prepared directly by dissolving a definite weight of it in a definite volume of solvent is known as primary standard, *e.g.*, crystalline oxalic acid, anhydrous Na_2CO_3, Mohr's salt, etc.

(viii) Secondary standard substance : The substance or reagent whose standard solution can not be prepared directly is called secondary standard, *e.g.* $KMnO_4$, NaOH, KOH, etc.

Number of equivalents = Normality × Volume (L)

Number of equivalents of titre = Number of equivalents of titrant

$$\boxed{N_1V_1 = N_2V_2}$$

Where N_1 = Normality of titre, V_1 = Volume of titre

N_2 = Normality of titrant, V_2 = Volume of titrant

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If volume is taken in ml

Then, Number of milliequivalents (m.eq.)

= Normality × Volume (in ml)

then also, $\boxed{N_1V_1 = N_2V_2}$

The above equation is known as normality equation.

Similarly molarity equation is also given but it is usually applicable for dilution of a solution.

$M_1V_1 = M_2V_2$

Normality = Molarity × n, where n = valency factor

Thus $N_1V_1 = N_2V_2$ can be written as

or $M_1V_1n_1 = M_2V_2n_2$

or $\boxed{\frac{M_1V_1}{M_2V_2} = \frac{n_2}{n_1}}$

Redox titrations

- Redox titrations involving $KMnO_4$ as oxidising agent are called *permanganometric titrations*. In these titrations reducing agents like Mohr's salt, $(NH_4)_2SO_4$. $FeSO_4 \cdot 6H_2O$, $FeSO_4$, H_2O_2, oxalic acid and oxalates are directly titrated against $KMnO_4$ as oxidising agent in acidic medium.

Indicator

- In these titrations, $KMnO_4$ acts as self indicator. In acidic medium, $KMnO_4$ reacts with reducing agent (like oxalic acid or Mohr's salt), when whole of the reducing agent has been oxidised the remaining $KMnO_4$ is not decomposed and imparts pink colour to the solution and thus acts as an indicator.

End point

- In $KMnO_4$ titration end point is from colourless to permanent light pink colour.

Titration of oxalic acid *vs* $KMnO_4$

- ***Indicator***
  $KMnO_4$ is a self indicator.
- ***End point***
  Colourless to permanent pink colour ($KMnO_4$ in burette).
- ***Chemistry of experiment***

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5[O]$$

$$\left[\begin{matrix}COOH\\|\\COOH_{(aq)}\end{matrix} \cdot 2H_2O + [O] \xrightarrow{60\text{-}70^\circ C} 2CO_2 + 3H_2O\right] \times 5$$

$$2KMnO_4 + 3H_2SO_4 + 5\begin{matrix}COOH\\|\\COOH\end{matrix} \cdot 2H_2O \longrightarrow K_2SO_4 + 2MnSO_4 + 18H_2O + 10CO_2$$

or $[MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O] \times 2$

$[C_2O_4^{2-} \longrightarrow 2CO_2 + 2e^-] \times 5$

$$2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \longrightarrow 2Mn^{2+}\ 8H_2O + 10CO_2$$

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It is clear from the above reactions that two moles of $KMnO_4$ react with five moles of oxalic acid.

- $KMnO_4$ accepts five electrons and gets reduced from MnO_4^- to Mn^{2+} whereas oxalic acid releases two electrons and gets oxidised from $H_2C_2O_4.2H_2O$ to CO_2.
- Oxalic acid solution is heated to 60-70°C before titrating with $KMnO_4$ because in cold, reaction is very slow due to slow formation of Mn^{2+}. When the solution is heated, liberation of Mn^{2+} speeds up which autocatalyses the reaction and therefore reaction proceeds rapidly. Heating of oxalic acid solution also expels the CO_2 evolved during the reaction which otherwise does not allow the reaction to go to completion.

- ***Autocatalysis***
  It is the process in which one of the reaction product catalyses the further reaction of the reactants.
- ***Calculations***
  We can apply normality equation to this titration as

$$\underset{\text{(Oxalic acid)}}{N_1V_1} = \underset{(KMnO_4)}{N_2V_2}$$

  Volume of both the solutions are known in the experiment. By knowing normality of one solution, normality of other solution can be calculated.
  We can also apply molarity equation to this titration. Since two moles of $KMnO_4$ react with 5 moles of oxalic acid

$$\frac{M_{KMnO_4} \times V_{KMnO_4}}{M_{\text{Oxalic acid}} \times V_{\text{Oxalic acid}}} = \frac{2}{5}$$

  By knowing the molarity of one solution that of the other solution can be calculated.
  Normality and molarity of a solution are related as
  Normality = Molarity × number of electrons gained or lost

– Strength of any solution can be calculated as
  Strength = Normality × Equivalent mass
  or Strength(g/L) = Molarity × Molecular mass
  Equivalent mass of oxalic acid

$$= \frac{\text{Molecular mass}}{2} = \frac{126}{2} = 63$$

  Equivalent mass of $KMnO_4$

$$= \frac{\text{Molecular mass}}{5} = \frac{158}{5} = 31.6$$

– Percentage purity of a given salt can also be calculated

$$\text{Percentage purity} = \frac{\text{Strength of pure sample}}{\text{Strength of given sample}} \times 100$$

Titration of Mohr's salt *vs* $KMnO_4$

- ***Indicator***
  $KMnO_4$ is a self indicator.
- ***End point***
  Colourless to permanent light pink ($KMnO_4$ in burette).

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- ***Chemistry of experiment***

$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5[O]$

$[2FeSO_4.(NH_4)2SO_4.6H_2O + H_2SO_4 + [O] \longrightarrow Fe_2(SO_4)_3 + 2(NH_4)2SO_4 + 13H_2O] \times 5$

$2KMnO_4 + 8H_2SO_4 + 10FeSO_4.(NH_4)2SO_4.6H_2O \longrightarrow K_2SO_4 + 2MnSO_4 + 5Fe_2(SO_4)_3 + 10(NH_4)_2SO_4 + 68H_2O$

or $MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

$[Fe^{2+} \longrightarrow Fe^{3+} + e^-] \times 5$

$MnO_4^- + 8H^+ + 5Fe^{2+} \longrightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$

It is clear from the above reactions that one mole of $KMnO_4$ reacts with five moles of Mohr's salt.

$KMnO_4$ accepts five electrons and reduces from MnO_4^- to Mn^{2+} whereas in Mohr's salt one electron is released so that Fe^{2+} is oxidised to Fe^{3+}.

- ***Calculations***

According to normality equation

$$\underset{\text{(Mohr's salt)}}{N_1V_1} = \underset{(KMnO_4)}{N_2V_2}$$

Volume of both solutions are known in the experiment. By knowing normality of one solution, that of other solution can be calculated.

Molarity equation can also be applied to this titration. Since one mole of $KMnO_4$ reacts with five moles of Mohr's salt, $\frac{M_{KMnO_4} \times V_{KMnO_4}}{M_{\text{Mohr's salt}} \times V_{\text{Mohr's salt}}} = \frac{1}{5}$

where M_{KMnO_4} = Molarity of $KMnO_4$ solution, V_{KMnO_4} = Volume of $KMnO_4$ solution

$M_{\text{Mohr's salt}}$ = Molarity of Mohr's salt, $V_{\text{Mohr's salt}}$ = Volume of Mohr's salt

By knowing the normality of one solution, that of other solution can be calculated.

– Strength of a solution can be calculated as

Strength(g/L) = Normality × Equivalent mass

or Strength(g/L) = Molarity × Molecular mass

Eq. mass of Mohr's salt $= \frac{\text{Molecular mass}}{1} = 392$

Eq. mass of $KMnO_4$ $= \frac{158}{5} = 31.6$

– Percentage purity of a given salt can also be calculated

$$\text{Percentage purity} = \frac{\text{Strength of pure sample}}{\text{Strength of given sample}} \times 100$$

Acid-Base titrations

In acid-base titration the amount of an acid or base is determined by titrating it against a standard solution of base or acid respectively.

Acid-base titration involves neutralization reaction.

$\underset{\text{(from acid)}}{H^+} + \underset{\text{(from base)}}{OH^-} \longrightarrow \underset{\text{(water)}}{H_2O}$

In acid base titration there is a sudden change in pH at the end point.

The point at which there is sudden change in pH with addition of very small amount of the titrant to the titrate (titre) is called point of inflection.

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Indicator

Acid-base indicators are generally complex organic molecules which are either weak acids or weak bases, *e.g.* phenolphthalein is a weak organic acid (represented as HPh) and methyl orange is a weak organic base (represented as MeOH). These indicators dissociate in aqueous solution such that the unionised indicator and its conjugate part (*i.e.* either conjugate acid or conjugate base) have different colours.

$$\underset{\text{(Colourless)}}{HPh} \rightleftharpoons \underset{\text{(Colourless)}}{H^+} + \underset{\text{(Pink)}}{Ph^-} \qquad \underset{\text{(Yellow)}}{MeOH} \rightleftharpoons \underset{\text{(Red)}}{Me^+} + \underset{\text{(Colourless)}}{OH^-}$$

The choice of an indicator for a particular acid-base titration should be made in such a way that indicator used shows change in colour in the same pH range as developed around the equivalence point.

To show the colour change by an indicator, $\boxed{pK \text{ indicator} = \text{pH at equivalence point}}$

– For strong acid and strong base titration, methyl orange, thymol blue or phenolphthalein can be used.
– For strong acid and weak base titration, methyl orange or methyl red can be used as an indicator.
– For weak acid and strong base titration, phenolphthalein is best suited indicator.

Some common acid-base indicators

Indicator colour change, from acidic to alkaline medium	**pK(ind)**	**pH range**	**Example of titration**
Methyl orange (red ⇒ yellow)	3.7	3.1 – 4.4	Weak base *vs* strong acid titration *e.g.* Ammonia titrated with hydrochloric acid
Bromocresol green (yellow ⇒ blue)	4.0	3.8 – 4.6	Weak base *vs* strong acid titration
Methyl red (red ⇒ yellow)	5.1	4.2 – 6.3	Weak base *vs* strong acid titration
Bromothymol blue (yellow ⇒ blue)	7.0	6.0 – 7.6	Strong acid *vs* strong base titration *e.g.* Hydrochloric acid with sodium hydroxide
Phenol red (yellow ⇒ red)	7.9	6.4 – 8.2	Strong acid *vs* strong base titration *e.g.* Hydrochloric acid with sodium hydroxide
Thymol blue (basic form), (yellow ⇒ blue)	8.9	8.0 – 9.6	Weak/strong acid *vs* strong base titration
Phenolphthalein (colourless ⇒ pink)	9.3	8.3 – 10.0	Weak acid vs strong base titration *e.g.* Ethanoic acid titrated with sodium hydroxide
Alizarin yellow (yellow ⇒ violet)	–	10.1 – 12.0	Weak acid *vs* strong base titration

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End